Raimondo Schettini · Alain Trémeau ·
Shoji Tominaga · Simone Bianco ·
Marco Buzzelli (Eds.)

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Computational Color Imaging

8th International Workshop, CCIW 2024
Milan, Italy, September 25–27, 2024
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Preface

The Computational Color Imaging Workshop (CCIW) has been at the forefront of color imaging research and innovation for the past 15 years. With a rich history of fostering collaboration and showcasing cutting-edge advancements in the field, CCIW has become a renowned platform for color imaging enthusiasts. Building on the successes of previous editions, we welcome you to the proceedings of CCIW 2024, held in Milan from 25th to 27th September 2024.

The 2024 edition of CCIW was proudly organized by the University of Milano-Bicocca, Italy, in cooperation with the Norwegian University of Science and Technology, Norway, Nagano University, Japan, and Université Jean Monnet, Saint-Étienne, France, with the endorsement of esteemed organizations such as the International Association for Pattern Recognition (IAPR), the Italian Association for Research in Computer Vision, Pattern Recognition, and Machine Learning (CVPL), Gruppo del Colore from Associazione Italiana Colore, and Centro Ricerche Patrimonio Storico Artistico e Culturale from University of Milano-Bicocca (BIPAC).

The primary objective of CCIW 2024 was to explore the integration of advanced Artificial Intellifence (AI) algorithms and deep learning models in color imaging, pushing the boundaries of what is possible in this dynamic field. To this extent, we welcomed contributions addressing a wide range of topics, including color image processing, analysis, and applications. We adopted a rigorous single-blind peer review process, ensuring that each paper received thorough evaluations from at least three reviewers. We are therefore pleased to present 18 high-quality papers that were accepted for presentation at the workshop out of 21 submissions, plus two invited papers and one invited extended abstract that accompany corresponding keynote talks. These papers were selected based on their scientific merit, originality, and relevance to the goals of CCIW 2024.

The program of CCIW 2024 was structured to provide a rich and diverse set of presentations, tutorials, and keynotes, reflecting the broad scope and latest trends in computational color imaging. The workshop commenced with a series of spotlight sessions and tutorials on the first day, followed by paper presentations and keynotes on the subsequent days.

The spotlights included:

- Simone Zini, from University of Milano-Bicocca: “Brightening the dark: Advances in Low-Light Image Enhancement and Night Photography Rendering” [1].
- Luigi Celona, from University of Milano-Bicocca: “Modern Approaches to Explainable Image Aesthetic Assessment” [2].
- Marco Buzzelli, from University of Milano-Bicocca: “Analysis of automatic white balance datasets and methods” [3].
- Flavio Piccoli, from University of Milano-Bicocca: “Personalizing White-Box Image Enhancement: Deep Learning and User-Centric Innovations” [4].
- Philippe Colantoni, from Université Jean Monnet: “The impact of lightness, color and geometry of illumination in Human Pose Estimation for Performing Arts” [5].

The tutorials included:

- Marcello Picollo from the Institute of Applied Physics "Nello Carrara" led a tutorial on "Color Measurements on Paintings: Methodologies and Issues".
- Miguel Ángel Martínez Domingo from the University of Granada conducted a session on "Advanced digital imaging techniques applied to cultural heritage".

We are honored to have had distinguished researchers delivering keynote speeches on various cutting-edge topics:

- Ján Morovič from HP Inc. discussed "From Gutenberg to Llamas: print optimization through first principles and AI".
- Peter Morovič from HP Inc. presented on "Atomic color: from points to probability distributions".
- Marcello Picollo from the Institute of Applied Physics "Nello Carrara" covered "VNIR and SWIR hyperspectral imaging for the study of Picasso paintings".
- Miguel Ángel Martínez Domingo from the University of Granada talked about "Spectral imaging for the study of artworks and historical documents."
- Giuseppe Claudio Guarnera from the University of York discussed "Advancements in Practical Facial Skin Reflectance Measurement and Realistic Age Transformation".

The papers and keynotes presented on the second and third day of the workshop are organized here in six broad categories:

Color Theory and Spaces. Papers in this category delve into the fundamental aspects of color formation, difference, and spaces, providing new insights and methodologies for understanding and utilizing color in various contexts. The keynote by Morovič et al. [6] offered a deep dive into the probabilistic nature of color formation and its implications for color science. Daneshvar et al. [7] examined the color properties of Planckian radiators and extended the Wien and Planck loci for negative temperatures. Basova et al. [8] developed a uniform color space that maintains perceptual uniformity and advanced hue linearity. Becatti et al. [9] explored the perceived color differences in various contexts and their implications for colorimetry.

Image Enhancement and Retinex Theory. This category features advancements in image enhancement techniques and applications of the Retinex theory, focusing on improving image quality and processing under varying lighting conditions. Lecca [10] exploited the Milano Retinex contrast to enhance images with strong changes of light intensity. Ulucan et al. [11] presented a method for intrinsic image decomposition based on Retinex theory, superpixel segmentation, and scale-space computations, improving reflectance estimation in images.

Hyperspectral and Spectral Imaging. Contributions here explore the latest hyperspectral and spectral imaging technologies, including applications in art, medical imaging, and material analysis, highlighting the versatility and depth of spectral imaging methods. The keynote by Picollo et al. [12] applied VNIR and SWIR hyperspectral imaging techniques for the study of Picasso paintings, revealing insights into the artist's techniques and materials. The keynote by Martínez Domingo explored spectral imaging for the study of artworks and historical documents. Vernet et al. [13] focused on stabilizing the spectral power distribution of a tunable multichannel LED lighting system,

improving color accuracy and consistency in various applications. Raza et al. [14] utilized low-light hyperspectral imaging methods using HDR techniques to capture detailed images in challenging lighting conditions. Tominaga et al. [15] presented spectral imaging methods for estimating fluorescence emission spectra from plant grains and leaves, enhancing agricultural analysis. Fsian et al. [16] demonstrated innovative stitching techniques for spectral filter array video sequences, enabling high-resolution spectral imaging from multiple frames. The keynote by Guarnera advanced practical facial skin reflectance measurement and realistic age transformation, improving the accuracy and realism of age progression models.

Printing and Machine Learning Applications. Papers in this section discuss innovations in printing technology and the application of machine learning to optimize color processes, enhance image quality, and achieve new levels of accuracy in color reproduction. Richard [17] utilized Pascal Cellular Automata for dithering, introducing a novel approach to halftoning and improving print quality. The keynote by Morovič et al. [18] explored print optimization through first principles and AI, bridging traditional printing techniques with advanced artificial intelligence methods. Díaz Estrada et al. [19] employed an instance-based CycleGAN for object segmentation with few annotations, advancing the field of segmentation in plant phenotyping. Ojeleye et al. [20] demonstrated the robustness of 2D pose estimation under different color illuminations with their ChromaPose technique, enhancing pose estimation accuracy in varying lighting conditions.

Perceptual Quality, Medical Imaging, and Art Applications. This diverse category covers research on perceptual quality assessment, medical imaging techniques, and applications in art conservation and authentication, demonstrating the broad impact of color imaging technologies. Tanaka et al. [21] investigated image features for perceptually equivalent gloss reproduction through a comparison of real objects and images, enhancing the accuracy of gloss reproduction in digital imaging. Bianconi et al. [22] explored the classification of lung nodules on CT via pseudo-colour images and deep features from pre-trained convolutional networks, improving diagnostic accuracy in medical imaging. Bianco et al. [23] conducted an exploratory study on painter verification using color palettes, applying color-based features to authenticate works of art.

Color Constancy. Papers in this category address the challenges of achieving color constancy in imaging, presenting new algorithms and models to maintain consistent color perception across different lighting environments. Cotogni et al. [24] developed illuminant equivariant networks for computational color constancy, enhancing the robustness of color constancy algorithms to varying lighting conditions. Ulucan et al. [25] revisited color constancy using CNNs, incorporating recent observations to improve the accuracy of color constancy models. Buzzelli et al. [26] proposed methods for enforcing temporal consistency for color constancy in video sequences, ensuring stable color perception over time in dynamic scenes.

We would like to express our gratitude to the many individuals and organizations who made CCIW 2024 possible. Special thanks go to our endorsers for their invaluable support. We are also grateful to the reviewers for their diligent work in evaluating the submissions, and to the authors for their high-quality contributions.

We are particularly excited to have included the Best Student Award, supported by the Italian Association for Research in Computer Vision, Pattern Recognition, and Machine Learning (CVPL), to recognize outstanding contributions from emerging scholars in the field.

In closing, we hope that CCIW 2024 proved to be a stimulating and rewarding experience for all participants, fostering new collaborations and inspiring future research in computational color imaging.

September 2024

Raimondo Schettini
Simone Bianco
Marco Buzzelli
Shoji Tominaga
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VNIR and SWIR Hyperspectral Imaging for the Study of Picasso Paintings (Abstract of Invited Talk)

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Abstract. In this communication, some applications of the hyperspectral imaging instrumentation created at IFAC-CNR will be presented for the study of paintings by Pablo Picasso belonging to the permanent collection of the Picasso Museum in Barcelona (Spain).

Keywords: Hyperspectral Imaging technique, Pablo Picasso, Data Processing

Introduction

Over the past decade, hyperspectral imaging methods, which belong to the broader group of spectroscopic imaging techniques, have been increasingly popular in museums and conservation laboratories for documentation purposes and to support art historical studies [1, 2]. In addition, this technique is increasingly being applied to obtain detailed information about the constituent materials of the works being analyzed, their state of preservation, and as support of conservation treatments on different types of artworks and archaeological objects. In recent years, numerous studies have been devoted to tailoring HSI instrumentation and methodologies to the specific needs of art conservation and applications in the museum context (Fig. 1). As measurement campaigns have progressed, issues have also arisen, such as the need to develop new image processing software, including image-cube mosaicking, new methods of pigment identification and mapping, and visualization and archiving of results [3–5].

Reflectance hyperspectral imaging (HSI) operates in the ranges of Visible and Near IR (VNIR, 400–1000 nm) and shortwave IR (SWIR, 1000–2500 nm). Since it is now recognized as one of the most advanced techniques for noninvasive examination of paintings (naturally assisted by the other imaging techniques and spots), the trend is to employ it for more systematic use to achieve simultaneous analysis and documentation of the surfaces and inner layers of paintings.

This communication reports on HSI data obtained on some of Picasso's early twentieth century paintings, currently in the permanent collection of the Museu Picasso in Barcelona. The study was conducted as part of two Spanish inter-institutional research projects, PROMESA and MIMO, aimed at understanding the failure mechanisms of modern and contemporary paintings.

Fig. 1. IFAC-CNR hyperspectral imaging scanner prototype during HSI VNIR data acquisition on the painting Woman in an armchair (1917) by Picasso.

The first case study concerns the painting "The Blue Glass" (La Copa Blava, 1903) from Pablo Picasso's Blue Period. This study was carried out as part of the interdisciplinary program "Picasso: Blue Project", which resulted in an exhibition held in 2022 at the Museu Picasso in Barcelona [6]. For this event, a selection of paintings from the Blue Period (1901–1904) collection was revisited through the eye of cutting-edge analytical techniques, including HSI. Beneath the surface, this oil on canvas painting presents a layering of different compositions that bear 'didactic' witness to the artist's modus operandi. The presence of another painting beneath the surface had been known for some time (thanks to X-ray radiography), but the HSI data allowed an unprecedented visualization of the artistic metamorphosis of this painting. Picasso's well-known creative practice of repainting on his own canvas was brought to light thanks to the HSI images, which provided unique material for visualizing the hidden layers of the painting, revealing later and yet unknown earlier versions, as well as the painting's artistic evolution.

As a second case study, analyses of some paintings by Picasso from 1917, also belonging to the Museu Picasso in Barcelona, are reported. The first part of this study focused on two works of art, and HSI data were used to document the artistic technique

and to define the chemical composition of the palette [7]. In addition, these data supported surface observations of the morphology and crack patterns of the pictorial layers in order to gain a better understanding of the damage found. Instead, the second part focused on the application of new tools for data processing, in addition to statistical and multivariate analysis. In recent years, in fact, increased awareness of the potential offered by HSI data and the ability to rely on increasingly powerful and efficient computing machines has led to the establishment of procedures for more systematic use of hyperspectral imaging in the field. This led to the first approaches toward Artificial Intelligence (AI), using Machine Learning (ML) and Deep Learning (DL) methods. In this exploratory study, several algorithms based on AI methods were applied to process HSI data acquired on three 1917 paintings using a "data-mining" approach, where the HSI data is analyzed to unveil new correlations and extract embedded information [4].

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Color Theory and Spaces

Atomic Color: From Points to Probability Distributions

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Abstract. Colorimetry can be computed for a given observer, illuminant and a surface, using the standard CIE formulae, resulting in a discrete point in three dimensional space, the CIE XYZ tristimulus value. While other representations are also possible (such as CIE LAB, LCh etc.), they all ultimately share one characteristic, that color is represented as a point in 3D space. However, there is known uncertainty and variation in every one of the stimuli, and quantities from which colorimetry is computed and perceptual attributes predicted. Likewise, there is uncertainty in perceptual evaluation – not only as a function of stimulus properties (size, context) but also fundamental variability between observers. The representation of color as a discrete point in 3D however obscures this underlying variability. In this paper an alternative is put forth, which claims that representing colors as points is both reductive and inaccurate when it comes to reflecting real-world performance and also – perhaps counterintuitively – introduces unnecessary complexity when designing optimization problems. Not only is it more correct to consider colorimetry in terms of ranges or probability distributions, it also leads to more realistic ways to talk about, e.g., whether or not two samples match, whether or not a colorimetry is in-gamut or not, and ultimately also helps in formulating color and machine learning models that open up new possibilities in using the latest AI techniques.

Keywords: Color formation · color difference · colorimetry · machine learning · AI

1 Introduction

Much of color science relies on standardized observers, illuminants, viewing conditions, states of adaptation, measurement conditions, etc. Each of these is based on empirical data collected from multiple observers or measured multiple times and subsequently analysed and processed to converge on a single representation. Establishing these standards results in a common language to interchange colorimetric information, which is an important step in color science and a cornerstone for industry to rely on. An implicit and sometimes overlooked assumption here is that an average or *mean observer* is represented, and an *average from multiple measurements* is taken for the standard representation of illuminants, for example. In other words, there is inherent variation in the data and therefore uncertainty behind every such representation. While the step of

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converging on a single specific set of values is needed in order to provide the single agreed representation, an analysis of the underlying uncertainty in the data is equally important and can not only shed light on the nature of the signals but also help in dealing with color imaging problems.

Knowing that every measurement (of surfaces, illuminants, observers, ambient lighting conditions, etc.) and every color reproduction event (capture, print, display) results in somewhat different colorimetries, implies that there are tolerances and thresholds that will inherently depend on the specific devices and conditions. This in turn defines what two colors can be considered different: take a pair of colorimetries that come from color measurements of some physical samples. Common color analysis would dictate that a color difference is computed between the two colorimetries and the difference is declared to be the result of this comparison, for example in terms of the CIE ΔE2000 color difference metric. However, what if the uncertainty of a single colorimetry is, say, within 0.2 ΔE2000 and the color difference between the two samples is 0.1 ΔE2000? Can it reliably be stated that the two samples are different, and if so then with what level of confidence and by how much? Each of the two colorimetries corresponds to a sequence of assumptions that get "compressed" into single fixed values, but that really come from distributions of values, of which any given sample is a valid example. Figure 1 shows such a case, for a patch that was printed at 3 times with 3 copies each and measured 3 times each, resulting in 9 prints and 27 measurements in total.

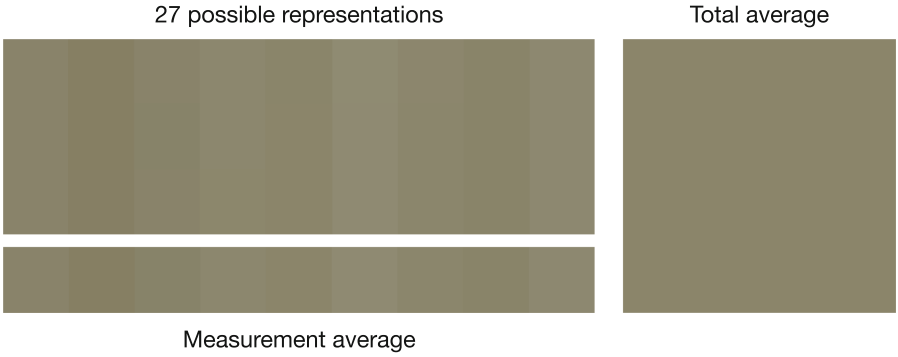

Fig. 1. Example colorimetries of 3 prints of 3 repetitions of a chart, measured 3 times each (left) and the average representation (right).

Each printing and measurement results in inherent differences, with the largest difference (the two examples that are furthest away) being 4.3 ΔE2000 and with an average of 1.4 ΔE2000 between any pair of colorimetries from among these 27. But each individual sample is a valid possibility for the event of printing a given input value (RGB or CMYK) on a given printer and measuring it with a given spectrophotometer. In this ecosystem, therefore, differences below these values cannot be considered as corresponding to different states of the system output, since the system itself is incapable of reliably differentiating among them. Note however that in this case both the sample (print) and the measurement vary. For a fixed physical sample, the measurement uncertainty is represented by the relation of the bottom row vs each of the columns in the

full data set (where each column represents the 3 measurements of the same physical sample). In this case the uncertainty is significantly lower, on the order of 0.3 ΔE2000 on average. This means that if a single physical sample is present, measurements of samples that differ below this level of uncertainty cannot be considered different, while if multiple physical samples are at play (e.g., due to repeated generation of a sample: e.g., multiple captures of an object or multiple prints for a given fixed input), uncertainty is greater.

The key takeaway here is that it is important to know the uncertainty in each of these steps to answer when two samples are different, and that this implies a move from thinking of colorimetry as discrete points in 3D space, to ranges or probabilities (Fig. 2). Knowing when two samples are different, in turn, impacts all other evaluation, processing or optimization, e.g., white balance, print color model, color reproduction, attribute optimization or other problems. While this insight may seem obvious and in some contexts is being taken into account and even quantified, what is argued here is that such uncertainty needs to be looked at from first principles, and that colorimetric comparisons and evaluations need to take it into account from start to finish. In doing so, optimization and machine learning problems can also be reformulated and regularized for more robust solutions and therefore improved performance, as will be shown later.

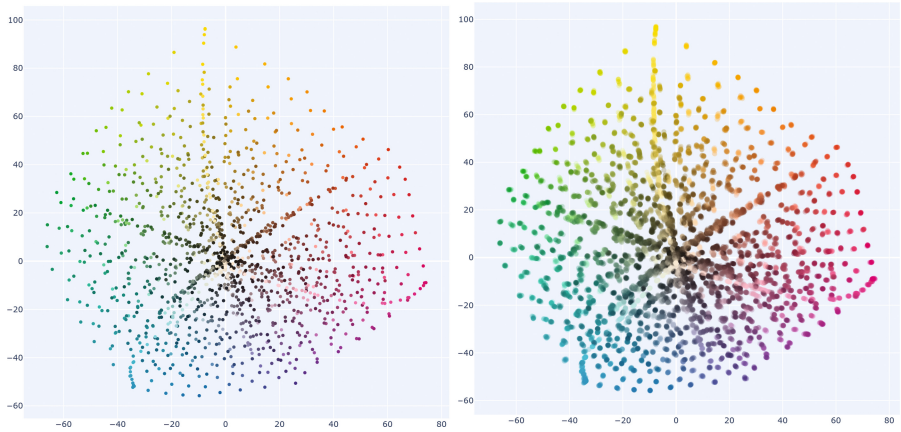

Fig. 2. CIE a*b* plot of 27 (3 copies of a chart at 3 times and measuring them 3 times each) times 1504 patches of an ECI profiling target plotting their average over all 27 print and measure instances (left) as well as each individual sample (right).

In summary, colorimetry needs to move from thinking about points (Fig. 2 left) in a coordinate system (such as CIE XYZ or CIE LAB) to distributions of possible instances that define local ranges or probabilities (Fig. 2 right), since, given all the uncertainty that underpins color formation, it is impossible to consider a fixed point in space and equally, to talk about color attributes such as color differences as single values instead of ranges or values with associated probabilities.

This article is organized as follows. First, a detailed overview of uncertainty will be presented – starting with the representation of spectral quantities and stimuli themselves

and various sources of error, through how color formation can be thought of (given uncertainty) all the way to uncertainty in perception. A brief overview of what this implies for color matching, inclusion and cardinality is presented next. Then, two cases of where using a simple model of uncertainty helps regularize and improve machine learning problems is highlighted, before presenting a discussion on further and more advanced applications of the presented concepts.

2 Uncertainty

In this section various sources of uncertainty will be described through a seemingly simple-looking question: when are two stimuli different? To consider two stimuli different, they have to be reliably generated, captured or represented as different as well as perceived (or predicted to be perceived) as different. First several sources of variation will be examined – both for stimuli and percepts – and finally color formation itself will be revisited from this perspective as it is key to whether or not two stimuli result in a difference.

2.1 Spectral Representation and Standards

The first source of uncertainty comes from how core quantities of color formation are measured and represented. Spectrophotometers, telespectroradiometers, colorimeters, cameras, displays, printers, etc., all have a certain level of encoding bit-depth as well as various forms of noise or repeatability that are non-zero. The level of noise or error then implies the threshold of what can be considered as different quantities or stimuli.

Quantization can be thought of as different densities of n-cube grids, where n corresponds to the sampling dimensionality. For spectral data this can be, e.g., $n = 16$ or 31 for a 400 to 700 nm range at 20 or 10 nm sampling respectively or for color capture this can be, e.g., $n = 3$ for RGB data. For example, an 8-bit quantization corresponds to a n-cube grid of side $1/2^8$, while a 16-bit quantization corresponds to a n-cube grid of side $1/2^{16}$. For a 31–sample spectral representation, using the SOCS dataset of 53,489 reflectances [1] as a ground-truth, at an 8-bit representation the median quantization error can be 0.1 and as high as 0.35 ΔE2000. At 16 bits the errors have a median below 0.05 and a maximum of 0.15 ΔE2000.

Measurement noise also results in uncertainty and considering a simple uniform random error ($\mu = 0$ and range of -0.5 to 0.5%) over the same SOCS spectral data set, its ΔE2000 color difference vs the theoretical noise–free original reference, even for low levels of spectral noise (less than 3%), can result in substantial colorimetric errors, with a median of 0.19 and a maximum of 6.5 ΔE2000 [2]. Furthermore, the above analysis holds for a single device used, however, different spectrophotometers measuring spectral reflectance can result in average errors between them of 0.5 to 3.0 ΔE2000 and a maximum of 2.5 to 8.0 ΔE2000 [3–12].

What this means is that even before considering how surfaces or illuminants interact and result in color stimuli, representation or encoding as well as (intra and inter) device noise already introduce a significant level of uncertainty. Surfaces or objects that are measured or represented within these noise thresholds, cannot be considered as different

with certainty to begin with. Furthermore, this noise may also be variable, both spectrally (as a function of wavelength) and colorimetrically (as a function of color attributes). Figure 3 shows an example of such a noise distribution in colorimetric terms, with the size of the square markers being proportional to the level of uncertainty. As can be seen in this case, in general, darker colors show larger uncertainty.

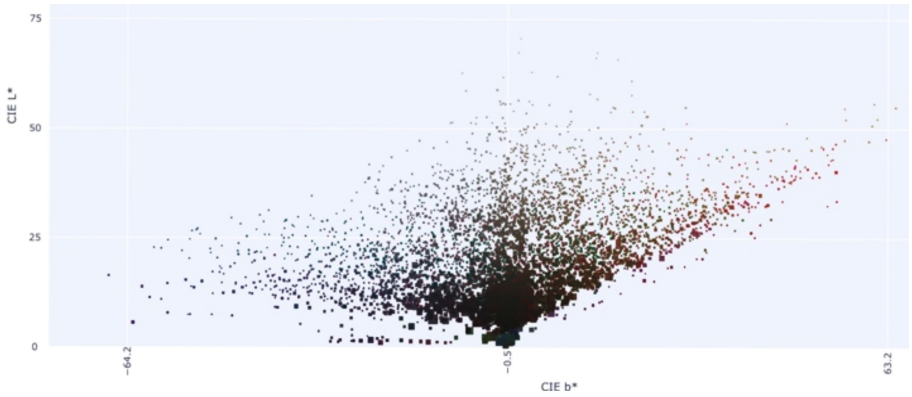

Fig. 3. Spectral noise interpreted in ΔE2000 color difference terms shown as CIE LAB plot (L*b* projection, note the cropped L* axis) to illustrate the distribution of largest errors (above 1.0 ΔE2000) – the values correspond to the errors analysed for a 3% per wavelength noise level.

This analysis relies on the assumption that spectral representation is meaningful in the domain provided by measurement devices, i.e., equidistant spectral sampling, such as at 1 nm, 2 nm, 5 nm, 10 nm or 20 nm intervals in the range of visible wavelengths, typically between 380 nm and 730 nm approximately. However, measured spectral data also shows that the regular, equidistant sampling has some inherently more important wavelengths or wavelength ranges than others [13, 14], this in turn may also imply higher or lower levels of uncertainty as a function of wavelength.

Standard illuminants, for example the D-series representing daylights, are also the result of multiple measurements [15–17] and a subsequent model to represent them [18], while measured light sources – like surface reflectances – also result in variability (likely of a higher magnitude still due to them inherently varying as a function of short time periods, unlike physical samples which in general vary more slowly).

Moving on to observer spectral sensitivities: the CIE 1931 Standard Colorimetric Observers themselves are based on 17 observers who performed color matching experiments [19–21] with the standard representing an average of these. Subsequent versions of cone sensitivities are also always derived from multiple different observers performing color matching experiments and then converging on some form of average observer. However, different observers can vary substantially even under the same experimental conditions, resulting in 2.5 to 3.0 ΔE2000 differences between different attempts at a color match and even the same observer repeating the experiments multiple times will differ between 1 and 1.5 ΔE2000 [22]. To illustrate some of this variability, Fig. 4 shows the spectral sensitivities of the Stiles and Burch color matching data-set from 1959.

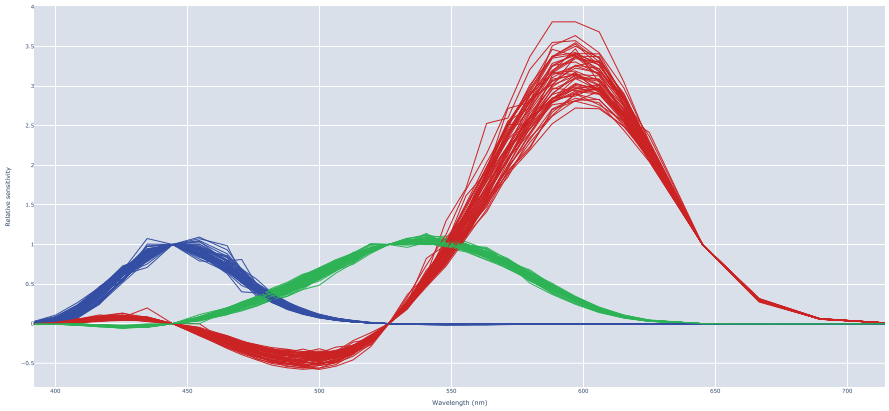

Fig. 4. Stiles and Burch's color matching experiment results ([23] via [24]) showing observer to observer variation of performing the same color matching experiment.

Taken together, these sources of variation show that behind every colorimetric representation, variability and uncertainty should be considered. While following the common representation, a surface, an observer and an illuminant combine to a single tristimulus value (e.g. CIE XYZ or LAB), instead, a neighbourhood of a tristimulus value or a distribution of likelihoods for a given tristimulus value should be considered. The next section builds on this analysis and proposes color formation changes that reflect uncertainty.

2.2 Color Formation

Color formation for a Mondrian world of Lambertian surfaces, as defined by the CIE, is the sum over the visible wavelength range of the product of a surface spectral reflectance $\boldsymbol{r}$, an illuminant spectral power distribution $\boldsymbol{l}$ and observer spectral sensitivities – such as the three CIE 1931 standard colorimetric observer – $\boldsymbol{S}$ (normalized to give a value of $Y = 100$ for a perfect white diffuser), written as follows:

$$\boldsymbol{x} = \boldsymbol{r} * (\boldsymbol{l} \bullet \boldsymbol{S}^T) \tag{1}$$

with • denoting the vector dot-product, and $\boldsymbol{x}$ the sensor response (i.e., a CIE XYZ). Both $\boldsymbol{r}$ and $\boldsymbol{l}$ are $1 \times n$ vectors (where n is the spectral sampling, e.g., 31 or 16 for a 400 nm to 700 nm sampling at 10 nm or 20 nm respectively) and $\boldsymbol{S}$ an $m \times n$ matrix where m is the number of sensors resulting in $\boldsymbol{x}$ being a $1 \times$ m vector of sensor responses (e.g., 1×3 for an RGB camera where $\boldsymbol{x}$ = RGB or the CIE color-matching functions where $\boldsymbol{x}$ = XYZ) [25]. A first, simple way to incorporate uncertainty is to acknowledge that each quantity in Eq. (1) comes from a possible range of values, hence:

$$\boldsymbol{x} \pm \boldsymbol{\varepsilon_x} = (\boldsymbol{r} \pm \boldsymbol{\varepsilon_r}) * ((\boldsymbol{l} \pm \boldsymbol{\varepsilon_l}) \bullet (\boldsymbol{S} \pm \mathrm{E}_S)^T) \tag{2}$$

with $\boldsymbol{\varepsilon_r}$, $\boldsymbol{\varepsilon_l}$ and E_S referring to the uncertainty or *epsilon* neighborhoods from the surface reflectance, illuminant and observer spectral sensitivities respectively, resulting in the tristimulus value likewise being in an $\boldsymbol{\varepsilon_x}$ neighborhood of $\boldsymbol{x}$.

Assuming known spectral $\boldsymbol{\varepsilon}$ and their impact on colorimetry, a further extension of this is to formulate Eq. (2) as a pair of inequalities, as follows:

$$\boldsymbol{x_L} \geq \boldsymbol{r} * \left(\boldsymbol{l} \bullet \boldsymbol{S}^T\right) \wedge \boldsymbol{x_U} \leq \boldsymbol{r} * \left(\boldsymbol{l} \bullet \boldsymbol{S}^T\right) \tag{3}$$

where $\boldsymbol{x_L}$ and $\boldsymbol{x_U}$ are the lower and upper bounds on the tristimulus values, in effect defining a bounding box around $\boldsymbol{x}$, such that $\boldsymbol{x_L} = \boldsymbol{min}(\boldsymbol{x} - \boldsymbol{\varepsilon_x})$ and $\boldsymbol{x_L} = \boldsymbol{max}(\boldsymbol{x} + \boldsymbol{\varepsilon_x})$. Bounding constraints are still a simplification so a convex polyhedron around the target tristimulus value, enabling more complex constraints that more closely describe the real data as compared to the simple cube constraints in Eq. 3 can be formulated. This case requires a characterization of the convex constraints first, in order to describe them as half-planes or linear inequalities – for example by computing the convex hull around multiple different measurements – canonically described by a matrix $\boldsymbol{A}$ and column vector $\boldsymbol{b}$: $[\boldsymbol{A}, \boldsymbol{b}]$ such that:

$$\mathbf{A}\left(\boldsymbol{r} * \left(\boldsymbol{l} \bullet \boldsymbol{S}^T\right)\right) \leq \mathbf{b} \tag{4}$$

Note that $[\boldsymbol{A}, \boldsymbol{b}]$ may be variable as a function of colorimetry, such as shown in the example in Fig. 3 where, e.g., for the same level of spectral noise or uncertainty, the resulting neighborhood varies as a function of lightness.

Finally, still in spatial terms, an equidistance constraint can be defined as well – for example if metric space uniformity is considered, then formulating the target colorimetry as lying within an epsilon neighborhood of a central point in space:

$$f\left(\sqrt{\sum \left(\boldsymbol{r} * \left(\boldsymbol{l} \bullet \boldsymbol{S}^T\right)\right)^2}\right) \leq \boldsymbol{\varepsilon} \tag{5}$$

where *f()* may express some transformation of the colorimetry to a perceptual space where equidistance is important (e.g., f() may be a transformation from CIE XYZ to CIE LAB).

A direct consequence of reformulating color formation in the above terms is the likelihood of the existence of a colorimetry and the ability to distinguish with certainty whether or not two colorimetries are the same. A colorimetry within an upper/lower (Eq. 3), convex (Eq. 4) or equidistant (Eq. 5) boundary, cannot be reliably distinguished from any other colorimetry that falls within the same boundary.

The above formulation follows a Boolean mindset such that answers to the question of whether two stimuli are the same or different is binary – two colors match if they are within the neighborhood, they mismatch otherwise – however, the sources of uncertainty should really be thought of as following some probability distribution.

Therefore, if colorimetry can be thought of as coming from a probability distribution centered around a central value $\boldsymbol{x}$, such that colorimetries around $\boldsymbol{x}$ will have some likelihood (or expectation) of those being observed or measured, then if each quantity, ***r, l, S***, comes from a random process, each has some underlying distribution, denoted as P_r, P_l, P_S, representing the probability distributions of reflectance, illuminant and observer spectral sensitivities, then the resulting colorimetry can be written as:

$$\boldsymbol{x} \sim g(P_r, P_l, P_S) \tag{6}$$

where ~ denotes "is distributed according to" and *g()* represents the resultant probability distribution for $\boldsymbol{x}$ as a function of the probability distributions for the constituent quantities. If colorimetry is defined in such probabilistic terms, as in Eq. 6, equality of two colorimetries also becomes a probabilistic expression $\mathcal{P}(\|\boldsymbol{x}_1 - \boldsymbol{x}_2\|)$ – what is the likelihood of two samples being the same – which in turn depends on the respective probability distributions *g()* for each case, instead of a binary comparison (whether or not $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ are strictly equal or fall within the various forms of boundary as described above). Figure 5 illustrates the concepts introduced in Eq. 1 to 6:

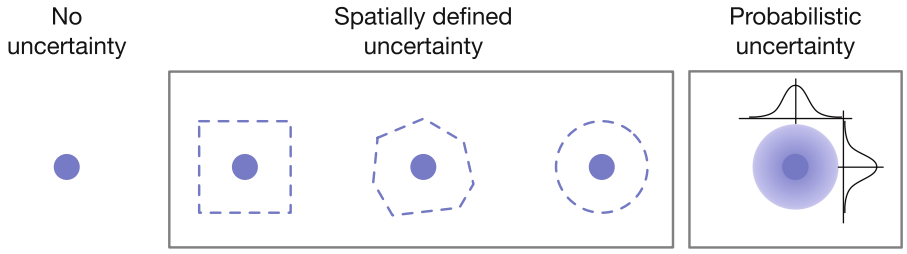

Fig. 5. Representations of uncertainty from none (left), to spatially defined uncertainty (bounding box, convex constraints, equidistance), to probabilistically defined uncertainty (right).

While this change of paradigm in color formation introduces various levels of additional complexity that each requires characterization of the sources of uncertainty – in terms of various boundaries or in terms of probability distributions – in doing so, the representation is both more accurate in that it more closely represents physical reality and comes with advantages for various forms of optimization and learning as shall be seen later.

2.3 Matching and Inclusion

The first direct consequence of considering uncertainty is on color matching: when do two colors match (i.e., when is their difference below a perceptibility threshold)? and by implication on inclusion: when is a color included in a volume of colors (i.e., when can it be represented or reproduced by a system that is characterized by a volume of colors)?

If colorimetries come from random processes, their differences also need to be considered as probabilistic [26]. To do so, the distributions for each stimulus need to be characterized first, before making comparisons between the respective distributions and making judgements about matching and inclusion. A first approach here is to assume normality – i.e., that color stimuli follow a Gaussian distribution around a mean – and compute difference via pooled variance [27], such that:

$$\Delta\overline{C}_{1,2} = \left|\overline{C}_1 - \overline{C}_2\right| \tag{7}$$

$$Var_p\left(C_{1,2}\right) = \frac{(n_1 - 1)Var(C_1) + (n_2 - 1)Var(C_2)}{n_1 + n_2 - 2} \tag{8}$$

where colorimetries are samples from a normal distribution, denoting ($\overline{C}$) their means and ($Var(C)$) their variances, for the two cases being compared, with n_1 and n_2 denoting the sample sizes of the two sets of colorimetries.

Taking the aforementioned set of target colorimetries that result from 3 printing events of 3 copies each, measured 3 times, resulting in 27 total colorimetries for a single digital input (CMYK value of a ECI 2002 chart [28]) and computing pair differences then yielded 729 ΔEs per CMYK. The 95% ranges had a coefficient of determination of 0.692 and the Kolmogorov-Smirnov test for normality [27] failed. Looking more closely at the first member of this set, Fig. 6 shows both the colorimetries from the two systems (as blue and green circles respectively) and the histogram of their pair differences. As can be seen, there are clusters among the colorimetries, which also correspond to clusters in the histogram that are not consistent with a unimodal, normal distribution.

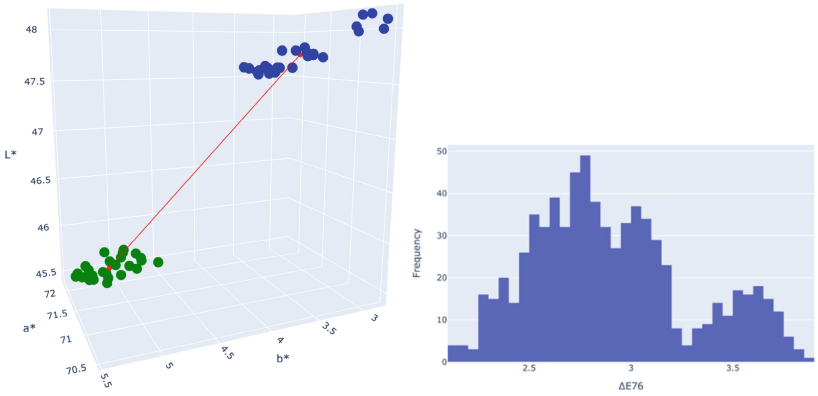

Fig. 6. Colorimetries of first member of large dataset from two systems (top) and the histogram of their pair differences (bottom).

Considering color differences probabilistically extends naturally to the question of inclusion. If a color is within some tolerance of the gamut volume, it can be considered as in-gamut (with some probability), as opposed to the more common binary Boolean approach of in-out. This requires constructing gamuts that take uncertainty into account, which can be done by computing the means and variances of all gamut-determining measured samples (such as the ECI profiling target), constructing a hull using alpha shapes to obtain a triangulated surface, then, given a test colorimetry (defined with its mean and variance) express the closest point on the gamut surface with its respective mean and variance (based on the triangulated data points). From the test colorimetry and the closest point colorimetry – both defined in terms of means and variances – compute the pooled variance using Eq. 7 and 8, resulting in a distance distribution from which e.g. interquartile ranges can be reported. Figure 7 shows an example of the triangulated gamut boundary as well as the norm per-dimension variances for each vertex.

Taking a test color, with CIE LAB values of [22.1, −7.8, −22.1], variances of [0.096, 0.015, 0.041] and an L2 norm of 0.11, the nearest point to it on a mean gamut is at a distance of 1.37 ΔE76, with CIE LAB coordinates of [23.2, −7.7, −21.3], a pooled

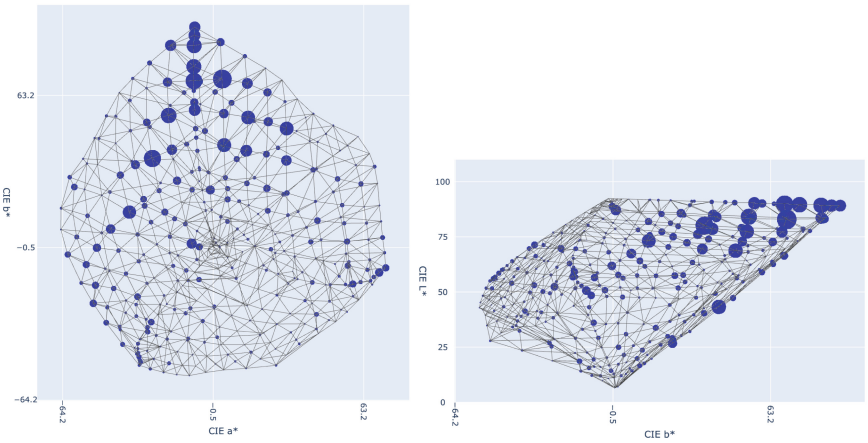

Fig. 7. The gamut boundary of one of the extensive data sets, showing the norm of per-dimension variances at each vertex. For ease of illustration, circle radii indicating variance are 50x the actual variance value.

variance based on the enclosing surface triangle's vertex variances of [0.085, 0.031, 0.103] and L2 norm of 0.14. Given these variances, the range of distances between the test color and the gamut with a 95% range is 0.70 to 2.04 ΔE76.

In other words, instead of thinking that the out-of-gamut distance is 1.37, a probabilistic treatment of the question predicts that the distance will be in the 0.70–2.04 ΔE76 range 95% of the time. Another way to express this relationship is to say that the probability of the test color being no more than 1.0 ΔE76 out-of-gamut is 16% in this case and that the probability of it being no more than 1.5 ΔE76 is 64%.

2.4 Perception

Finally, perception itself – the result of the interactions of surfaces, illuminants and observers – taking both the complexity of viewing conditions into account as well as the flow of perception beyond Lambertian color formation models assumed so far, is also subject to uncertainty and variability. Taking again the example of when two colors are different, but now posing it in its fullness of "when are two colors *perceived* as being different?", is subject to a host of conditions which change thresholds or tolerances significantly and hence present an element of uncertainty.

To begin with, color difference metrics are always defined under specific conditions for which they apply, based on the experimental data and conditions under which they were derived. Be it uniform patches viewed with separation [29], or changes in angular subtense [30] and separation of the samples [31], all are models that aim to approximate experiments conducted by a group of observers repeating a task multiple times with multiple samples. Once derived, these models are often used in much broader contexts than they were originally intended for and – like for Lambertian color formation – also derive from inherent uncertainty that is ultimately "compressed" into the single model or metric.

Two recent examples of exploring variability here focus on evaluating perceived color difference for thin lines (versus area fills) [32] as well as thresholds for continuous transitions in complex imagery [33]. Figure 8 shows the results of psychovisual experiments for the latter two examples. As can be seen, the thresholds of Just-Noticeable-Differences vary greatly depending both on content (thin lines vs area fills) as well as what context a stimulus is in (continuous synthetic transition or complex imagery).

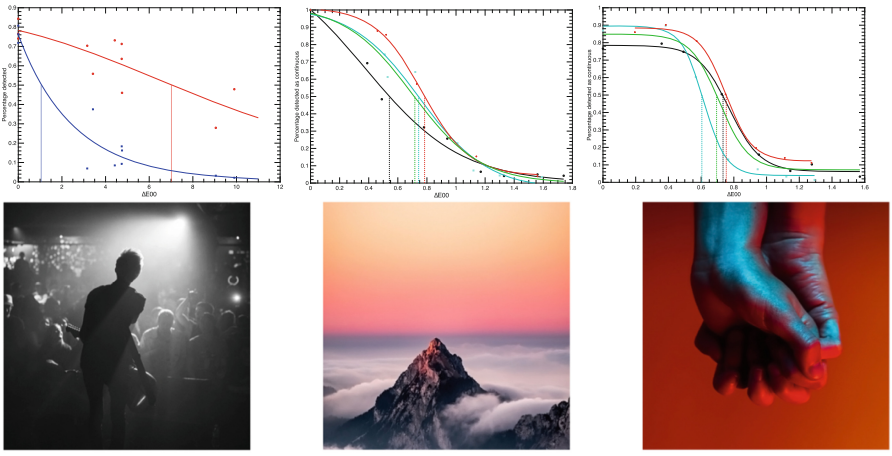

Fig. 8. Perceptibility thresholds for same (blue) and different (red) thickness line pairs (top left), continuity perceptibility thresholds for gray, cyan, red and combined (green) transitions (top middle), continuity perceptibility thresholds (top right) for the images shown in the bottom row.

Whether or not two colors can be considered same or different therefore greatly depends on the content and viewing conditions (other than the observer, illuminant variability already mentioned) and furthermore is again subject to statistical uncertainty. Even for fixed conditions and fixed content, whether or not differences can be perceived requires again a probabilistic answer of the form "stimulus X vs stimulus Y under conditions Z viewed by observer A has a likelihood P of being perceived as different".

3 Using Uncertainty in Optimization and AI

The insights and analysis shared so far is of a descriptive nature: finding more complete ways to understand and express the inherent uncertainty in color formation and color perception, in the elements involved and therefore also in how to respond to questions of equality or inclusion which rely on it. However, appropriate models of uncertainty also play an important role in machine learning and AI. In this section examples are shown where considering uncertainty unlocks new possibilities of formulating optimization problems followed by a discussion of possible further implications of this paradigm.

3.1 From Metamerism to Paramersim

Metamerism is defined as the phenomenon where two perceptual experiences are indistinguishable from each other under one set of conditions, while becoming distinguishable under a change in some or all the conditions. A typical example is that of a change of illuminant where objects (to a fixed observer) appear to match in color under one light source, but show a clear color difference under a different light source. The underlying reason for this is that colorimetric matching and spectral matching are not the same and while colorimetric matching can be achieved more easily (many different primaries can result in the same colorimetry, but not the same spectral composition), it only holds under the conditions it was designed for.

Picking up the strict definition of color formation from Eq. 1, for a fixed observer described by $\boldsymbol{S}$, two surfaces $\boldsymbol{r}_1$ and $\boldsymbol{r}_2$ are said to be metameric if and only if there exists an illuminant $\boldsymbol{l}_\mathrm{A}$ and illuminant $\boldsymbol{l}_\mathrm{B}$ such that:

$$\boldsymbol{r}_1 * \left(\boldsymbol{l}_A \bullet \boldsymbol{S}^T\right) = \boldsymbol{r}_2 * \left(\boldsymbol{l}_A \bullet \boldsymbol{S}^T\right) \text{ and} \tag{9}$$

$$\boldsymbol{r}_1 * \left(\boldsymbol{l}_B \bullet \boldsymbol{S}^T\right) \neq \boldsymbol{r}_2 * \left(\boldsymbol{l}_B \bullet \boldsymbol{S}^T\right) \tag{10}$$

All reflectances $\boldsymbol{r}$ that satisfy Eq. (9) form the metamer set [34]:

$$\mathcal{M}(\boldsymbol{x}) : \left\{\boldsymbol{r_i} : \boldsymbol{r}_i * \left(\boldsymbol{l}_A \bullet \boldsymbol{S}^T\right) = \boldsymbol{x}\right\} \tag{11}$$

And the degree of mismatch (Eq. 10) is then the projection of the metamer set $\mathcal{M}(\boldsymbol{x})$ onto the mismatch condition (illuminant $\boldsymbol{l}_B$ in this case). The metamer set can be computed without knowing $\boldsymbol{l}_B$ by simply computing all reflectances that result in the same colorimetry. Some of these will then not result in a mismatch under the test illuminant $\boldsymbol{l}_B$ but it is always possible to design such and illuminant under which a mismatch will take place.

While it can be shown that this set is convex, it is also a rank-deficient or degenerate set. The reason for this is that the linear subspace defined by $\left(\boldsymbol{l}_A \bullet \boldsymbol{S}^T\right)$ (the illuminant and the observer), has a unique solution in relation to the tristimulus $\boldsymbol{x}$, via the pseudo-inverse. However, being under-determined (if spectral data is represented at n spectral samples, and tristimulus values are 3D, the system of linear equations has $n - 3$ degrees of freedom, hence the solution resides in an $n - 3$ dimensional subspace of n-space), there are infinitely many possible solutions. Even introducing constraints such as physical realisablity (i.e., that $\boldsymbol{r}$ has values between 0 and 1), the use of a linear model bases (i.e. that $\boldsymbol{r}$ can be written as a weighted some of some basis functions, typically obtained from Principal Component Analysis) or the introduction of a naturalness constraint (that each solution be a convex combination of some existing spectral data), which reduce the dimensionality of the problem and the degrees of freedom, the equation remains ill-posed and requires a solution of the form of a fundamental (the pseudo-inverse) and metameric blacks (the orthogonal space, invisible to the illuminant x observer combination) [35].

In the context of color correction – mapping device RGBs to colorimetry – a first level of uncertainty-aware approach has been formulated by computing the full metamer

set and then picking one of the possible metamers (the likeliest given a maximum-likelihood approach) leading not only to improved performance but also to an explicit characterization of error-bars [36]. However, this approach still focused on an exact match under the source conditions.

Instead, departing from an exact color match and switching to a color formation model that accounts for uncertainty, enables a reformulation of the problem from one of matermism – a strict match – to paramerism – an approximate match (as described in Eq. 3):

$$\mathcal{P}(\boldsymbol{x}) : \left\{ \boldsymbol{r_i} : \boldsymbol{r}_i * \left(\boldsymbol{l}_A \bullet \boldsymbol{S}^T \right) \leq \boldsymbol{x_U} \wedge \boldsymbol{r}_i * \left(\boldsymbol{l}_A \bullet \boldsymbol{S}^T \right) \geq \boldsymbol{x_L} \right\} \tag{12}$$

Or more generally:

$$\mathcal{P}(\boldsymbol{x}) : \left\{ \boldsymbol{r_i} : \boldsymbol{A} \left(\boldsymbol{r_i} * \left(\boldsymbol{l} \bullet \boldsymbol{S}^T \right) \right) \leq \boldsymbol{b} \right\} \tag{13}$$

where *[**A, b**]* describe the various constraints or neighborhood that describe the uncertainty of color formation.

Here, a key insight is that while Eq. (9) is under-determined and whose solution lies in an *n-3* dimensional subspace of reflectance space, Eq. (12) or (13) are fully formed, non-degenerate and directly solvable using the tools of computational geometry. Since $\boldsymbol{A} * \boldsymbol{x} \leq \boldsymbol{b}$ defines a half-plane intersection problem or convex hull problem (both Eq. 12 and 13 can be written in this form) this can be solved using standard algorithms and packages such as qhull [37] or cdd [38]. Note that this specific solution (of directly computing the convex hull) – while valid in its own right – currently has some practical limitations due to memory and processing power, and is practically feasible for up to 20–22 spectral samples (or dimensions if using a linear model basis), but becomes infeasible for some of the more common representations such as, e.g., 31 spectral samples or higher.

Figure 9 show a comparison of the approximate (constrained) solutions to metamer sets as well as the direct, exact and unconstrained solution of the paramer set (for a small neighborhood threshold) as well as applying this approach to the Macbeth ColorChecker chart and visualizing paramer mismatch volumes for a change in illuminant from CIE D50 ($\boldsymbol{l}_A$) to CIE A ($\boldsymbol{l}_B$) – indicating the color mismatch of reflectances that are parameric under D50 (i.e. within a small threshold of a target CIE XYZ) under CIE A where the near-match breaks and results in full, n-dimensional convex spectral polytopes projected on the 3D space as convex polyhedra.

3.2 Learning Ink-Use

Another simple but powerful example of using uncertainty in regularizing optimization problems is that of optimizing ink-use in the HANS domain [39].

Halftone Area Neugebauer Separation (HANS) is a paradigm where print is controlled in terms of Neugebauer Primary area coverages. Key to this idea is the principle of inks interacting non-linearly (therefore NPs being the native components) and color formation being additive and optically averaging (therefore if NP colorimetries are known and their relative area coverages are controlled, colorimetry is well behaved). Since the

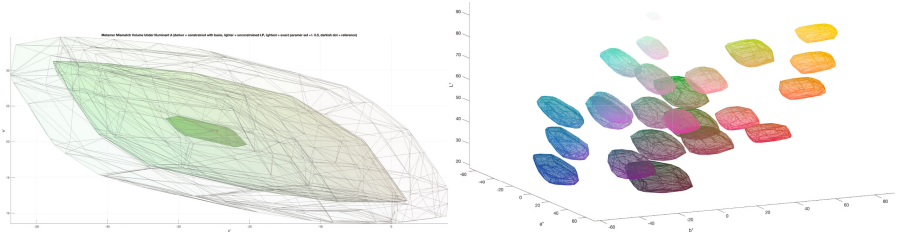

Fig. 9. Comparison of metamer sets (that require additional assumptions about the dimensionality of spectral data) and paramer sets (that can be computed directly and exactly, within some introduced tolerance, here ±0.5 in CIE XYZ terms) on the left, and the paramer mismatch volumes – projections of paramer sets onto a different illuminant (CIE A), shown as convex polytopes in CIE LAB space.

space of NPacs is significantly higher dimensional than that of inks and results in exponential redundancy (even enabling metamers with 3 inks alone since a CMY system with binary drop deposition is 8D in HANS terms [40]), it allows defining optimizations of the form: "find the NPac that optimizes an attribute" (such as the amount of ink used by the NPac, or attributes like grain or color inconstancy, etc.…) and matches a given color. One of the most direct applications of this idea is that of ink-use optimization since the objective function is simply the amount of ink used and relates linearly to the NPac. It turns out that, assuming a simple color model (i.e. predicting colorimetry from an NPac), this problem can be formulated using Linear Programming (LP) [41]. This problem can be written as:

$$\sum_{i=1}^{N} \boldsymbol{w}^{\boldsymbol{i}} * \boldsymbol{P}_{\boldsymbol{C}}^{\boldsymbol{i}} = \boldsymbol{x} \tag{14}$$

where $\boldsymbol{P}_{\boldsymbol{C}}^{\boldsymbol{i}}$ refers to the i-th NP colorimetry ($\boldsymbol{P_C}$ is an N × 3 matrix), $\boldsymbol{w}^{\boldsymbol{i}}$ refers to the i-th convex weight (such that their sum adds up to 1 and they are from a [0, 1] interval) – here $\boldsymbol{w}$ also defines the area coverages of the aforementioned NPac – with $\boldsymbol{x}$ denoting the target colorimetry. The set of all possible $\boldsymbol{w}^{\boldsymbol{i}}$ vectors represents the metamers that the system with $\boldsymbol{P_C}$ NPs can reproduce. The objective function is then simply the $\boldsymbol{w}^{\boldsymbol{i}}$ weighted ink-use of each of the NPs (for example in drops the NP of "C" will be 1 drop while the NP of "CCM" will be 3 drops). Then, let $\boldsymbol{P_I}$ be an N × 1 vector of ink-uses for each of the Neugebauer Primaries, then LP minimizes:

$$\min_{\boldsymbol{w}} \boldsymbol{w}^{\boldsymbol{T}} * \boldsymbol{P_I} \tag{15}$$

subject to the equality constraint in (14) and coupled with several additional linear constraints on the weights themselves.

However, due to precision and the consequent fragility, the aforementioned optimization in Eq. (15) may result in no solution found – because no exact match of NP colorimetries to the target colorimetry need exist. Furthermore, some NPacs (combinations of NPs) may not be considered for a given colorimetry, again due to not resulting in an exact match.

Taking the uncertainty as described before – underlying all color formation as well as all quantities involved in computing or capturing it as described earlier – into account,

regularizes the problem and results in more matching colors and better results overall. Elegantly, this change remains within the formulation of linear programming, since instead of the equality constraint in Eq. (14), two sets of inequality constraints are used instead:

$$\begin{aligned} \sum_{i=1}^{N} w^i * P_C^i &\leq x + \epsilon \\ \sum_{i=1}^{N} w^i * P_C^i &\geq x - \epsilon \end{aligned} \tag{16}$$

Using this change in the optimization results in more colorimetries being matched (within a given tolerance) and ultimately better optimality (in terms of median and 95$^{\text{th}}$ %tile ink-use over a uniform sampling of a CIE LAB reference gamut). Table 1 shows the tolerances (expressed in CIE XYZ, where a tolerance of 1.0 means $\epsilon = 0.5$), median and 95$^{\text{th}}$ %tile ink-use over a uniformly sampled set of LABs.

Table 1. Ink-use optimization performance as a function of colorimetric uncertainty – i.e., a tolerance for the target colorimetry for which ink-use is being minimized, in order to regularize the optimization problem.

Uncertainty (CIE XYZ)	**0.0**	0.5	1.0	1.5	2.0	3.0	4.0	5.0	**6.0**
Median ink-use (ng)	**10.42**	10.39	10.31	10.24	10.17	10.01	9.99	9.77	**9.62**
95$^{\text{th}}$ %tile ink-use (ng)	**22.08**	21.64	21.20	20.56	20.54	19.52	18.90	18.33	**17.90**

Taking uncertainty into account, even with a simplified model such as that of Eq. (16), allows reducing the maximum ink use from 22.08 to 17.89 – a 22% relative improvement, with the median ink-use also reducing an extra 8% vs the reference case with zero tolerance. The reason for this is that while matching a colorimetry exactly may not be possible with a certain combination of NPs, a better choice may exist in a close neighborhood of that colorimetry and therefore more candidate NPacs are considered for a given colorimetric neighborhood and better choices can be found. This also explains why the results in Table 1 are monotonically decreasing since anything that is a valid solution with 0 tolerance is also a valid solution with a > 0 tolerance.

However, other than improved performance, key to this approach is its robustness since more target colorimetries will have solutions (will have convex combinations of NP colorimetries within the neighborhood of the target CIE XYZ). In this context, even a relatively large level of uncertainty (a cube of side 6.0 in CIE XYZ), does not affect the final precision, since ultimately printing and measurement would follow the optimization step and would compensate for any differences [41].

3.3 A Probabilistic Framework of Uncertainty for AI

The above examples are just some of those that show how appropriately introducing uncertainty into the optimization formulation enables solving problems that until now were either ill-posed, fragile or could not be solved directly at all. However, the move

from considering an exact match to targeting a neighborhood and in the future also considering the underlying probability is a shift in how to approach color and imaging problems not only at the level of formulation but also (as shown before) in terms of how to evaluate performance. These insights, together, lend themselves to some common principles when approaching color algorithm design:

- **Problem formulation**
  Characterize variability and uncertainty in source stimuli of the problem first. For example, for computational photography, understand the distribution of captured signals for a constant scene and characterize it in all relevant domains (spectral, colorimetric, perceptual, device contone) depending on the domain of target optimization. Build probabilistic models (static or variable throughout color space) in order to take them into account when designing the problem. Once understood, formulate the problem in probabilistic or neighborhood terms.
- **Performance evaluation**
  Given the previous characterization of variability and uncertainty, once it comes to evaluating the goodness of performance, also do so taking uncertainty into account. This means both reporting ranges instead of single values, or probabilities instead of yes/no answers. For example, once the color gamut of a printer is characterized in probabilistic terms, report gamut volume as a range and evaluate inclusion in terms of the probability of inclusion, instead of binary inclusion.
- **Algorithm options**
  Finally, also consider what algorithms could be most suitable to a particular problem. Color models, color correction and reflectance estimation etc. often focus on regression approaches. These are formulated in terms of minimizing error over some training data. Here, using all data instead of the average data will already result in a more robust fit for real world data. However, if the objective is that of solving the problem with uncertainty, other approaches may become relevant, such as classification approaches since instead of minimizing error, classifying the input into color neighborhoods could yield more robust results (depending on the target application). Likewise statistical learning approaches that depart from deterministic models to probabilistic ones and provide measures of confidence in predictions are more representative for uncertainty.

4 Conclusions

The color of a stimulus can be expressed by a single set of colorimetric or appearance attributes – as a point in 3D space – if an only if the inherent uncertainty in all quantities, conditions and observers is assumed to be represented by some average or representative case. However, this does not reflect the fundamental underlying uncertainty of color formation and color perception which has variability, error and noise every step of the way. In this paper an overview of various sources of uncertainty was provided and a framework for taking such uncertainty into account was presented, with an argument for using this uncertainty to the advantage of both a more accurate representation of reality as well as a computationally more robust vehicle for color imaging problem formulation, evaluation and application. A move from thinking about color as a point in 3D space to a

probability distribution, surprisingly, results in new possibilities and advantages where future work will focus in particular on continuing to build up a probabilistic framework and using statistical learning algorithms.

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Extended Wien and Planck Loci

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Abstract. In colour science we are familiar with the Planckian locus, a curve in chromaticity space that is indexed by the temperature T of a blackbody radiator, that approximately delineates the colours of typical lights. In the CIE (u', v') chromaticity diagram the reddest (lowest-T) light intersects the Spectral locus at 830 nm; as T rises, the lights become progressively orange, yellow, white and blue with the bluest light being defined by an infinite colour temperature. Interestingly the bluest blue Planckian chromaticity is far from the spectral locus. The Wien locus is parameterised by a simpler equation than Planck and runs almost parallel with the Planckian locus.

The loci are so close together that a temperature conversion brings corresponding chromaticities into a most complete coincidence. Though, the Wien locus is longer - extends more towards the short-wave part of the chromaticity diagram - than the Planck locus, for an infinite temperature.

In this paper, we extend the Wien formula to allow negative temperatures. When we do this the Wien locus naturally extends all the way to intersect the spectral locus (at 360 nm). The extension to negative temperatures allows the Wien locus to model any reasonable light's chromaticity within its arc. We show that the extended Wien locus is continuous: negative and positive infinite colour temperatures converge to the same point. However, there is a substantial discontinuity at $T = 0$ evidenced by the large (u', v') difference between the blue and red ends of the Wien locus. A practical application of this result is discussed (via the theory of Locus Filters) to improve design of color-correction filters in photography.

Keywords: Planckian Locus · Color of negative temperature · Wien Locus

1 Introduction

The Planckian locus, also known as the blackbody locus, describes the path that the colour of a blackbody takes as the blackbody temperature changes. This locus often presents in a standard chromaticity space as shown in Fig. 1(a). Throughout the paper, we provide all figures in CIE (u', v') chromaticity diagram

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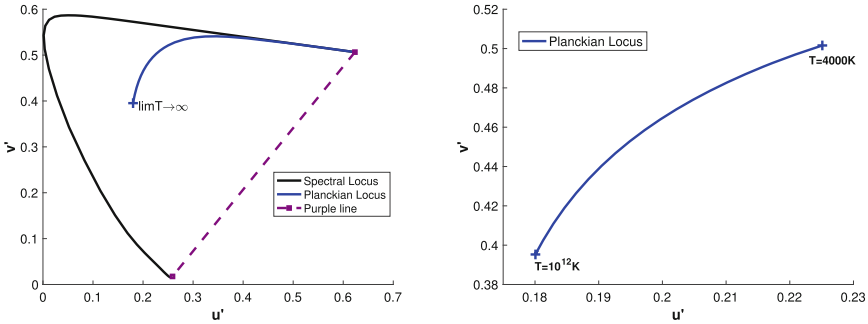
as it is more perceptual. From right to left, the locus (see Fig. 1(a)) starts from a deep red at low temperatures, progressing to orange, yellow, reaching a white hue, and ultimately ends in a bluish-white at high temperatures. However, it is not yet clear what the high temperatures mean, as indicated by $+\infty$ in diagram. So, this paper starts with the question of what the temperature of $+\infty$ Kelvin signifies and where the Planckian locus ends.

Robertson [1] reported the chromaticity endpoint of the Planckian locus in CIE 1960 (u, v) diagram to be at $(0.18006, 0.26352)$, corresponding to a temperature of $+\infty$ K (or 0 Mired [2]). However, Robertson [1] did not explain how the point was derived. For all practical purposes we suppose that 10^{12}Kelvin as sufficiently close to the endpoint of Planckian locus (so we can generate spectra and plot the corresponding (u', v') corrdinates. A zoomed-in view of the Planckian endpoint is shown in Fig. 1(b). Notice that we transformed this point to the CIE 1976 (u', v') chromaticity as $(0.18006, 0.39528)$. In Sect. 2 we will analytically determine the Planckian endpoint and show that our endpoint is equal to the Robertson endpoint (potentially meaning that we have re-invented Robertson's method). In the same chromaticity space, the purple dashed line (See Fig. 1(a)) that connects the monochromatic red (right point) to the monochromatic blue (left point) is shown. Also, the reddest light is defined over the Planckian locus while the bluest blue's definition using the Planckian locus remains an open question.



(a) (b)

Fig. 1. (a) Common Planckian locus and its endpoint in CIE(u', v') chromaticity diagram, (b) Exact (u', v') coordinate of endpoint $(+\infty$ K) of Planckian locus.

It is tempting to ask whether the chromaticity at infinite T is really the end of the useful Planckian locus. Indeed, there is potentially a vehicle for extending the locus, and that vehicle is to allow negative values of T. The infinite positive-T limit would have to match (in chromaticity) the infinite negative-T limit, and the join could have at most a removable discontinuity. As we will show in this paper, the negative-T extension meets the above criteria, but only in the Wien approximation of the Planck spectrum—not for the Planck spectrum

itself. Thus, we show that the Wien locus is in a general sense a continuous locus over any chromaticity diagram, such that it starts from the monochromatic red and ends in the monochromatic blue. Also, we show that the Planckian locus does not extend like Wien. For completeness we will determine the limit of Planckian locus at infinite temperature. Finally, we will describe an application for the negative-T Wien extension in the design of color-correction filters used by photographers.

The rest of the paper is organized as follows. In Subsect. 1.1 we present the background and related equations to the Wien locus. In Sect. 2 we define Wien locus and prove its continuity over chromaticity diagrams, and in Sect. 3 we show some experiments to show the usefulness of our Wien locus. The paper ends with a short conclusion.

1.1 Related Work - Planck and Wien Loci -

For a blackbody radiator, the spectral radiant emittance E_P is calculated using Planck's formula as a function of wavelength λ and colour temperature T [3]:

$$E_P(\lambda, T) = kc_1\lambda^{-5}(e^{\frac{c_2}{T\lambda}} - 1)^{-1} \tag{1}$$

where c_1 and c_2 are constants equal to $3.74183 \times 10^{-16}\,\mathrm{Wm}^2$ and $1.4388 \times 10^{-2}\,\mathrm{mK}$, respectively. Additionally, the scalar k modulates the intensity of the light spectrum. Another formula is the Wien displacement law that also describes the blackbody radiation E as [3]:

$$E(\lambda, T) = kc_1\lambda^{-5}e^{-\frac{c_2}{T\lambda}} \tag{2}$$

The Wien and Planck functions both describe similar spectra for low (say < 4000 K) color temperatures. However, the generated spectra become more different the higher the temperatures become. We show this in Fig. 2 by plotting two loci in the CIE(u', v') diagram. Here, to calculate the chromaticity of radiators with different temperatures (4000 to $+\infty$K), we use Planck's and Wien's laws and refer to them as Planckian and Wien loci. The (u', v') point for 4000K is almost coincident for Wien and Planck and so is plotted as a single black point. When temperature increases, two loci diverge such that their infinite temperatures are far from each other, resulting in a visually significant difference in chromaticity [4]. In general, we see mismatch between the Planck and Wien Loci for a given temperature, such that each point on the Wien locus is cooler than its corresponding point on the Planck locus.

To resolve this problem, Deeb et al. [5] modeled Wien temperature from Planckian temperature. They defined a Look-up-table correction function that takes any Planckian temperature and gives its closest Wien counterpart through minimizing the angular error between the two spectra (i.e., Planck and Wien spectra).

To facilitate numerical calculation, let's assume that the infinite-temperature chromaticity of the Planck locus is effectively equal to the chromaticity at 10^{12}K

(see Fig. 1 for details). This very high temperature converted, using the method of [5], to the Wien domain, it is just 32033 K.

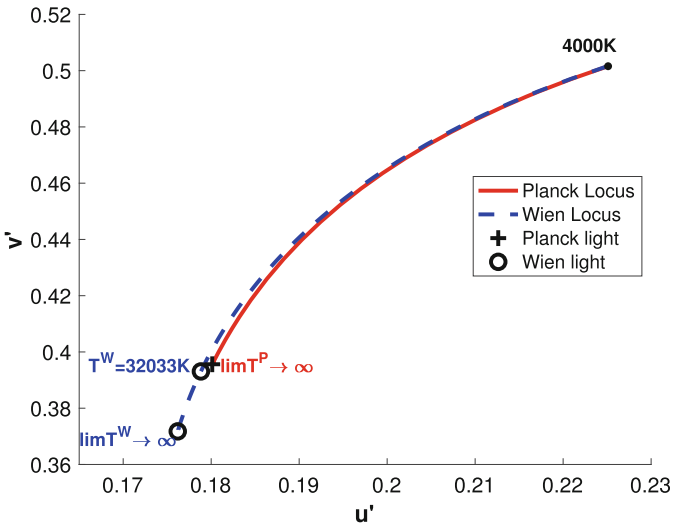


Fig. 2. Wien Locus compared to the Planckian Locus in (u', v') diagram.

Many studies have sought to design a specific filter that changes a given illuminant in the scene to another chosen one e.g., [6–10]. The most related theory to our current research is the Locus Filter theory, which functions similarly to a photography filter, making filtered light to lie on the Wien-Planckian locus [6,11].

In Locus Filter theory, if we have a Wien illuminant with a temperature T_1, applying the Locus filter to that obtains another Wien illuminant with a temperature T_2. By dividing the spectra of these two Wien lights provided by Eq. 2 we obtain a filter transmittance function $F_{Locus}\left(\lambda, T_{lf}\right)$ that has the desired property:

$$F_{Locus}\left(\lambda, T_{lf}\right) = e^{-\frac{c_2}{T_f \lambda}} \tag{3}$$

where the T_{lf} parameter is called *Locus Filter Temperature* (LFT), and is equal to:

$$T_{lf} = \frac{1}{\frac{1}{T_2} - \frac{1}{T_1}} \tag{4}$$

From Eq. 4, the LFT is dependent on two temperatures, T_1 and T_2. Now, lets apply this filter F_{Locus} to a third light that has a temperature T_3 to make a new light spectrum as E^{filt},

$$E^{filt}(\lambda, T) = E(\lambda, T)F_{Locus}(\lambda, T) \tag{5}$$

then the new light E^{filt} is also lies on the Wien locus with temperature T_4 calculated as (see [6] for details):

$$T_4 = \frac{1}{\frac{1}{T_{lf}} + \frac{1}{T_3}} \tag{6}$$

Interstingly, immediate from Eq. 4 is that LFTs can be negative. It follows that when we apply a Locus Filter with negative temperature, then T_4 could be negative dependant on T_3 value. Fortunately Wien law 'works' for negative temperatures. When T_4 in Eq. 6 is negative, we show that the corresponding Wien light spectrum is bluer than the bluest Planckian light possible. An important - and necessary - property of Locus Filter theory - when negative colour temperatures are admissible - is that any given Locus filter must map the entire Wien locus onto itself.

In the next section, we propose a new path - call it Wien Locus - based on the Wien's displacement law to describe such *filtered* lights as well as some available commercial blue lights. The interesting property of this locus is that despite admitting negative temperatures, it is continuous over chromaticity diagrams (except at T = 0).

2 Extending Wien Locus

Let us begin by inserting negative temperatures into the Wien equation. The form of Wien's equation readily admits negative temperatures (see Eq. 2). As an example, we plot the relative energy of a Wien light with the colour temperature of −3000 K in Fig. 3. It shows light with blue colour as its dominant energy is in the shorter wavelengths. This light is bluer than the infinite temperature Planckian and Wien lights.

Now, let's consider the behavior of chromaticities for temperatures with a general domain of $(-\infty, +\infty)$ in Fig. 4. We plotted the spectrum locus according to the CIE colour matching functions, so it starts from left to right from 360 nm and ends at 830 nm. The Colour matching functions are only available for this range of visible spectrum and are available at 1 nanometre sampling [12].

Positive temperatures make the main Wien and Planck loci which start from right to left from monochromatic red (say at 830 nm, as this is the last wavelength of CIE color matching functions [12]), past the orange, white until reaches to the infinite color temperature.

However, for temperatures larger than 32033 K, the Wien equation produces lights even bluer than Planck's for Wien's law. Moreover, and the key concern of this paper, negative colour temperatures produce an extended Wien locus. The extended locus starts from $-\infty$ - at a chromaticity point coincident with $+\infty$ - and ends at the monochromatic blue (or 360 nm that is the first wavelength of CIE color matching functions). Here, a question arises: is this locus *continuous*

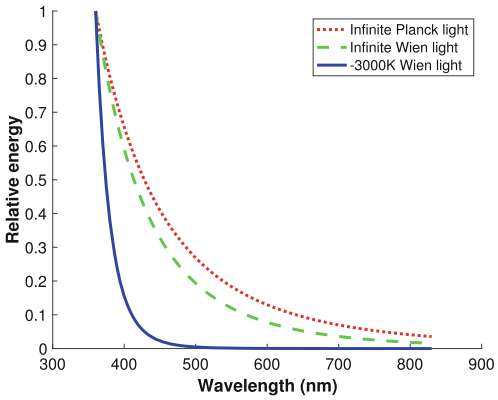


Fig. 3. A Wien light with −3000 K temperature compared to infinite Planck and Wien lights.

everywhere? If yes, what is the intersection point of *positive* (main Wien), and *negative* (extended Wien) loci?

In Subsect. 2.1, we will prove that negative and positive infinite temperatures induce the same point in the chromaticity space. In addition, we will analytically determine the chromaticity point of the infinite temperature of Wien locus.

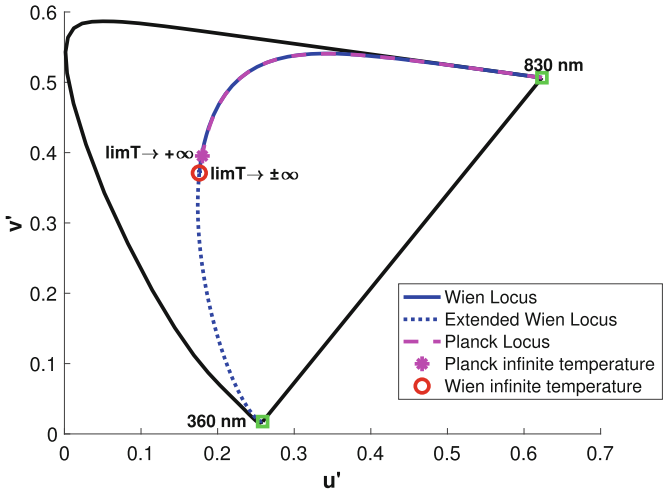


Fig. 4. Extended Wien Locus compared to Planck Locus.

2.1 Wien Locus Continuity

By continuity, we refer to the absence of disruption in the chromaticity path that Wien's law generates. Although the Wien locus is, by this token, mostly well-ordered in T, there are a few exceptions. Indeed, while $\lim_{T\to 0^+}$ and $\lim_{T\to 0^-}$ make two different points (two ends of locus), $\lim_{T\to\infty^+}$ and $\lim_{T\to\infty^-}$ make the same point.

Theorem 1. *Wien locus is a continuous path in an arbitrary chromaticity diagram.*

Proof. Let's rewrite Wien's spectrum (Eq. 2) in terms of absolute inverse temperature:

$$E\,(U,\lambda) = kc_1\lambda^{-5}e^{\frac{-c_2U}{\lambda}} \tag{7}$$

where $U = \frac{1}{T}$.

Now, we calculate the colour signal of a Wien light, E, as:

$$P_i = \int_\omega R_i(\lambda)E(U,\lambda)d\lambda \tag{8}$$

where $R_i(\lambda)$ is the response function of the ith sensor for $m-dimensional$ vision system.

A chromaticity is a colour projected into a $m-1$ dimensional space such that colour signal normalizes to the amount of colour vector $\underline{P}$.

$$p_i(U) = \frac{P_i}{\sum\limits_{j=1}^{m} P_j} \tag{9}$$

By substituting Eq. 7 and Eq. 8 in Eq. 9, we have

$$p_i(U) = \frac{\int_\omega R_i(\lambda)c_1\lambda^{-5}e^{\frac{-c_2U}{\lambda}}d\lambda}{\int_\omega [\sum\limits_{j=1}^{m} R_j(\lambda)\,]\,c_1\lambda^{-5}e^{\frac{-c_2U}{\lambda}}d\lambda} \tag{10}$$

Now, we are interested in finding $\lim\limits_{T\to\infty^{\pm}} p_i$, and this statement is simply equal to find $\lim\limits_{U\to 0^{\pm}} p_i$. Let's assume that color-matching functions - defined as $R(\lambda)$ - of any arbitrary kind of observer are bounded positive functions of visible wavelength over a compact domain that does not include $\lambda = 0$, where Wien shows troublesome behavior (i.e., infinity). Therefore, it is sufficient for continuity arguments that troublesome points (i.e., U = 0) be investigated only for the *general ratio* of Eq. 10. Thus, we define G as the general ratio of Eq. 10:

$$G(U) = \frac{\int_\omega M(\lambda)e^{\frac{-c_2U}{\lambda}}d\lambda}{\int_\omega N(\lambda)e^{\frac{-c_2U}{\lambda}}d\lambda} \tag{11}$$

where $M(\lambda)$ and $N(\lambda)$ are positive weighting functions over the *visible* range. The factor λ^{-5} has been folded into $M(\lambda)$ and $N(\lambda)$ functions to make it clear that we expect no trouble from wavelength. Note that $M(\lambda)$ will be different for each chromaticity coordinate.

To evaluate the limit of Eq. 11, the important part to consider is the exponential, *exp*, in Wien's equation. Remembering we are using U (absolute inverse temperature), the infinite-temperature limit can be conveniently evaluated in the U domain, at U = 0. There is no pathology, such as an indeterminate form or directionality in the limit at U = 0, so direct evaluation at U = 0 suffices. Thus, let's determine the limit of Eq. 7 at U = 0, so we have:

$$\begin{aligned} \lim_{U\to 0^{\pm}} e^{\frac{-c_2 U}{\lambda}} &= 1 \\ \lim_{U\to 0^{\pm}} E\left(U,\lambda\right) &= kc_1\lambda^{-5} \end{aligned} \tag{12}$$

Now we simply calculate the limit of Eq. 11 as

$$\lim_{U\to 0^{+}} G(U) = \frac{\int_{\omega} M(\lambda)d\lambda}{\int_{\omega} N(\lambda)d\lambda} \tag{13}$$

Also the limit of general ratio for $U \to 0^{-}$ is equal to

$$\lim_{U\to 0^{-}} G(U) = \frac{\int_{\omega} M(\lambda)d\lambda}{\int_{\omega} N(\lambda)d\lambda} \tag{14}$$

Finally, for every chromaticity space, we have:

$$\lim_{U\to 0^{+}} p_i = \lim_{U\to 0^{-}} p_i \tag{15}$$

Based on our first assumption, this equation is equal to

$$\lim_{T\to +\infty} p_i = \lim_{T\to -\infty} p_i \tag{16}$$

As both limits are equal, we can almost say that the extended-Wien Locus is continuous in U everywhere in the domain of $(-\infty, +\infty)$ over the chromaticity space. The exception is the infinite limit of U, which is the absolute zero of T. In that limit, there is a jump discontinuity between the $+T$ and $-T$ sides of $T = 0$. In chromaticity space, that jump comprises the ends of the extended-Wien locus.

Now we know that the Wien locus is continuous at the location of infinitely positive and negative color temperatures, let us determine the corresponding chromaticity coordinate.

Theorem 2. *The chromaticity coordinate of the infinite point of Wien Locus is equal to the chromaticity of the light spectrum* λ^{-5}.

Proof. As we calculated the limit of the ratio in Eq. 13 and Eq. 14, now we write the special ratios, at $U = 0$ that are the chromaticity coordinates of the infinite temperature limit of the Wien blackbody. Now, rather than use the M and N functions, we rewrite the ratio in terms of the color matching functions:

$$\lim_{T\to\infty^{\pm}} p_i = \frac{\int_\omega R_i(\lambda)\,\lambda^{-5}d\lambda}{\int_\omega [\sum_{j=1}^{m} R_j(\lambda)\,\lambda^{-5}]\,d\lambda} \tag{17}$$

So, the chromaticity of infinite temperature Wien light is equal to the chromaticity of light spectrum - defined by $\underline{\omega} = [\lambda_1, \lambda_2, ..., \lambda_n]$, a $1 \times n$-vector in the visible range - to the power of -5. Notice that this theorem holds true in any system, even a non-trichromatic one.

When we evaluate this integral we find that the infinite Wien color temperature chromaticity point is (0.17613, 0.37104) in CIE (u', v') diagram.

2.2 Chromaticity of Infinite Temperature Planckian Light

As we determined the chromaticity for an infinite temperature Wien light, we are also interested in defining the infinity point for Planckian black-body radiators.

Theorem 3. *The chromaticity coordinate of the infinite temperature of Planckian black body radiator is equal to the chromaticity of the light spectrum* λ^{-4}.

Proof. In order to find the chromaticity of infinite temperature light, we replace Wien's formula in Eq. 10 with Planck's equation.

$$p_i = \frac{\int_\omega R_i(\lambda)c_1\lambda^{-5}(e^{\frac{c_2U}{\lambda}} - 1)^{-1}d\lambda}{\int_\omega [\sum_{j=1}^{m} R_j(\lambda)\,]\,c_1\lambda^{-5}(e^{\frac{c_2U}{\lambda}} - 1)^{-1}d\lambda} \tag{18}$$

As in the Wien proof, we can further simplify Eq. 18 by writing the general ratio, G as:

$$G(U) = \frac{\int_\omega M(\lambda)(e^{\frac{c_2U}{\lambda}} - 1)^{-1}d\lambda}{\int_\omega N(\lambda)(e^{\frac{c_2U}{\lambda}} - 1)^{-1}d\lambda} \tag{19}$$

where $M(\lambda)$ and $N(\lambda)$ are bounded positive weighting functions over the *visible* range. But equation is not as simple as Wien limit. Thus, one way to simplify this equation is replacing *exp* function by its Maclaurin series expansion (we can use Maclaurin and not Taylor as the point of interest is $U = 0$). Lets write the Maclaurin expansion of *exp* function as

$$e^q = 1 + q + \frac{q^2}{2!} + \frac{q^3}{3!} + ... \tag{20}$$

where $q = \frac{c_2 U}{\lambda}$, also the zero'th-order term is 1, and the first-order term q dominates and remains (as $U \to 0$, as temperature increases). Thus, using Eq. 20 near $U = 0$, we can write

$$e^q \approx 1 + q = 1 + \frac{\lambda}{c_2 U} \approx e^{\frac{c_2 U}{\lambda}} \tag{21}$$

It follows by substituting the Maclaurin series

$$\lim_{U \to 0^+} (e^{\frac{c_2 U}{\lambda}} - 1)^{-1} = \frac{\lambda}{c_2 U} \tag{22}$$

Now, remembering that $U = \frac{1}{T}$ and substituting in Eq. 2 (where we ignore the scaling term i.e. $k = 1$)

$$\lim_{U \to 0^+} E_P\,(U, \lambda) = \frac{c_1 \lambda^{-4}}{c_2 U} \tag{23}$$

Now, we calculate the chromaticity coordinates of the infinite-temperature limit of the Planck black-body equation:

$$\lim_{T \to \infty^+} p_i = \frac{\int_\omega R_i(\lambda)\, \lambda^{-4} d\lambda}{\int_\omega [\sum_{j=1}^{m} R_j(\lambda)\, \lambda^{-4}]\, d\lambda} \tag{24}$$

The final result is similar to the final Wien expression (Eq. 17), except the Wien exponent -5 becomes -4 in the Planck case. Accordingly, the chromaticity (in any system, even a non-trichromatic one) of the infinite-temperature Planck radiator is the chromaticity of λ^{-4}.

When we evaluate this integral we find that the infinite Planckian color temperature chromaticity point is $(0.18006, 0.39528)$ in CIE (u', v') diagram. Notice that we only show 5 decimal numbers to demonstrate that our derivation recovers the same chromaticty as Robertson [1]. It is entirely possible that Robertson followed a similar line of reasoning to ourselves. In this case, we are happy we can make his derivation available to the community.

3 Experiments

3.1 Finding Limit Points of the Wien Locus

Based on our proof, we determine the chromaticity coordinates of both positive and negative infinite temperature in the 2-dimensional CIE (u', v') diagram and show in Fig. 5. We see that positive and negative loci connects at the same (u', v') point (0.17613, 0.37104), shown as a red circle. So, the locus continuously starts from right at monochromatic red and ends at the left side to achieve a monochromatic blue. Indeed, Wien locus from right to left the temperature monotonically increases from $\lim_{T \to 0^+}$ to $\lim_{T \to \infty^+}$. Also, in the extended Wien Locus, from left to right the temperature decreases from $\lim_{T \to 0^-}$ to $\lim_{T \to \infty^-}$

to reach the original Wien Locus at infinity. Regarding the chromaticities for $T \to 0^+$ and $T \to 0^-$, these are indicated as points respectively at 830 and 360 nanometres. We will prove elsewhere in the future that as the temperature decreases towards 0 the limits must be the chromaticities recorded for Delta functions at the max and min wavelengths over which the spectral sensitivities of the visual system are defined.

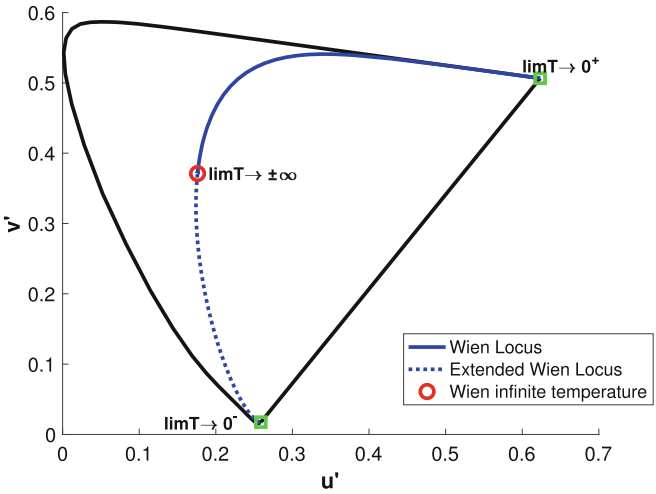


Fig. 5. Wien Locus and its extension.

3.2 Locus Filtered Lights

In Fig. 6, we show an example of chromaticity shift in the (u', v') diagram after applying a Locus Filter to a real light. To do this, we select a random real light from the Granada Daylights dataset [13]. Then we apply a Locus Filter with a temperature of $-5000\,\mathrm{K}$ (a negative locus filter temperature, see Eq. 4) to this Granada light using Eq. 5. We plot the chromaticities of the original light (red circle) and the filtered counterpart (red cross) in (u', v') diagram. It is clear that the filtered light goes beyond the Planck locus (endpoint for an infinite temperature shown as a blue square), while it can still be accounted for by our extended-Wien locus. Also shown in the (u', v') diagram are chromaticities on the Wien locus that are closest to the actual and filtered Granada lights. In Fig. 7, we show the same Granada light modulated by a Locus filter (with a positive) $5000\,\mathrm{K}$ locus filter temperature.

3.3 Describing the Colour of Commercial Lamps

As a case study, we selected the spectral power distribution (SPD) of a blue lamp from a dataset of spectra of commercial lamps [14–16]. This dataset includes

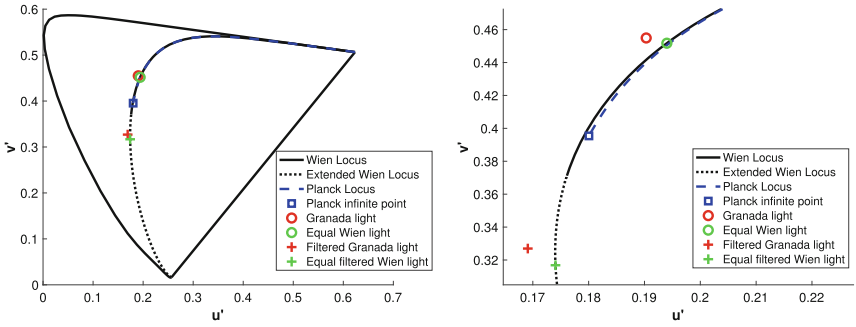


Fig. 6. (a) Change of correlated color temperature with a Locus filter temperature of -5000 over Wien Locus in CIE (u', v') diagram, (b) Zoomed-in. (Color figure online)

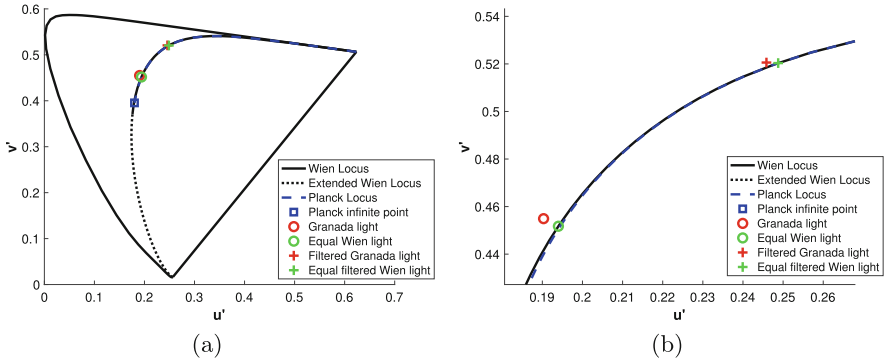
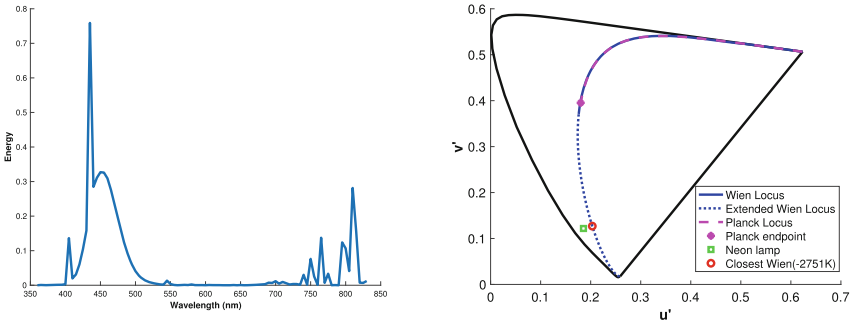


Fig. 7. (a) Change of correlated color temperature using a Locus filter with temperature of 5000 K over Wien Locus in CIE (u', v') diagram, (b) Zoomed in. (Color figure online)

various types of real lights such as LED, Metal Halide, and Neon lights. Figure 8 (a) shows the energy of a Neon blue lamp; its peak indicates that this is a blue light. Now, we plot this lamp on the CIE (u', v') diagram as green square. It is clear that this point is very close to the extended Wien Locus, while being far from the Planckian locus. To elaborate on this, we found the closest Wien light to this lamp according to the minimum $D_{u'v'}$ metric. Thus, the closest Wien light (red circle in Fig. 8(b)) to this Neon light has only $D_{u'v'} = 0.0184$ that is good approximation. We also found the closest Planckian to this lamp, which is the endpoint of the Planckian locus (i.e., $\lim_{T\to\infty^+}$), with $D_{u'v'} = 0.2739$, indicating noticeable color difference [4].

Generally, real lights are manufactured so their integrated (u', v') chromaticities fall near the Planckian locus. Commonly filters are applied to change the colour of the light and amongst these filters are those that are bluish in colour (and in terms of locus filter theory can be thought of as having a negative colour

temperature). The extended locus provides an elegant, single parameter, description of the colour of the filtered light (the positive or negative temperatures).



(a) (b)

Fig. 8. (a) Spectral power distribution of a blue commercial lamp, (b) CIE (u', v') chromaticity point of lamp. (Color figure online)

4 Conclusion

The Planckian locus 'stops' for infinite colour temperature. The infinite temperature Planckian light spectrum, though bluish, is far from being either the bluest blue commercially available or theoretically admissible light. The Wien locus - which runs (almost) coincident to the Planckian locus - only accounts for slightly bluer lights than those allowed by Planck's equation. In this paper we developed the Wien extended locus which can account for almost all typical lights from monochromatic red, through oranges, yellows whites, blues to monochromatic blue. A key notion in our work is the idea of negative colour temperature. The equation defining the Wien displacement law naturally admits positive and negative temperatures in a way that concomitantly extends the locus. The extended locus allows us to describe many more commercially available lights by their colour temperatures than is currently possible and, moreover, allows us to always determine the colour temperature of a light that is filtered by a bluish correction filter (our extended locus is a natural requirement of Locus Filter Theory).

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Uniform Color Space with Advanced Hue Linearity: PCS23-UCS

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Abstract. In device-dependent color spaces, the Euclidean distance often fails to accurately represent perceived color differences, making these color spaces unsuitable for image similarity measurements. Conversely, certain device-independent color spaces, known as uniform color spaces (UCS), do align the Euclidean distance with perceived color differences. Among these, CAM16-UCS is currently considered the leading model. However, applications such as gamut mapping require additional properties, such as the preservation of planes of constant hue, which CAM16-UCS lacks. We present PCS23-UCS, a novel UCS that addresses this limitation. PCS23-UCS is specifically designed to preserve the central bundle of planes in the color space while maintaining state-of-the-art uniformity for small color differences.

Keywords: Uniform color space · Perceptual uniformity · Hue linearity

1 Introduction

Manufacturers of televisions, smartphones, and cameras prioritize precise color reproduction, highlighting the importance of evaluating image quality in the development of visualization algorithms. This evaluation aims to establish a metric for reproduction accuracy. Quality assessment methods must explicitly incorporate characteristics of the human visual system, which is crucial for human end-users. Utilizing uniform color spaces (UCS) is common in modeling human color perception, as they ensure that Euclidean color distances linearly reflect perceived color differences. In a UCS, the lengths of threshold (just noticeable) difference vectors should be consistent across all points and directions.

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The CIE XYZ color space, derived from human cone spectral sensitivity functions, is standard in colorimetry but highly non-uniform. Although CIELAB, established in 1976 as a UCS [20], remains widely used, it exhibits imperfect uniformity. Numerous efforts have been dedicated to creating more uniform color spaces [2,5,13,15,25,27]. CAM16-UCS, introduced in 2016 [14], is currently the leading UCS [18].

In 2021, proLab emerged as a novel UCS [12], though its uniformity is significantly lower than that of CAM16-UCS. The main advantage of proLab UCS is that it maintains additional geometric properties besides uniformity. Defined by a projective transformation of CIE XYZ, proLab preserves the linearity of color manifolds.

This study aims to develop a novel UCS with geometric properties similar to proLab and uniformity comparable to CAM16-UCS. The insufficient uniformity of proLab arises from the strict requirement to preserve all linear manifolds. Improving its uniformity necessitates relaxing this geometric constraint. We identify the bundle of color planes passing through the UCS origin as pivotal manifolds to preserve in UCS practical applications. Our novel model, PCS23-UCS, extends proLab, achieving state-of-the-art uniformity for small color differences while maintaining the linearity of these key manifolds.

1.1 UCS Requirements

Beyond perceptual uniformity, UCSs must possess specific geometric properties to address practical challenges in contemporary television, smartphone, and camera manufacturing industries. Safdar and Luo's paper summarizes essential criteria for modern UCSs, including local and global uniformity, as well as geometric requisites such as perceptual hue linearity, neutral point convergence for unique hues, and minimal neutral point error [28].

Additional UCS requirements stem from the necessity to perform various color transformations within these spaces. One significant type of such transformations is gamut mapping, which involves converting colors from a wider gamut to a smaller gamut of a target display while preserving the original content's information and colorfulness. Gamut mapping typically involves a fixed transformation rule on the chromaticity diagram or the lightness-chroma plane of constant hue in specific UCSs [32,33], such as altering hue along a line passing through the neutral point. Xu et al. [33] explore gamut mapping algorithms employing various UCSs and identify issues like hue shifts and non-orthogonality of lightness axis to chroma axes. Consequently, gamut mapping requires additional UCS properties like lightness scalability, hue linearity, independence of saturation and lightness, and convexity of the gamut. Hue linearity means that colors of the same perceived hue form a straight line on the chromaticity diagram. These properties are crucial not only for gamut mapping but also for other color reproduction tasks, such as tone mapping. Hue linearity is also beneficial for color camera calibration [19] and image enhancement via saturation improvement [9].

In this paper, we focus on two fundamental properties of UCSs: uniformity and hue linearity, while foregoing the preservation of all linear manifolds.

The paper is organized as follows. Section 2 provides a review of relevant psychophysical datasets. Section 3 outlines measures for assessing UCS quality using these datasets. Section 4 introduces our proposed UCS model, PCS23-UCS, while Sect. 5 details the optimized parameters of PCS23-UCS. Section 6 presents quantitative results for the PCS23-UCS model with optimized parameters. Finally, Sect. 7 summarizes the results of the study.

2 Psychophysical Datasets

Psychophysical datasets are crucial for evaluating the perceptual properties of UCS. For this study, we consider two types of datasets: constant hue datasets and color difference datasets. The former is used to estimate the uniformity of UCSs, while the latter is used to assess the hue linearity of UCSs. We will now discuss each type in detail.

2.1 Color Difference Datasets

Psychophysical experiments on perceived color differences aim to ascertain metric parameters of color spaces. These experiments measure the threshold changes in stimuli perceived by the human eye across various points in the color space. Typically, such datasets consist of triplets: two colors defined by CIE XYZ coordinates and the value of the perceived distance between them.

The widely utilized color difference dataset, COMBVD meta-dataset [16], comprises small color differences (less than 5 CIELAB units). It consists of four sub-datasets: BFD-P [17], Witt [30], Leeds [11], and RIT-DuPont [1]. The number of color pairs differs across these datasets: 2779 in BFD-P, 418 in Witt, 312 in RIT-DuPont, and 307 in Leeds.

Another widely used color difference dataset is the Munsell dataset [10,23,24]. The Munsell dataset captures large color differences across a wider color gamut compared to COMBVD. In this study, the COMBVD dataset is employed for model tuning and validation, while the Munsell dataset is utilized to regularize the model behavior in color regions not covered by COMBVD.

Table 1. Viewing conditions parameters for experimental CD datasets.

Dataset	Illuminant	L_a	Y_b	Surround
BFD-P	D65, M, C	100	20	average
RIT-DuPont	D65	127.3	10.9	average
Leeds	D65	100	20	average
Witt	D65	82.8	24.9	average
Munsell	C	64	20	average

Table 1 summarizes the viewing conditions (illuminant, the adapting luminance L_a, the luminance factor of the neutral background Y_b, and the surround) for each dataset.

2.2 Constant Hue Datasets

To evaluate the perceptual hue linearity of UCSs, datasets consisting of colors with constant perceived hue are employed. Three widely used and publicly available datasets include those by Hung and Berns [8], Ebner and Fairchild [3], and Xiao et al. [31].

The Hung and Berns dataset [8] comprises 132 test stimuli with different saturation levels, of which only 48 have constant lightness (12 different hues at 4 saturation levels). However, due to significant observer variations in the subset with variable lightness, studies commonly utilize only the subset with constant lightness [27,35].

The Ebner and Fairchild dataset [3] consists of colors with 15 different hues and 21 lightness-saturation combinations per hue. The stimuli used in this study had low and medium saturation levels.

Xiao et al. [31] employed color stimuli with consistent saturation and lightness levels, using 9 different combinations of lightness and chroma to scale each unitary hue. The authors noted discrepancies between their dataset and others (Hung and Berns, Ebner and Fairchild) when testing UCS performance. This difference may be attributed to Xiao et al. using a different procedure that allowed observers to provide more consistent responses [35].

In our current study, we use these three datasets – Hung and Berns, Ebner and Fairchild, and Xiao et al. – to assess the hue linearity property of different UCSs. Table 2 summarizes the viewing conditions (illuminant, the adapting luminance L_a, the luminance factor of the neutral background Y_b, and the surround) for these datasets.

Table 2. Viewing conditions parameters for experimental hue linearity datasets.

Dataset	Illuminant	L_a	Y_b	Surround
Hung and Berns	C	10	20	dark
Ebner and Fairchild	D65	14.2	35	dark
Xiao et al.	CIE XYZ = (98.0, 100, 139.7)	23	20	dim

3 UCS Quality Measures

3.1 Perceptual Uniformity Measure

Perceptual uniformity in a color space refers to how accurately Euclidean distances within that space approximate perceived color differences. The STRESS

measure is commonly used to quantify nonuniformity: higher STRESS values indicate poorer uniformity [6,26,29]. However, when working with multiple datasets like the COMBVD meta-dataset, the scales of CDs may vary, and their ratio is unknown. Consequently, the original STRESS measure cannot be directly applied. Various modifications have been proposed by different authors [14,21], but these do not match with the classic STRESS when CD scales are consistent. In a recent paper [24], a modified version of STRESS, $\text{STRESS}_{\text{group}}$, was introduced, which additionally satisfies this property.

3.2 Hue Linearity Measure

To determine the overall error of hue prediction in a UCS, one commonly employed measure is the mean standard deviation (SD) of hue angles [27,35]. The SD measure is computed between the hue angles of all colors with the same perceived hue and is expressed in degrees.

4 Proposed Color Space

The proposed model combines two frameworks: the projective transformation underlying proLab [12] and the polynomial expansion used by Finlayson for color correction [4].

4.1 ProLab Model Preserving All Linear Manifolds

In [12], the proLab UCS, preserving the linearity of manifolds, was introduced. The proLab model is defined by a zero-preserving projective transformation of CIE XYZ:

$$\begin{bmatrix} L_p \\ a_p \\ b_p \end{bmatrix} = \mathrm{T_c}(P\mathrm{T_h}(\mathbf{c_x})), \tag{1}$$

where $\begin{bmatrix} L_p & a_p & b_p \end{bmatrix}^T$ represents color coordinates in proLab space, $\mathbf{c_x} = \begin{bmatrix} X & Y & Z \end{bmatrix}^T$ denotes a vector of color coordinates in CIE XYZ preprocessed using a simplified von Kries chromatic adaptation transform (as in CIELAB). $\mathrm{T_h}$ and $\mathrm{T_c}$ are functions for transforming between Cartesian and homogeneous coordinates:

$$\mathrm{T_h}(\mathbf{c}) = \begin{bmatrix} \mathbf{c} \\ 1 \end{bmatrix}, \mathrm{T_c}(\mathbf{h}) = \begin{bmatrix} I_n & \mathbf{0} \end{bmatrix} \frac{\mathbf{h}}{h_{n+1}}, \ \mathbf{c} \in \mathbb{R}^n, \mathbf{h} \in \mathbb{R}^{n+1}, \ h_{n+1} \neq 0, \tag{2}$$

where I_n represents the 3×3 identity matrix, and P is a projective transformation matrix:

$$P = \begin{bmatrix} p_{11} & p_{12} & p_{13} & 0 \\ p_{21} & p_{22} & p_{23} & 0 \\ p_{31} & p_{32} & p_{33} & 0 \\ p_{41} & p_{42} & p_{43} & 1 \end{bmatrix}. \tag{3}$$

The proLab model is a low-parameter model and quite rigid, preserving all linear manifolds, which restricts its perceptual uniformity. We propose a model that generalizes proLab to more accurately describe perceived color differences while preserving only the most relevant linear manifolds for practical applications.

4.2 PCS23-UCS Model Preserving the Central Bundle of Planes

Let us examine a category of transformations preserving a bundle of planes passing through the origin. Such bundle in CIE XYZ coordinates is transformed into planes parallel to the Y-axis in CIE xyY coordinates. Therefore, a general transformation preserving the bundle of planes passing through the origin in CIE XYZ can be represented as follows in CIE xyY:

$$\begin{bmatrix} L^+ \\ x^+ \\ y^+ \end{bmatrix} = \begin{bmatrix} \mathrm{f}(x, y, Y) \\ \mathrm{H}\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) \end{bmatrix}, \tag{4}$$

$$\begin{bmatrix} L^+ \\ a^+ \\ b^+ \end{bmatrix} = L^+ \begin{bmatrix} 1 \\ x^+ \\ y^+ \end{bmatrix}, \tag{5}$$

where $\begin{bmatrix} x\ y\ Y \end{bmatrix}^T$ are CIE xyY color coordinates, $\mathrm{H}\colon \mathbb{R}^2 \to \mathbb{R}^2$ is a projective transformation, and $\mathrm{f}\colon \mathbb{R}^2 \times \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$.

We propose employing projective transformations of polynomial expansions as the f function. Polynomial expansions of various degrees were initially proposed as calibration transformations from camera color space to the standard observer color space [4,7]. For instance, the polynomial expansion of the third degree for the vector $\begin{bmatrix} x\ y\ y/Y \end{bmatrix}^T$ is as follows:

$$\begin{aligned} \boldsymbol{\rho}_{3,3}(x, y, \frac{y}{Y}) = [&x,\ y,\ \frac{y}{Y},\ x^2,\ y^2,\ \left(\frac{y}{Y}\right)^2,\ xy,\ x\frac{y}{Y},\ y\frac{y}{Y},\ x^3,\ y^3,\ \left(\frac{y}{Y}\right)^3, \\ &xy^2,\ x\left(\frac{y}{Y}\right)^2,\ yx^2,\ y\left(\frac{y}{Y}\right)^2,\ \frac{y}{Y}x^2,\ \frac{y}{Y}y^2,\ xy\frac{y}{Y},\ 1]^T. \end{aligned} \tag{6}$$

With the introduced notation, the proposed class of transformations in the CIE xyY space can be expressed as:

$$\begin{aligned} &\begin{bmatrix} L^+ \\ x^+ \\ y^+ \end{bmatrix} = \begin{bmatrix} \mathrm{T_c}(V\mathrm{T_h}(\boldsymbol{\rho}_{3,3}(x, y, y/Y))) \\ \mathrm{T_c}(H\mathrm{T_h}(\begin{bmatrix} x\ y \end{bmatrix}^T)) \end{bmatrix}, \begin{bmatrix} L^+ \\ a^+ \\ b^+ \end{bmatrix} = L^+ \begin{bmatrix} 1 \\ x^+ \\ y^+ \end{bmatrix}, \\ &V = \begin{bmatrix} v_{1,1}\ v_{1,2}\ \dots\ v_{1,20} \\ v_{2,1}\ v_{2,2}\ \dots\ 1 \end{bmatrix}, H = \begin{bmatrix} h_{11}\ h_{12}\ h_{13} \\ h_{21}\ h_{22}\ h_{33} \\ h_{31}\ h_{32}\ 1 \end{bmatrix}, \end{aligned} \tag{7}$$

Equation (7) defines a transformation class PCS23-UCS (Polynomial Conical Space of 2023, with UCS denoting Uniform Color Space of general purpose,

similar to CAM16-UCS), which preserves the central bundle of planes. Notably, the proLab model is a special case of the PCS23-UCS model.

This property allows for the introduction of chromaticity coordinates a^+ and b^+ in the PCS23-UCS color space in a very natural way:

$$\begin{bmatrix} x^+ \\ y^+ \end{bmatrix} = \frac{1}{L^+} \begin{bmatrix} a^+ \\ b^+ \end{bmatrix}. \tag{8}$$

Additional Restrictions on PCS23-UCS. Firstly, PCS23-UCS must preserve the origin, achieved when the V matrix element $v_{1,12}$ associated with the $(y/Y)^3$ monomial, is zero. Secondly, the general PCS23-UCS transformation does not ensure invertibility and continuity. Let us outline conditions ensuring its invertibility and continuity within the visible color gamut.

Continuity and invertibility of PCS23-UCS demand both $\mathrm{f}(x, y, Y)$ and $\mathrm{H}(x, y)$ to be continuous and invertible. The function $\mathrm{f}(x, y, Y)$ is continuous and invertible on a segment $Y \in [a, b]$ for specified (x, y) if it is monotonic. For the homography H to be continuous and invertible, its Jacobian determinant must be positive. Let Ω represent the set of chromaticities within the visible color gamut under a specific illuminant. Thus, the PCS23-UCS transformation is continuous and invertible within the visible color gamut if two conditions are met:

$$\begin{cases} \forall (x, y) \in \Omega,\ \forall\ Y_1, Y_2 \in [0, 1] : Y_1 < Y_2 \Rightarrow f(x, y, Y_1) < f(x, y, Y_2), \\ \forall (x, y) \in \Omega : \det(\mathbf{J}_{\mathrm{H}(x,y)}) > 0. \end{cases} \tag{9}$$

We verify the first condition from system (9) on a uniform grid defined in the CIE xyY space. The grid is constructed as follows: let N_c points be specified on the plane $Y = Y_{min}$ in the CIE xyY space within the color triangle, such that the distance between two neighboring points along the x or y axis is s_c. Y_{min} is chosen close to zero. From each such point, $N_l - 1$ points are marked along the Y axis up to the value $Y = 1$, with a step $s_l = (1 - Y_{min})/(N_l - 1)$.

We introduce a quantity characterizing the monotonicity of the function f, the sum of negative numerical derivatives θ, normalized by the number of nodes in the grid:

$$\theta = \frac{1}{N_c N_l} \sum_{i=1}^{N_c} \sum_{j=1}^{N_l} \max\left(0, -\frac{L^+_{i,j+1} - L^+_{i,j}}{Y_{j+1} - Y_j} + \epsilon\right). \tag{10}$$

where $L^+_{i,j} = \mathrm{f}(x_i, y_i, Y_j)$ and $\epsilon > 0$ penalize the derivatives close to zero. This helps to rigorously specify the requirement of increasing $\mathrm{f}(x, y, Y)$ during optimization. We will consider the function f to strictly increase with Y for any $(x, y) \in \Omega$ if $\theta = 0$.

Orientation of the PCS23-UCS Model. Hereafter, we assume that the vector $\mathbf{c_x}$ has been normalized to the white point using the chromatic adaptation

transform (simplified von Kries, CAT16, or another), so the normalized white point coordinates are $\mathbf{c}^*_{\mathbf{x}} = \begin{bmatrix}1\ 1\ 1\end{bmatrix}^T$.

To ensure that the white point $\mathbf{c}^*_{\mathbf{x}}$ in the PCS23-UCS space has coordinates $\begin{bmatrix}100\ 0\ 0\end{bmatrix}^T$, the PCS23-UCS transformation (denoted as $\mathrm{C}(\mathbf{c_x})$) can be complemented with scaling and axis rotation L^+, a^+, b^+ which do not alter the metric properties of the space:

$$\hat{\mathrm{C}}(\mathbf{c_x}) = R_b R_a K \mathrm{C}(\mathbf{c_x}), \tag{11}$$

where K is a scaling matrix, R_a, R_b are rotation matrices around axes a^+ and b^+, respectively. Let the coordinates of the white point after the transformation C be denoted as $\mathbf{c}^* = \begin{bmatrix}L^w\ a^w\ b^w\end{bmatrix}^T$. Then the matrices K, R_a, R_b are defined as follows:

$$K = \frac{100}{\|\mathbf{c}^*\|_2} I_3, \quad R_a = \begin{bmatrix}\cos\alpha & 0 & \sin\alpha \\ 0 & 1 & 0 \\ -\sin\alpha & 0 & \cos\alpha\end{bmatrix}, \quad R_b = \begin{bmatrix}\cos\beta & \sin\beta & 0 \\ -\sin\beta & \cos\beta & 0 \\ 0 & 0 & 1\end{bmatrix}, \tag{12}$$

where $\alpha = \arctan 2(b^w, L^w)$, $\beta = \arctan 2(\tilde{a}^w, \tilde{L}^w)$, $\tilde{a}^w$ and $\tilde{L}^w$ are a^+ and L^+ components of vector $R_a\mathbf{c}$, respectively.

Chromatic Adaptation in PCS23-UCS. The prepossessing step in PCS23-UCS, akin to proLab and CIELAB models, involves a chromatic adaptation transform. Specifically, the PCS23-UCS model employs the CAT16 model [14] for this purpose.

5 Optimized Parameters for PCS23-UCS

The PCS23-UCS model comprises a total of 49 parameters: 40 in matrix V and 9 in matrix H. However, to maintain the origin, $v_{1,12}$ must be zero. Additionally, since the projective transformation is defined up to a nonzero scaling factor, we set the lower right elements in the matrices V and H ($v_{2,20}$ and $h_{3,3}$, respectively) to one. This leaves 46 parameters for optimization, hereinafter denoted as $\mathbf{p}$.

To optimize PCS23-UCS for perceptual uniformity, we employed the COMBVD dataset. The set of experimental color differences in COMBVD is denoted as $\Delta V = \langle \mathbf{\Delta V_1} \dots \mathbf{\Delta V_6} \rangle$, where $\mathbf{\Delta V_i}$ represents the color difference vector for the ith subset ($i \in \overline{1,6}$). Similarly, the set of differences computed by the PCS23-UCS model is denoted as $\Delta E = \langle \mathbf{\Delta E_1} \dots \mathbf{\Delta E_6} \rangle$. The $\mathrm{STRESS}_{\mathrm{group}}$ between the sets ΔV and ΔE is denoted as $U[\Delta V, \Delta E]$.

COMBVD primarily comprises low saturated colors within the sRGB triangle and lacks coverage of the entire color triangle. To mitigate the overfitting, we enriched the training dataset with chains of equally spaced saturated colors from the Munsell dataset [24]. These chains varied in brightness (Value) while maintaining constant perceived Hues and Chromas. A total of 65 chains with Chroma in the range [18, 32] were selected. Similarly, let $U[\Delta V_m, \Delta E_m]$ denote the $\mathrm{STRESS}_{\mathrm{group}}$ between the sets of experimental color differences in Munsell, ΔV_m, and those computed by PCS23-UCS, ΔE_m.

To ensure the continuity and invertibility of PCS23-UCS, conditions (9) must be satisfied. We substitute the first condition with the requirement that θ (10) equals zero. θ was computed on a grid in the CIE xyY space with steps $s_c = 0.0344$ and $s_l = 0.0204$ (total of 13650 nodes). During optimization, the invertibility of H was not enforced but was numerically validated for the final model iteration.

The parameters of PCS23-UCS $\mathbf{p}^{opt}$ were optimized using the following equation:

$$\mathbf{p}^{opt} = \underset{\mathbf{p}\in\mathbb{R}^{46}}{\operatorname{argmin}}\, U[\Delta V, \Delta E(\mathbf{p})] + \; w_m U[\Delta V_m, \Delta E_m(\mathbf{p})] + w_\theta \theta(\mathbf{p}, \epsilon). \tag{13}$$

Weights were selected as $w_m = 0.8$, $w_\theta = 0.1$, and the minimum value of the numerical derivative at grid nodes as $\epsilon = 0.02$. Multi-start optimization employed the Nelder-Mead method [22], selecting the best optimization from multiple runs with different initial approximations. The initial guesses for parameters in each optimization were chosen from a uniform distribution in the range $[-100; 100]$.

The optimized parameters for matrices V and H are presented in Table 3 and expression (14), respectively.

Table 3. Parameters for matrix V from the lightness transformation $\mathrm{f}(x, y, Y)$. The column headings indicate the monomials where the factors from the projective transformation matrix V occur. The parameters are divided into two groups: for the first and second rows of the V matrix.

1st row	x	y	$\frac{y}{Y}$	x^2	y^2	$(\frac{y}{Y})^2$	xy	$y(\frac{y}{Y})$	$x(\frac{y}{Y})$	x^3
	−10.359	0.865	−0.179	72.352	58.373	0.178	−82.239	5.374	−0.109	−34.927
	y^3	$(\frac{y}{Y})^3$	xy^2	$x(\frac{y}{Y})^2$	$y(\frac{y}{Y})^2$	$y^2(\frac{y}{Y})$	x^2y	$x^2(\frac{y}{Y})$	$xy(\frac{y}{Y})$	1
	−6.896	0	−28.230	0.020	−0.075	−7.943	18.296	−0.317	0.626	0.913
2nd row	x	y	$\frac{y}{Y}$	x^2	y^2	$(\frac{y}{Y})^2$	xy	$y(\frac{y}{Y})$	$x(\frac{y}{Y})$	x^3
	0.831	9.619	3.543	41.052	−15.804	0.621	−111.849	−15.445	−2.655	24.607
	y^3	$(\frac{y}{Y})^3$	xy^2	$x(\frac{y}{Y})^2$	$y(\frac{y}{Y})^2$	$y^2(\frac{y}{Y})$	x^2y	$x^2(\frac{y}{Y})$	$xy(\frac{y}{Y})$	1
	110.343	0.126	0.727	0.359	−1.713	116.235	20.871	71.751	−145.501	1

$$H = \begin{bmatrix} 1.741 & -6.825 & 1.544 \\ 5.480 & -1.227 & -1.160 \\ -0.436 & -0.475 & 1 \end{bmatrix}. \tag{14}$$

The matrices responsible for the subsequent scaling and rotation of the space, along with the parameters required for their computation, are provided in (15)–(18).

$$\mathbf{c}^* = \begin{bmatrix} 5.880 & -1.272 & 2.176 \end{bmatrix}, \quad K = \begin{bmatrix} 15.631 & 0 & 0 \\ 0 & 15.631 & 0 \\ 0 & 0 & 15.631 \end{bmatrix}, \tag{15}$$

$$\alpha = 0.354, \quad R_a = \begin{bmatrix} 0.938 & 0 & 0.347 \\ 0 & 1 & 0 \\ -0.347 & 0 & 0.938 \end{bmatrix}, \tag{16}$$

$$\beta = -0.200, \quad R_b = \begin{bmatrix} 0.980 & -0.199 & 0 \\ 0.199 & 0.980 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \tag{17}$$

$$R_b R_a K = \begin{bmatrix} 14.367 & -3.109 & 5.316 \\ 2.915 & 15.319 & 1.079 \\ -5.425 & 0 & 14.660 \end{bmatrix}. \tag{18}$$

6 Performance of PCS23-UCS

The proposed PCS23-UCS model offers the following advantage compared to the state-of-the-art models, CAM16-UCS or CAM16-SCD. By design, PCS23-UCS maintains the central bundle of planes, including constant-hue planes, and preserves color cone as convex cone.

We evaluate PCS23-UCS against various UCS models on psychophysical datasets, including the widely used CIELAB [20], the current state-of-the-art CAM16-UCS and CAM16-SCD [14], and proLab [12], which boasts good geometric properties. During the testing of UCS on various datasets, the corresponding white points from Table 1 and Table 2 were used. The adapting luminance L_a, the luminance factor of the neutral background Y_b, and the surround were

Table 4. UCS Uniformity ($\text{STRESS}_{\text{group}}$ values, lower is better).

	COMBVD	BFD-P D65	BFD-P M	BFD-P C	RIT-DuPont	Witt	Leeds
CAM16-SCD	0.295	0.256	0.353	0.304	0.221	0.311	0.228
CAM16-UCS	0.305	0.274	0.357	0.3	0.197	0.323	0.254
PCS23-UCS	0.311	0.289	0.325	0.379	0.300	0.381	0.332
CIELAB	0.426	0.41	0.432	0.55	0.334	0.517	0.4
proLab	0.441	0.452	0.431	0.488	0.302	0.519	0.394

Table 5. UCS Hue Nonlinearity (SD units, lower is better).

	Mean	Xiao et al.	Ebner and Fairchild	Hung and Berns
PCS23-UCS	2.45	2.73	3.83	3.24
proLab	2.48	2.89	3.70	3.35
CIELAB	2.68	2.67	3.59	3.75
CAM16-SCD	3.28	3.96	3.96	4.15
CAM16-UCS	3.28	3.96	3.96	4.15

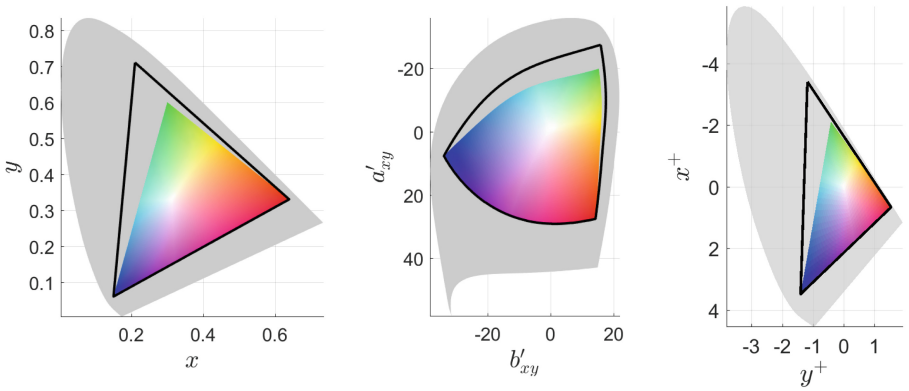

Fig. 1. Chromaticity diagrams showing the gamut of the standard illuminant D65 across various color spaces. The sRGB color gamut is depicted in colors, while the Adobe RGB 1998 gamut is outlined in black. The gray background represents the remaining colors within the color cone.

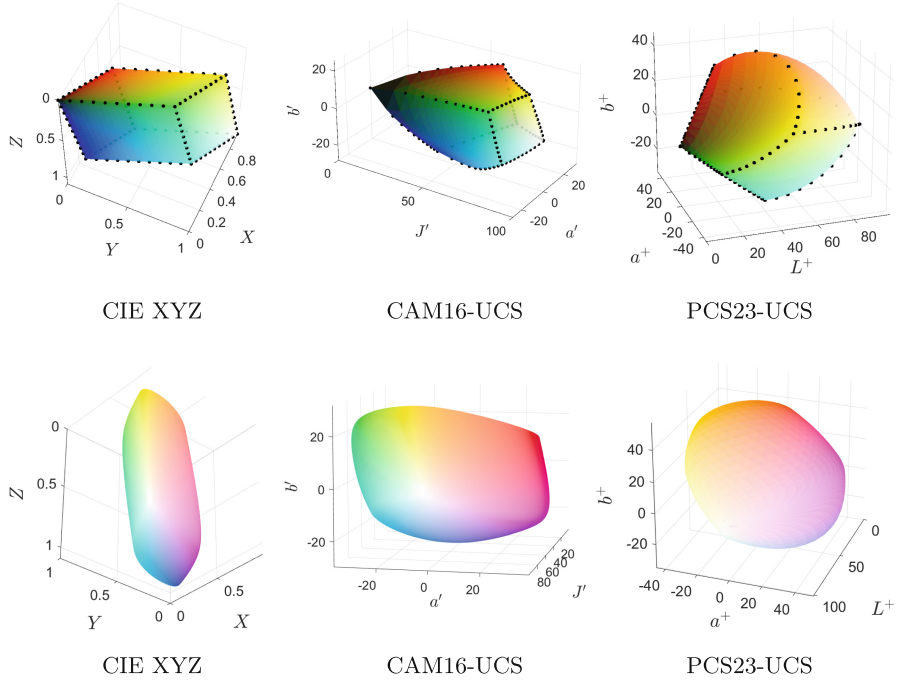

Fig. 2. Gamuts of sRGB (top) and D65 illuminant (bottom) in various color spaces.

also set according to these tables for CAM16-UCS, CAM16-SCD, and PCS23-UCS. Other observation conditions under which the datasets were obtained were not additionally considered, following the practice of previous researchers [34]. The uniformity and hue linearity of these UCS models are presented in

Tables 4 and 5, respectively. PCS23-UCS demonstrates higher hue linearity and nearly equivalent perceptual uniformity compared to the state-of-the-art CAM16-UCS. We have demonstrated that although achieving high perceptual uniformity conflicts with the projectiveness of the proLab color space transformation, it remains consistent with preserving the hue linearity of PCS23-UCS.

Additionally, we compared the chromaticity diagrams of several color spaces (see Fig. 1): the classical CIE xy, the chromaticity diagram of CAM16-UCS, and PCS23-UCS. In PCS23-UCS, chromatic coordinates can be introduced in the standard way: $x^+ = a^+/L^+, y^+ = b^+/L^+$). In CAM16-UCS, chromatic coordinates cannot be introduced in the same manner because straight lines do not represent sets of colors with the same chromaticities. The most straightforward method to introduce chromatic coordinates in CAM16-UCS is as follows: $a'_{xy} = a'(t\mathbf{c_x})$, $b'_{xy} = b'(t\mathbf{c_x})$, where J', a', and b' are coordinates in CAM16-UCS, $\mathbf{c_x} = \begin{bmatrix} x & y & 1-x-y \end{bmatrix}^T \in C_x$, and the scalar $t > 0$ is such that $J'(t\mathbf{c_x}) = 50$. The PCS23-UCS chromaticity diagram preserves the convexity of the color triangle

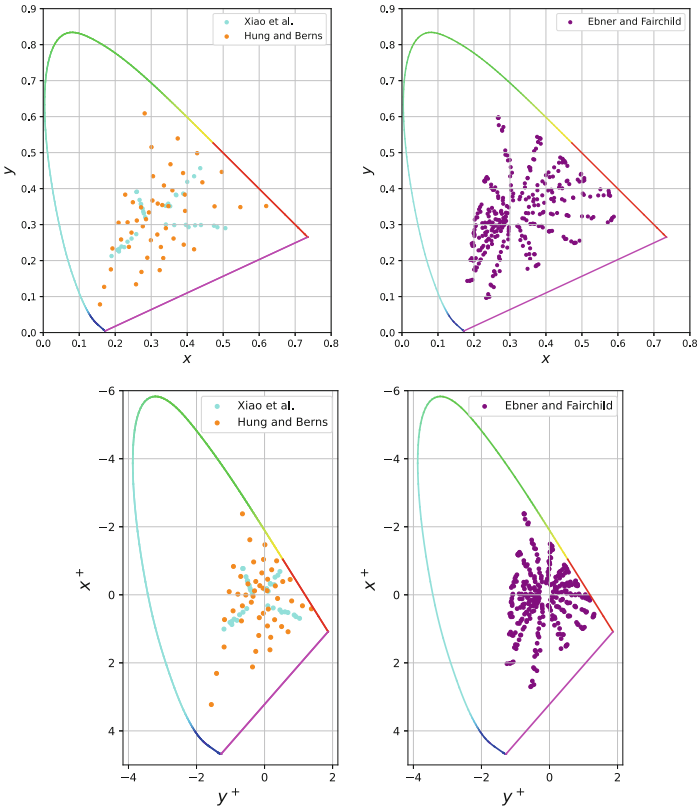

Fig. 3. Visualization of constant hue datasets on CIE xy (top) and PCS23-UCS (bottom) chromaticity diagrams.

(unlike CAM16-UCS) as well as the straight lines of color gamut triangles (like CIE xy), while also offering greater uniformity. Moreover, PCS23-UCS preserves the central bundle of lines from CIE XYZ and has a convex gamut (see Fig. 2). Hue linearity in PCS23-UCS compared to linear color space CIE xy is illustrated in Fig. 3.

7 Conclusion

In this work, we propose PCS23-UCS, a novel uniform color space developed through a projective transformation of a polynomial extension of CIE XYZ coordinates. Compared to the leading CAM16-UCS model, PCS23-UCS offers superior geometric properties, preserving the central bundle of planes, lines, and constant hue planes. While it exhibits a slight decrease in perceptual uniformity relative to CAM16-UCS, PCS23-UCS emerges as the optimal solution for practical applications such as tone mapping and gamut mapping.

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Color Difference in Context: an Experiment

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Abstract. Color difference evaluation is an important topic in colorimetry, and it has led to many methods of calculations over the years, which still aim to improve accuracy.

This study aims to explore the efficacy of existing color difference measures in evaluating perceived color differences under varying contextual conditions.

A novel experiment was conducted wherein participants evaluated color pairs against different backgrounds. This contribution presents the preliminary results of this study which compare the perceived color on varying backgrounds with some of the most used color differences.

Overall, the results show that the precision gained by improving pointwise color metrics is pointless due to the high variability of color perception in varying contexts caused by spatial mechanisms of the human visual system.

In conclusion, this study highlights the importance of creating alternative methods to measure color differences that reflect how we see colors in real life. Such a challenge could improve color research and its applications across different fields.

Keywords: Color difference · Perceived color difference · Color in context · Color appearance

1 Introduction

Evaluating the difference between two colors has always been one of the hot topics in colorimetry, and many ways of calculating it have been developed and changed over the years to obtain better results.

Classical colorimetry works under the strict basic constraint of *aperture mode*, in which color is seen isolated through an aperture that prevents its association with a specific object or source [3]. The basic CIE XYZ color space of colorimetry is based on this fundamental and it has quickly proven to be not suitable for measuring color differences due to its non-uniformity, limited correlation with human perception, and dependence on illuminant (the contribution of the illuminant source and the objects are indeed not distinguishable by each other).

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This problem was formally addressed in 1976, with the definition of the so-called *perceptual uniform spaces*: CIE L*a*b* and CIE L*u*v* spaces [2]. These color spaces added the effect of a reference illuminant to the system, moving the color description of color appearance from *aperture mode* to *object mode*.

The space addresses successfully the perceptual uniformity, even if not perfectly. Despite this, the *aperture mode* constraint remains, since the so-called *object mode*, still remains a point-wise description of color stimulus, with the color considered as a phenomenon isolated from a context and its spatial complexity.

The very first attempt to include a bit of context is represented by the Color Appearance Models (CAMs). The CIECAM spaces were developed to take the background into consideration, but in a limited way: the stimulation is linked to a single background color, and only its lightness (without chroma) is used in the calculations. It is a strong simplification of all the possible visual configurations and in fact CIECAM spaces have significant limits in real-condition applications [7] so that they are not accepted as standards [9].

Given all these considerations, in this contribution we want to present a test to compare the various chromatic distances with the changes that the context causes in the final appearance sinthetized by our vision system. This work does not want to make any rank among chromatic distances but rather to show how these measures relate to changes caused by spatial configuration. In other words, this experiment does not aim to discuss or modify any distance measure; rather we are interested in showing the paramount changes between color differences within and without any form of context, here we have chosen the simplest one, that is the change in color background.

2 Experiment Setup

The experiment consists of placing a pair of colors on different backgrounds to first attempt to compare the variation between the color appearance measured by the various chromatic distances and the generated color perception.

In particular, different images like the one represented in Fig. 1 were shown to 10 participants, who were asked to rate the perceived difference between the two color patches for each background in a magnitude scale from 1 to 5 of a nominal Likert scale:

- 1 – Imperceptible difference;
- 2 – Slightly perceptible difference;
- 3 – Perceptible difference;
- 4 – Very perceptible difference;
- 5 – Definitely perceptible difference.

As can be seen from Fig. 1, we selected four different backgrounds, keeping the same lightness ($L^* = 50$): red (R), blue (B), green (G), and black and white noise (N). All the configurations were shown at the same time to keep the difference evaluation consistent.

The test was performed making sure that any possible condition-changing variable was kept constant: only one monitor was used, and it was calibrated before starting the test. The subjects were all kept at the same distance from the screen (50 cm) and the room was kept in darkness (Table 1).

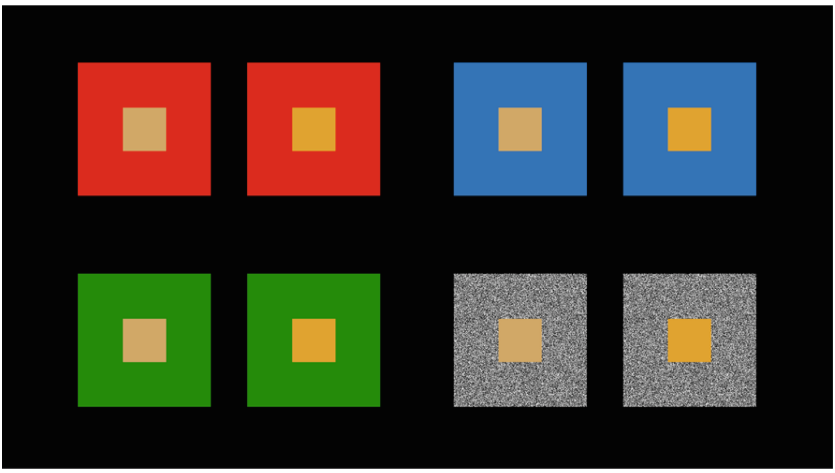


Fig. 1. Example of pair of colors (1C-01) used in the test.

Table 1. Coordinates of the backgrounds, calculated using D65 as reference with a standard observer of 10°.

Background	Code	L*	a*	b*	Color
Red	R	50	80	70	
Green	G	50	50	50	
Blue	B	50	5	-50	
White Noise	N	100	0	0	
		0	0	0	

The experiment involved 10 participants who had no visual impairments or color blindness. Among them, 60% were female and 40% were male; 80% of the participants were aged between 20 and 30, the oldest was 60 years old.

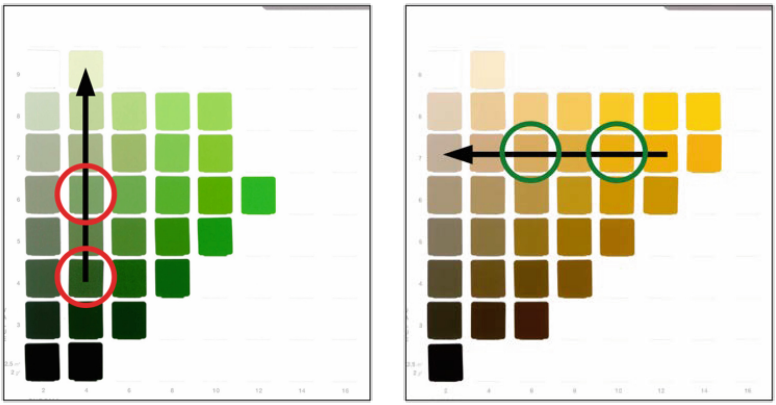
The images were shown sequentially, each separated by a 5-s black screen to avoid the afterimage effect. Each image was displayed for 10 s.

For this work, 20 pairs of patches were extracted from the Munsell Book of colors Matte Collection (X-Rite) to represent changes in its three possible variables:

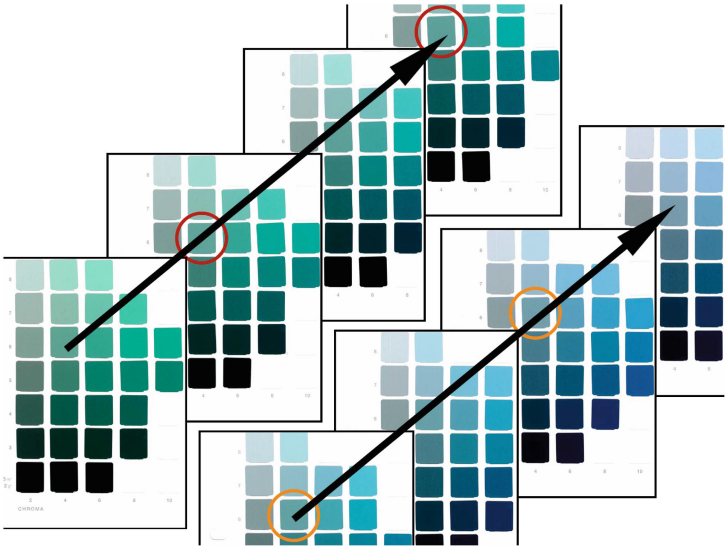
– 7 pairs for *value* variation. For the choice of the pairs, a distance of two tiles was used, see for example Fig. 2a.

– 4 pairs *chroma* variation. For the choice of the pairs, a distance of two tiles was used, see for example Fig. 2b.
– 9 pairs *hue* variation. For the choice of the pairs, a distance in hue of five units (equivalent to two sheets for our Munsell atlas) was used, see for example Fig. 2c.

Table 2 summarizes the color difference considered in this study. These color differences were compared to the rating of color difference perceived by the users.



(a) Value variation. (b) Chroma variation.



(c) Hue variation.

Fig. 2. Example of chosen pairs varying the three parameters of the Munsell color space: value, chroma and hue.

Table 2. ΔE values for the pairs used in the test

Model	Year	Color space	Formula	Reference
ΔE_{76}	1976	CIE L*a*b*	$\Delta E_{76} = \sqrt{(\Delta L^{*2} + \Delta a^{*2} + \Delta b^{*2})}$	[12]
ΔE_{94}	1993	CIE L*a*b*	$\Delta E_{94} = \sqrt{\left(\frac{\Delta L^*}{k_L S_L}\right)^2 + \left(\frac{\Delta H^*_{ab}}{K_H S_H}\right)^2 + \left(\frac{\Delta C^*_{ab}}{K_C S_C}\right)^2}$ with $C = \sqrt{a^2 + b^2}$, $\Delta H = \sqrt{\Delta a^2 + \Delta b^2 - \Delta C^2}$	[1]
ΔE_{00}	2001	CIE L*a*b*	$\Delta E_{00} = \sqrt{\left(\frac{\Delta L'}{k_L S_L}\right)^2 + \left(\frac{\Delta C'}{K_C S_C}\right)^2 + \left(\frac{\Delta H'}{K_H S_H}\right)^2 + R_T \left(\frac{\Delta C'}{K_C S_C}\right)\left(\frac{\Delta H'}{K_H S_H}\right)}$ with $C' = \sqrt{a^2 + b^2}$, $\Delta H' = 2\sqrt{C'_1 C'_2} sin(\Delta h'/2)$	[8]
ΔE_{CMC}	1984	CIE L*a*b*	$\Delta E_{CMC} = \sqrt{\left(\frac{\Delta L^*}{l S_L}\right)^2 + \left(\frac{\Delta C^*_{ab}}{c S_C}\right)^2 + \left(\frac{\Delta H^*_{ab}}{S_H}\right)^2}$ with $C' = \sqrt{a^2 + b^2}$, $\Delta H = \sqrt{\Delta a^2 + \Delta b^2 - \Delta C^2}$	[4]
ΔE_{DIN99}	2002	DIN99	$\Delta E_{99} = \frac{1}{k_E}\sqrt{\Delta L^2_{99} + a^2_{99} + b^2_{99}}$ with $L_{99} = 105.51 ln(1 + 0.0158 L^*)$, $a_{99} = C_{99} sin(h_{99})$, $b_{99} = C_{99} cos(h_{99})$	[5]
ΔE_{WLab}	2015	WLab	$\Delta E_W = \sqrt{\Delta L^2_W + \Delta a^2_W + \Delta b^2_W}$	[6]

3 Results

The medium value and standard deviation of the difference in the participants' color perception between each pair of colors in the different backgrounds are reported in Table 3, Table 4 and Table 5.

Table 3. Medium value and standard deviation of perceived difference for each pair of colors varying in value and background.

	Color pair		Background	Average difference	Standard deviation	Total sandard deviation
			R	4.7	0.48	
			B	4.8	0.42	
			G	4.8	0.42	
V1_01	2.5R 3/4	2.5R 5/4	N	4.7	0.48	0.44
			R	3.4	0.97	
			B	3.8	0.79	
			G	3.2	0.79	
V2_01	10B 6/2	10B 8/2	N	4.4	0.70	0.91
			R	4.4	0.70	
			B	4.5	0.71	
			G	4.6	0.70	
V3_01	7.5GY 4/4	7.5GY 6/4	N	4.6	0.84	0.72
			R	4.4	0.84	
			B	4.9	0.32	
			G	4.8	0.42	
G1_01	5YR 4/1	5YR 6/1	N	4.6	0.70	0.62
			R	3.5	0.97	
			B	3.6	1.07	
			G	2.7	0.95	
G1_02	5YR 7/1	5YR 9/1	N	4.2	0.92	1.09
			R	4.0	0.82	
			B	4.5	0.71	
			G	4.0	0.82	
G2_01	5G 4/1	5G 6/1	N	4.7	0.48	0.76
			R	4.3	0.82	
			B	4.7	0.48	
			G	4.2	0.79	
G3_01	5P 4/1	5P 6/1	N	4.5	0.53	0.68

Figure 3, Fig. 4 and Fig. 5 show a comparison between perceived differences among the various configurations, with the color differences summarized in Table 2. In particular, in each image, the top graph shows the average ratings of participants along with the standard deviation, for each pair of colors in each background; while the bottom graph displays the chosen color differences. It is important to note that the formulas used in Table 2 do not consider the configuration against which the colors are viewed. As a result, the four markers are aligned at the same value for each background. While ΔE remains constant through the different backgrounds, the average of perceived color keeps changing.

In general, the results obtained in the test showed a great gap in the evaluation according to the background, with the highest disagreement for the pair 1G-02 (that shows a total standard deviation of 1.09 across all the answers of all

Table 4. Medium value and standard deviation of perceived difference for each pair of colors varying in hue and background.

	Color pair		Background	Average difference	Standard deviation	Total sandard deviation
			R	1.6	0.52	
			B	1.8	0.63	
			G	1.8	0.79	
H1_01	5P 3/6	10P 3/6	N	2.8	0.63	0.78
			R	1.6	0.52	
			B	1.7	0.67	
			G	1.4	0.52	
H1_02	5RP 3/6	10RP 3/6	N	2.4	0.84	0.73
			R	1.0	0.00	
			B	1.0	0.00	
			G	1.0	0.00	
H2_01	5BG 6/4	10BG 6/4	N	2.0	0.00	0.00
			R	1.1	0.32	
			B	1.2	0.42	
			G	1.3	0.48	
H2_02	2.5B 6/4	7.5B 6/4	N	1.1	0.32	0.38
			R	1.1	0.32	
			B	1.7	0.48	
			G	1.3	0.48	
H3_01	2.5RP 7/6	7.5RP 7/6	N	1.7	0.67	0.55
			R	1.3	0.67	
			B	1.8	0.42	
			G	1.3	0.48	
H3_02	5R 7/6	10R 7/6	N	2.0	0.67	0.63
			R	1.0	0.00	
			B	1.1	0.32	
			G	1.0	0.00	
G4_01	5GY 5/1	10GY 5/1	N	1.3	0.48	0.30
			R	1.0	0.00	
			B	1.1	0.32	
			G	1.0	0.00	
G5_01	5P 3/1	10P 3/1	N	1.0	0.00	0.16
			R	1.0	0.00	
			B	1.1	0.32	
			G	1.0	0.00	
G6_01	10G 7/1	5BG 7/1	N	1.0	0.00	0.16

the participants for each background). The only color pair that kept the standard deviation null is 2H-01, where the two colors were always perceived as equal.

For what concerns the pairs of colors varying in hue (see Fig. 3), all the hue couples are perceived as identical or only slightly different. Pair 1H-01 has the highest standard deviation between the ratings ranging from imperceptible to very perceptible according to the background used. In particular, the background that generates the highest difference perception is the white noise, with a mean value of perceptible (2.8, which is the highest rating value) while all the other backgrounds gave an average between imperceptible and slightly perceptible (1.6 to 1.8). The pair with the lowest rating values is 2H-01 with a mean value of 1 which, as stated before, is the only one whose colors were perceived as identical in every answer.

Table 5. Medium value and standard deviation of perceived difference for each pair of colors varying in chroma and background.

	Color pair		Background	Average difference	Standard deviation	Total sandard deviation
			R	3.9	0.74	
			B	3.6	1.17	
			G	3.7	1.06	
C1_01	10RY 7/6	10RY 7/10	N	4.6	0.84	1.01
			R	1.2	0.42	
			B	1.1	0.32	
			G	1.6	0.52	
C2_01	7.5PB 5/6	7.5PB 5/10	N	1.5	0.53	0.48
			R	1.2	0.42	
			B	1.6	0.70	
			G	1.6	0.52	
C2_02	7.5PB 5/8	7.5PB 5/12	N	1.2	0.42	0.55
			R	3.8	0.63	
			B	3.7	0.67	
			G	3.5	0.71	
C3_01	10Y 8/6	10Y 8/10	N	4.5	0.53	0.72

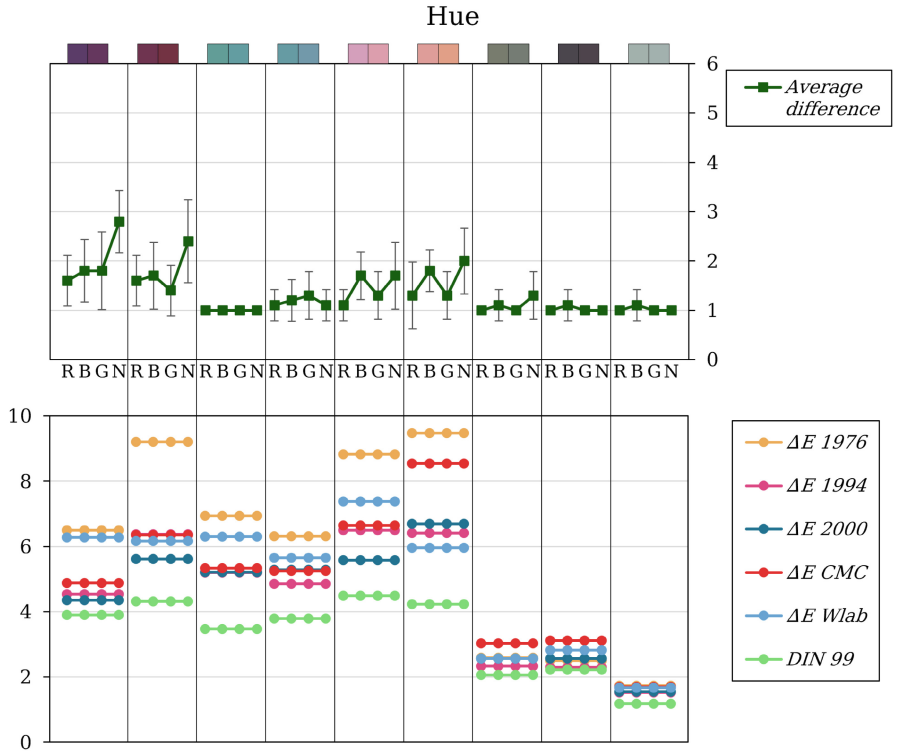


Fig. 3. Pair of colors varing for hue: ΔE values compared to the results of the test.

The color pairs that vary in Munsell value (see Fig. 4) show the highest perceived difference, ranging from the perceptible difference of 3.2 of pair 2V-01

on the green background to the definitely perceptible difference of 4.9 of pair 1G-01 on the blue background. As stated before, 1G-02 has the higher standard deviation between the participants' answers, which on the contrary are well aligned on pair 1V-01 with a standard deviation of 0.44.

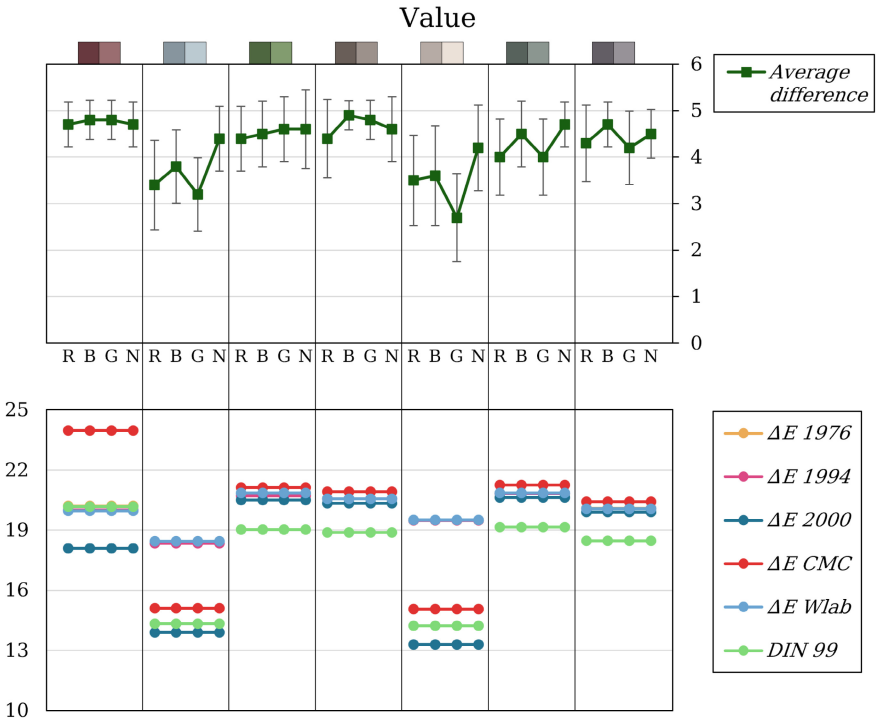


Fig. 4. Pair of colors varing for value: ΔE values compared to the results of the test.

While value and hue couples can be grouped in a range of answers, the same cannot be said for the pairs that vary in *chroma* (see Fig. 5) because, even though the perceived difference should be constant according to Munsell, the two color pairs 2C are perceived as more similar than the other two pairs, with an average answer of imperceptible to slightly perceptible difference (1.4), while the other couples are perceived as very different (average of 3.9). These results show that the Munsell atlas, while quite accurate in some parts, is not a good representation of the perceived differences when the parameter of chroma changes.

To compare our results with the color difference measures in Table 2, we looked at how each color difference correlated with the average of test answers across different backgrounds. Not considering the variation in the perceived difference and using only its total average among all the backgrounds, we found a high correlation (at least 0.82) with color differences, confirming the reliability of the users rating.

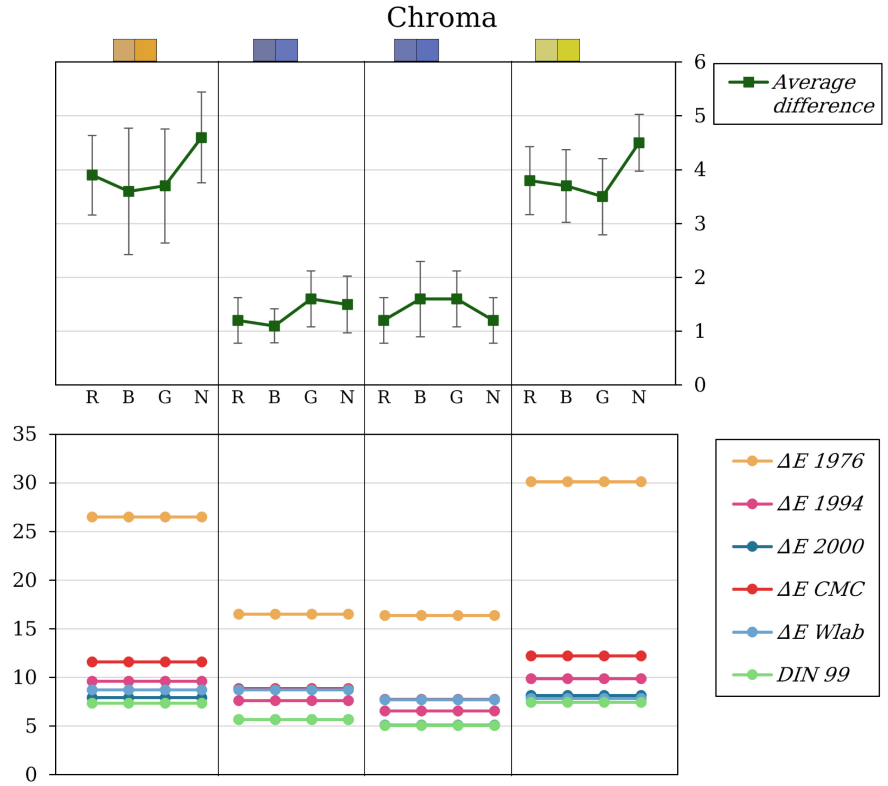


Fig. 5. Pair of colors varing for chroma: ΔE values compared to the results of the test.

This confirms that color differences generally are in line with users evaluations if we cancel the variability of spatial configuration. In other words, if we do not consider the spatial computation mechanism of the human visual system.

Looking at the backgrounds, a significant subset of the color pairs, about 45%, got the highest average perceived difference when shown against the white noise background; 30% on the red background (mainly the pairs with grey colors), and the remaining 20% to the green background.

As expected, the perceived difference of the participants varied noticeably across all backgrounds. The answer remained consistent only about 36% of the time, challenging the constant effect assumption of backgrounds at the base of CIECAM.

The results suggest that the search for precision in the quest for the "perfect" color difference measure, in the "perfect" color space, retains its meaningfulness only in every possible pointwise application, that is out of any context. In other words, the precision gained by adopting the ultimate color difference calculation (measure and/or space transform) is not worth the effort when changes in context alter it significantly. Thus, context complexity puts functional classic point-wise

color differences out of game. To deal with such complexity, computational model of human visual system [10,11] or alternatively other mean of vision experiments generalization [13] could be used.

4 Conclusions

In this study, we ran a test where participants rated the perceived color differences between pairs of colors placed on different backgrounds. The color pairs were chosen to represent changes in the three variables of the Munsell color space (value, chroma, and hue), keeping the same numeric distance between their properties. The answers of the participants were then compared to some of the classical color difference measures.

When examining pairs of colors with varying hues, participants generally perceived them as identical or only slightly different. For pairs of colors varying in value, perceived differences were higher. Pairs of colors varying in chroma presented both high and low differences. While Munsell theoretically predicts consistent differences across all directions in space, our results showed discrepancies when used in varying contexts.

Regarding color in context, some backgrounds influence perceived color differences more than others. The white noise background reached the highest average ratings, followed by the red and green backgrounds. Participant ratings changed 64% of the time with the backgrounds, showing that the fixed constraints of the existing models like CIECAM are not sufficient to correctly describe the human visual system behavior.

In conclusion, the study proves that the context in which a color is placed affects its perception. The results highlight that the current point-wise approach of colorimetry and its classical color differences that do not consider the background and the context in which the colors are placed is deceiving. Given this, rather than research for more and more improved color difference measures, since the context causes a high variance, it would be more interesting for the scientific community to explore new methods able to consider complex spatial configurations.

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Image Enhancement and Retinex Theory

Exploiting Milano Retinex Contrast to Enhance Images with Strong Changes of Light Intensity

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Abstract. Improving the quality of non uniformly lighted images is particularly hard due to the presence of regions that, having different brightness, requires different degrees of enhancement. The recently published algorithm REK proposes an interesting solution for enhancing images with abrupt changes of light intensity. REK linearly up-scales the image brightness to increase the quality of dark regions and combines the image with up-scaled brightness with the input one to preserve the quality of the bright regions. The up-scaling parameter α is estimated unsupervisedly based on the segmentation of brights and dark regions. This estimate has two main disadvantages: first, it makes REK dependent on the segmentation algorithm; second, the segmentation may be adversely affected by noise often present in badly illuminated images. To overcome these issues, this work proposes a new estimation of α based on the comparison of image Sobel gradient with an enhanced contrast, specifically the Milano Retinex contrast, which - as the name suggest - is inspired by the principles of Retinex theory. The new estimation of α provides good results, with reduced artifacts and saturation effects.

Keywords: Image Enhancement · Non Uniform Illumination · Retinex Theory

1 Introduction

Images of unevenly lit scenes include regions with different brightness levels. This is the case of images captured under back-light or including intense light sources not spreading over the acquired scene. In these images, the intensity light strongly changes across the scene and generates very dark regions, with illegible content and details, and very bright areas, with clearly readable content. Improving the overall quality of such images is really challenging, since dark and bright regions require different levels of enhancement. This problem has been addressed by several methods, like, e.g. histogram equalization approaches [3,16], deep learning techniques [5,19], Retinex-inspired algorithms performed at

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multiple scales [1,2], Retinex inspired bilateral filtering [8], multi-level enhancement algorithms [9,17] and algorithms combining multi-exposed images [15,18]. Although these methods generally improve the quality of dark areas, problems like false colors, artifacts along the edges, low naturalness and saturation (i.e. one or more color intensities are cast to 255) are often reported.

In this work we focus on the algorithm REK [9] that proposes a computationally efficient solution to enhance images with abrupt changes of light intensity. REK up-scales the brightness of the input image by a linear map that increases the quality of dark regions. This operation relights the image according to the von Kries model [6]: the name REK is just an acronym of the key-concepts of image *re*lighting and von *K*ries model. Being applied to the whole image, the REK linear brightness up-scaling may over-enhance the bright regions, introducing undesired artifacts. To mitigate these effects, REK combines the input image and that with up-scaled brightness by a weighted sum (see Fig. 1). The up-scaling parameter α defines the extent of the image enhancement and is an user-defined parameter. However, REK also provides an unsupervised estimate of α working as follows: the image brightness is partitioned in dark and bright areas by a threshold-based segmentation and α is set to the ratio of the mean brightness of the bright regions (slightly diminished to mitigate saturation effects and preserve image edges and naturalness as much as possible) and mean brightness of the dark ones. This estimation of α has two main drawbacks: first, it makes the enhancement dependent on the segmentation; second, the segmentation can be adversely affected by noise usually present in badly illuminated images.

In this work, we replace the segmentation-based estimation of α with a new approach, based on the comparison between image edges detected by a Sobel filter and by the light invariant *Mi*lano *R*etinex *Co*ntrast MiRCo described in [13] and grounded on the Retinex theory [7]. The rationale behind the new estimate is that, while edges in dark areas are poorly detected by Sobel filter, they are clearly highlighted by MiRCo. Based on this consideration, (1) we detect the edges needing for enhancement (i.e. edges in dark areas) as the pixels where Sobel gradient magnitude (normalized to range over $[0, 1]$) is much lower than the corresponding MiRCo value; (2) we set α to the mean value of the ratios between MiRCo and the normalized Sobel gradient magnitude on the edges detected in (1). Although within estimate, the enhanced images are generally darker than those given as output by the original REK, they exhibit less saturation and artifacts, while have still clear content and details.

The paper outlines as follows: Sect. 2 presents the workflow of REK; Sect. 3 describes the original and the new estimates of α; Sect. 4 reports experiments and results; finally, Sect. 5 draws our conclusions.

2 REK

As mentioned in Sect. 1 and depicted in Fig. 1, REK takes as input a color image I and up-scales each R, G, B color component of I (thus the brightness B of I) by a linear map with coefficient $\alpha > 1$, obtaining a new image $K = \alpha I$. This linear

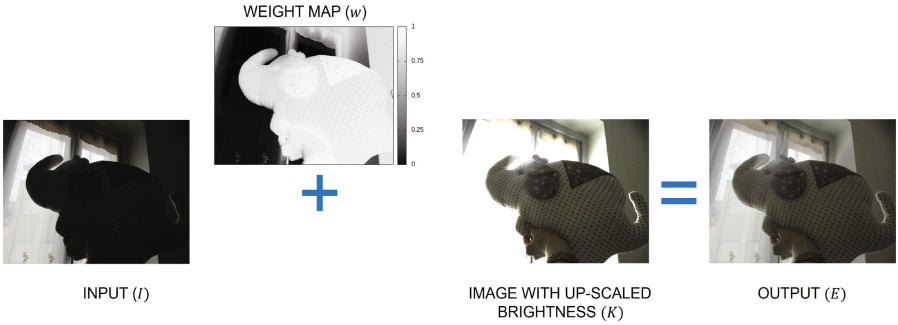

Fig. 1. Workflow of REK: the color input image (I) is summed to a linearly over-enhanced version (K) with weights (w). The result (E) is an enhanced version of I.

operation increases the brightness (thus the quality) of the dark regions, but, since it is applied on the whole image, it may over-enhance the bright regions, causing undesired artifacts, like e.g. saturation and false colors. To overcome this issue, REK sums I and K pixel by pixel with weights w that increase when B decreases. In this way, w awards the contribution of K from the enhanced dark areas and that of I from the original bright areas. Mathematically, the enhanced image E is computed by the following equation:

$$E(x) = (1 - w(x))I(x) + w(x)K(x) \tag{1}$$

where w is a function of the brightness B of I and is defined as follows:

$$w(x) = \left(1 - \frac{B(x) - m}{M - m}\right)^p. \tag{2}$$

Here $B(x)$ is computed as the mean value of the red, green and blue components of $I(x)$; m and M are the minimum and maximum values of B and p is an integer parameter. The higher p, the weaker the enhancement is. In general, recommended values for p are 3 or 5 (see [9]).

In the original REK, the unsupervised estimate of α requires to segment B in dark and bright regions. This partition is performed by clustering the pixels of B according to their value with respect to the threshold

$$T = \frac{M + m}{2}. \tag{3}$$

Precisely, pixels with B smaller than T are labeled as dark, while the others as bright. The value of α is defined as follows:

$$\alpha = \max\left(1, \frac{\mu_b - \delta_b}{\mu_d}\right) \tag{4}$$

where μ_b and δ_b are the mean value and the standard deviation of B on the bright regions and μ_δ is the mean value of B on the dark areas. Figure 2 shows

two examples of badly lighted images, the segmentation of their brightness based on T and the enhancement results reached by REK (see the boxes with the blue borders).

3 The Proposed Estimate of REK's Up-Scaling Parameter

According to the Retinex theory [7], edges represent key elements for human color vision system and in particular for human color constancy. Therefore, their use in machine color vision appears particularly worthy of studying. Based on this consideration, we estimate α by comparing standard and enhanced edges. Specifically, we implement three steps: first, we compute the edges of B by the Sobel filter and by MiRCo; second, we compare the magnitude of the Sobel gradient with MiRCo to detect the set Q of edges needing for enhancement (i.e. edges placed in dark areas); third, we set the parameter α to the mean ratio of the magnitudes of MiRCo and Sobel gradient on Q. These steps are described in details in the following paragraphs and illustrated in Fig. 2 on two examples.

Computation of the Magnitude of the Sobel Gradient of B - We compute the Sobel gradient ∇B of B and we measure its magnitude $|\nabla B(x)|$ at each pixel x by the L^∞ norm, normalized to range over [0, 1].

Computation of the Milano Retinex Contrast of B - We compute MiRCo as described in [13]. MiRCo is a contrast measure derived from the path-based Milano Retinex color image enhancer [12] (PMR for short), which is a variant of Retinex [7]. As Retinex, PMR takes as input a color image, enhances its color channel independently, and merges the enhanced channels (called *the channel lightnesses*) in a new RGB image (i.e. the algorithm output), where content and details are more readable, while possible chromatic dominants of the light are lowered or even removed.

MiRCo is defined as a function of the PMR lightness L computed on the image brightness B instead of on a color channel, precisely:

$$\text{MiRCo}(x) = 1 - L(x) \qquad \forall\, x \in\ S(B) \tag{5}$$

where $S(B)$ indicates the set of pixels composing B. According to PMR and [13], the lightness L is obtained by sampling the neighborhood of x with n short, random paths $\gamma_1, \ldots, \gamma_n$ starting at x, and by processing the intensities along each path by the so-called *ratio-threshold-product-reset mechanism*, which implements the following steps:

- *Path Scanning* - Each path $\gamma_k : \{1, \ldots, l\} \rightarrow S(B)$ with length $l > 1$ is scanned from its end $\gamma_k(l)$ to $x = \gamma_k(1)$;
- *Ratio-threshold-product-reset* - For any $t = 1, \ldots, l-1$, the ratio $R_k(t) := \frac{B(\gamma_k(t+1)}{B(\gamma_k(t))}$ is computed (division by zero is of course prevented) and the *threshold rule* is applied, i.e. any ratio R_k belonging to the interval $[1-\varepsilon, 1+\varepsilon]$ (with ε close to 0) is cast to 1. This operation discards from the lightness

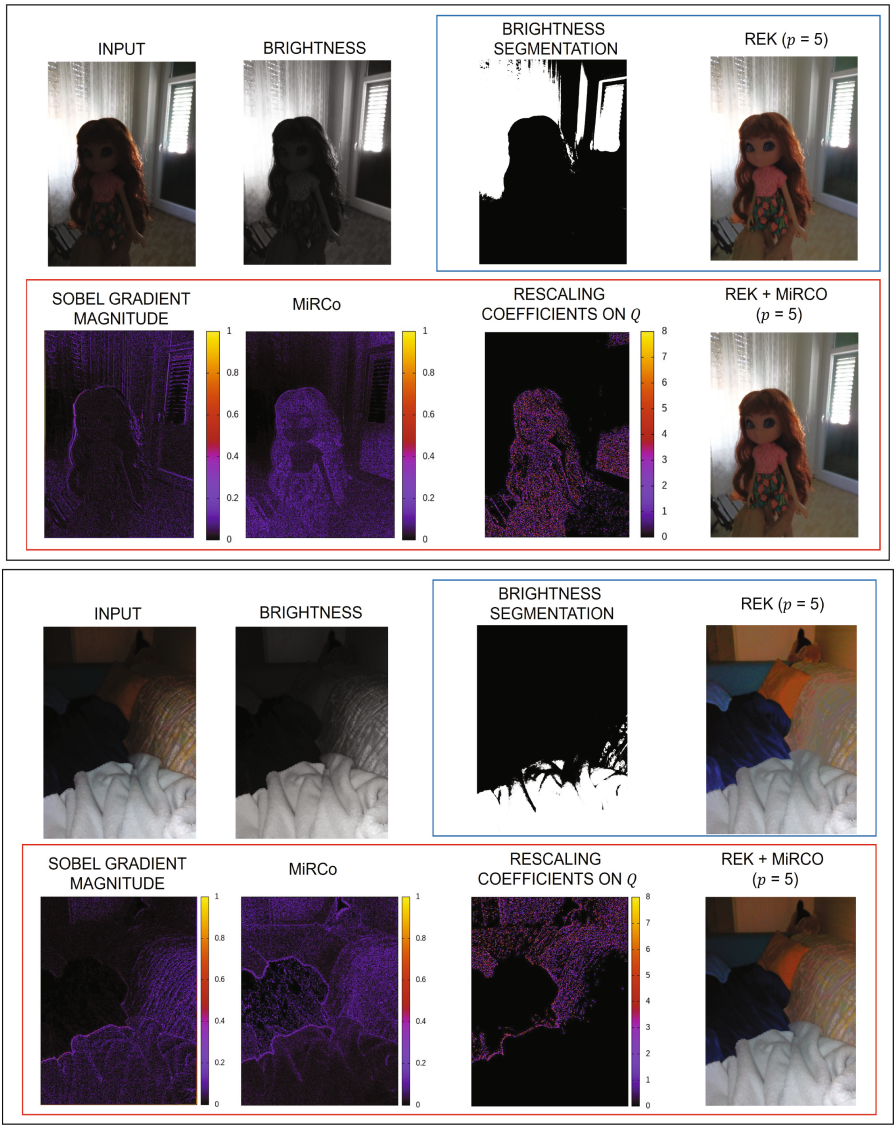

Fig. 2. Two examples of image enhancement by REK and by REK+MiRCo (i.e. REK equipped with the new estimate of α). Each example shows the input image and its brightness, the image segmentation in bright and dark regions, the enhancement performed by REK (in the blue box on top) and the outputs of the main steps of REK+MiRCo (in the red box at bottom): the magnitude of the Sobel gradient, MiRCo and the map with the values of ρ used for estimating α (see Sect. 3 for more details). (Color figure online)

computation the slight edges, which are proved to be irrelevant to human vision [7]. Then, the R_k's are multiplied cumulatively along the path and the so-called *reset rule* (also related to human vision system [7]) is implemented, i.e.: any cumulative product exceeding 1 is cast to 1;

– *Averaging results* - The cumulative products at x, computed over the n paths γ_ks (let's say the $L_k(x)$s) are averaged together, giving the lightness $L(x)$, i.e.

$$L(x) = \frac{1}{n}\sum_{k=1}^{n} L_k(x). \tag{6}$$

In MiRCo, the path length l must be small, since - as a contrast measure - MiRCo wants to capture local intensity variations. The most appropriate value of l depends on the image size and content. In general, the proper value of l increases with the image resolution. In fact, in large images, narrow neighborhoods around pixels do not contain relevant information and slight variations of intensity often occur even in areas that are perceptually uniform: for these images, large neighborhoods (thus long paths) must be considered for detecting edges relevant to the enhancement process. Typically, for images with more than 100 mega pixels, l is set to 5 or 7, while for smaller images $l = 3$ generally suffices. For these values, MiRCo is close to an edge map. In the current implementation, based on [13], ε is set to 0.004.

Thanks to its definition, MiRCo provides a contrast measure robust to in-plane rotations and - most importantly for our work - to light dimming, i.e. *MiRCo detects image edges regardless of the light intensity.*

Detection of Edges Needing for Enhancement - Sobel gradient poorly detects edges in dark areas, because it basically computes edges as differences of intensity of adjacent pixels: if these intensities are low, the gradient magnitude is low too. On the contrary, MiRCo computes edges by ratios of intensities and this enables an accurate edge detection regardless of the illumination, therefore even in dark areas. In particular, MiRCo is greater than the normalized Sobel gradient $|\nabla B|$ on edges in dark areas, smaller or equal to $|\nabla B|$ on edges in bright areas, and null on (almost) flat regions. Therefore, we define the set of edges *needing for enhancement* (i.e. edges placed in dark areas) as:

$$Q_e = \left\{ x \in B(s); \rho(x) := \frac{\text{MiRCo}(x)}{|\nabla B(x)|} > 1, \text{ with } |\nabla B(x)| \neq 0 \right\}. \tag{7}$$

Estimate of α - To make the estimate of α more accurate and computationally efficient, we compute α as the mean value of ρ on a subset Q of Q_e, obtained by removing from Q_e noisy or irrelevant pixels. Specifically, we exclude from Q_e the pixels with ρ too close to 1 (in this case MiRCo and ∇B are very similar), i.e. we retain only pixels with $\rho(x) > 1.05$. Then, we compute the distribution D_ρ of ρ and we remove from Q_e all the edges whose ρ is above the τ-percentile of D_ρ, usually with τ set close to 100%: in fact, these edges are usually outliers.

Finally, we exclude from Q_e isolate, small aggregates of pixels, which are likely noise, and we estimate of α as the mean value of the ρs on Q, i.e.:

$$\alpha = \frac{1}{|Q|} \sum_{x \in Q} \rho(x). \tag{8}$$

4 Experiments and Results

The performance of REK within the new estimate of α have been measured both quantitatively and qualitatively on the dataset BSIMAGE [11]. This dataset contains 70 real-world color images acquired under backlight or under intense lights not spreading over the scene. We set the parameters of REK+MiRCo (i.e. of REK equipped with the new estimate of α) taking into account the resolution of these images. Precisely, for computing MiRCo, we consider 10 paths with length $l = 5$ per pixel, while for the estimate of α, we set $\tau = 95\%$ and we remove from Q_e any isolated connected component containing less than 80 pixels (for the smallest image from BSIMAGE, this threshold corresponds to exclude less than 0.01% of the image size). Experiments on REK and REK+MiRCo have been repeated for $p = 3$ and 5. Examples of images from BSIMAGE and of their enhancement by REK+MiRCo are displayed in Figs. 2, 3, 4 and 5.

Table 1. Objective Evaluation of REK+MiRCo (for $p = 3, 5$) in comparison with REK and other seven enhancers.

Algorithm	m_1 (↑)	$m_2[\times 10^{-2}]$ (↑)	$m_3[\times 10^{-3}]$ (↓)	NIQE (↓)
INPUT	63.46	7.34	4.27	3.34
REK+MiRCo ($p = 3$)	96.09	9.31	3.71	2.85
REK+MiRCo ($p = 5$)	87.37	8.55	3.95	2.92
REK ($p = 3$) [9]	109.93	10.53	3.83	2.82
REK ($p = 5$) [9]	99.01	9.49	4.05	2.86
FUSION [17]	88.73	13.36	3.45	3.27
CLAHE [20]	83.21	15.57	3.10	3.14
MSR+AGCWD [4]	112.07	16.17	2.53	3.10
MS-LRSR [1]	120.39	16.66	2.24	2.96
SuPeR-B [8]	117.17	11.75	3.15	2.90
EnlightenGAN [5]	132.44	6.90	3.03	6.02
ExCNET [19]	109.63	13.30	3.21	3.06

The quantitative assessment is performed by analysing the trend of four measures before and after enhancing. These measures are the mean value of the image brightness B (m_1), the mean value of the normalized magnitude of the Sobel gradient of B (m_2), the L^1 distance of the probability distribution of B

Fig. 3. Examples of image enhancement by REK + MiRCo.

from an uniform one (m_3) and the naturalness level measured by NIQE [14]). The m_is are related to human perception, namely to the visibility of the image content (m_1) and details (m_2), to the image colorfulness (m_3) and to the image naturalness (NIQE). An enhancer is expected to increase m_1 and m_2, while decreasing m_3 and NIQE.

Table 1 reports these values, averaged over BSIMAGE, for the input images and for their versions enhanced by the following algorithms: REK with the original estimate, REK within the new estimate (REK+MiRCo), the multi-level enhancer in [17] (FUSION), the contrast adaptive histogram equalization in [16] (CLAHE), the multi-scale Retinex approaches in [4] (MSR+AGCWD) and in [1] (MS-LRSR), the Retinex inspired bilateral filter in [8] (SuPeR-B) and the deep learning based enhancers in [5] (EnlightenGAN) and in [19] (ExCNET). Examples of images from BSIMAGE and of their enhancement by REK+MiRCo compared to the methods listed above are displayed in Fig. 4. The trend of the m_is is as expected for all the enhancers, apart from EnlightenGAN whose values of m_2 and NIQE are respectively lower and higher than those measured on the input images. Actually, EnlightenGAN increases the brightness very much and compresses the variability range of the edge magnitude, yielding the worst value of m_2. EnlightenGAN also introduces some color distortions that make the enhance images less natural than the original ones. In general, REK+MiRCo outputs images darker and slightly less contrasted than REK, but achieves better results in terms of colorfulness, while naturalness values output by the two methods are very close to each other. As in REK, also in REK+MiRCo the enhancement effect is slightly stronger when $p = 3$ than when $p = 5$, i.e. for $p = 3$ we obtain brighter and more cotrasted images, with a better colorfulness and naturalness, than for $p = 5$. In general, the brightness level of REK+MIRCo is close to that of FUSION and CLAHE, that however grant better values of m_2 and m_3, but have

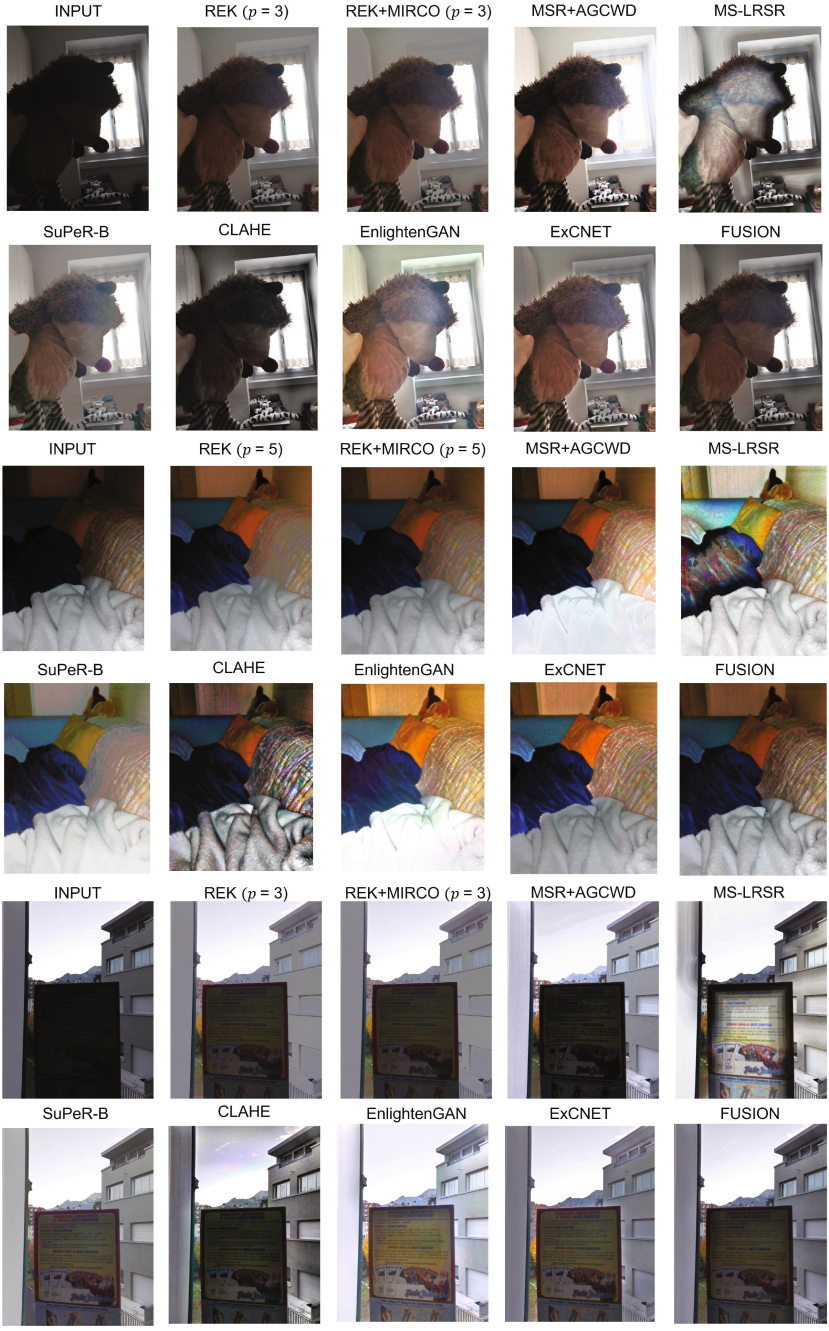

Fig. 4. Some images from BSIMAGE processed by REK, REK+MiRCo and other seven image enhancers.

worse results in terms of NIQE. Good values of m_is are reached also by ExC-NET, SuPeR-B and, in particular, by MSRSR+AGCWD and MS-LRSR. These latter two algorithms have the highest values of m_2, while the lowest value of m_3, but are outperformed by other methods in terms of NIQE.

To provide a comprehensive picture of the enhancers performance, this quantitative analysis is coupled with a qualitative analysis, that evaluates additional features not considered by the objective measures. Precisely, in the qualitative assessment proposed here, images were observed by humans to detect possible unpleasant effects, like saturated pixels, false colors, noise, halos, and to judge the overall naturalness of the image (see the examples inf Fig. 4). This analysis shows that:

1. saturation effects, causing a loss of details, occur sometimes in REK, meaning the enhancement is too aggressive (i.e. α is too high); this behaviour is more evident for $p = 3$ than for $p = 5$ and is mainly due to a non adequate image segmentation and/or to the presence of almost white bright regions. As illustrated in Fig. 2, the threshold-based segmentation of the input image may be critic: while on the image on top, REK obtains a very good result, on the image at bottom some details of the sofa at right are lost. This is because the segmentation identifies as bright region only the lightest, almost white area (depicting a cover) whose value μ_B is close to 255. Consequently, the dark intensities are thus rescaled towards a very high value, proximate to 255, causing saturation. This does not happen in REK+MiRCo, where the details in the bright regions are better preserved, although the overall brightness is lower than in the images enhanced by REK;
2. both REK+MiRCo and FUSION do not generate artifacts, the images content and details are readable, but the brightness (especially for FUSION and for $p > 3$) is quite low; in general, edges are enhanced slightly better by FUSION;
3. CLAHE and MS-LRSR generally return poorly natural images, where dark regions are still dark and many artifacts occur; CLAHE reports artifacts along edges and several dark or bright halos are present over the image, making this latter unpleasant and unrealistic (see for instance, the second and third examples in Fig. 4); MS-LRSR has exaggerate, thick edges that makes the image like cartoonized, as shown by all the examples in Fig. 4;
4. MSR+AGCWD tends to over-enhance bright regions (see e.g. the first example in Fig. 4) and in some cases generates false edges;
5. SuPeR-B often returns washed-out images, with some saturated areas (see, e.g., the second example in Fig. 4); these effects are partially due to a too aggressive enhancement action and partially to the removal of the chromatic color dominant of the light;

6. EnlightenGAN reports often false colors and an overall yellowish coloration that makes images poorly natural; some foreground objects present a whitish halo around the boundary (see, e.g., the first and third examples in Fig. 4);
7. ExCNET provides in general very good results, although in some cases it generates a white halo around the enhanced dark regions, as illustrated by the third example of Fig. 4.

Taking into account both quantitative and qualitative analysis, we can say that the best results have been obtained by REK+MiRCo and FUSION, since they enhance image quality without introducing artifacts, and by ExCNET that reports white halos only in few cases. In general, FUSION and ExCNET provide a better edge enhancement than REK and REK+MiRCO. As an advantage with respect to machine learning approaches like, e.g., ExCNET, REK+MiRCo does not require a specific hardware to be executed. Finally, we notify that identifying the *best* enhancer in an absolute way is practically impossible: different applications may require different enhancement levels, for instance images that appear unrealistic for humans may be suitable for machine tasks.

REK+MiRCO has been currently implemented in C++. In this, not optimized implementation, the computation of MiRCo for an image with size 3120 $\times$ 4160 requires on average about 3.5 min on a standard PC with Intel(R) Xeon(R) CPU E3-1245 v6 @ 3.70 GHz, while less than 1 s is needed to estimate α and enhance the image. This computational time can be reduced by computing MiRCo on a down-scaled version of the input image and up-scaling the contrast map back to the original image size for estimating α. We observed that halving the size to 1560 $\times$ 2080 and - consequently - using less and shorter paths (specifically, 6 paths per pixel with $l = 3$), the execution time decreases to 17 s, while the enhancement level remains still satisfactory, especially in terms of colorfulness and artifact production. However, brightness, contrast and NIQE are on average slightly worse than those obtained when the original image size is used (see Table 2 and Fig. 5 for an example). As a conclusion, REK+MiRCo appears an effective tool for *offline* image enhancement.

Table 2. Objective Evaluation of REK+MiRCo using half-scaled input images for the computation of MiRCo.

Algorithm	m_1 ($\uparrow$)	$m_2[\times 10^{-2}]$ ($\uparrow$)	$m_3[\times 10^{-3}]$ ($\downarrow$)	NIQE ($\downarrow$)
REK+MiRCo ($p = 3$)	92.55	9.10	3.65	2.89
REK+MiRCo ($p = 5$)	84.59	8.40	3.88	2.96

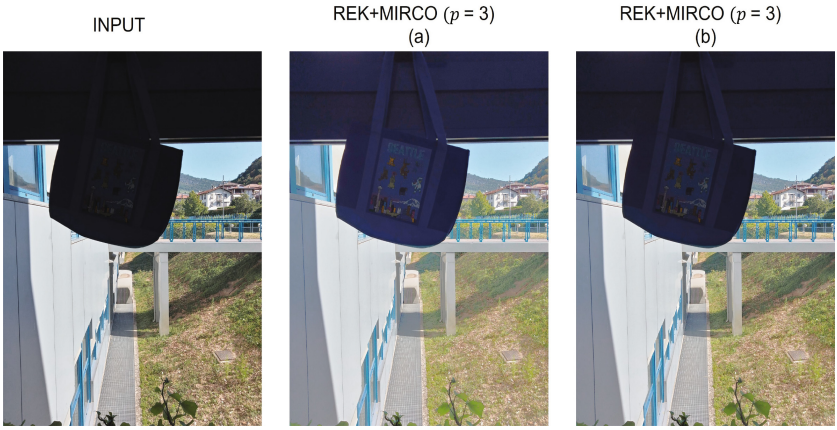

Fig. 5. An example of image enhancement by REK + MiRCo with MiRCo computed on the original image (a) and with MiRCo computed on a half-scaled version of the input image (b).

5 Conclusions

In this work, we considered the algorithm REK, which enhances images captured in highly uneven light conditions such as background and spotlight, and we proposed a novel, unsupervised, edge-based estimation of the REK parameter α. The use of edges for image enhancement appears worthy of interest: in fact, as the Retinex theory states, edges play an important role in human color vision and in particular in human color constancy. Based on this observation, the proposed estimate of α is performed by using the Retinex-inspired contrast measure MiRCo. This new estimate enables a good level of image enhancement, especially in terms of colorfulness and naturalness: while the enhanced images are darker and slightly less contrasted than those enhanced by the original REK, they do not present artifacts like saturation, halos and false colors. Compared to other methods qualitatively, the performance of REK+MiRCo (like that of REK) is close to that of FUSION, however this latter reports a better edge enhancement. In this respect, we note that the work in [10], where REK is used for enhancing badly illuminated archaeological images, suggests to couple REK with CLAHE to enhance better the edges. Future work will consider alternative techniques for improving the edge visibility by REK+MiRCo, for instance by using the illuminant invariant information provided by MiRCo. A possible code optimization will also investigate to reduce the current execution time of MiRCo.

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Intrinsic Image Decomposition Based on Retinex Theory, Superpixel Segmentation and Scale-Space Computations

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Abstract. Intrinsic image decomposition enables us to estimate the low-level features of images. Due to the benefits it provides and the challenges it holds, intrinsic image decomposition has been extensively studied over more than five decades. It can be utilized in various computer vision and computer graphics pipelines to improve the efficiency of tasks such as object classification and recoloring, and image segmentation. In this study, we introduce an algorithm for reflectance and shading estimation, offering a simple yet effective solution to the ill-posed intrinsic image decomposition problem. Our learning-free method leverages a combination of the fundamentals of the Retinex theory, scale-space computations, and superpixel segmentation. The assumptions of the Retinex theory enable us to provide a straightforward solution to a complex problem, while scale-space computations allow us to highlight low-level features and superpixel segmentation helps us to preserve local information. We evaluated our algorithm that mainly focuses on single objects on three benchmarks, namely, MIT Intrinsic Images, Bonneel, and MPI Sintel, which either consist of single objects or complex scenes having different characteristics. According to our experiments, our algorithm provides competitive results compared to the other methods.

Keywords: Intrinsic image decomposition · Reflectance estimation · Shading estimation

1 Introduction

The human visual system estimates the true colors of the objects unconsciously by discounting the effects of the light source present in the environment [21]. However, machine vision systems have difficulty performing this task effectively due to the presence of occlusion, reflections, glare effect, and over- and under-saturated regions [42,45]. To assist machine vision systems and increase their performance in numerous applications, we can benefit from the field of intrinsic image decomposition (IID) [3]. The main aim of intrinsic image decomposition is to recover a scene's low-level features such as reflectance, and shading, which are also referred to as "family of intrinsics" [3]. While reflectance carries information about the actual color of the objects irrespective of the scene's lighting conditions and the position of the capturing

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device, the shading component includes features related to the illumination, shadows, and geometry of the scene. Hence, each family member contains distinct characteristics that carry cues about the input scene which might not be directly observable in a single scene. Therefore, using intrinsics rather than the input image itself can benefit both high- and low-level computer vision tasks, including but not limited to object recoloring [8], surface re-texturing [48], exposure correction [52], and object classification [5].

Intrinsic image decomposition is framed as a computational challenge, since the problem is severely under-constraint, i.e., we only have a single input image but multiple unknown intrinsics. Due to this reason, studies in this field generally relax the ill-posed nature of the problem by assuming that any scene can be represented using the Lambertian image formation model. The under-constraint nature of the problem is further relaxed by assuming that the sensor responses of the capturing device are narrow-band, and there is a point light source illuminating the scene uniformly. Consequently, any input scene I at the spatial location $(\mathbf{x}, \mathbf{y})$ can be formulated as follows:

$$I(\mathbf{x}, \mathbf{y}) = R(\mathbf{x}, \mathbf{y}) \cdot S(\mathbf{x}, \mathbf{y}), \tag{1}$$

where R is the reflectance that represents the ratio between the total reflected and total incident illumination, and S is the shading that presents the interaction between the illumination and the surface [3].

Even though these relaxations simplify the intrinsic image decomposition problem, it is still quite challenging to estimate the reflectance and shading of a scene. Therefore, this field has gained significant attention over more than five decades, and various studies have been conducted. Both traditional algorithms relying on image statistics [2,3,11,16,17,22–24,26,28,36,37,41,47,53] and learning-based methods relying on neural networks [4–7,19,30,32,33,35,39,49–51] have been introduced to provide an effective solution to the problem. These algorithms may have different necessities, i.e., they may require user scribbles, different focal distances, a depth map, an image sequence captured with different viewing conditions, an input sequence where the light source position varies in each scene, an image stack captured under distinct illumination conditions, or a time-varying image sequence [12,43]. Algorithms performing decomposition by using only a single input image are generally considered to be more beneficial since in real-world applications we may not have any other information than the input scene, requiring user interaction might be inefficient, and creating image stacks in the appropriate format is troublesome [43].

Several of the intrinsic image decomposition algorithms build their strategies on the fundamentals of the assumptions of the Retinex theory [28]. The utilization of this theory is not surprising since the Retinex method is based on the analysis of the human visual system, and it is known that algorithms making use of biological findings tend to provide efficient solutions with a low computational complexity [45,53]. In this study, we propose a Retinex-based learning-free intrinsic image decomposition method operating in scale-space and relying on superpixel segmentation. We carry out our computations at multiple scales since the effectiveness of using scale-space is already proven in the field of intrinsic image decomposition [20,38]. Furthermore, we prefer this approach since in our previous studies, we explicitly demonstrated that the effectiveness of methods relying on color information can be significantly enhanced using scale-space,

which is sensitive to low-level features [46]. Moreover, we integrate superpixel segmentation, whose advantages are also proven in this field [18,27,40], into our algorithm to better preserve local information and reduce possible noise [1]. Our algorithm requires a single RGB image as input and we mainly focus on images containing a single object similar to previous works, i.e., Barron and Malik [2], and Baslamisli *et al.* [6]. Contrary to these studies, we demonstrate our method's outcomes also on more complex scenes.

The remainder of this paper is organized as follows. In Sect. 2, we introduce our method. In Sect. 3, we present the experimental results of our algorithm, and in Sect. 4, we provide a brief summary and discuss possible future directions of our study.

2 Proposed Method

We present an approach based on the fundamentals of the Retinex theory while utilizing scale-space computations, and superpixel segmentation to estimate the reflectance and shading of a single image. As aforementioned, we benefit from the Retinex theory, since it is based on the investigations of the human visual system. Numerous studies trying to solve ill-posed problems related to the image formation model utilize the Retinex theory as a building block of their strategy since algorithms based on the human visual system tend to provide simple yet effective solutions [45,53]. We prefer to use the Retinex theory in multiple scales since in our previous investigations, we explicitly demonstrated that simply carrying computations into scale-space improves the efficiency of color feature-based methods which is not surprising considering that colors are low-level features of the scene and the scale-space is sensitive to such features [46]. At each scale, we also use superpixel segmentation to better preserve local features [1]. In the following, we explain the steps of our strategy.

In our algorithm, we assume that the input image may contain noise, therefore we first apply a 3×3 median filter to the input image I. Then, we form an image pyramid containing the scale-space representation of I. We adaptively determine the number of levels of the image pyramid based on the image resolution as $N = \lfloor \frac{log(min(r,c))}{log(2)} \rfloor - 2$, where r and c are the numbers of rows and columns of the image, respectively. We do not take the coarsest two levels into account since locality degrades substantially at these layers which may negatively affect the preservation of the fine details in the scene [44,46]. Also, while forming the image pyramid, at each scale, we use a 5×5 Gaussian kernel during the downsampling operation to prevent aliasing in the downsampled image as much as possible.

At each scale, we utilize the assumptions of the Retinex theory [28] to estimate the reflectance, in particular, we are following the assumptions of the color Retinex method introduced by Grosse *et al.* [24]. The Retinex algorithm enables us to differentiate between image variations that are either caused by reflectance or shading by investigating the local derivatives of the image [53]. While large derivatives are associated with reflectance changes $\hat{r}$, small derivatives are related to changes in shading. Given a certain threshold t, the horizontal gradient ($\mathbb{G}_x$) and the vertical gradient ($\mathbb{G}_y$) can be classified as follows:

$$\hat{r}_x = \begin{cases} \mathbb{G}_x, & \text{if } |\mathbb{G}_x| \geq t_x \\ 0, & \text{otherwise} \end{cases} \qquad \hat{r}_y = \begin{cases} \mathbb{G}_y, & \text{if } |\mathbb{G}_y| \geq t_y \\ 0, & \text{otherwise} \end{cases}, \tag{2}$$

where $\hat{r}_x$, and $\hat{r}_y$ represent the classified horizontal and vertical gradients for the reflectance.

We assume that there is a single light source illuminating the scene, thus all illumination changes can be represented by a single direction in RGB color space. We can denote this direction as the brightness subspace b, which can be represented by the span of the white vector $[1, 1, 1]^T$ [24]. This vector demonstrates the uniform changes in all three color channels equally, i.e. it controls the changes in brightness. On the other hand, the changes in chromaticity, i.e., the color quality irrespective of brightness, are available in the chromaticity subspace ch which is the null space of the white vector. We denote the projection of the input image into these subspaces as $\mathbb{G}^b$ and $\mathbb{G}^{ch}$, and utilize separate thresholds for the brightness and chromaticity changes [24]. In the color Retinex method, specific thresholds are chosen for images by utilizing cross-validation. In other words, for a given dataset, all parameter combinations are used for all images, and then for each image, the parameters that provide the lowest mean error on the other images are chosen. Although this approach provides satisfying outcomes on a specific benchmark, its computational complexity is high, and in case the scenes' statistical distributions vary significantly across the dataset the efficiency of the algorithm may decrease. Therefore, we prefer to rely on the statistical information of each image itself to determine effective thresholds adaptively, and compute the gradients for the reflectance $\hat{r}_x$, and $\hat{r}_y$ according to these thresholds as follows:

$$\hat{r}_x = \begin{cases} \mathbb{G}^b_x, & \text{if } \left|\mathbb{G}^b_x\right| \geq \alpha t^b_x \text{ or } \left|\mathbb{G}^{ch}_x\right| \geq \alpha t^{ch}_x \\ 0, & \text{otherwise.} \end{cases} \tag{3}$$

$$\hat{r}_y = \begin{cases} \mathbb{G}^b_y, & \text{if } \left|\mathbb{G}^b_y\right| \geq \alpha t^b_y \text{ or } \left|\mathbb{G}^{ch}_y\right| \geq \alpha t^{ch}_y \\ 0, & \text{otherwise.} \end{cases} \tag{4}$$

where t^b_x and t^b_y, are associated with the thresholds for brightness with corresponding gradient directions. Similarly, t^{ch}_x and t^{ch}_y are the thresholds controlling the chromaticity information. All thresholds are computed by taking the means of $\mathbb{G}^b$ and $\mathbb{G}^{ch}$. α is a scaling factor which is determined as 0.5 based on practical experiments. Thanks to this adaptive thresholding operation, at this stage, we obtain a lower computational cost than the color Retinex method.

After classifying the gradients, we solve an optimization problem to find a reflectance image $\hat{R}$ that best matches these target gradients at each spatial location as follows:

$$\begin{aligned} \arg\min_{\hat{R}(\mathbf{x},\mathbf{y})} & [\hat{R}(\mathbf{x},\mathbf{y}) - \hat{R}(\mathbf{x}-1,\mathbf{y}) - w_y(\mathbf{x},\mathbf{y})\hat{r}_y(\mathbf{x},\mathbf{y})]^2 \\ & + [\hat{R}(\mathbf{x},\mathbf{y}) - \hat{R}(\mathbf{x},\mathbf{y}-1) - w_x(\mathbf{x},\mathbf{y})\hat{r}_x(\mathbf{x},\mathbf{y})]^2, \end{aligned} \tag{5}$$

where w_x and w_y are the weights emphasizing regions containing significant gradient changes that contribute the most to the overall gradient. We assume that in a local window, the closest pixels contribute more than the pixels that are far away from the region having a gradient change, and this can be controlled by a Gaussian function. To compute the weights, first, we find the local changes Φ by applying a Prewitt filter to

the input scene and then, we fit them into a Gaussian curve as, $w = \left|\exp\left(-\Phi/2\sigma^2\right)\right|$ with a scaling factor σ of 3 which is determined based on practical experiments.

After computing an initial reflectance at each scale, we utilize superpixel segmentation to prevent color ambiguities at local regions caused by possible noise and color variations. We divide the image into its superpixels by using simple linear iterative clustering (SLIC) [1]. We prefer to adopt SLIC among other methods such as LSC [31] and SEEDS [10] since it preserves the boundaries well and it has low computational complexity. We compute local average color of each superpixel and consider it as the reflectance of that specific cluster, thus we obtain a coherent reflectance estimate. Subsequently, we further enhance this coherency by using an edge-aware smoothing filter. We apply a guided filter as a post-processing step [25], and obtain the reflectance for each scale of the image pyramid.

Afterwards, we collapse the pyramid $\mathcal{P}\{\hat{R}\}$ to obtain the final reflectance. We collapse the pyramid by first upsampling the coarsest level so that it matches the size of the reflectance image in its consecutive finer scale as follows:

$$\mathcal{R}' = upsample(\mathcal{P}\{\hat{R}\}^N, size(\mathcal{P}\{\hat{R}\}^{N-1})), \tag{6}$$

where $\mathcal{R}'$ is the upsampled image, $upsample$ is an operation that matches the size of $\mathcal{P}\{\hat{R}\}^N$ to the size of $\mathcal{P}\{\hat{R}\}^{N-1}$.

Subsequently, we linearly combine the reflectance image $\mathcal{P}\{\hat{R}\}^{N-1}$ with $\mathcal{R}'$ as follows:

$$\mathcal{R} = average(\mathcal{R}', \mathcal{P}\{\hat{R}\}^{N-1}), \tag{7}$$

where $\mathcal{R}$ is the new reflectance at the scale $N-1$, and $average$ is a function that performs linear combination.

Then, we upsample $\mathcal{R}$ to match the size of the reflectance at scale $N-2$ and linearly combine the resulting image with $\mathcal{P}\{\hat{R}\}^{N-2}$. We continue to perform Eq. 6 and Eq. 7 until we reach the finest scale of the pyramid. The resulting image is our final reflectance. After estimating the reflectance component of the scene, we obtain the shading component by applying Eq. 1.

3 Experiments and Discussions

In this section, we first briefly explain our experimental setup by introducing the datasets and the error metric we utilize. Then, we discuss our algorithm's performance while providing both statistical and visual results. It is worth mentioning that the baseline method given in our results corresponds to the baseline algorithm given in the study of Bonneel *et al.* [12] where the chromaticity image is considered as reflectance, and the square root of the direct average of the RGB channels is taken as shading.

3.1 Datasets and Evaluation Strategies

To benchmark our algorithm we utilize three datasets having different characteristics, such as containing single objects and complex scenes.

The MIT Intrinsic Images dataset [24] is created by Grosse *et al.* in 2009, and since then, it is one of the notable intrinsic image decomposition benchmarks in this field. The MIT Intrinsic Images dataset includes a total of 220 images, where the scenes are created with 20 real objects having different shapes and texture details. Each image contains a single object placed in front of a plain black background, and each scene is illuminated via a single light source positioned at different locations. The MIT Intrinsic Images dataset considers the reflectance, shading, and diffuse components, alongside the specularity information and the binary masks highlighting the positions of the objects. It is worth to note that similar to other works [4,7,19], we statistically evaluate our method on the images utilized in the evaluation of the original study of this dataset [24].

To analyze how well our algorithm performs on complex scenes, we utilize two datasets namely, the Bonneel dataset [12], and the MPI Sintel dataset [15]. The dataset introduced by Bonneel *et al.* [12] consists of photorealistic scenes. The images contain challenging features such as transmissive surfaces, defocus blur, subsurface scattering and specular elements. The reflectances of the scenes are created using computer graphics, while the shading is obtained by applying Eq. 1. The MPI Sintel dataset contains 23 different scenes, and 1064 rendered images obtained from an open-source 3D animated short film called Sintel [15]. The dataset is created for the evaluation of optical flow but it has also become a widely used benchmark for intrinsic image decomposition tasks. For each synthetic scene, MPI Sintel provides the ground truth reflectance component of the scene. Since the images in this dataset contain repetitive scenes, in our experiments we utilize random frames from each scene.

To provide a statistical investigation we follow the common practice of performing an objective evaluation through the widely used quality metric in the field of intrinsic image decomposition, namely, local mean squared error (LMSE) which is particularly designed for this task [24]. LMSE is based on the calculations of the classical mean squared error, and it is computed by averaging the MSE scores over overlapping windows. While reporting the results, we calculate LMSE by using a window size of 20 as in the original work of the metric. It is important to note that we report the results of the intrinsic image decomposition methods either according to recent comprehensive studies [4,12,19] or running the codes of the methods published by the authors in their default settings.

3.2 Discussion

In Table 1, we present our statistical results for the MIT Intrinsic Images dataset where we compare our algorithm with 19 different traditional and learning-based methods. Our algorithm is among the top-three best performing methods, while the two neural networks-based models achieving lower LMSE scores are fine-tuned on this dataset. Compared to the reflectance, we obtain a lower mean error for the shading component where we greatly preserve the structures in the objects (Fig. 1). It is worth mentioning that we conducted an ablation study on our algorithm where we analyzed the contributions of the individual parts of our method. First, we removed both the scale-space computations and the superpixel segmentation, then we removed either the scale-space computations or the superpixel segmentation and we noted that in case we remove one

Table 1. Statistical comparison of the performance on the MIT Intrinsic Images, Bonneel, and MPI Sintel datasets. For each dataset, LMSE scores are given. The top results are highlighted using color coding as follows, best: blue, second-best: cyan, third-best: green, forth-best: yellow, and fifth-best: orange. The methods highlighted with * are fine-tuned on the corresponding datasets. To follow the original work of Bonneel *et al.* [12], we only report the mean LMSE score on this dataset. For the MPI Sintel dataset, we do not render the shading component, and we prefer to use the actual ground truth reflectance provided by the benchmark to evaluate our results. Also, for the MPI Sintel dataset, we set a fixed threshold for the method of Grosse *et al.* [24].

MIT Intrinsic Images dataset [24]

Algorithm	Reflectance	Shading	Average	Algorithm	Reflectance	Shading	Average
Baseline	0.0704	0.0685	0.0694	Yuan *et al.* (2019) [51]	0.0462	0.0537	0.0499
Grosse *et al.* (2009) [24]	0.0529	0.0774	0.0652	Ma *et al.* (2020)* [33]	0.0212	0.0192	0.0202
Gehler *et al.* (2011) [23]	0.0393	0.0282	0.0338	Xu *et al.* (2020) [49]	0.0614	0.0672	0.0643
Shen *et al.* (2011) [37]	0.0523	0.0522	0.0523	Liu et al (2020) [32]	0.0640	0.0474	0.0557
Zhao *et al.* (2012) [53]	0.0512	0.0253	0.0383	Ren *et al.* (2020) [36]	0.0486	0.0737	0.0611
Barron and Malik (2014) [2]	0.0491	0.0281	0.0386	Baslamisli *et al.* (2021) [4]	0.0390	0.0447	0.0419
Shi *et al.* (2017) [39]	0.0606	0.0595	0.0601	Baslamisli *et al.* (2021) [7]	0.0438	0.0418	0.0428
Baslamisli *et al.* (2018) [6]	0.0854	0.2038	0.1446	Qian *et al.* (2021) [35]	0.0291	0.0319	0.0305
Lettry *et al.* (2018) [29]	0.0436	0.0472	0.0454	Das *et al.* (2022)* [19]	0.0210	0.0220	0.0215
Yu and Smith (2019) [50]	0.0573	0.0765	0.0669	**Proposed**	0.0336	0.0266	0.0301

The Bonneel dataset [12]

Algorithm	Average	Algorithm	Average
Bousseau *et al.* (2009)* [14]	0.0299	Barron and Malik (2014)* [2]	0.0352
Grosse *et al.* (2009)* [24]	0.0289	Bell *et al.* (2014)* [9]	0.0262
Shen *et al.* (2011)* [37]	0.0201	Narihira *et al.* (2015)* [34]	0.0212
Gehler *et al.* (2011)* [23]	0.0217	Zhou *et al.* (2015)* [54]	0.0190
Zhao *et al.* (2012)* [53]	0.0301	Lettry *et al.* (2018) [29]	0.0260
Garces *et al.* (2012)* [22]	0.0405	Ren *et al.* (2020) [36]	0.0287
Bonneel *et al.* (2014)* [13]	0.0275	**Proposed**	0.0244

MPI Sintel dataset [15]

Algorithm	Reflectance	Algorithm	Reflectance
Baseline	0.0415	Lettry *et al.* (2018) [29]	0.0464
Grosse *et al.* (2009) [24]	0.0450	Ren *et al.* (2020) [36]	0.0393
Shen *et al.* (2011) [37]	0.0399	**Proposed**	0.0402
Zhao *et al.* (2012) [53]	0.0676		

or two of our main steps, our algorithm's error increases from 0.0301 up to 0.0351 on average. Thus, each part of the algorithm has a contribution to its performance.

As given in Table 1, our algorithm focusing on single objects also performs well on the Bonneel dataset which contains significantly more complex scenes than the MIT Intrinsic Images dataset. Without any modification, we outperform several algorithms on this benchmark, while some of the existing methods achieve lower LMSE scores. However, it is important to note that most of the results utilized for comparison are taken from the study of Bonneel *et al.*, where the authors fine-tuned all methods on this dataset [12]. In other words, several distinct parameter combinations are used for each method, and for each image, the best-performing parameters are kept, e.g., the lowest LMSE achieved for each scene is reported. As shown in Fig. 2, overall, our algorithm recovers the colors in the reflectance well, while strong illumination conditions can leak into the reflectance component as in other algorithms.

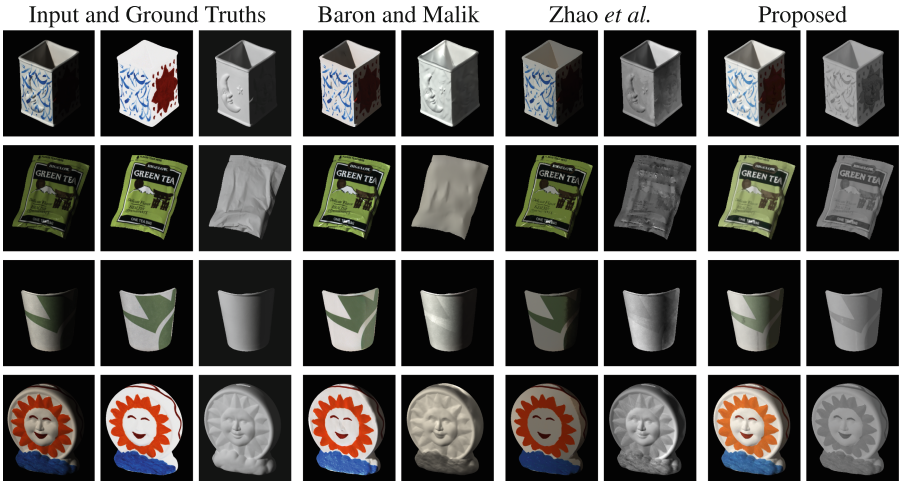

Fig. 1. Visual comparison of the algorithms on random examples on the MIT Intrinsic Images dataset.

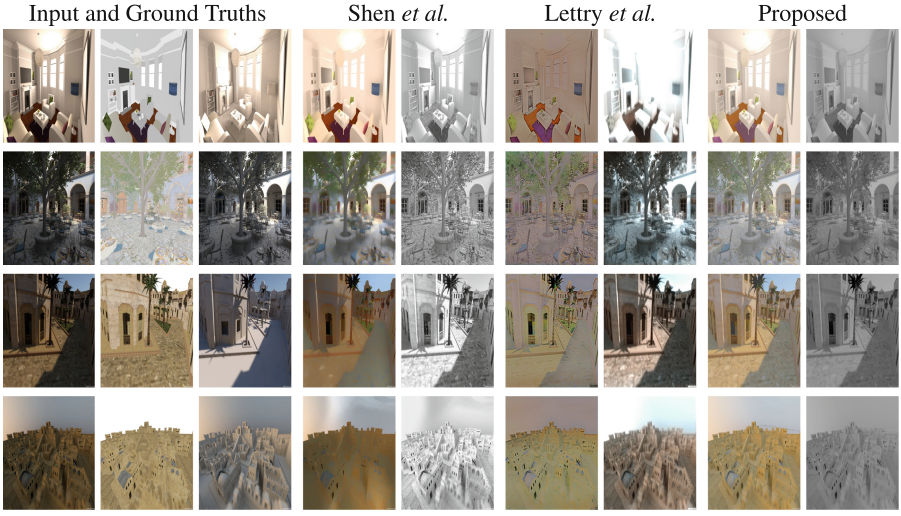

Fig. 2. Visual comparison of the algorithms on random examples on the Bonneel dataset.

We present our algorithm's statistical analysis on a subset of the MPI Sintel dataset in the last part of Table 1. Our method designed for single objects, can perform efficiently on this dataset and preserve the structures in the images well. The visual results for MPI Sintel are provided in Fig. 3.

As a final note, our algorithm achieves competitive results compared to other methods yet it suffers from widely recognized issues such as shadow, illumination and texture leakage, which are challenges shared by other algorithms in the

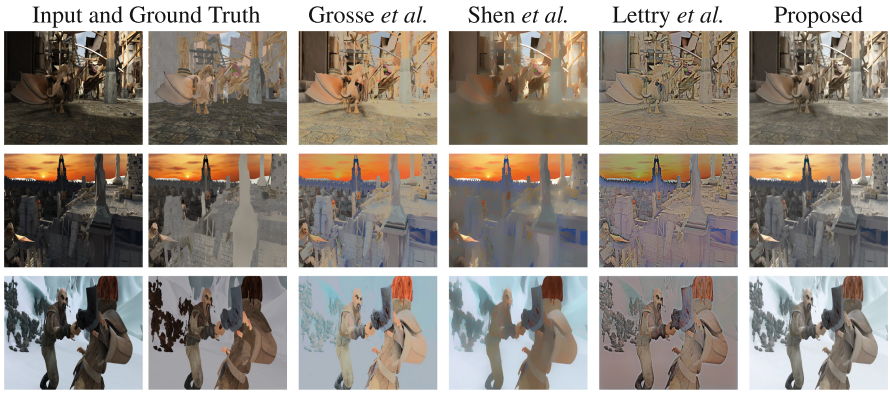

Fig. 3. Visual comparison of the algorithms on random examples on the MPI Sintel dataset.

field [6,17,29,37,42]. For instance, challenges due to shadows can be seen on the third row in Fig. 2 where shadows leaked into the estimated reflectances of the algorithms. Also, since we utilize the commonly used Lambertian image formation model, our method's applicability in real-world scenarios where non-Lambertian surfaces could be present should be further investigated.

4 Conclusion

Intrinsic images present different features of images and they can be used in various computer vision and computer graphics applications such as surface re-texturing and exposure correction to improve the outcomes. We have introduced a reflectance and shading estimation strategy that relies on the Retinex theory, scale-space computations, and superpixel segmentation. All components of our method contribute to its efficiency by giving importance to different aspects of the intrinsic image decomposition problem. We demonstrate that our simple yet effective method can achieve competitive results compared to the state-of-the-art without requiring a huge amount of data which we see as a strength of our algorithm. As future work, we will focus on the limitations of our algorithm such as shadow, illumination, and texture leakage to further improve our outcomes, particularly on complex scenes from new datasets. Furthermore, we will evaluate our method's performance on non-Lambertian surfaces.

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Hyperspectral and Spectral Imaging

Stabilization of the Spectral Power Distribution of a Tunable Multichannel LED Lighting System

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Abstract. The advancements in Light-Emitting Diodes (LEDs) have allowed spectrally tunable light sources to gain attention in many fields of research thanks to their ability to produce a specific light output. However, LED outputs can fluctuate with temperature, and aging components can lead to noticeable discrepancies in light characteristics. This study thoroughly examines the Telelumen Dittosizer light player LED panel to exemplify a commercially available device and the associated challenges in predicting and stabilizing its output. Then, we introduce an innovative algorithm aimed at addressing such a stabilization challenge, based on a straightforward characterization procedure along with an external spectrometer. The accuracy of the algorithm was validated with different inputs, achieving a $\Delta_{E,2000}$ lower than 0.5. Our findings demonstrate the ability to stabilize the spectral power distribution for a minimum of 30 min. The proposed algorithm is hardware-independent and adaptable to any combination of spectrally tunable light sources and spectrometers.

Keywords: Spectrally tunable light · LED lighting system · Dynamic adjustment · Luminous efficiency · Multichannel

1 Introduction

Light-Emitting Diodes (LEDs) have become a firmly established technology, dominating various lighting applications such as indoor and outdoor lighting [8,15], automotive lighting [12] and displays [7]. This is primarily due to their significant energy savings, ongoing reduction in manufacturing costs, durability, and compact size [25,26]. Furthermore, there is now a diverse array of LED options available from UV to IR through the visible range, allowing spectrally tunable light sources to gain attention in research works. Their ability to output a specific light spectrum in real time makes them valuable tools in many

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fields. For example they are used in psychophysical experiments related to human vision [14,18,24], light simulation [5], or for the construction of image datasets dedicated to machine vision [20]. Extensive work has been dedicated to optimizing these devices to produce light spectra with optimal characteristics, including color rendering, texture visibility, and energy efficiency [1,9,22,23].

While the light emitted from a multichannel LED panel can be finely adjusted by varying the drive level of each channel, ensuring that the device produces the expected spectrum is a significant challenge. Accurately predicting the output of a tunable LED device is complicated by the fact that the output of a light-emitting diode depends on its junction temperature [4,13,19]. This can result in decreased light efficiency and a slight shift towards longer wavelengths. Moreover, these fluctuations can vary significantly depending on the type of LEDs used. Another issue arises from the gradual degradation of LEDs over time, leading to a decrease of luminous efficacy [16].

In the literature, LEDs have been studied in order to model their behavior as a function of their junction temperature. Chhajed et al. [3] measured the junction temperature of 4 types of LEDs and reviewed their optical properties at different temperatures. Authors showed that increasing the temperature of LEDs induced a noticeable difference in light rendering. In order to fix such an issue, various approaches have been proposed. For example, Qu et al. [17] used the measurement of junction temperature to stabilize the output of a RGB LED lamp. Chen et al. [2] proposed a model using optical, thermal and electrical parameters. Li et al. [10] integrated photodetectors into the light-emitting diodes when Llenas et al. [11] integrated a compact spectrometer to stabilize the output of a tunable LED device using a closed-loop feedback system.

The goal of the work presented in this paper is to propose a new approach based on an algorithm able to dynamically adjust the drive levels of the different channels of any tunable multichannel light source in order to match a given spectrum and maintain it over a long period of time using a spectrometer. The following is divided into two main parts. Firstly, a full characterization of the spectrally tunable LED device used in our study will be presented. This characterization will highlight the technological difficulties faced when trying to stabilize the output of such a device. After then, the algorithm we propose will be described and the results it provides will be discussed. Since the proposed algorithm is independent of hardware the code will be made available.

2 Characterization of a Tunable Multichannel LED Panel

The measurements carried out to characterise a tunable multichannel LED lighting system used in this study were made on the basis of a range of power supply configurations applied over a long period of time to the lighting system. The goal was to quantify the evolution of the light output until reaching thermal stability and to identify the difficulties in getting a stable spectral power distribution over time.

2.1 Technical Specifications

The spectrally tunable LED panel used in this work is a Dittosizer light player from Telelumen (see Table 1 for its main characteristics). The unit was suspended to the ceiling of a 250 × 250 cm light booth with white walls. Nineteen out of the 24 available channels were retained for characterization purposes. The two ultraviolet (UV) and two infrared (IR) channels were excluded as they fall outside the visible range. Additionally, one channel was omitted due to its non-linear behavior in comparison to its drive level.

The measurements of the Spectral Power Distributions (SPD) of light sources were performed using a JETI spectraval 1511 spectrometer. The characteristics of this spectrometer are summarized in Table 2. The measured SPD of the 24 LEDs of the Telelumen panel are shown in Fig. 1. The range selected for the measurement was from 380 nm to 780 nm with 5 nm spectral resolution.

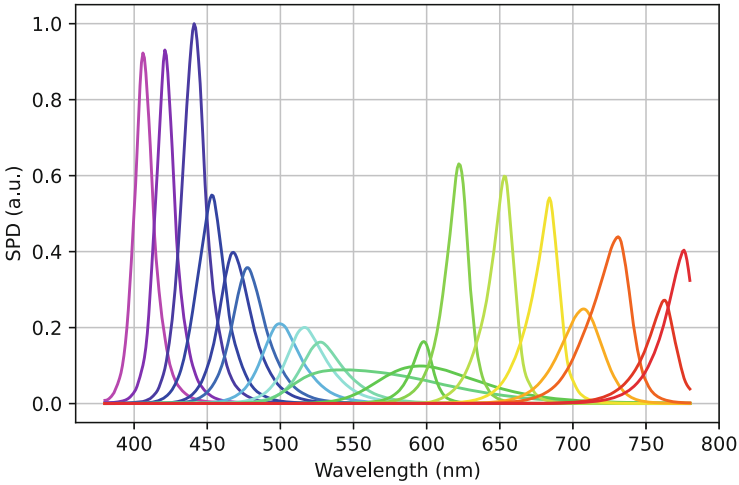


Fig. 1. Spectral power distribution of the 19 selected channels within the visible range of the Telelumen LED panel measured with a spectral resolution of 5 nm and interpolated to 1 nm with akima interpolation.

Both the tunable LED panel and the spectrometer were connected to a PC (i7, 32 GB RAM) using their respective Application Programming Interfaces (APIs), Python 3.10 and the Luxpy library [21]. The Telelumen panel can be controlled by sending directly an array of 24 float values between 0 and 1 representing the drive levels for each channel. The API of the Telelumen panel can also be used to get the readings of the internal temperature sensors.

Table 1. Characteristics of the Telelumen Dittosizer light player

Size	602 × 602 mm
Power consumption	∼100 W
N° of channels	24 (365–940 nm)
Max luminous output	∼5000 lm
Precision	250:1

Table 2. Characteristics of the JETI spectraval 1511 spectrometer

Spectral range	350. . . 1000 nm
Spectral resolution	5 nm
Measuring range	0.2. . . 140000 cd/m^2

2.2 Influence of the Temperature on the Light Output

The output of a LED depends on the temperature of its junction. Temperature fluctuations induce two primary effects on the output of a LED: dimming and spectral shift. As temperature rises, the luminous efficacy of LEDs decreases, which can be mitigated by gradually increasing the drive levels. Conversely, the spectral shift presents a challenge as it results in a noticeable color deviation. Both these effects are more pronounced for LEDs emitting longer wavelengths.

Measuring the junction temperature of a LED is not obvious. In an experimental setup, indirect estimation can be attempted by gauging the temperature of the heatsink on which the LED is mounted on. However, in a real device containing multiple LEDs of different nature and no control over the positioning and the number of temperature sensors, estimating the individual junction temperatures becomes unfeasible.

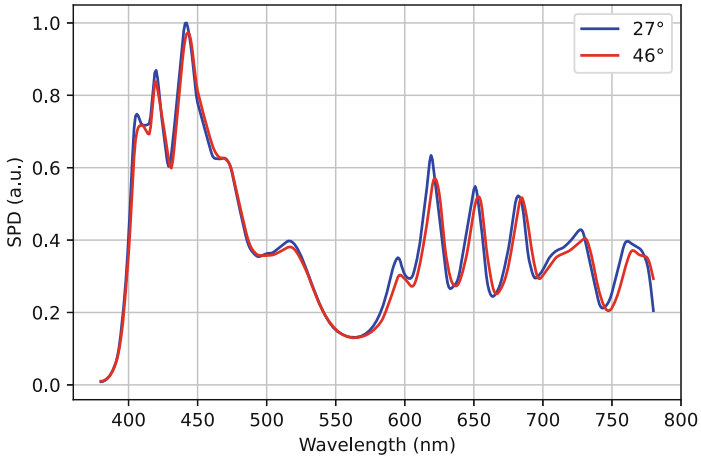


Fig. 2. Variations in spectral output with increasing temperature of LEDs.

To address this issue, the temperature reading from a sensor on the heatsink of the device, was used. Such a reading is easily accessible through the API without requiring disassembly of the device and it is considered to be closely correlated with the junction temperature.

Figure 2 illustrates the impact of temperature on the output of a mulichannel LED device. In this example, the drive level of all channels was maintained at 15% until the device reached thermal stability. A measure of the SPD was performed at internal temperatures of the luminaire of 27 °C and 46 °C. A significant variance in spectral output can be observed once the device stabilizes at its maximum temperature.

2.3 Characterization Procedure

As mentioned before, the objectives of the characterization were both to assess how temperature affects the light output of a multichannel LED lighting system and to identify the difficulties involved in stabilising over time the emitted spectrum. A method was devised to consistently measure the light output across various temperatures of the device. Since the 19 individual channels chosen are entirely independent, the characterization of the Telelumen LED panel was performed by measuring the light output from each channel separately. This was achieved across 100 different drive levels and over 15 temperature settings ranging from 30 °C to 45 °C. To regulate the device's temperature, all LEDs were turned on until the device reached the desired temperature just before each measurement. The detailed procedure for characterization is outlined in Fig. 3. Throughout the measurements, the spectrometer was positioned directly under the LED panel at a distance of about 55 cm.

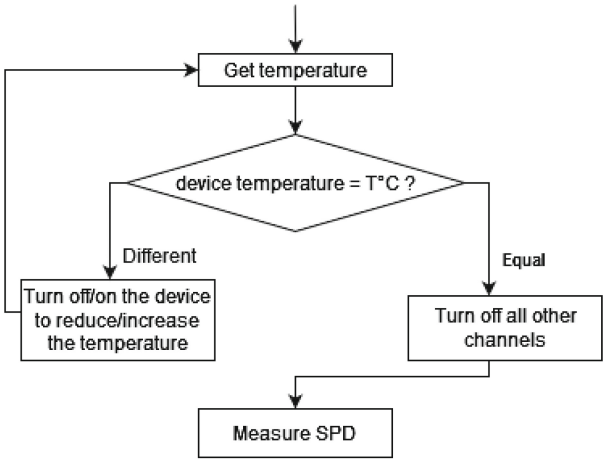


Fig. 3. Flowchart of the characterization procedure

2.4 Characterization Results

Measurement values obtained at minimum (30 °C) and maximum (45 °C) temperature of the heatsink are summarized in Table 3.

Measurement results clearly show that temperature affects all channels to varying degrees. Channels V1 to B1 are less sensitive exhibiting a reduced loss of luminous efficacy and no discernible spectral shift. B2 to G2 have a more pronounced loss of luminous efficacy along with a slight spectral shift. Channels L to FR3 exhibit a significant loss of intensity, reaching a maximum of 30% on channel R1, accompanied by a significant spectral shift towards the longest wavelengths of the visible range and is gradual with increasing temperature (noted as "grad" in Table 3).

In addition, the peak amplitude was measured to ascertain its proportionality with the drive level at different temperatures. It was confirmed that all selected channels exhibit almost linear properties, although the slope decreases with the decrease in luminous efficacy.

Table 3. Characterization of the 19 channels of the Telelumen LED panel in the visible range.

	V1	V2	RB1	RB2	B1	B2	C	G1	G2	L
Peak wavelength (nm)	406	421	441	454	468	478	500	517	528	544
Amplitude attenuation	7%	6%	4%	3%	4%	5%	16%	1%	15%	2%
Peak shift (nm)	no	no	no	no	no	<10	<10	<10	<10	grad

	PC-A	OR	R1	R2	DR1	DR2	FR1	FR2	FR3
Peak wavelength (nm)	594	624	636	654	686	708	732	765	777
Amplitude attenuation	8%	27%	30%	28%	6%	21%	28%	10%	7%
Peak shift	grad	grad	grad	grad	grad	grad	grad	grad	grad

3 Spectral Stabilization Algorithm

The characterization conducted in the previous section illustrates the challenges encountered in stabilizing the SPD of the light emitted by a tunable multichannel LED panel. In this section, we propose an algorithm able of determining the optimal drive levels for a specified output and maintaining its stability over an extended duration. This algorithm is based on a simplified characterization procedure consisting in only measuring the maximum output of each channel.

3.1 Theoretical Optimization

In theory, with a comprehensive characterization of the device, as outlined in the preceding section, and a thorough understanding of all parameters, it would be possible to precisely predict the input drive levels required to replicate a target spectrum.

Given that the output of a LED device is the sum of its individual channels, we could predict the precise output by performing the following calculation:

$$\forall \lambda \in [380, 780], S_{T,I}(\lambda) = \sum_{i=0}^{L-1} S^i_{T^i,I^i}(\lambda) \tag{1}$$

with L the number of channels, T^i and I^i the temperature and drive level of the i^{th} channel.

Then, a simple minimization of the mean squared error (MSE) can be performed to find the optimal drive levels:

$$I^{new} = \min_{I} \int_{380}^{780} \left(Ta(\lambda) - \sum_{i=0}^{L-1} S^i_{T^i,I^i}(\lambda) \right)^2 d\lambda \tag{2}$$

with Ta the target SPD to match.

However, without a full characterisation we cannot have access to the values of $S^i_{T^i,I^i}(\lambda)$. Therefore, we propose an approximation using only the measurements taken at the maximum drive level at a single temperature.

3.2 Approximation of the Impact of the Temperature

It is possible to approximate the output of a LED at any drive level using the spectrum of the output at the maximum drive level with a linear approximation [6]:

$$S_{T_0,I_0}(\lambda) = I_0 \times S_{T_0,1}(\lambda)$$

with S_{T_0,I_0} the SPD of an LED at the temperature T_0 and drive level I_0, and $S_{T_0,1}$ the SPD at its maximum drive level.

The primary limitation of this approximation is its viability solely under constant temperature conditions.

The impact of temperature on the output of the channels can be represented by incorporating a deviation in intensity dI and a shift in wavelength $d\lambda$ such as:

$$S^i_{T_1,I_0}(\lambda) = (I^i_0 + dI^i) \times S^i_{T_0,1}(\lambda + d\lambda^i)$$

We can obtain these deviations by minimizing the MSE criterion with a single measurement of the output of the device.

$$(dI_{min}, d\lambda_{min}) = \min_{dI,d\lambda} \int_{380}^{780} \left(M(\lambda) - \sum_{i=0}^{L-1} (I^i + dI^i) \times S^i_{T_0,1}(\lambda + d\lambda^i) \right)^2 d\lambda \tag{3}$$

with M the measured spectrum.

3.3 Optimization of the Drive Levels

Ultimately, to determine the updated drive levels, we simply substitute the theoretical values in Eq. 2 with the approximation obtained with Eq. 3:

$$I^{new} = \min_{I} \int_{380}^{780} \left(Ta(\lambda) - \sum_{i=0}^{L-1} (I^i + dI^i_{min}) \times S^i_{T_0,1}(\lambda + d\lambda^i_{min}) \right)^2 d\lambda$$

with Ta the target spectrum.

4 Experimental Results

To validate the accuracy of the proposed algorithm, various inputs were compared with and without stabilization following this protocol:

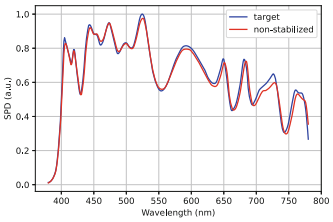
1. Turn on the LED panel at the selected drive levels.
2. Measure and record of the output as the target.
3. Let the device cool down.
4. Record the output **with** stabilization for 30 min.
5. Let the device cool down.
6. Record the output **without** stabilization for 30 min.

As an illustration, Fig. 4 and Fig. 5 show the contrasting behavior of the LED panel with and without the stabilization algorithm. Figure 4a/5a depicts the output after 30 min without stabilization while Fig. 4b/5b showcases the output with stabilization. This example illustrates that, as the device heats up, the LEDs experience a decrease in luminous efficacy, particularly those emitting longer wavelengths. This induces both a reduced illuminance and a color shift. With the stabilization algorithm, the loss of luminous efficacy is compensated by a gradual increase of the drive levels.

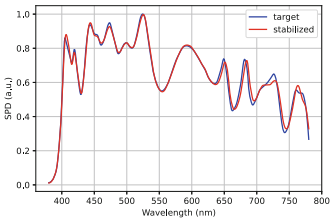
After 30 min, the stabilized output is kept under a $\Delta_{E,2000}$ lower than 0.5 compared to the target, whereas the non-stabilized output exhibits a $\Delta_{E,2000}$ difference exceeding 1. Additionally, the illuminance remains stable around 705 lux whereas it decreases to under 700 lux without stabilization.

The evolution of the difference to the target can also be compared using spectral similarity metrics such as the Mean Square Error as shown in Fig. 6.

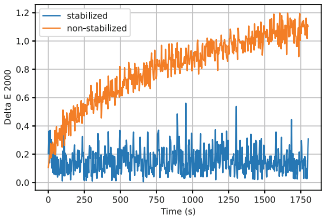
In order to quantify the color difference induced by the rising temperature of the device, a photo of an X-Rite ColorChecker was taken after one hour under a D65 simulation with and without stabilization. The resulting ΔE_{2000} color difference can be seen on Fig. 7. The background and over half of the squares of the ColorCheker display a ΔE_{2000} over 1, meaning that most of the image is perceivably shifted in colors.



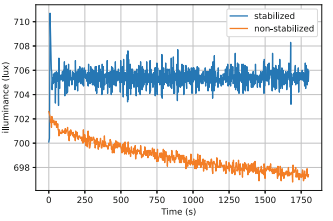
(a)



(b)

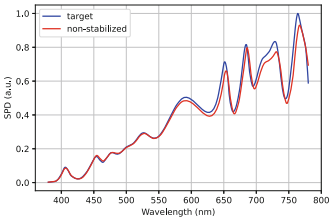


(c) Evolution of the $\Delta_{E,2000}$ differences.

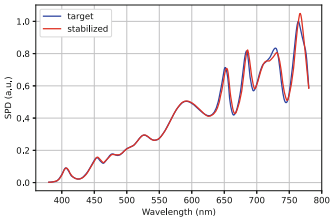


(d) Evolution of the illuminance.

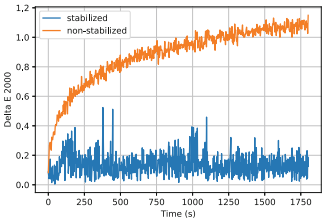
Fig. 4. Comparison of the output with and without stabilization (D65).



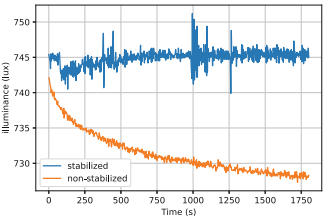
(a)



(b)

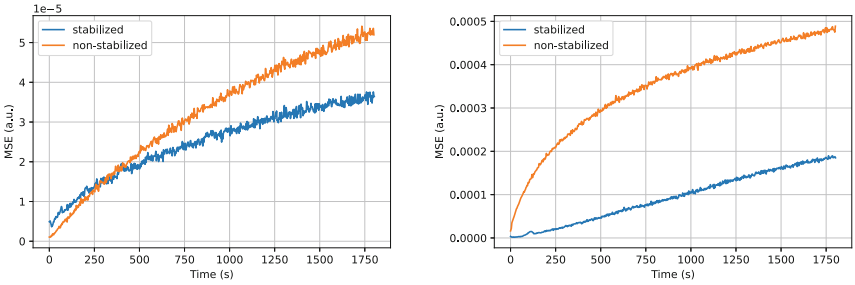


(c) Evolution of the $\Delta_{E,2000}$ differences.



(d) Evolution of the illuminance.

Fig. 5. Comparison of the output with and without stabilization (Illuminant A).



(a) D65 (b) Illuminant A

Fig. 6. Evolution of the MSE for the two examples

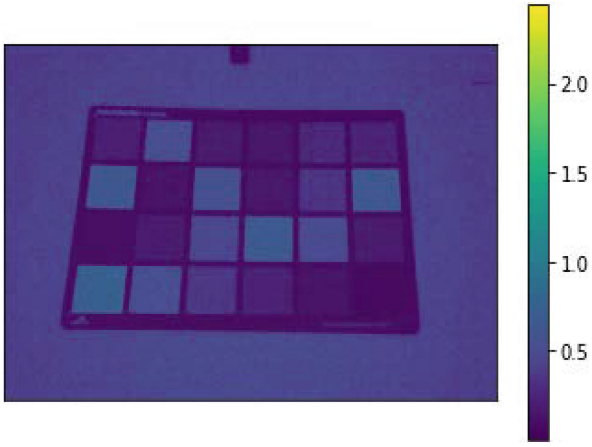


Fig. 7. ΔE_{2000} color difference between stabilized and non-stabilized light

5 Conclusion

In our study, we conducted a comprehensive characterization of the Telelumen Dittosizer light player LED panel, illustrating the challenges faced when trying to predict and stabilize the output of a commercially available spectrally tunable light. Then, we developed a simplified characterization procedure as the foundation for a stabilization algorithm. The proposed algorithm successfully achieved real-time stabilization of the output of a multichannel LED panel, both in terms of color and illuminance. Moreover, it effectively keeps the variations of the light output imperceptible to the human observer, making it usable in the context of human vision psychophysical experiments. Furthermore, the algorithm's hardware independence allows for adaptation to various combinations of spectrally tunable light sources and spectrometers.

There exist opportunities for improvement in this algorithm. Specifically, the computation time could be reduced optimizing the code, using a different optimizer and better positioning of the spectrometer. Different cost functions could also be investigated to better match specific light characteristics such as Color Rendering Index (CRI), Correlated Color Temperature (CCT), etc. This would improve both the precision and versatility of the proposed algorithm. This work focused on light in the visible range, but with minor changes it could be adapted to a wider spectral range.

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Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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Low Light Hyperspectral Imaging Using HDR Methods

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Abstract. Hyperspectral imaging provides spectral information for every pixel in an image, though the data is less reliable for low levels of illuminance. Merging hyperspectral imaging with High Dynamic Range (HDR) imaging can improve the loss of accuracy in low-lit environments by compensating for dark pixels with high exposure values. A study involving a light-booth with a maximum of 135 cd/m^2 shows a significant improvement in spectral and photometric accuracy of the spectral data using HDR techniques. The sRGB image quality also improves significantly. The HDR merging method using a mean or median value of valid pixels also significantly improves results as compared to the first correct pixel. The results from this study provide a promising outlook for the use of hyperspectral imaging in poorly lit scenes and should be further investigated.

Keywords: HDR imaging · low-lit hyperspectral imaging

1 Context

Accurate reproduction in digital imaging is strongly influenced by the characteristics of the photodetectors used in the capturing instrument, by the display device and the color management profile [1]. In digital imaging, any colored pixel is characterized by its RGB value representative of the chromaticity coordinates. HyperSpectral Imaging (HSI) combines 2D imaging sensors with spectroscopy and provides image data containing spatial and spectral information. For each pixel of the image, the Spectral Power Distribution (SPD) is measured to generate datasets with three dimensions (row, column, of the pixel and corresponding spectral information per wavelength), see Fig. 1. The use of spectral data (spectral reflectance for objects combined with spectral distribution for light sources) instead of RGB values makes it possible to determine, for all the elements of a scene, the spectral information reaching the eye.

Recently, hyperspectral cameras have seen a fast development with the availability of high spectral/spatial resolution devices on the market. Though, the HSI devices that currently exist perform poorly under low light conditions (<50–100 cd/m^2). Spectral images of low-lit scenes show a dramatic reduction in spectral,

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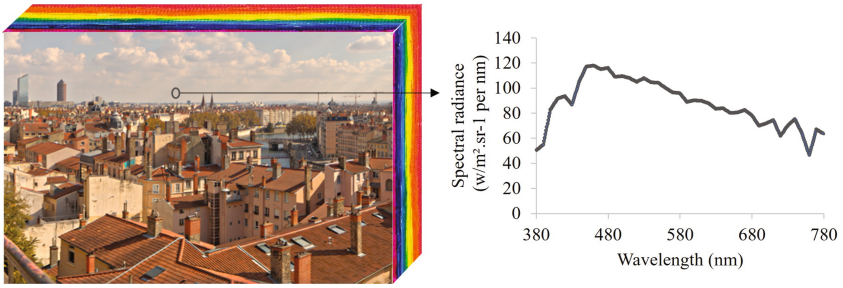

Fig. 1. A sRGB image obtained from an urban hyperspectral image (left); and the SPD of the circled pixel (right)

colorimetric and photometric accuracy [6]. This is particularly problematic for nighttime scenes that could benefit from a spectral characterization, for example to analyze and improve nighttime road lighting.

High Dynamic Range (HDR) imaging offers the possibility to accurately capture dark pixels as well as bright pixels by combining images with different exposure values. This technique can be extended to hyperspectral images also by taking hyperspectral captures at varying exposure levels to accurately obtain the spectral information for every pixel, dark as well as bright, for low-lit environments. Such a combination of imaging technique could perhaps improve the accuracy of hyperspectral images for scenes with low levels of lighting. This fusing technique provides a unique opportunity to compensate a low spectral accuracy in non-uniformly lit scenes with the help of HDR imaging's improved pixel accuracy, whether poorly lit or not. Such a technique has already been used for improving spectral imaging of artworks illuminated by artificial lamps [3], though its application on low-lit scenes is not yet known. In the context of this study, instead of scenes with equally present dark and bright pixels, we aim to concentrate on scenes with a majority of dark pixels. In contrast to non-uniform yet brightly lit scenes, to which most of the hyperspectral cameras are less sensitive, overall dark scenes are a critical issue for spectral imaging. In the future, if hyperspectral cameras can optimally capture dark scenes, such a fusing technique would always be relevant for low-cost spectral devices, which compromise on spectral precision at the expense of lower prices.

2 Material and Setup

2.1 Instruments

A Specim push broom hyperspectral camera (Specim FX10, S/N 1200061) was used to establish a generalized protocol. It uses a sCMOS sensor and works in the VIS-NIR range (400 nm–1000 nm). It can also apply binning on spectral and spatial data by reducing sampling intervals and thus saving memory required for each capture. Other specifications are detailed in Table 1.

A JETI Specbos 1211-UV spectroradiometer (S/N 2011273) was used for reference spectral measurements. Its spectral accuracy is 0.5 nm, chromatic accuracy is ±0.002 for the chromaticity coordinates under illuminant A and luminance accuracy is of 2% at 100 cd/m^2 under illuminant A. Table 1 gives a summary of all heading levels.

Table 1. Specifications of the Specim FX10 hyperspectral Imaging System

Characteristics	Value
Aperture	f/2.1
Spectral resolution	5.5 nm (FWHM)
Spatial resolution	1024 pixels
Focal length	15 mm
Binning	2×1
Frame rate	15 fps

2.2 Scene Acquisition Setup

This study aims to capture different low-light scenes with a hyperspectral camera using HDR imaging and to identify if there is a significant interest in doing so. In the context of nighttime road lighting, maximum luminance does not normally go above 15–30 cd/m^2. This could be a potential testing scene, as nighttime roads are a major challenge for spectral imaging.

Thus, we decided to identify two scenes: (proof of concept) a classic light-booth scene illuminated with a low-power lamp; and (concept testing): a tunnel road with non-uniform and low levels of lighting. Two measurement scenes, each containing a Macbeth Color Checker Chart (MCC) were chosen for testing, see Fig. 2 and Fig. 3. A light-booth with low-light which was illuminated by a single LED light source (CCT = 2800K, Ra =82, max luminance 115 cd/m^2) and a low-lit reduced-scale model of an automobile tunnel, illuminated with 24 LED chips (max luminance 35 cd/m^2, CCT = 5000K, Ra = 85). Their respective luminance maps (in cd/m^2) are also shown in Fig. 2 and Fig. 3. Each scene was illuminated for a minimum 5 min before any measurement to ensure light source stability, which was stable within 5 s.

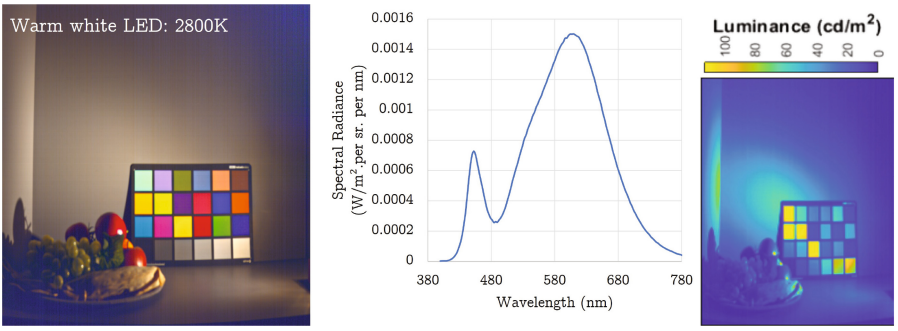

Fig. 2. The low-lit light booth (left), the SPD curve of the light source measured on MCC 19 (center) and the luminance map of the whole scene (right)

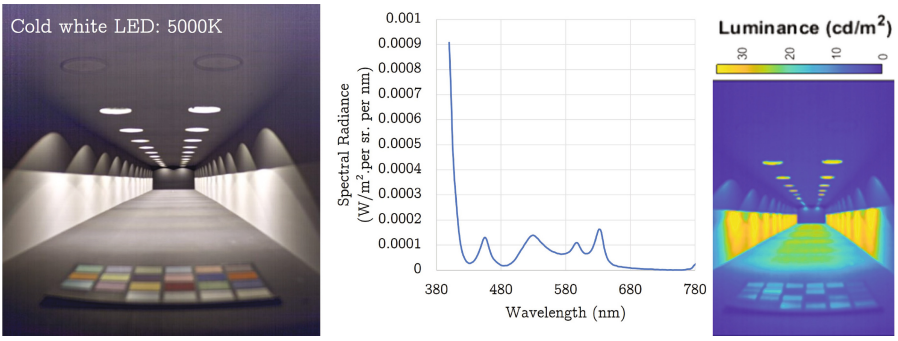

Fig. 3. The reduced-scale tunnel model (left), the SPD curve of the light source measured on MCC 19 (center) and the luminance map of the whole scene (right)

3 Methodology

High-Dynamic-Range (HDR) imaging fuses different images captured at multiple exposure values to create a single image, ensuring that each pixel is adequately exposed. In hyperspectral imaging, most of the new generation line-scanner devices, 'scan' and measure the spectral data of the whole scene 'line-by-line', which is repeated for each target wavelength. A line here generally translates to the array of spectral sensors embedded in the device as per the spatial resolution. To fuse the two imaging techniques, a streamlined process is necessary. The process involves the identification of a central Exposure Value (middle EV). Followed by the identification of two or more EVs, higher and lower to the middle EV, and finally capturing the spectral images at each EV.

To properly expose the spectral radiance of each pixel of the spectral image, the hyperspectral image obtained from each EV can be combined to form a single spectral image. In order to determine the EV which optimally exposes a pixel, we also need to identify the upper and lower limits of the spectral imaging system (spectral radiances corresponding to noise or saturated data).

To identify the raw radiance values of the HSI device that over/under expose the camera optics, hyperspectral captures were done under an equi-energy light source with increasing exposures of 5–10–20–40–80 ms. The captures were done in dark ($0\,cd/m^2$, camera lid on) to identify noise and at $2791\,cd/m^2$ to identify saturation (measured on reference white Spectralon). These measurements helped us in identifying:

a) The raw radiance value which corresponds to over-exposure, characterized by flat signals (3438 raw radiance units).
b) The raw radiance value which corresponds to under-exposure, characterized by unstable raw radiance with values comparable to noise (1.58 raw radiance units).

By using the upper and lower limits of raw radiance values for properly exposed pixels, the spectral data from different EVs can be fused to create a single spectral image. The merging techniques include: (i) hdrFIRST, which identifies the first radiance value (starting from the smallest EV) within the camera's limits and discards spectral data from other EVs; and (ii) hdrMEAN/hdrMEDIAN, which identifies all radiance values within the proper exposure limits and calculates either a mean (hdrMEAN) or median (hdrMEDIAN) radiance value per pixel. The complete methodology is outlined in a flowchart in Fig. 4.

Dark measurement (noise) was subtracted from each spectral capture before any further analysis. After dark subtraction, either type of merged HDR hyperspectral output requires further calibration to produce radiance in SI units. For calibration, captures were taken in a light booth covered with a white cloth of uniform reflectance ($\rho = 0.8$), illuminated by a bright-cold incandescent source, and containing a Macbeth ColorChecker (MCC) and a Spectralon white standard. The raw hyperspectral data corresponding to the Spectralon was extracted from the uncalibrated output file and divided by the reference spectral radiance measured in the same geometry with a spectroradiometer to obtain the calibration curve. This calibration technique is detailed in a previous publication [6].

For the purpose of data analysis, the spectral radiances of the MCC patches were extracted from each spectral image. These radiances were further converted to color coordinates (XYZ and CIELAB) values with the CIE 2° standard observer Color Matching Functions (CMFs). sRGB images were obtained from the hyperspectral images by using the XYZ tristimulus values of the entire image, for each pixel using the iCAM06 image color appearance model [2], with gamma = 1.5, p = 0.75, and F = 1 for average surround. D value for chromatic adaptation was fixed at 0.9 for real-world perception [7].

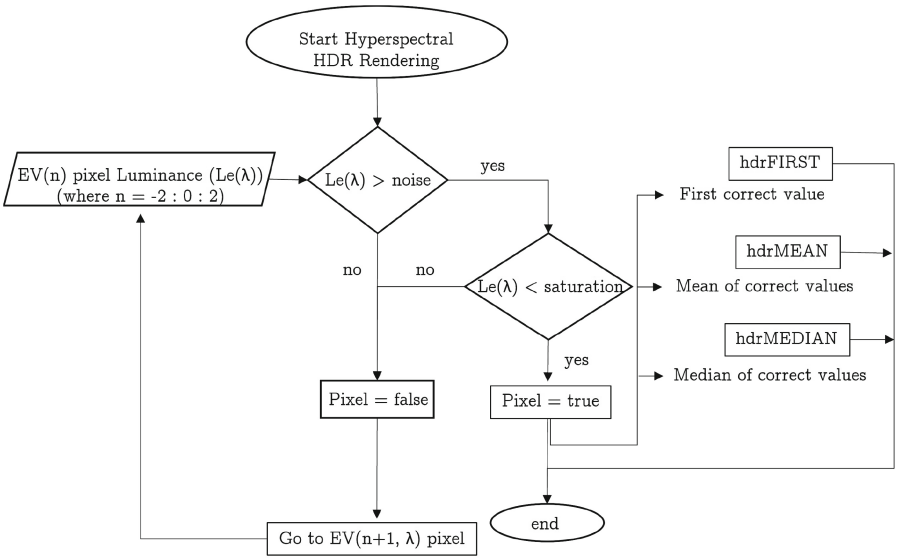

Fig. 4. Flowchart for hyperspectral HDR imaging acquisition using different fusion methods

4 Results

After obtaining the calibrated HDR hyperspectral images for the different merging methods, comparisons were done on the 24 MCC color patches to identify the method that produces the most accurate spectral image (for radiometric-photometric-colorimetric accuracy and image quality). Assessments were also done to compare the HSI-HDR data with HSI data obtained from a single optimum EV (HSI).

Radiometric comparisons were done using the Normalized-Root-Mean-Square-Deviation (NRMSD) scores between the spectral data acquired from the spectroradiometer (Eq. 1) and that of the different HSI methods. The normalization of RMS errors ensured that the differences in magnitude of the spectral radiances is taken in account. Photometric comparisons were done using Mean-Absolute-Percentage-Error (MAPE) between the luminance values (Eq. 2) obtained from the reference spectral data and the HSI method. Colorimetric comparisons were similarly performed using the DE*00 color difference formula. Images were evaluated with the Naturalness-Image-Quality-Evaluator (NIQE) that produces values between 0–1 [4]. It makes use of measurable deviations from statistical regularities observed in "natural" images [4]. In this context, natural images mean images that underwent no image enhancement, and a lower score indicates a higher image quality.

$$NRMSD\ (\%) = \frac{\sqrt{\frac{1}{N}\sum_{\lambda i}^{N}\left(s_{\lambda i,t} - s_{\lambda i,r}\right)^2}}{\max(s_{\lambda i,t}\ ;\ s_{\lambda i,r})} \times 100 \tag{1}$$

Where, $s_{\lambda i,t}$ and $s_{\lambda i,r}$ are the spectral radiances measured with the tested HSI system and with the reference (spectroradiometer) respectively, and N is the number of spectral samples (=144 wavebands).

$$MAPE = \frac{100}{n} \sum_{i=1}^{n} \left| \frac{Y_{i,t} - Y_{i,r}}{Y_{i,r}} \right| (\%) \tag{2}$$

Where $Y_{i,t}$ and $Y_{i,r}$ are the luminance values of the i^{th} color patch measured with the tested HSI system and with the reference (spectroradiometer) respectively, and n is the number of MCC patches.

The raw results for NRMSD, MAPE and DE*00 for the low-lit light booth are shown in Fig. 5, and that for the tunnel are shown in Fig. 6. The NIQE scores are given in Fig. 7, alongside each image obtained with the respective method.

As it can be seen from the figures, HSI-HDR methods show a clear enhancement in spectral and photometric accuracy for the low-lit light booth, but not for colorimetric accuracy. The NRMSD and MAPE values are lower (thus better) for either of the HSI-HDR method, and the one for HSI (single EV) method are higher (thus worse). For DE*00 scores, the colors seem to be better reproduced for the HSI method, while they degrade for HSI-HDR methods. For the tunnel, neither of the HSI-HDR method changed/improved the accuracy, for either of the metric from NRMSD, MAPE or DE*00 scores. If we look at the image quality scores, HSI-HDR methods systematically have a lower NIQE scores (thus better). Notably, hdrMEAN and hdrMEDIAN sRGB images have a lower amount of noise as compared to the HSI or hdrFIRST method, which seems to be relatively noisier and artifact ridden.

To analyze the results, a two-way ANOVA (ANalysis Of VAriance) was performed followed by a Tukey's HSD posthoc test to assess the significance ($p < 0.05$), see Table 2. The HSI imaging method (HSI, hdrFIRST, hdrMEAN and hdrMEDIAN) was the dependent variable while the various metrics identified above for accuracy measurements were the independent variables.

Table 2. ANOVA results for the spectral, photometric and colorimetric accuracy comparison for the two setups, under four different hyperspectral techniques.

Setup	METRIC	df	F-value	p-value
LIGHT-BOOTH	NRMSD	3	65.69	<2.2e−16
LIGHT-BOOTH	MAPE	3	249.94	<2.2e−16
LIGHT-BOOTH	DE*00	3	17.45	1.56e−08
TUNNEL	NRMSD	3	2.19	0.97
TUNNEL	MAPE	3	0.35	0.79
TUNNEL	DE*00	3	0.46	0.71

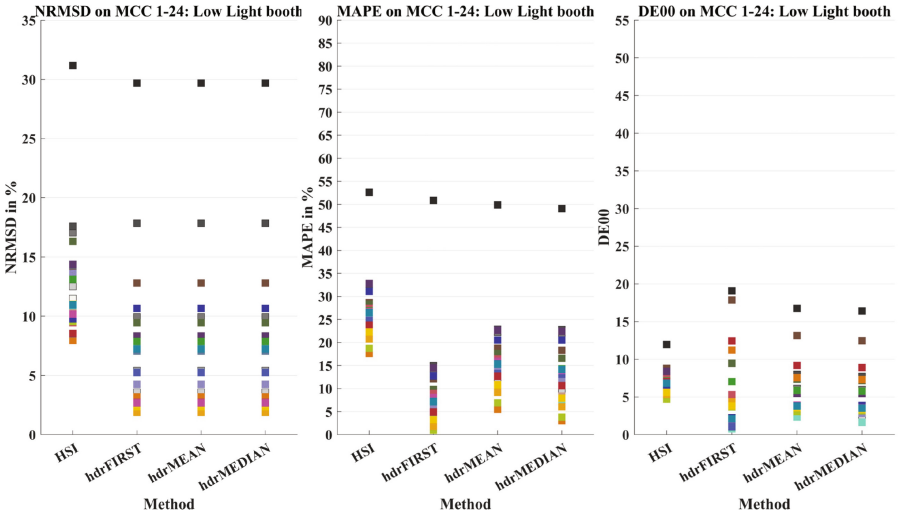

Fig. 5. Spectral, photometric and colorimetric accuracy comparison in the low-lit light booth for different hyperspectral techniques. The marker color represents the color of the MCC patch (Color figure online)

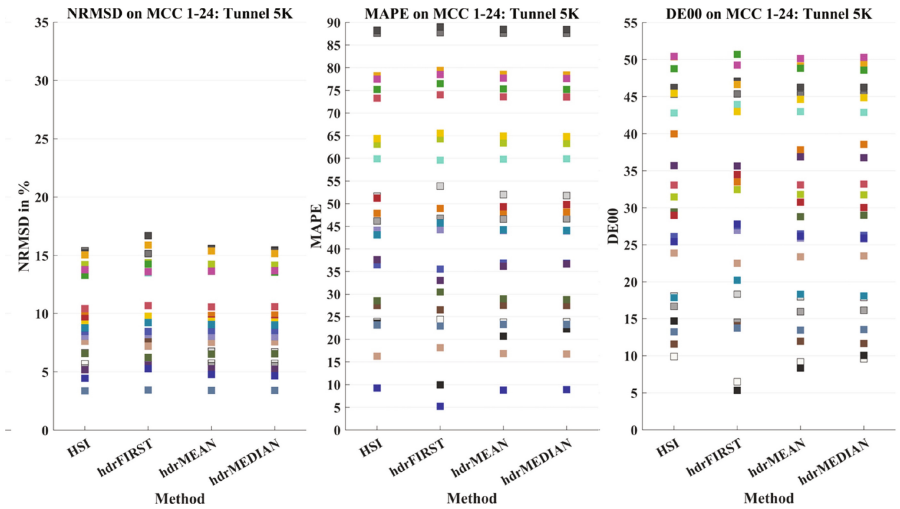

Fig. 6. Spectral, photometric and colorimetric accuracy comparison in the reduced-scale tunnel for different hyperspectral techniques. The marker color represents the color of the MCC patch (Color figure online)

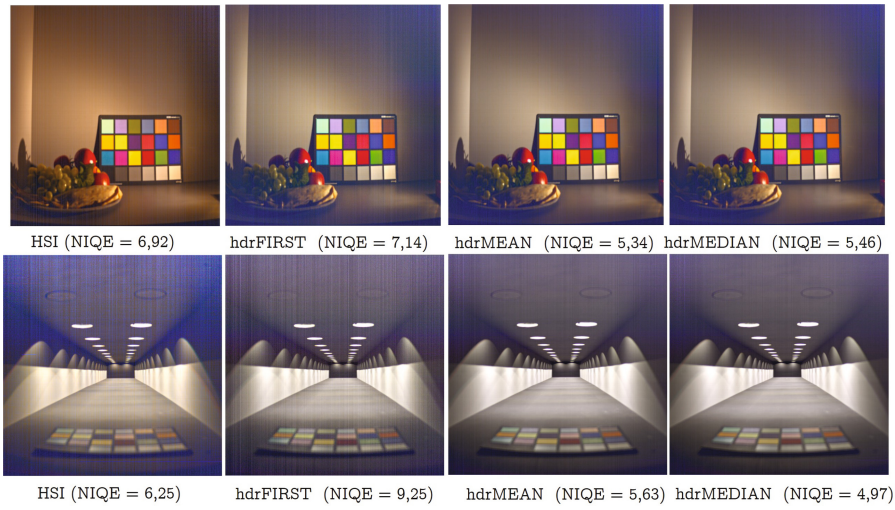

Fig. 7. Image quality comparison in the low-lit light booth (top) and reduced-scale tunnel (bottom) for different hyperspectral techniques. The NIQE value is provided under each image

The ANOVA results also show that, for the low-lit light booth, radiometrically and photometrically, a significant improvement was identified for either of the HSI-HDR method, see Fig. 8. Colorimetrically, the DE*00 values for the low-lit light booth data were significantly better for HSI (=single EV) method. The tunnel was similarly reproduced for all the methods (HSI or HSI-HDR), see Fig. 9.

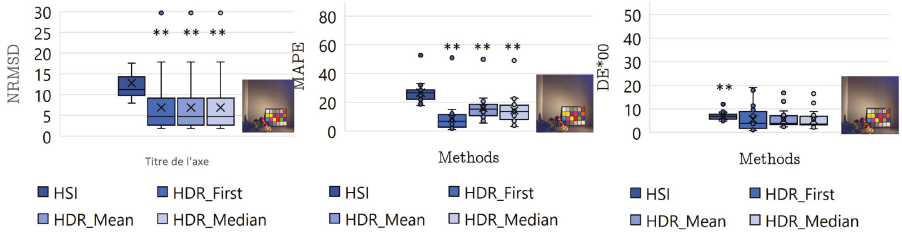

Fig. 8. ANOVA graphs for the spectral, photometric and colorimetric accuracy comparison in the low-lit light booth for different hyperspectral techniques

In terms of image quality, for either of the scenes, hdrMEAN and hdrMEDIAN methods produced the best quality images (visually and also as per the NIQE scores). No statistical test was done for unitary values of NIQE images per method.

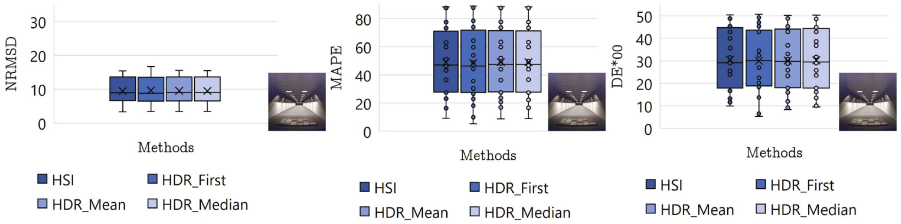

Fig. 9. ANOVA graphs for the spectral, photometric and colorimetric accuracy comparison in the reduced-scale tunnel for different hyperspectral techniques

5 Discussion

The use of HDR techniques for hyperspectral imaging shows a significant improvement in the spectral and photometric accuracy for a light booth illuminated by a low power LED lamp (max luminance = 135 cd/m^2). Interestingly, the colorimetric errors show a significant degradation for the HSI-HDR methods.

This type of behavior has already been reported in the literature, where different HSI systems compared against each other sometimes have relatively a poor colorimetric accuracy despite having a good spectral and photometric accuracy [5]. This phenomenon, which is out of the scope for this study, is still not understood and should be further studied. Perhaps, the use of a biased HDR bracketing could resolve this issue where only EVs higher than the central EV are selected instead of higher and lower EVs.

For a small-scale model of a tunnel, no impact is observed for the use of HSI-HDR method. This is probably due to the positioning of the Macbeth color chart. The chart is positioned horizontally in the very beginning of the tunnel, and the amount of noise is too high for any method to improve the radiance extracted from the spectral image. When looking at the image quality scores (NIQE), it can be seen that HDR techniques improve the overall image quality. Since the NIQE scores are calculated for the entire image, the impact of the MCC positioning is compensated by the improvement of accuracy in other parts of the image.

Normalized scores from the above tested metrics are shown in Fig. 10 to provide an overall analysis of the measurement accuracy using different methods. A lower aggregated score is considered to be better. It can be seen that, generally, the use of HSI-HDR techniques show an improvement for low-light hyperspectral imaging, at least for the low-lit light booth.

Since, the aggregate results of the three HSI-HDR methods are similar, the use of hdrMEDIAN method is preferred as it produces sRGB images of a higher quality compared to the other methods.

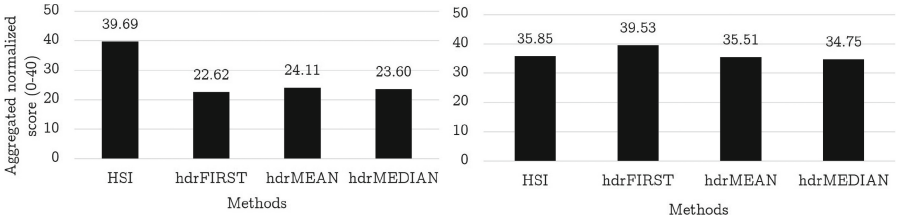

Fig. 10. Aggregated results for the different hyperspectral techniques for the low-lit light booth (left) and the reduced-scale tunnel (right)

6 Conclusion

Hyperspectral imaging provides a non-invasive method to analyze the spectral, photometric and colorimetric content of a complex scene through a single capture. Though, low-lit environments are prone to a reduction in accuracy due to a high noise. Extending the HDR imaging technique to hyperspectral imaging has demonstrated a significant improvement for the radiometric/photometric data and image quality for low-lit light-booth scene. The improper positioning of the MCC chart in the tunnel perhaps rendered the comparison of accuracy for different methods impossible in the small-scale model of a tunnel illuminated with LED chips. This highlights the importance of object positioning in controlled measurement scenes. A slightly tilted vertical positioning of the MCC could perhaps improve data acquisition and is planned in a future study. Globally, the technique shows promising results for hyperspectral imaging at low light levels.

This technique could also be extended for improving the spectral accuracy of hyperspectral cameras, which have a poor accuracy irrespective of the light levels. A further study is planned that aims to capture the abovementioned scenes with a low-cost hyperspectral camera with high inaccuracies using the median HDR-HSI method.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article

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Spectral Imaging Methods for Estimating Fluorescence Emission Spectra from Plant Grains and Leaves

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Abstract. We propose spectral imaging methods for estimating fluorescence spectra emitted from plant grains and leaves. Two types of multiband imaging systems with six channels are constructed using ordinary off-the-shelf cameras; the first of these is for rice grains, wherein a mobile phone camera is rendered multiband to detect fluorescence emission in the visible wavelength region; the second one is for plant leaves, wherein a monochrome camera is used with additional optical filters to detect chlorophyll fluorescence in the red to far-red wavelength region. A statistical approach, inspired by the Wiener estimator is developed to estimate fluorescence emission from the image data. Here, the observations are modeled by accounting for noise in the imaging system, and the incident fluorescence spectra to the system are estimated from the observed image data. The noise variance is estimated on the basis of the fluorescence spectrum measurement. In the experiments, UV-light is used to illuminate real rice grains and green leaves; the emitted fluorescence spectra are estimated, and the reliability of the proposed methods is demonstrated.

Keywords: multiband imaging system · fluorescence spectral estimation · plant grain and leaves · fluorescence emission

1 Introduction

Fluorescence is an optical phenomenon, in which a material is first excited by light radiation in a specific wavelength region, and the excited state upon relaxation emits light radiation in another longer wavelength region [1, 2]. When the excitation wavelengths of the material are in ultraviolet region, the emission wavelengths are mostly in visible region. When the emission occurs within the visible range, the fluorescent effect improves the visual appearance of the object surfaces. This is because the surfaces of most fluorescent objects appear brighter and more vivid compared with the original

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color surface without the fluorescent emission, based on the surface light reflection. Fluorescence substances with such a characteristic are often incorporated into man-made objects, such as paper, paint, plastic, dye, and cloth to improve their visual appearance. Till date, although most studies on fluorescence measurement, imaging, and analysis in color image science and technology have often focused on artificial fluorescent objects [3–7], also in medical, food, and plants, fluorescence imaging have been used and a several studies were reported [8–11].

In addition to man-made objects, a variety of natural objects exist, which emit fluorescence, for example, minerals, deep-sea fish, and plants. Among these, plants have recently attracted significant interest as a research problem owing to the need to increase food production and food security as populations increase. In this study, we consider estimation of fluorescence emission from plant grains and leaves.

Rice and wheat are the most widely consumed grains on a daily basis. Rice is suitable for the hot and humid climates of Asia, while wheat is suitable for regions with dry summers, such as Europe and North America. Deterioration of rice after harvest, manifests in loss of freshness and flavor. Recently, fluorescence detection methods have been found to be effective for evaluating rice quality. When UV light illuminates a rice grain, usually the absorbed photon is in the UV range and the fluorescent is emitted in the visible range. The fluorescent intensity increases as the storage temperature and period increase. An imaging system was proposed for capturing images of the fluorescence emitted from rice grains and measuring the brightness of that image [12, 13].

Fluorescence from plant leaves is mostly originated by chlorophyll pigments. The measurement of chlorophyll content at the leaf is of great importance for stress detection of plants, growth status diagnosis, and studying the interaction between plants and the environment [14]. Additionally, detecting fluorescence is important for investigating photosynthetic activity in leaves [15]. Imaging systems that measure chlorophyll fluorescence often detect fluorescence emissions in specific wavelength regions by using a large-scale system for remote sensing or a specialized system built in a laboratory.

This study proposes spectral imaging methods for estimating fluorescence spectra emitted from plant grains and leaves. As currently available spectral cameras are expensive, and have low resolution and poor usability in spectral measurements, we constructed multiband imaging systems to estimate the fluorescence spectra with six channels based on ordinary off-the-shelf cameras. Two types of multiband systems are built in this study because the fluorescence emission from rice grains and plant leaves have spectral distributions in different wavelength regions. The first of these systems, meant for rice grains, makes a mobile phone camera multiband to detect the fluorescence emission in the visible wavelength region; the second system, meant for plant leaves, uses a monochrome camera with additional optical filters to detect chlorophyll fluorescence in the red to far-red wavelength region.

In terms of an algorithm to estimate the fluorescence emission spectra from image data, we adopt an approach inspired by the Wiener estimation method (see [16, 17]). The spectral estimation of fluorescence emission reconstructs a high-dimensional spectral function from low-dimensional image data. Because of the mapping from low to high dimensions, this reconstruction may trigger an ill-posed problem. The Wiener method is a statistical approach where the observations are modeled by accounting for noise in

the imaging system; this method is widely used to estimate the spectral reflectance of reflective objects as a solution for ill-posed problems. In the previous paper [18], we proposed an improvement of the Wiener method for surface spectral reflectance. In this study, we are facing to estimate spectral distribution of a light source.

In our spectral estimation of fluorescence emission, the light source is not the UV-light used for excitation, but the emitted light, which is incident on the imaging system. Therefore, the estimation problem can be reduced to that of estimating the incident light spectrum from the observed image data.

Experiments are conducted using real rice grains and green leaves to validate the proposed spectral imaging methods for estimating the fluorescence emission spectra. A UV-light source is used to illuminate the target objects and the emitted fluorescence is observed by the imaging systems. A spectro-radiometer is also used for measuring the emission spectra. The reliability of the proposed methods is confirmed via comparisons of the estimates from the image data with direct measurements.

2 Imaging Systems

A UV LED light (NCSU276A, NICHIA, Tokushima, Japan) was used to excite fluorescence from plants. Figure 1 shows the spectral power distribution of the UV LED light. The peak wavelength of the UV-light was 365 nm. The global spectral distribution of the fluorescence light was measured using a spectro-radiometer (CS-2000, Konica Minolta, Tokyo, Japan).

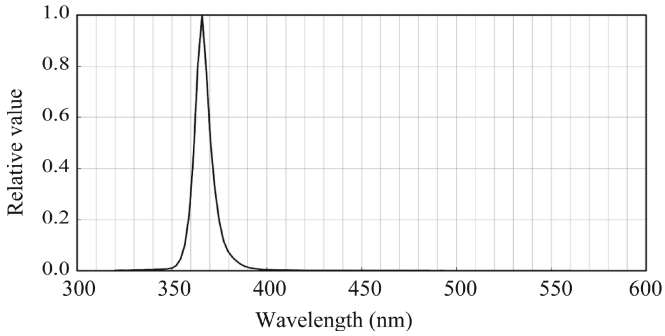

Fig. 1. Spectral power distribution of the UV LED light.

2.1 Multiband System Using a Mobile Phone Camera

As fluorescence emission from rice grains has the spectral distribution in the visible wavelength range, we used a mobile phone camera with RGB channels (iPhone 6s, Apple Inc, Cupertino, CA, USA) and made it multiband. A similar multiband method was proposed for surface spectral reflectance estimation [19]. The bit depth of the camera employed is 12 bits. Figure 2 depicts the entire system for multiband imaging. The curves of spectral sensitivity functions of the RGB camera are shown in Fig. 3 [20]. We also confirmed that the relationship between RGB outputs and luminance values is linear.

For multiband image acquisition, we selected two color filters from a set of commercial color filters (Fujifilm Optical Filter, Fujifilm, Tokyo, Japan). Figure 3 shows the spectral transmittance curves of these filters. Combining these transmittances with the original spectral sensitivities leads to different sets of trichromatic spectral sensitivity functions. The SP-6 filter is effective for shifting the spectral sensitivities to the short and long wavelengths in the visible range, while the SP-7 filter is effective in shifting the spectral sensitivities to the middle wavelength. Thus, two sets of the modified trichromatic spectral sensitivities would result in an imaging system with six spectral bands in the visible wavelength region. Figure 5 shows the overall spectral sensitivity functions of the present multiband imaging system with six channels, constructed using an RGB mobile phone camera and two color filters. It is noteworthy that this multiband imaging system is available for fluorescence image acquisition over the entire visible range. Each filter was attached to the front of the mobile phone camera lens. The image of each channel was acquired in Adobe's digital negative (DNG) format (Fig. 4).

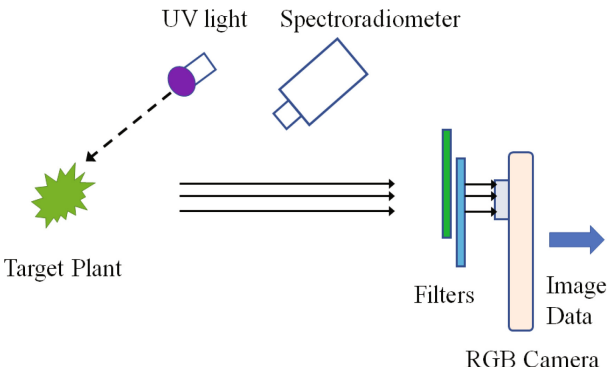

Fig. 2. Schematic of the overall system for multiband imaging using a mobile phone camera.

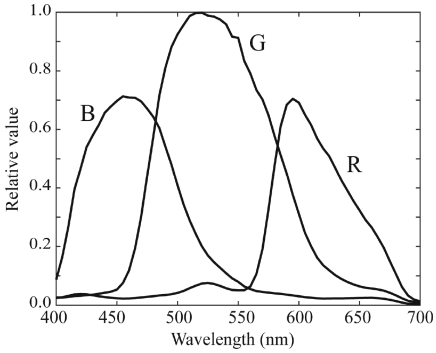

Fig. 3. Spectral sensitivity functions of the RGB camera

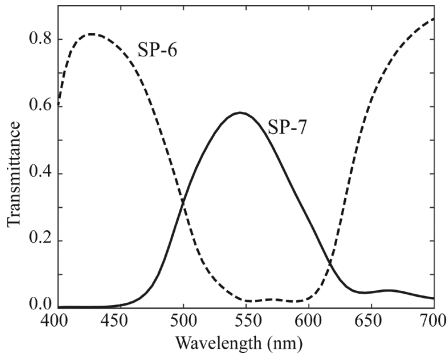

Fig. 4. Spectral transmittance curves of the two filters used.

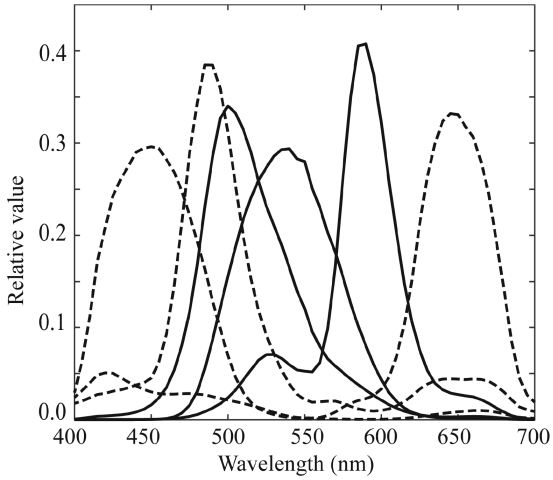

Fig. 5. Overall spectral sensitivity functions of the multiband imaging system constructed using an RGB mobile phone camera and two color filters.

2.2 Multiband System Using a Monochrome Camera

Chlorophyll fluorescence emission from plant leaves has the spectral distribution in the red to far-red wavelength region. As the mobile phone camera does not have sufficient sensitivity in this wavelength region, as shown in Fig. 3, we used a monochrome camera and made it multiband with additional filters. Figure 6 shows the overall multiband imaging system. The camera is a monochrome CCD camera with 12-bit dynamic range and Peltier cooling (QImaging, Retiga 1300, Shoshin EM, Aichi, Japan). Additional optical filters were selected so that the combination of the camera sensitivity and filter transmittances would result in an overall spectral sensitivity in the red to far-red region. Figure 7 shows the spectral sensitivity function of the monochrome camera, which has sensitivity up to nearly 800 nm. Figure 8 shows the spectral transmittance curves of

six sharp-cut filters SC-64, SC-66, SC-68, SC-70, SC-72, SC-74 (Fujifilm Optical filter, Fujifilm, Tokyo, Japan) selected for multiband image acquisition. Each filter was attached to the front of the camera lens. Figure 9 shows the resulting overall spectral sensitivity functions of the present multiband imaging system with six channels, constructed using a monochrome camera and six sharp-cut filters. Note that this imaging system is specialized for detecting fluorescence spectra in the red to far-red region. The image data were acquired in 16-bit tiff (tag image file format) format.

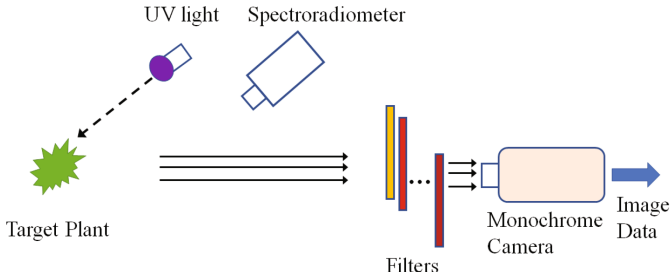

Fig. 6. Schematic of the overall multiband imaging system using a monochrome camera.

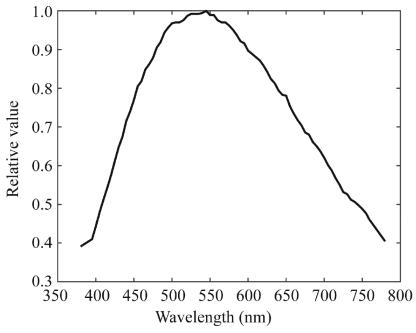

Fig. 7. Spectral sensitivity function of chrome camera.

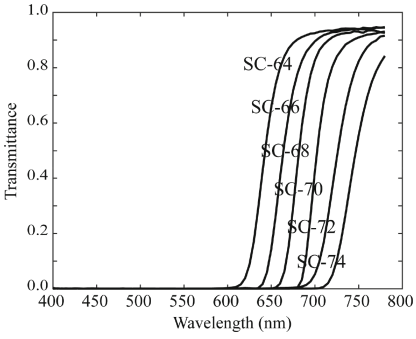

Fig. 8. Spectral transmittance curves of the mono the six sharp-cut filters.

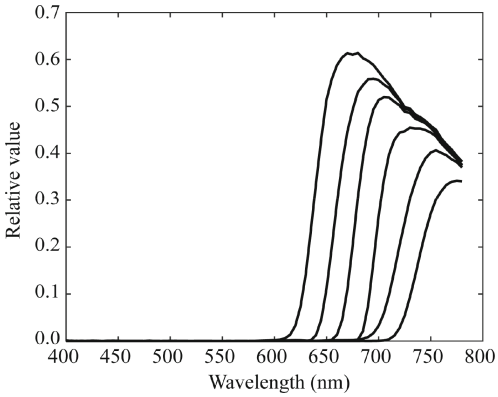

Fig. 9. Overall spectral sensitivity functions of the multiband imaging system constructed using a monochrome camera and six sharp-cut filters.

3 Estimation Method

3.1 Observation Model

We express the six observations y_i (i = 1, 2, …, 6) as

$$y_i = \int x(\lambda) r_i(\lambda) d\lambda + n_i, \tag{1}$$

where $x(\lambda)$ is the spectral-power distribution of fluorescence emitted from the target plant, and $r_i(\lambda)$ (i = 1, 2, …, 6) denote the spectral sensitivity functions of the multiband imaging systems. Note that Eq. (1) does not include the UV light source because the emission light from the plant is the light source.The wavelength range of integration is in the visible range (400–700 nm) for the first imaging system, and in a wider range (400–780 nm) including red to far-red for the second imaging system. The sensitivity functions $r_i(\lambda)$ are shown in Figs. 5 and 9 for the first and second systems, respectively, which are defined by the multiplication of the RGB spectral sensitivity functions of the mobile phone camera and spectral transmittance of the additional color filters for the first system, and by the multiplication of the spectral sensitivity functions of the monochrome camera and spectral transmittance of the additional sharp-cut filters for the second system. Furthermore, n_i denote the noise in the respective imaging system. We assume that n_i are the white noise with zero mean and the variance a, and were uncorrelated with $x(\lambda)$.

For digital representation, the fluorescence spectra and spectral sensitivities are sampled at N wavelength points at equal intervals, and represented by N-dimensional column vectors as

$$\mathbf{x} = \begin{bmatrix} x(\lambda_1) \\ x(\lambda_2) \\ \vdots \\ x(\lambda_N) \end{bmatrix}, \qquad \mathbf{r}_i = \begin{bmatrix} r_i(\lambda_1) \\ r_i(\lambda_2) \\ \vdots \\ r_i(\lambda_N) \end{bmatrix} (i = 1, 2, ..., 6). \tag{2}$$

The discrete representation of the observation model is then expressed in a matrix form as follows:

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{n}, \tag{3}$$

where $\mathbf{A}$ is a $(6 \times N)$ matrix defined by the spectral sensitivities and is written as

$$\mathbf{A}^t = [\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_6]\Delta\lambda \tag{4}$$

and $\mathbf{y}$ and $\mathbf{n}$ are 6-dimensional column vectors written as

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_6 \end{bmatrix}, \quad \mathbf{n} = \begin{bmatrix} n_1 \\ n_2 \\ \vdots \\ n_6 \end{bmatrix}. \tag{5}$$

The superscript t and $\Delta\lambda$ in Eq. (4) represent the matrix transposition and wavelength sampling interval, respectively. When the continuous spectra are sampled with $\Delta\lambda = 5$, the discrete spectral functions are represented by 61-dimensinal column vectors with $N = 61$ in 400–700 nm for the first system, and 77-dimensional column vectors with $N = 77$ in 400–780 nm for the second imaging system.

3.2 Estimation Algorithm

We estimate the fluorescence spectrum $\mathbf{x}$ from the observation $\mathbf{y}$ at every pixel point, based on the observation model Eq. (1). As in the Wiener estimator, the estimate $\hat{\mathbf{x}}$ of $\mathbf{x}$ was found in the form [18]

$$\hat{\mathbf{x}} = \mathbf{B}\mathbf{y}, \tag{6}$$

where $\mathbf{B}$ is an $(N \times 6)$ constant matrix. Note that $\mathbf{x}$ is the spectral distribution, not the spectral reflectance. Matrix $\mathbf{B}$ is determined to minimize the mean-square error (MSE) between the estimate $\hat{\mathbf{x}}$ and original x. This error is described as

$$J = \mathbf{E}\Big[\|\mathbf{x} - \hat{\mathbf{x}}\|^2\Big], \tag{7}$$

where the expectation operator $\mathbf{E}[\mathbf{x}]$ denotes the mean or average of $\mathbf{x}$, and symbol $\|\mathbf{x}\|^2$ indicates the L2 norm of $\mathbf{x}$.

The error minimization for J results from matrix $\mathbf{B}$ being solved as

$$\mathbf{B} = \mathbf{R}\mathbf{A}^t\big(\mathbf{A}\mathbf{R}\mathbf{A}^t + a\mathbf{I}\big)^{-1}, \tag{8}$$

where $\mathbf{R}$ is the autocorrelation matrix $\mathbf{E}\big[\mathbf{x}\mathbf{x}^t\big]$, and $\mathbf{I}$ is an $(N \times N)$ identity matrix with ones in the diagonal and zeros elsewhere. Thus, the optimal estimate $\hat{\mathbf{x}}$ of the fluorescence spectrum can be obtained as

$$\hat{\mathbf{x}} = \mathbf{R}\mathbf{A}^t\big(\mathbf{A}\mathbf{R}\mathbf{A}^t + a\mathbf{I}\big)^{-1}\mathbf{y}. \tag{9}$$

The autocorrelation matrix should be determined using a dataset of various fluorescence spectra. However, it is noteworthy that fluorescence spectra emitted from various plants are limited to specific wavelength ranges, and currently no database of fluorescence spectra with sufficient data exists. Therefore, in this study, we set $\mathbf{R} = \mathbf{I}$, that is, every spectrum other than itself is uncorrelated.

Furthermore, in estimating spectral reflectance, the noise variance a can be determined using a white reference standard with known reflectance [18]. However, in our estimation problem of fluorescence emission, such standardization is not available for noise estimation. Therefore, in this study, the noise variance a is determined so that the error between the averaged fluorescence spectra estimated over the target image area and the measured fluorescence spectrum directly by the spectro-radiometer with a wide aperture size is minimized as follows:

$$\|\mathbf{E}[\hat{\mathbf{x}}] - \mathbf{x}_0\|^2 \rightarrow \min, \tag{10}$$

where $\hat{\mathbf{x}}$ and $\mathbf{x}_0$ are the normalized vectors of the estimated fluorescence spectrum and direct measurement, respectively.

4 Experimental Results

4.1 Fluorescence Estimation for Rice Grains

We tested rice grains of the cultivar "Akitakomachi" harvested in 2023 in Akita Prefecture, Japan. The polished rice grains were purchased at the market and measured. The rice grains were placed in a transparent bag, and while it was illuminated with UV light from the outside, the spectral-power distribution of the fluorescence emitted from the rice grains was measured using the spectro-radiometer. Figure 10 shows the measured fluorescence spectrum of the rice, wherein the rise in the spectral curve at 400 nm was suspected to be due to the effect of the illuminated UV light.

We captured multiband images of the surface of rice grains using the first imaging system from the same direction as the spectro-radiometer. Since fluorescence emission

is Lambertian, it does not matter if the viewing angles of the radiometer and the camera are slightly different. We observed that the estimates of the fluorescent spectra from the image data varied due to the noise variance in the imaging system. Therefore, as an important step, we estimated the parameter a of the noise variance. The fluorescence spectra were first estimated at all pixel points with a proper value of a, and all the estimates were averaged. Next, we varied a as a variable, and searched the most proper a value that minimized the MSE between the estimates and direct measurement. Figure 11 compares the averaged fluorescence spectrum estimated with the most proper value $a = 0.4$ from the image data and direct measurement, shown in Fig. 10.

The intensity of the emitted fluorescence was calculated as a norm of the 61-dimensional vector of the estimated spectrum at each pixel. Figure 12 shows the intensity image of the rice grains estimated with $a = 0.4$. Furthermore, to know the fluorescence color of the rice grains, we applied the CIE color matching functions to the estimated fluorescence spectrum to get the tristimulus values and converted those into sRGB values. Figure 13 demonstrates the sRGB image with fluorescence color at all pixel points of the rice image.

The same rice grains were measured three month later. The measurement setup was different so the increase in fluorescence emission intensity could not be confirmed. However, the normalized spectral distribution of the estimated fluorescence emission was nearly the same.

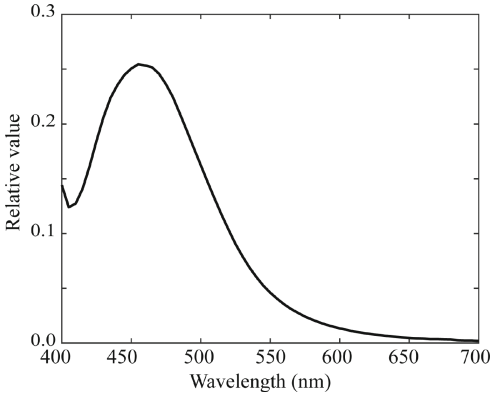

Fig. 10. Measured fluorescence spectrum of rice grains by spectro-radiometer.

4.2 Fluorescence Estimation for Plant Leaves

"Ohba", which refers to the leafy part of "Green Shiso" (Perilla frutescens var. Crispa), shown in Fig. 14, is edible and sold in markets. Figure 15 shows the measured fluorescence spectrum from a leaf of the Ohba, whose surface was illuminated directly by UV light. The spectral distribution had a high intensity in the red to far-red wavelength region. It was clearly evident that the fluorescence is originated by chlorophyll pigments.

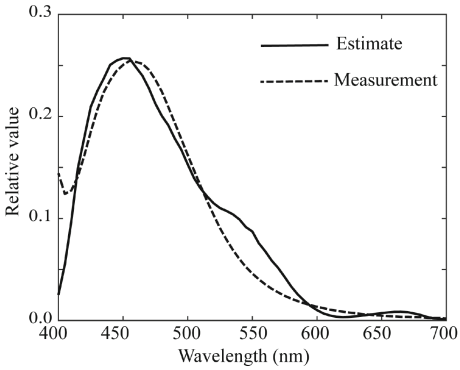

Fig. 11. Comparison of averaged fluorescence spectrum estimated with the noise variance $a = 0.4$ from the image data and direct measurement (shown in Fig. 10).

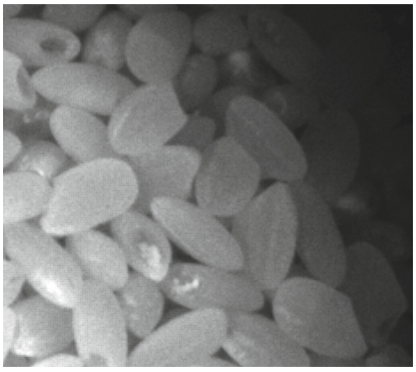

Fig. 12. Intensity image of fluorescence emitted by rice grains.

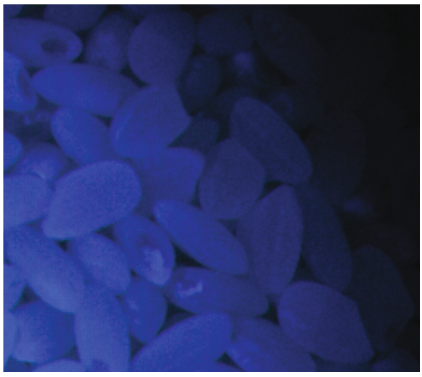

Fig. 13. sRGB image with fluorescence color at all pixel points of the rice grain image.

Two peaks are observed in the distribution at approximately 685 and 740 nm. It is known that the shape changes depending on the chlorophyll content [15].

Multiple images were captured of the leaf surface using the second imaging system by sequentially changing the optical filters, which were then combined as a 6-dimensional multiband image. The intensity of the chlorophyll fluorescence emission fluctuated over time; therefore, we calibrated the camera data so that the intensity remained constant. Figure 16 compares the averaged curve of the fluorescence spectra estimated with the most proper value, $a = 0.005$ from the image data with the spectral curve obtained from the direct measurement in Fig. 15.

The intensity distribution of the emitted fluorescence was calculated from the 77-dimensional vectors of the estimated spectra in the range of 400–780 nm. Figure 17 shows the intensity image of the chlorophyll fluorescence observed from the Ohba leaf. Furthermore, we can observe the fluorescence color of the leaf only in the visible wavelength region. Figure 18 demonstrates the sRGB image with fluorescence color at all pixel points of the leaf image.

Fig. 14. Photographic image of "Green Shiso" plant.

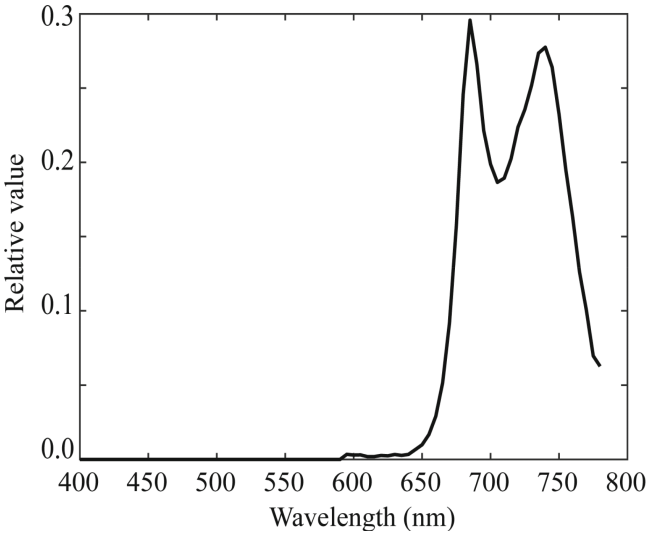

Fig. 15. Measured fluorescence spectrum of Ohba leaf by spectro-radiometer.

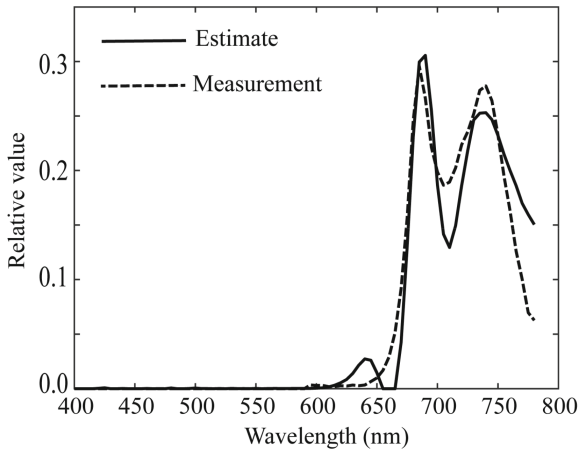

Fig. 16. Comparison of average curve of fluorescence spectra estimated from image data and spectral curve from direct measurements (shown in Fig. 14)

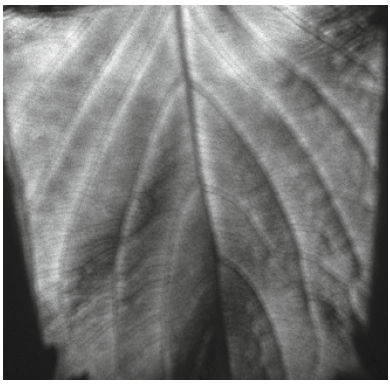

Fig. 17. Intensity image of chlorophyll fluorescence observed from Ohba leaf.

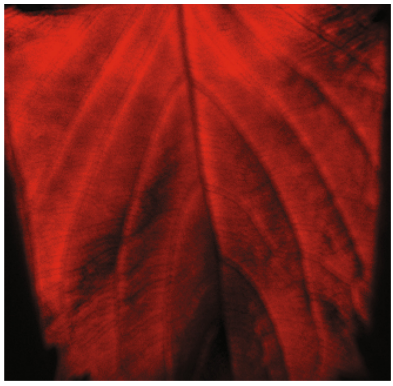

Fig. 18. sRGB image with fluorescence color at all pixel points of the leaf image.

5 Conclusions

In this study, we have proposed spectral imaging methods for estimating fluorescence spectra emitted from plant grains and leaves. We constructed two types of multiband imaging systems with six channels based on ordinary off-the-shelf cameras: one for rice grains by making a mobile phone camera multiband to detect the fluorescence emission in the visible wavelength region, and another for plant leaves, using a monochrome camera with additional optical filters to detect chlorophyll fluorescence in the red to far-red wavelength region.

The two systems targeted different wavelength ranges, so it is not possible to determine which is better. The spectral sensitivity of the mobile phone camera is only in the visible range. On the other hand, the spectral sensitivity of the monochrome camera is

relatively wide, beyond the visible range. Therefore, if many optical filters are available in the short wavelength range, it would be possible to construct a multiband imaging system for both plant grains and leaves using only a monochrome camera system. However, the former system is much easier to use than the latter and is noteworthy in that it allows measurements of plants to be taken outdoors.

We adopted a statistical approach inspired by the Wiener estimator to estimate fluorescence emission from the image data. An algorithm was presented to solve the mapping from six to high dimensions in the spectral estimation of fluorescence emission. The observations were modeled by accounting for noise in the imaging system, and the incident fluorescence spectra was estimated from the observed image data. The noise variance also was estimated based on a direct measurement of the fluorescence spectrum.

In the experiments using real rice and green leaves, UV-light illuminated the plant targets, and the emitted fluorescence was observed by the imaging systems. The reliability of the proposed method was confirmed by comparing the estimates from the image data to direct measurements.

Our future work will involve applications to evaluating rice quality and investigating chlorophyll content in leaves.

It should be noted that combining two spectral imaging methods proposed here can help estimate fluorescence spectra in a wide wavelength range from visible and far-red. Furthermore, the combined system can estimate not only fluorescence spectra, but also general spectral power distribution of non-fluorescence, and even surface spectral reflectance of any object.

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Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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Stitching from Spectral Filter Array Video Sequences

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Abstract. Hyperspectral imaging offers high spectral and spatial resolution, but its high costs and time-consuming nature make it difficult to use. Spectral Filter Array (SFA) imaging presents an alternative, offering high spectral resolution, user-friendliness, and affordability, but at the cost of limited spatial resolution. This paper presents an approach to address this trade-off, starting with raw overlapping frames from spectral videos, followed by a demosaicking network process before tackling the stitching problem. Our experiments on various spectral videos, supported by image quality metrics and qualitative demonstrations, indicate that this approach effectively enhances the spatial resolution of spectral images while reducing artifacts. The integration of the demosaicking and the stitching provides a robust solution for spectral video applications, paving the way for further advancements in panoramic spectral image stitching.

Keywords: Spectral Imaging · Stitching · Demosaicking · Super Resolution

1 Introduction

Spectral Imaging holds immense potential for applications such as food safety inspection [26], land cover categorization [16] and object tracking [12], offering a nuanced and comprehensive perspective that extends beyond the capabilities of traditional imaging techniques. In the realm of cultural heritage, the utilization of spectral videos can prove instrumental in capturing intricate details of artifacts, manuscripts, and historical sites [24]. But due to hardware limitations, the spatial resolution is limited, compared to Color Filter Array (CFA) [3] imaging. Moreover, demosaicking which refers to the process of creating a fully-defined spectral image free of spatial and spectral distortions, is a crucial stage for Spectral Filter Array (SFA) camera [13]. Through the seamless integration of demosaicking and stitching algorithms, spectral videos can be transformed into large panoramas by combining multiple spectral images with overlapping areas. Demosaicking, in particular, plays a pivotal role in reconstructing a fully-defined

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image from a mosaic image. This process enhances the visual clarity and accuracy of the final image, enabling the preservation of spectral informations [18].

In the context of remote sensing, spectral video holds the key to unlocking a wealth of information about the Earth's surface [29]. By capturing data across multiple spectral bands, these videos provide a richer and more comprehensive understanding of the environment. The combination of demosaicking and stitching techniques in remote sensing applications facilitates the creation of high-resolution, spectral panoramas. This panoramic view can be instrumental in monitoring land cover changes, assessing vegetation health, and identifying environmental anomalies.

By stitching together multiple images of the same location taken from various angles, a panorama image is created through the process of image stitching. In general, when dealing with an issue involving just two images, one of the images is used as the reference, while the other is used to estimate a 3×3 homography, also known as a camera matrix [11]. However, the more images that are included, the more complex this problem becomes [22]. Another pivotal consideration in the construction of panoramas involves not only the spatial arrangement of images but also the harmonization of visual attributes across individual frames. It is imperative to adjust the appearance of each image meticulously to eliminate discernible boundaries in the resultant panorama. Even when images are captured simultaneously, variations in exposure, contrast, and the introduction of effects such as vignetting may necessitate careful calibration to achieve seamless integration [31]. Given the context of working with spectral images, the meticulous harmonization of visual attributes to achieve seamless transitions between images can result in a reduction of the spectral fidelity. This unintended consequence is undesirable, emphasizing the need for caution when undertaking the harmonization of visual attributes.

This paper aims to contribute to the field of spectral imaging by exploring the potential of spectral stitching to enhance the quality and spatial resolution of spectral images. By increasing the number of spectral images used in the stitching process, it is possible to achieve a more detailed and accurate representation of the target object or scene. This approach has the potential to be particularly useful in applications such as remote sensing, cultural heritage preservation, where the ability to capture highly detailed spectral information is crucial. This paper presents our findings on how the number of spectral images impacts the quality and spatial resolution of stitched images.

This paper is structured as follows: Sect. 2 offers an overview of existing literature on stitching algorithms and spectral demosaicking. In Sect. 3, we delve into our proposed methodology, providing a detailed explanation of our approach. Section 4 outlines our experimental setup and presents our analysis of the obtained results. Finally, we conclude with a brief summary of our findings and discuss avenues for future research.

2 Related Work

SFA camera emerged as a good option to captures well-aligned spectral images using a one-shot acquisition [13]. An SFA is made by arranging several bandpass optical filters in a pattern that repeats spatially, covering the monochromatic imaging sensor, inspired from the Bayer CFA [3]. Since all spectrum bands in the SFA are spatially subsampled, only sparse data from a single band is preserved in each individual pixel of the collected original spectral image. Consequently, the missing spectral values must be approximated from the gathered sparse spatial data in order to obtain fully-defined (demosaicked) spectral images (see Fig. 1). This process is referred to as SFA demosaicking [5], provides a critical component for reconstructing spectral images for SFA based camera [21].

Several handcrafted strategies have been investigated in previous research on SFA image demosaicking [25,27]. The weighted bilinear (WB) interpolation approach was presented by Brauers and Aach [5], extending CFA demosaicking to SFA. Chini et al. [7] first introduced the idea of the pseudo-panchromatic image (PPI), which is the average image of all spectral channels. The PPI difference (PPID) approach was developed by Mihoubi et al. [17,18] to improve SFA demosaicking. It sharpens the PPI by taking the channel residual structure and spatial-spectral correlation of SFA images into account. Convolutional neural networks (CNNs) have demonstrated potential in demosaicking tasks in recent years. Using mosaic convolution-attention networks, Feng et al. [10] developed a deep CNN that focuses on first feature extraction from unprocessed mosaic images. Nevertheless, checkerboard distortion results from their mosaic convolution module ignoring for spatial location variations. The demosaicking architecture was updated by Pan et al. [15] by utilizing a conventional two-pass residual interpolation approach and adding PPIs computed by a CNN. Despite its effectiveness, this method might have halo effects when guided filters are used. By introducing a Residual Network and improving PPI generation, Zhao et al. [30] built upon the work of Pan et al. [15]. They achieved state-of-the-art metrics and enhanced demosaicking outcomes by including edge-related data and using adaptive spatial and spectral correction inside the network.

Furthermore, image stitching has been an active area of research for many years, with methods continuously becoming more robust and yielding better results. Early methods relied on user-provided information to produce visually appealing panoramas. A major paper named AutoStitch by Matthew Brown and David G. Lowe [6], along with their proprietary software tool that could allow anyone to produce great panoramas simply by uploading their images in any order. Their algorithm was based on the features detector and descriptor SIFT [8], proposed by Lowe in 2004, and the RANSAC algorithm [4]. Since then, multiple other methods have emerged, each one proposing better results in edge cases by computing the stitching in a slightly different manner. Among the most popular recent methods, we can mention the APAP [28] and the AANAP [14] algorithms. We propose in this paper to implement a python version of the AutoStitch [6] algorithm specifically for spectral imaging. Since it is a well-established algorithm in the stitching community.

3 Methodology

Starting with the initial phase, our approach integrates a demosaicking network to obtain fully defined spectral images from mosaic images. This foundational step ensures refined inputs for the subsequent image stitching process. Transitioning to image stitching, the absence of ground truth image makes it impossible to apply metrics like PSNR or SSIM, as there is no reference for comparison between the stitched image and a target image. To this end SOTA stitching solution [14,28], often uses blind image quality assessment metrics such as BRISQUE [19] or NIQUE [20].

Our sequential procedure begins with the application of our Deep Spectral Demosaic network to obtain fully-defined (demosaicked) spectral images. Next, we apply the SIFT [9] and RANSAC [4] algorithms to the PPI for feature matching. To address variations in gain, we integrate gain compensation techniques and perform multi-band blending to seamlessly merge the images. To obtain a spectral stitched image, we apply the same computed parameters for the other bands without any new parameter computations, as all the bands have the same field of view.

3.1 Deep Demosaic Net

The proposed deep learning model comprises two integral modules: the Deep PPI module (PPI-net) and the Deep Residual Demosaic module (DRDm). The PPI-net is dedicated to extracting a pseudo-panchromatic image (PPI) from a raw mosaic image, serving as an intermediate representation capturing spatial information from multiple channels. Pre-trained to reconstruct a pseudo-panchromatic image from the raw SFA image, the PPI-net establishes a foundational step (see Fig. 1).

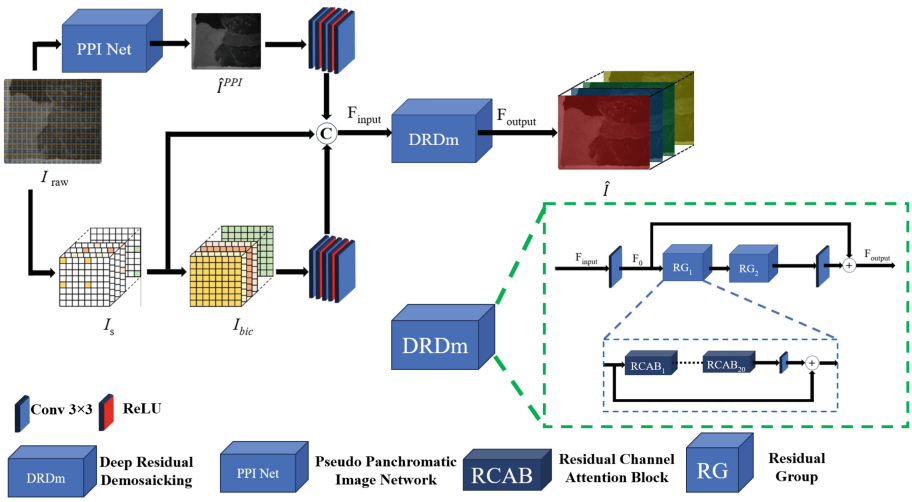


Fig. 1. Deep Spectral Demosaicking network architecture

The proposed deep demosaicking model comprises a two-branch CNN, the first branch consists of a deep pseudo-panchromatic network (PPI-net) in order to estimate the pseudo-panchromatic image from raw image, drawing inspiration from the successful methodology proposed by [23,30]. Additionally, the second branch serves as a preliminary demosaicking outcome achieved through bicubic interpolation. These branches are seamlessly combined into a cohesive module, namely the deep residual demosaicking (DRDm).

PPI-net. The determination of the pseudo-panchromatic image (PPI) at each pixel (p) involves computing the average value across all channels of spectral channels, as outlined in prior research [18].

$$I_p^{PPI} = \frac{1}{N} \sum_{n=1}^{N} I_p^{GT}, \tag{1}$$

where N denote the number of spectral bands.

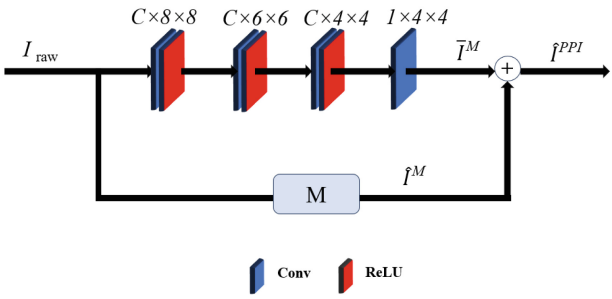


Fig. 2. Illustration of PPI-net module. Where $\oplus$ denotes elementwise sum.

Given the clear positive linear correlation found between the reconstructed spectral image and the preserved high-frequency details in the pseudo-panchromatic image (PPI) [23], we propose employing elementwise summation between $\bar{I}^M$ and $\hat{I}^M$, as depicted in Fig. 2. Here, M represents a filter matrix tailored for a 3×3 SFA pattern [15,18].

$$M = \frac{1}{36} \begin{bmatrix} 1 & 2 & 2 & 1 \\ 2 & 4 & 4 & 2 \\ 2 & 4 & 4 & 2 \\ 1 & 2 & 2 & 1 \end{bmatrix}. \tag{2}$$

Additionally, the PPI-net module's loss function, represented by Eq. 3, is meant to minimize the difference between the estimated PPI produced by the PPI-net and the ground truth PPI determined by Eq. 1.

$$L_{PPI} = \sum_{w=1}^{W} \sum_{h=1}^{H} \left|\left| \hat{I}^{PPI}(h,w) - I^{PPI}(h,w) \right|\right|, \tag{3}$$

where L_{PPI} is using MSE. h and w represents respectively the height and width of the image. Moreover, $\hat{I}^{PPI}$ is the estimated PPI and I^{PPI} is the ground truth PPI.

DRDm Module. Following the calculation of the pre-demosaicked image, using bicubic interpolation denoted as I_{bic} and the estimated PPI denoted as $\hat{I}^{PPI}$, both outputs are passed on into two identical blocks. Following this, they are concatenated and used as the DRDm module's input jointly with the sparsity image. The reconstruction part, shallow feature extraction, residual in residual (RIR) deep feature extraction, and deep feature extraction are the main components of our DRDm module.

Furthermore, the loss function of the DRDm module aim to minimize the difference between the estimated demosaicked spectral image and the ground truth image I^{GT}. Specifically, this loss function is the Mean Squared Error (MSE), which is defined as:

$$L_{DRDm} = \frac{1}{P} \sum_{p=1}^{P} \left|\left| I_p^{GT} - \hat{I}_p \right|\right|_2^2, \tag{4}$$

here, P refers to the total count of pixels, where p denotes the index of the pixel as indicated in Eq. 4.

3.2 Training Details

Our demosaicking model was trained using the ARAD-1K dataset [2]. This framework is implemented based on PyTorch, and trained using a NVIDIA RTX 2080 GPU equipped with 20 GB of VRAM and CUDA 11.8 for a total of 2000 epochs. We employed the Adam optimizer to update model parameters, specifying $\beta_1 = 0.9$ and $\beta_2 = 0.999$. Data augmentation techniques were applied to enhance the robustness of the training process, encompassing random cropping of spectral images from ARAD-1K datasets to 128×128 patches, as well as rotations by 90°, 180°, and 270°. Moreover, the batch size was set to 16. To facilitate convergence, we initialized the learning rate to 10^{-4} and adopted a halving strategy every 250 epochs. Moreover, All testing experiments are implemented using the same machine: Intel Core i9-11900k CPU 2.40 GHz, NVIDIA GPU RTX 3070 and 16 Gb of RAM.

3.3 Image Stitching

Feature Matching. The first step is to characterize each image by extracting local features. To do so, we use the SIFT extractor [8], which for each image gives

us the locations of a certain number of keypoints, along with a descriptor for each of these keypoints. The main reason for using SIFT instead of other methods such as the Harris Corner Detector [9] is that SIFT features are invariant under rotation and scale changes, which means that our model can handle images with varying orientation and zoom. Please note that, from now on, images will be indexed as i and j to differentiate between the two images we are trying to stitch.

Once the features have been extracted from all images, a match is being calculated for each pair of images. To do so, each feature of one image is matched to its two nearest neighbors in feature space from the other image. Then, the Lowe's ratio test is used to keep only relevant matching features. This test consists in comparing the first nearest neighbors to the second, and keeping only the matches where the distance with the closest neighbor is significantly smaller than the distance with the second closest one (significantly being determined by the value of the used scaling factor). This test performs a soft thresholding that eliminates most of the wrong matches. After this step, we have for each pair of images a list of matches, each one composed of a feature from each images. The next step is to use those matches to identify the images that would overlap each other in the panorama, and to compute homography matrices (Fig. 3).

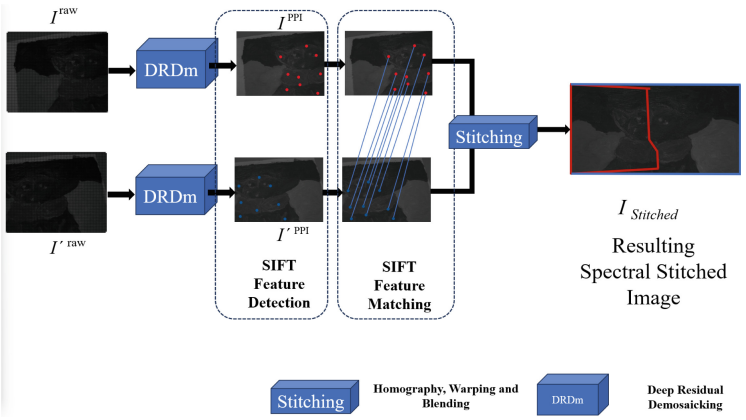


Fig. 3. Overview of our stitching pipeline, where we use only the resulting PPI image to compute stitching parameters. The rest of the spectral bands do not require computing stitching parameters from scratch; instead, the computed stitching parameters from the PPI band can be directly applied to them.

Robust Homography Estimation Using RANSAC. Using a minimum collection of randomly selected correspondences, RANSAC (random sample consensus) [4] is a robust estimation process that estimates images transformation parameters and selects the solution that best corresponds with the data. It works with multiple iterations. At each iteration, a set of four feature matches is randomly selected, and is used to compute the homography H between them using the direct linear transformation method [28]. This operation is repeated a certain number of times, each with a different randomly selected set of four samples.

Then, the homography with the highest number of inliers (i.e. matches whose projections are consistent with H within a tolerance ϵ pixels) is selected. It can be shown that, for a pair of images where half of the matches are corrects, the probability to have not found the correct homography after 500 iterations is approximately 1×10^{-14}, thus making it a pretty robust method.

Probabilistic Model for Verifying Image Matches. At this stage, each pair of images consists of a list of features matches and a homography for one image relatively to the other. However, we do not know yet which pairs of images should indeed overlap in the final panorama, and which pairs actually do not have any area in common. To verify the image matches, we compare the features matches that are consistent with the homography (RANSAC inliers) with the features matches inside the area of overlap between the two images but not consistent (RANSAC outliers). For a given image we denote the total number of features in the area of overlap n_f and the number of inliers n_i. Then, by making small assumptions on the distributions of the feature matches, we can estimate the probability that the image match is correct using these two numbers, and obtain a condition for an image match to be correct:

$$n_i > \alpha + \beta \times n_f. \tag{5}$$

The detailed calculus can be found in the AutoStitch original paper [6]. The parameters α and β could theoretically been adjusted based on the data, but we used the provided values, that is $\alpha = 8$ and $\beta = 0.3$

Using the above condition, we can keep only the correct image matches, and divide the data into connected components (i.e. sets of images that are linked together by matches). This allows us to recognize multiple panoramas in a set of images and reject noisy images that do not match any other images.

3.4 Gain Compensation

We discussed how to calculate each camera's geometric characteristics in earlier sections. Here, we demonstrate how to get the total gain across images, a photometric parameter. Similar configuration applies here, with an error function established across all images. For all overlapping pixels, the error function is the sum of the gain-normalized intensity errors [6]:

$$e = \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} N_{ij} \left(\frac{(g_i I_{ij} - g_j I_{ji})^2}{\sigma_N^2} + \frac{(1 - g_i)^2}{\sigma_g^2} \right). \tag{6}$$

where $N_{ij} = |R(i,j)|$ is the number of pixels in image i that overlap in image j , $\bar{I}_{ij}$ is the mean intensity of the image i in the overlap region with image j, defined as

$$\bar{I}_{ij} = \frac{\sum_{u_i \in R(i,j)} I_i(u_i)}{N_{ij}}, \tag{7}$$

where g_i is the gain associated with image I. The parameters σ_N and σ_g denote the standard deviations of the normalized intensity error and gain, respectively. It is noteworthy that an increase in these parameters is expected to result in a reduction of spectral fidelity. This phenomenon is explained by the relatively low standard deviations, specifically $\sigma_N = 1.5$ and $\sigma_g = 0.1$, underscoring the nuanced impact of parameter settings on the fidelity of spectral representations (Fig. 4).

Fig. 4. Left: Stitch Image with multi band blending. Right: Stitch Spectral Image without blending.

3.5 Blending

Multi-band Blending. The idea behind multi-band blending is to blend low-frequency information over a large spatial range, and high-frequency information over a short range. This approach ensures that small details are preserved while still allowing for a smooth transition between images. To achieve this, we first initialize blending weights for each image by finding the set of points for which image i is primarily responsible:

$$W^i_{max} = \begin{cases} 1 & \text{if } W^i = \arg\max W^j \\ 0 & \text{otherwise} \end{cases} \tag{8}$$

in other words, W^i_{max} will be 1 when image i has the maximum weight and 0 where any other image has a higher weight. These maximum-weight maps are progressively blurred to create the blending weights for each band, along with a similar blurring performed on the image itself to build each band. For the first band, a high pass version of the rendered image is formed:

$$I^i_\sigma = I^i * g_\sigma, \tag{9}$$

$$B^i_\sigma = I^i - I^i_\sigma, \tag{10}$$

where $*$ indicates convolution and g_σ is a Gaussian of standard deviation σ. In doing so, the spatial frequencies in the range of $\lambda \subset [0, \sigma]$ are represented by B_σ. The same convolution is used to construct the weight matrix linked to this band

$$W_\sigma^i = W_{max}^i * g_\sigma, \tag{11}$$

here, W_σ^i represent the blend weights Then, other bands are computed iteratively, using a standard deviation $\sigma' = \sqrt{(2k+1)}\sigma$ which is getting larger, with $k > 0$

$$I_{(k+1)\sigma}^i = I_{k\sigma}^i * g_{\sigma'}, \tag{12}$$

$$B_{(k+1)\sigma}^i = I_{k\sigma}^i - I_{(k+1)\sigma}^i, \tag{13}$$

$$W_{(k+1)\sigma}^i = W_{k\sigma}^i * g_{\sigma'}, \tag{14}$$

this way, each band k represents spacial frequencies in the range of wavelengths $\lambda \subset [\sqrt{(2k-1)}\sigma, \sqrt{(2k+1)}\sigma]$. Finally, overlapping images are merged linearly for each band using the associated blend weights.

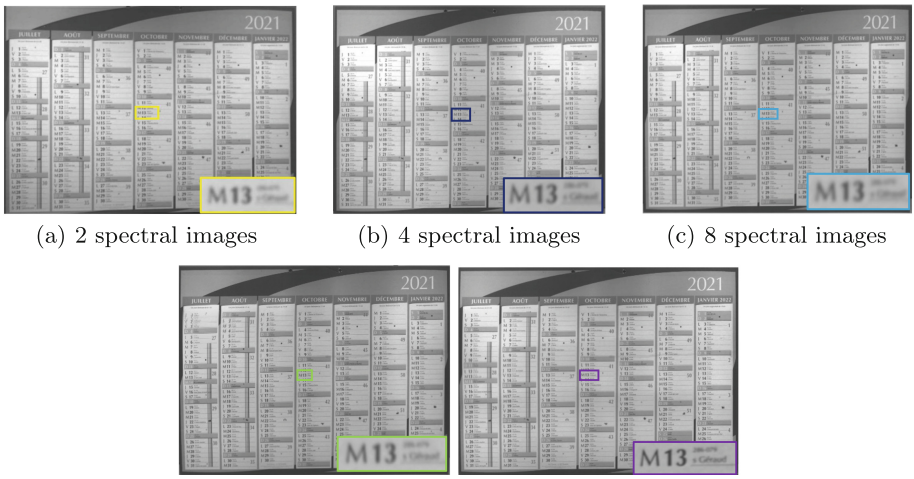
$$I_{k\sigma}^{multi} = \frac{\sum_{i=1}^n B_{k\sigma}^i W_{k\sigma}^i}{\sum_{i=1}^n W_{k\sigma}^i}. \tag{15}$$

More details and illustrations about this method can be found in the AutoStitch paper [6].

4 Experiment

To evaluate the effectiveness of our method, we designed a series of experiments aimed at discerning the impact of both viewing distance and the number of spectral images used on the visual quality of the stitched images. The experiments were conducted under controlled conditions to ensure consistency and reliability of the results. We began by selecting a calibrated object and placing it under uniform D65 illumination. Using a 9 band SFA camera from SILIOS Technologies [1], we captured raw images of the object from a considerable distance manually controlling the focus of the camera. The images were taken in a handheld manner, which introduced small rotations and translations. Initially, only two images were acquired, each covering a portion of the object's field of view. Subsequently, we systematically decreased the distance between the camera and the object, thereby increasing the number of raw images captured for stitching purposes. This approach allowed us to examine the impact of viewing distance on the stitching process and the resulting image quality (see Figs. 5 and 6).

The captured images were processed using our implemented stitching algorithm, which combines multiple images to create a panoramic view of the object. The algorithm utilizes the multi-band blending technique to seamlessly merge



(a) 2 spectral images (b) 4 spectral images (c) 8 spectral images

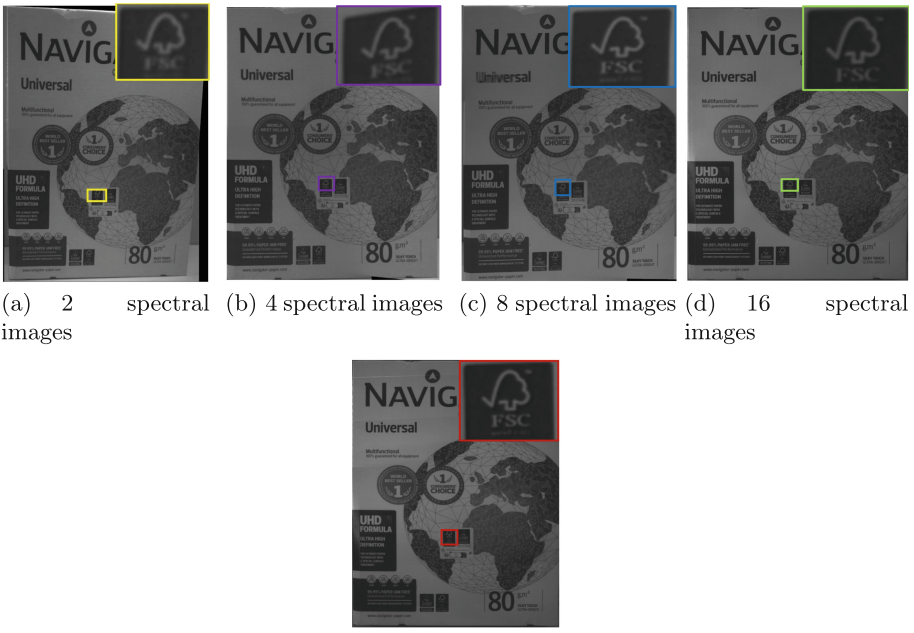
(d) 16 spectral images (e) 32 spectral images

Fig. 5. Spectral Stitched Images of the Same Object, a Calendar: Each stitching composed of various images from different distances—(a) from a distance of 130 cm, (b) from 93 cm, (c) from 76 cm, (d) from 58 cm, (e) from 35 cm.

adjacent images while preserving the spectral information captured by each image.

We assessed the visual appearance of the stitched images across different numbers of images to investigate their impact on image quality. Our analysis aimed to discern whether increasing the number of images enhances the quality of the stitched output or amplifies the probability of distortion. By comparing results obtained from varying numbers of images, we sought to identify trends regarding the influence of image count on image quality. Additionally, we examined the effects on image quality metrics based on natural image statistics such as BRISQUE and NIQUE to determine whether increasing the number of images correlates with improved objective measures of image quality. This investigation aimed to provide insights into the optimization of spectral imaging techniques, particularly in understanding the trade-offs between the number of images used and the resulting image quality.

Initially, it might seem intuitive that moving closer to the object and capturing more images would inherently lead to superior stitching results. However, our experiments challenge this assumption. We observed that as the distance between the camera and the object decreased and the number of images increased, the likelihood of encountering distortion and high-frequency blur in the stitched images also increased (see Fig. 7). This phenomenon suggests that proximity and image abundance do not guarantee improved stitching outcomes; instead, they may exacerbate issues related to image alignment, blending, and, notably, focus.



(a) 2 spectral images (b) 4 spectral images (c) 8 spectral images (d) 16 spectral images

(e) 32 spectral images

Fig. 6. Spectral Stitched Images of the Same Object, a Calendar: Each stitching composed of various images from different distances: (a) from a distance of 130 cm, (b) from 93 cm, (c) from 76 cm, (d) from 58 cm, (e) from 35 cm.

Indeed, one significant factor contributing to reduced stitching quality at closer distances is the increased risk of the camera losing focus. As the camera moves closer to the object, achieving and maintaining optimal focus becomes increasingly challenging, particularly when capturing multiple images to be stitched together. This potential loss of focus further compounds the challenges associated with proximity and image abundance, contributing to the degradation of stitching quality.

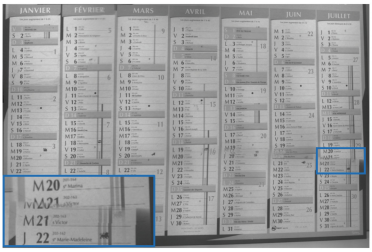
Furthermore, our analysis uncovered a interesting trend regarding the relationship between the number of images used for stitching and the resulting performance metrics and visual quality. While increasing the number of images and decreasing the distance to the object initially led to notable enhancements in stitching quality, we identified a critical threshold beyond which the marginal gains began to diminish. Beyond this threshold, the improvements in performance metrics and visual fidelity stagnated as depicted in the Table 1.

These findings underscore the importance of carefully balancing the trade-offs between proximity, image quantity, and stitching quality in spectral image stitching applications. While closer distances and more images offer the potential for enhanced detail and fidelity, they also introduce challenges related to

Table 1. Average Performance based metrics for various objects.

Nb of Images	BRISQUE ↑	NIQUE ↓
2	32.17	6.07
4	41.81	5.04
8	52.04	3.72
16	62.08	2.49
32	62.84	2.30

focus maintenance that can compromise the final stitching results. Understanding the nuanced dynamics at play in the stitching process is crucial for optimizing performance and achieving the desired balance between image quality and computational efficiency.



(a) 64 spectral images

Fig. 7. Unwanted artifacts appearing. Displayed spectral image at 420 nm.

5 Conclusion

In this paper, we proposed a sequential demosaicking and stitching algorithm for spectral imaging using a hand-held SFA camera from SILIOS Technologies [1]. Our approach involves capturing multiple raw images of a single object and stitching them together to obtain a higher spatial resolution and more detailed images. Through a series of experimentation and analysis, we have demonstrated the efficacy of our algorithm in enhancing the quality of spectral image stitching.

Moreover, Our study shows that capturing spectral images at a close distance to an object results in more images to stitch together, leading to highly detailed panoramas. However, this close proximity also increases the likelihood of distortion problems. Furthermore, The potential applications of this research include fields such as remote sensing, agriculture, and cultural heritage, where having both high spatial and spectral resolution in images is crucial. By leveraging these insights, future advancements in spectral image stitching can be driven towards more robust and reliable solutions that meet the diverse needs of the imaging community.

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Printing and Machine Learning Applications

Dithering with Pascal Cellular Automaton

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Abstract. In this article, I present two versions of a cellular automaton (CA) that evolve according to a set of rules derived from a well-known combinatorial structure: Pascal's Triangle. These CAs produce point sampling commonly observed in high-quality digital dithering and halftoning techniques. The first version is a probabilistic cellular automaton (PCA), which samples discrete values from Pascal's Triangle as a probability distribution. The second version is deterministic, where tone-dependent lattice paths are selected from Pascal's Triangle by analyzing the local points distribution in the halftone pattern. This deterministic version also offers significant computational gains over its probabilistic counterpart.

Keywords: Halftoning and Dithering · Cellular Automata · Error Diffusion · Image Rendering and Quality · Pascal Triangle · Lattice Paths · Quantization · Discrete Sampling

1 Introduction and Overview

I found that a simple Cellular Automata (CA) coupled with—Pascal's Triangle—as probabilistic rules, was able to produce high-quality halftone images. These images presents a point sampling signature that effectively simulates the pleasing texture and detail typically achieved with Frequency Modulation (FM) screening and specifically, Error Diffusion (ED) techniques. While cellular automata and error diffusion operate on different principles, they both share similarities in their spatial and temporal dependencies but also in their predictability, to some extent.

In cellular automata, each cell's state depends on its neighbours, showing spatial dependencies, and the system evolves in discrete steps according to predefined rules, indicating temporal dependencies. Similarly, in error diffusion, pixel quantization error spreads to neighbour pixels, showing spatial dependencies, and the process iterates over the image in a stepwise manner, with each pixel's quantization affecting subsequent ones, demonstrating temporal dependencies.

In general, cellular automata are governed by a set of predefined rules leading to a deterministic evolution. However, by introducing element of randomness or probability into their ruleset, probabilistic cellular automata (PCA) exhibit

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more complex dynamics and simulate stochastic processes with greater fidelity. Similarly, error diffusion algorithms are usually deterministic, but the addition of random noise to the quantization error or diffusion weights reduce the visibility of quantization artefacts, leading to more visually pleasing outputs.

However, beyond these spatial and temporal similarities, cellular automata and error diffusion differs in a fundamental aspect, the way data are quantified. Typically, cellular automata computations are performed using binary or integer values, reflecting the discrete nature of the states. Whereas error diffusion utilizes fractional weights to spread the quantization error of pixel values across neighbour pixels, resulting in continuous floating-point data calculations.

By recognising these characteristics, we can effectively assess how this novel approach utilising Cellular Automata and Pascal's Triangle compares to the established effectiveness of Error-Diffusion technics in generating high-quality halftone images. The combination of these elements in the context of digital dithering and halftoning not only enhances our current understanding but also opens up further research opportunities in both digital imaging techniques and computational theory.

1.1 Error Diffusion

Error diffusion is a digital image processing technique commonly employed in halftoning—a process that converts continuous-tone images into binary images while approximating the original appearance. It is classified as a frequency modulated (FM) dithering technique that produces halftone images with higher spatial resolution and superior image quality by varying the pixel density (frequency) based on the pixel values of the underlying image. Widely embraced across industries such as printing, publishing, and display technologies, error diffusion methods are pivotal in generating high-quality halftone images and find application in devices like inkjet printers and e-ink displays.

Error-Diffusion can be described as follows: the algorithm scans the image, row by row, and pixel by pixel, the current pixel value plus a feedback error is compared to a threshold, the result of the comparison assigns an output value, e.g. 0 or 1 to the halftone image and the quantization error is distributed to neighbour pixels using predefined coefficients. By distributing the error across the image, the error diffusion process tends to smooth out abrupt transitions between different intensity levels, maintaining the overall tonal balance and detail in the final output. Below are the Floyd and Steinberg coefficients, the ($\square$) indicates the currently processed pixel and ($-$) indicates the previous pixel.

$$\frac{1}{16}\begin{bmatrix} - & \square & 7 \\ 3 & 5 & 1 \end{bmatrix}$$

Over the past decades, researchers and specialists have explored numerous variations and extensions of error diffusion algorithms to address specific challenges, such as reducing artefacts, improving performance, and adapting to different types of input images and outputs, encompassing multilevel, color and

tone dependencies. These variations include modifications to the diffusion coefficients, error weighting schemes, scan path, threshold modulation, noise shaping and incorporation of perceptual models, leading to a substantial body of literature, articles and patent filings, [1–12].

Blue-Noise—Poisson Disc Sampling. To analyze the local points distribution in the halftone pattern, Ulichney [3] introduces a statistical tool in the frequency domain called the Radially Averaged Power Spectrum density or RAPSD [3]. This tool helps to identify blue-noise point sampling, which represents an "ideal" power spectrum for dither patterns (Fig. 1). Unlike other dithering patterns, such as ordered or clustered-dot dithering, a blue-noise dither pattern is a spatial arrangement of pixel values that minimizes low-frequency structures, meaning that any regular or repetitive patterns are avoided. This results in a more natural and visually pleasing pattern.

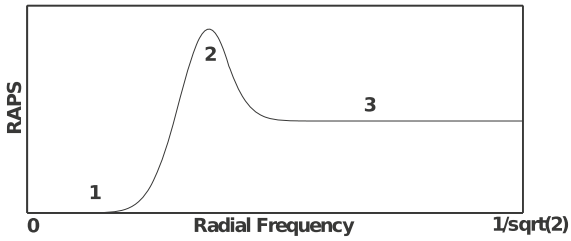
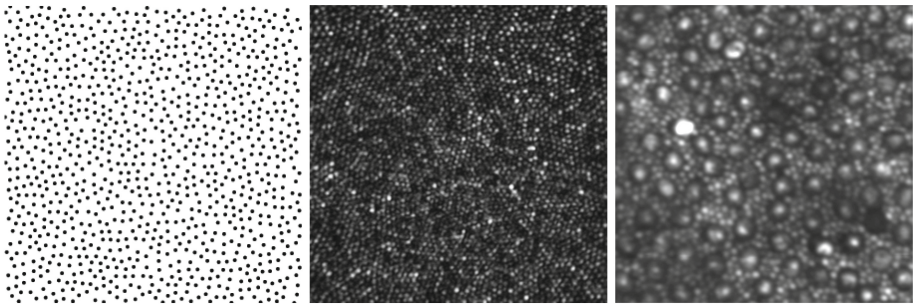


Fig. 1. Ideal Radially Averaged Power Spectrum of a blue-noise halftone pattern: (1) Pulse at origin (DC component = 0), (2) surrounded by a peak at the principal frequency and (3) surrounded by white noise. (Color figure online)

The concept of blue-noise dithering is closely related to Poisson Disc Sampling in the field of statistical sampling, which also aims to distribute points (or samples) with a minimum distance between them. It is generally used as an antialiasing technique, where unwanted artefacts introduced by regular sampling can be attenuated by non-uniform sampling [13] and [14].

Indeed, Yellott [15] discovered in 1984 that the cone photoreceptors in the mammalian rhesus retina are arranged following a Poisson disc distribution. Recently, the first photographs of rod receptors from a living human eye (Fig. 2b and 2c) have been taken using a noninvasive imaging technique [16].

A typical Poisson disc sampling algorithm adds points randomly, at each new added point it checks that the distance to neighbours points is beyond a fixed threshold, otherwise the point is rejected. The algorithm iterates until it can no longer add points (Fig 2a). A Comparison of methods for generating Poisson Disk Distributions is presented in [17].



(a) (b) (c)

Fig. 2. Poisson Disc Sampling Generated generated by Dart throwing (a), Human Cones Photoreceptors mosaic at the foveal center (b), Human Cones and Rods Photoreceptors mosaic at the foveal center - cones are the bigger, brighter points, while rods are the little dots that surround them (c) "Courtesy of the Dubra lab at the Medical College of Wisconsin".

1.2 Cellular Automata

Cellular automata were first introduced and developed by mathematicians John von Neumann [18] but it was the work of John Conway [19] that brought this computational model into the spotlight with his creation of the Game of Life. Since then, numerous researchers have contributed to the advancement of cellular automata theory and applications over the years. Stephen Wolfram extensively studied and popularized them in his book "A New Kind of Science" [20].

Typically, they consist of a grid of cells, each of which in a finite number of states. The state of each cell evolves over discrete time steps based on a set of rules that determine how the state of a cell changes based on its current state and the states of its neighbour cells.

Cellular automata can be classified into two main types:

- In deterministic cellular automata, the rules for state transitions are strictly defined and do not involve any randomness. This means that for a given initial configuration of cell states, the evolution of the system is completely predictable and reproducible.
- In probabilistic cellular automata (PCA), the rules for state transitions include elements of randomness. This means that the evolution of the system can vary even if the initial configuration is the same.

And the evolution of cells can occur in two different ways:

- In synchronous updating, all cells evolve simultaneously at each time step.
- in asynchronous updating, cells evolve sequentially rather than simultaneously.

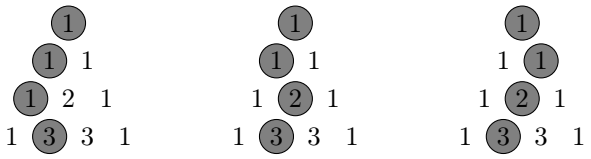
1.3 Pascal's Triangle

The Pascal's Triangle is a fascinating mathematical structure named after the French mathematician Blaise Pascal [21], although it was known to many mathematicians before him. It finds application across various fields due to its remarkable properties and connections to different areas of mathematics, ranging from combinatorics to algebra, from probability theory to geometry and extending into computer science and physics [22].

It's a triangular array of numbers where each number is the sum of the two numbers directly above it. The triangle starts with a single 1 at the top, and each row thereafter is constructed by adding adjacent numbers in the previous row. Here's an example of the first few rows:

$$\begin{array}{c} 1 \\ 1 \quad 1 \\ 1 \quad 2 \quad 1 \\ 1 \quad 3 \quad 3 \quad 1 \\ 1 \quad 4 \quad 6 \quad 4 \quad 1 \end{array} \qquad \begin{array}{c} \binom{0}{0} \\ \binom{1}{0}\ \binom{1}{1} \\ \binom{2}{0}\ \binom{2}{1}\ \binom{2}{2} \\ \binom{3}{0}\ \binom{3}{1}\ \binom{3}{2}\ \binom{3}{3} \\ \binom{4}{0}\ \binom{4}{1}\ \binom{4}{2}\ \binom{4}{3}\ \binom{4}{4} \end{array}$$

Binomial Coefficients and Lattice Paths Counting. Each element of Pascal's Triangle, denoted as $\binom{n}{k}$[1], represents the number of ways to choose k elements from a set of n elements. This is directly related to counting paths in a lattice grid. If you interpret the rows and columns of Pascal's Triangle as coordinates, you can see that moving from the top of Pascal's Triangle (0, 0) to any position (n, k) involves a series of steps right (increasing k) and down (increasing n). This mirrors moving in a lattice grid where you can only move right or up. Here's an example of the three different paths for $n = 3$ and $k = 1$, and respective lattice path.



n \ p	1	2	3
0	0	0	0
1	0	0	1
2	0	1	1
3	1	1	1

Finally, the sum of the elements in a row of Pascal's Triangle corresponds to the total number of lattice paths from the top of the triangle to that row. This sum is equal to 2^n for the n-th row, which encompasses all possible combinations of steps in the lattice grid.

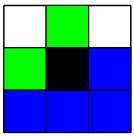
[1] $\frac{(n)!}{k! \times (n-k)!}$.

2 Proposed Methods

In its simplest form, I showcase two new dithering methods leveraging cellular automaton principles and Pascal's triangle, which are:

- Pascal's rows as Probabilistic rules
- Pascal's paths as Deterministic rules

In contrast to error diffusion, which typically reads one pixel ■ and updates (writes) the values of the next neighbour pixels ■, these two new methods read the values of the preceding neighbour pixels ■ and update (write) the value of the current pixel ■ accordingly.



Both methods involve generating a discrete value based on the preceding neighbour pixels' values, which is then added to the current pixel value before thresholding. The resulting quantization error is updated for the current pixel, and the process repeats iteratively and asynchronously, following a serpentine or zigzag scan pattern on the raster image. However, they diverge in the way the discrete value is generated.

- In the *probabilistic* version, we sample a discrete value from the rows of the Pascal cumulative distribution function (CDF), where the preceding pixels' values index the corresponding row to be sampled.
- In the *deterministic* version, for each contone value, we attribute a specific Pascal lattice path where the discrete value is indexed in the corresponding path by the preceding pixels' values.

Whereas the probabilistic version does not provide direct control over the points distribution pattern, the deterministic version allows selecting a specific lattice path for each contone value to minimize visual artefacts and enhance the perceived quality of the dithered image. This approach is similar to tone-dependent error diffusion techniques [9], where the diffusion weights are adapted based on the local tone of the image to produce the *blue-noise* dither pattern.

For each method, a visual assessment, the processing time, and the 2D power spectral density (PSD) estimation, along with its radially averaged power spectral density (RAPSD), are provided and compared with:

- Ulichney [3] - [FS with 50% random weights (4), fixed, non-deterministic]
- Ostromoukhov [9] - [3 diffusion weights, tone-dependent, deterministic]

2.1 Pascal's Rows as Probabilistic Rules

In this section, I present the probabilistic version for two preceding neighbour pixels' values ■ or Left/Right - Top.

Pascal's Cumulative Distribution Function. Given an 8-bit continuous-tone grayscale image, where each pixel's value ranges from 0 (representing black) to 255 (representing white), we generate Pascal's Triangle up to 256 rows, corresponding to the available levels of grayscale intensity.

Then, the binomial coefficients $\binom{n}{k}$ for each row of Pascal's Triangle are summed and normalized[2] to create a cumulative distribution function (CDF). This function allows for the sampling of new discrete values between 0 and 255 by leveraging the fact that the n-th row of Pascal's Triangle (starting from 0) has n+1 entries. Below, the Pascal's triangle cumulative distribution function generation (left-aligned form), up to the 5-th row.

Pascal's Triangle Coefficients	Cumulative sum of Coefficients	Normalized Cumulative sum (CDF)
Row 0 : 1	1	1
Row 1 : 1 1	1 2	$\frac{1}{2}$ 1
Row 2 : 1 2 1	1 3 4	$\frac{1}{4}$ $\frac{3}{4}$ 1
Row 3 : 1 3 3 1	1 4 7 8	$\frac{1}{8}$ $\frac{4}{8}$ $\frac{7}{8}$ 1
Row 4 : 1 4 6 4 1	1 5 11 15 16	$\frac{1}{16}$ $\frac{5}{16}$ $\frac{11}{16}$ $\frac{15}{16}$ 1

Sampling Pascal's CDF. Given the values of the two preceding neighbour pixels ■, A and B, and the value of the current pixel ■, C, the discrete sampling process follows these steps:

1. Calculate the absolute difference, Diff = $|A - B|$
2. Retrieve the corresponding row from Pascal's CDF, PCDF(Diff + 1)
3. Generate a random value from the uniform distribution, Rand = $U(0, 1)$
4. Find the first PCDF(Diff + 1) value that is greater than or equal to Rand.
5. Retrieve the discrete Index of this PCDF value
6. Calculate the new value for C, New ■ = Index + min(A,B)

$$\text{In summary: New} = \begin{cases} A & \text{if } A = B \\ [A, B] & \text{otherwise} \end{cases}$$

Dithering with Pascal's Probabilistic Rules. Finally, the current pixel value ■ plus the New value ■ is compared to a threshold (e.g. 255), the result of the comparison assigns an output value, e.g. 0 or 1 to the halftone image and the current pixel value, C, is updated with the resulting quantization error.

[2] The sum of the coefficients in the n-th row of Pascal's Triangle is 2^n.

Algorithm 1. Dithering with Pascal Probabilistic CA

```
1: procedure DitherImage(input_image)
2:     Initialize an empty dithered image
3:     Initialize PascalCDF
4:     for each pixel (x, y) in the input image do                ▷ Serpentine Scan
5:         pixel_A ← input_image(x, y − 1)
6:         pixel_B ← input_image(x − dir, y)
7:         pixel_C ← input_image(x, y)
8:         discrete_Value ← SamplePascalCDF(pixel_A, pixel_B, PascalCDF)
9:         new_pixel_C ← pixel_C + discrete_value
10:        dithered_image(x, y) ← new_pixel_C > Threshold          ▷ Thresholding
11:        quant_error ← new_pixel_C − 255
12:        input_image(x, y) = quant_error                         ▷ Update current pixel value
13:    end for
14:    return dithered image
15: end procedure
16: procedure SamplePascalCDF(A, B, PCDF)
17:    diff_AB ← abs(A - B)
18:    Rand ← Random number between 0 and 1                        ▷ Uniform distribution
19:    Index ← PCDF(diff_AB, Rand)
20:    return Index + min(A,B)
21: end procedure
```

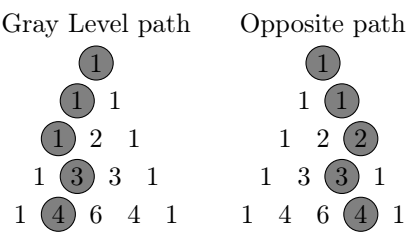
2.2 Pascal's Paths as Deterministic Rules

In this section, I present the deterministic version to halftone an 8-bit continuous-tone grayscale image based on two preceding neighbour pixels' values ■ or Left/Right - Top. To select, for each contone value, the Pascal path that best mimics a *blue-noise* dither pattern, we explore and analyze all the paths given by Pascal's triangle up to N rows, where N represents the maximum contone value allowed. From the radially averaged power spectrum density (RAPSD), we identify the path for which the spectral energy is concentrated above the principal frequency B_f, which is defined as a function of the input contone value C as:

$$B_f(C) = \begin{cases} \sqrt{C} & \text{for } 0 < C \leq \frac{1}{2} \\ \sqrt{1-C} & \text{for } \frac{1}{2} < C \leq 1 \end{cases}$$

However, an 8-bit continuous-tone grayscale image represent $2^{255} \simeq 5.8 \times 10^{76}$ paths to explore, which is infeasible due to the immense computational complexity and time required, even with GPU acceleration. To overcome this difficulty, from the pair factors of 255 [1,3,5,15,17,51,85,255], we select [15, 17] and generate all the possible paths for Pascal's triangle up to row 15 and 17. This results in $2^{15} = 32,768$ and $2^{17} = 131,072$ paths, respectively.

We select the paths that present a *blue-noise* dither pattern and combine them to generate the lattice paths of up to 256 entries for each gray level from 0 to 255. It is not necessary to find the respective lattice path for each gray level from 0 to N among the 2^N possible lattice paths in Pascal's triangle up to row N. Indeed, from the lattice paths of gray levels from 0 to $\lfloor \frac{N}{2} \rfloor$, we can generate the corresponding lattice paths for the gray levels from $\lfloor \frac{N}{2} \rfloor + 1$ to N by taking advantage of Pascal's lattice path symmetry.



n \ p	C	C'
0	0	0
1	0	1
2	0	2
3	1	2
4	1	3

Algorithm 2. Generate Pascal Triangle Paths

```
function GENERATE_PASCAL_PATHS(N)
    Input: N (number of rows)
    Output: paths (array of paths with indices)
    num_paths ← 2^N                                   ▷ Total number of paths
    paths ← cell array of size num_paths
    for i ← 0 to num_paths − 1 do
        binary_str ← dec2bin(i, N)                    ▷ Convert number to binary string
        row ← 0
        col ← 0
        path_indices ← [(row, col)]                   ▷ Initialize with the top node indices
        for j ← 1 to N do
            row ← row + 1
            if binary_str[j] =' 0' then
                col ← col                             ▷ Column stays the same for left child
            else
                col ← col + 1                         ▷ Column increments for right child
            end if
            path_indices ← append (row, col)
        end for
        paths[i + 1] ← path_indices                   ▷ Store the path with indices
    end for
end function
```

Algorithm 3. Dithering with Pascal Deterministic CA

```
procedure DITHERIMAGE(input_image, generated_paths)
    Initialize an empty dithered image
    for each pixel (x, y) in the input image do                ▷ Serpentine Scan
        pixel_A ← input_image(x, y − 1)
        pixel_B ← input_image(x − dir, y)
        pixel_C ← input_image(x, y)
        discrete_value ← DiscreteValueFromPaths(pixel_A, pixel_B, pixel_C, generated_paths)
        new_pixel_C ← pixel_C + discrete_value
        dithered_image(x, y) ← new_pixel_C > Threshold         ▷ Thresholding
        quant_error ← new_pixel_C − 255
        input_image(x, y) = quant_error                        ▷ Update current pixel value
    end for
    return dithered image
end procedure
procedure DISCRETEVALUEPATH(A, B, C, PATHS)
    diff_AB ← abs(A - B)
    Value ← PATHS(diff_AB, C)
    return Value + min(A,B)
end procedure
```

3 Results

3.1 Pascal's Rows as Probabilistic Rules

(See Figs. 3, 4 and 5).

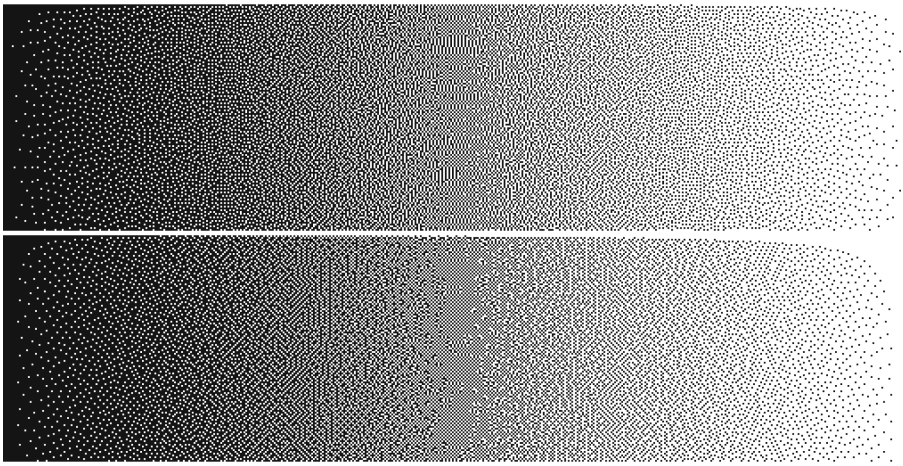


Fig. 3. Dithered gray ramp: (Top) Ulichney - FS with 50% random weights (4 neighbour pixels - non-deterministic); (Bottom) Our method (2 neighbour pixels - non-deterministic) (Color figure online)

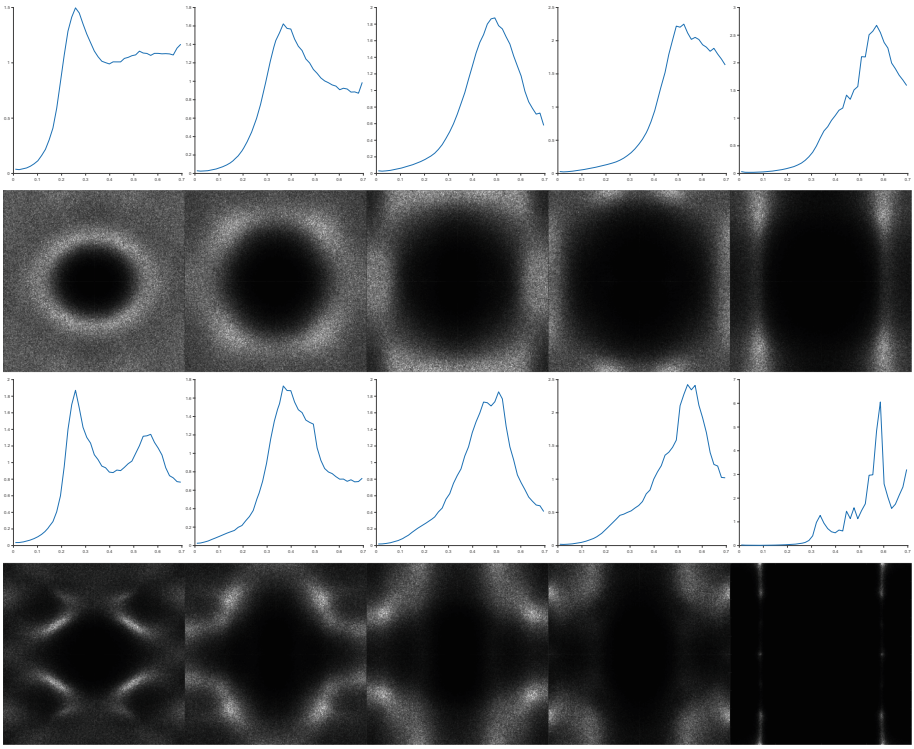


Fig. 4. RAPSD and PSD for gray levels [17, 34, 51, 68, 85]: (Top) Ulichney - FS with 50% random weights (4 neighbour pixels - non-deterministic); (Bottom) Our method (2 neighbour pixels - non-deterministic) (Color figure online)

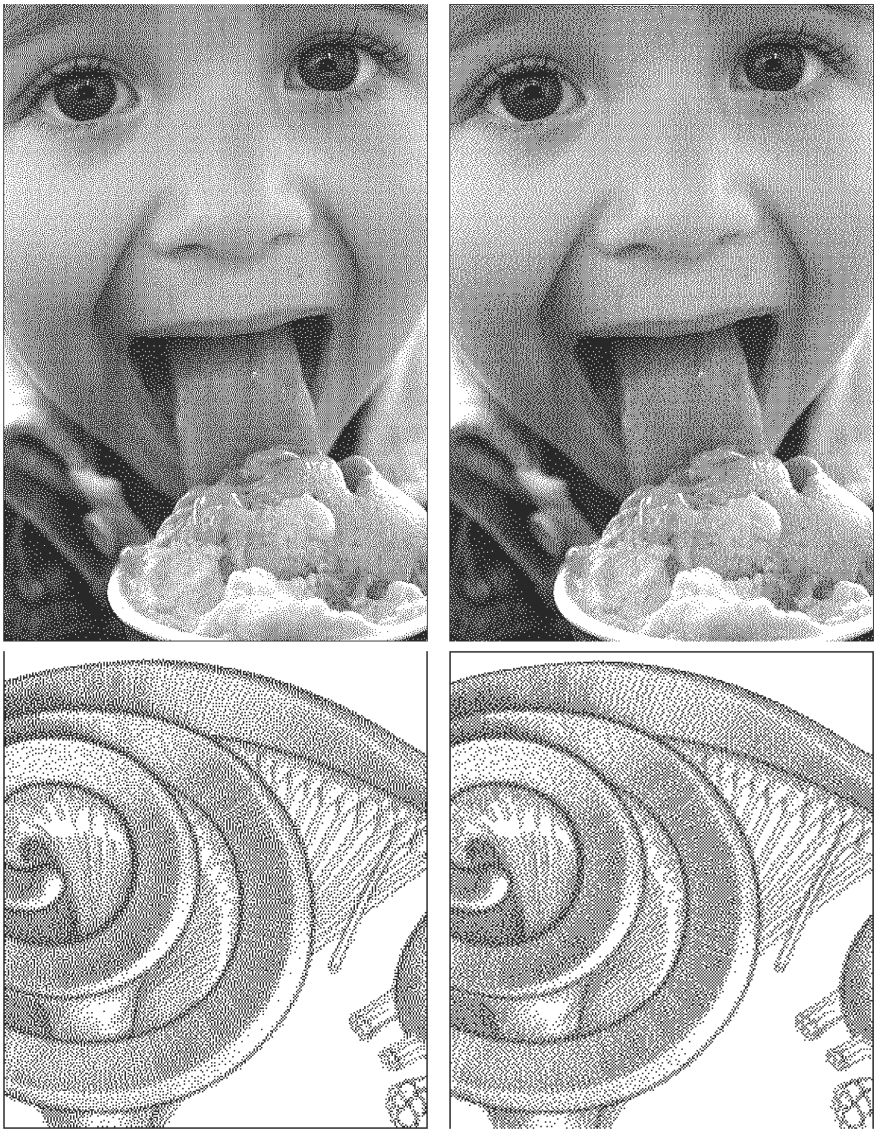


Fig. 5. (Left) Ulichney - FS with 50% random weights (4 neighbour pixels - non-deterministic); (Right) Our method (2 neighbour pixels - non-deterministic)

Remarks : With two preceding neighbour pixels, our probabilistic method presents higher intensities along the diagonals, indicating directional frequency distribution. Dithered images produced by this method have more structured and anisotropic textures. In contrast, Ulichney method presents a circular symmetry with a bright ring around the center, pointing to an isotropic frequency distribution where frequencies are uniformly spread in all directions.

3.2 Pascal's Paths as Deterministic Rules

(See Figs. 6, 7 and 8).

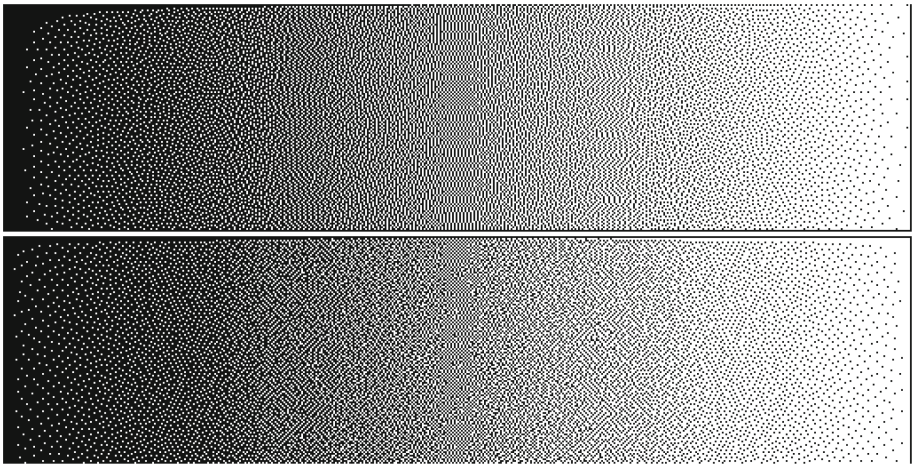


Fig. 6. Dithered gray ramp: (Top) Ostromoukhov - Tone-dependant weights (3 neighbour pixels - deterministic); (Bottom) Our Tone-dependent method (2 neighbour pixels - deterministic) (Color figure online)

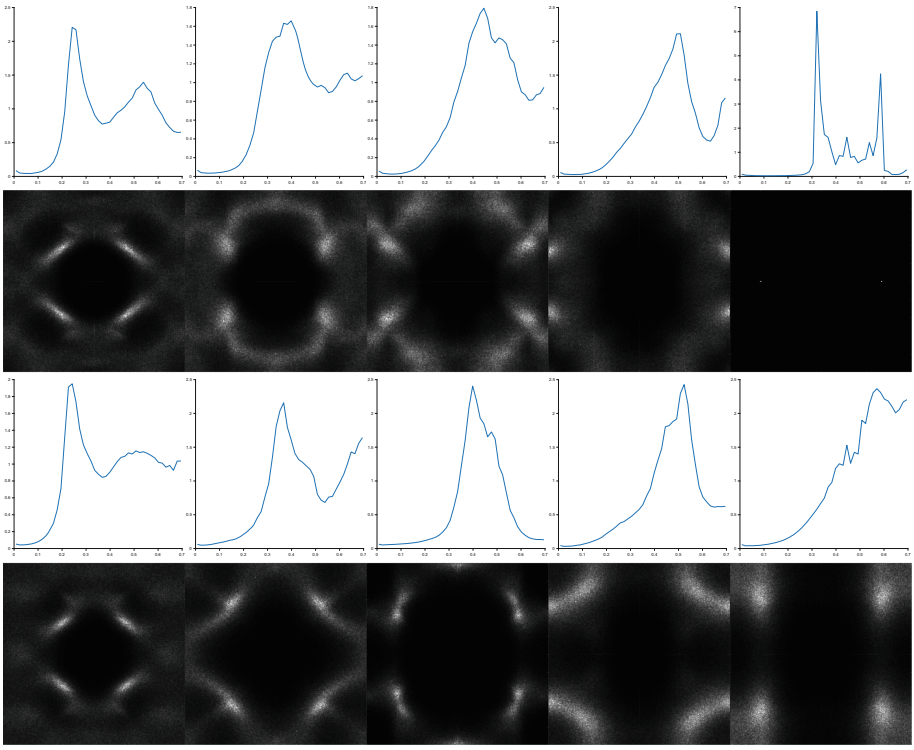


Fig. 7. RAPSD and PSD for gray levels [17, 34, 51, 68, 85]: (Top) Ostromoukhov - Tone-dependant weights (3 neighbour pixels - deterministic); (Bottom) Our Tone-dependent method (2 neighbour pixels - deterministic) (Color figure online)

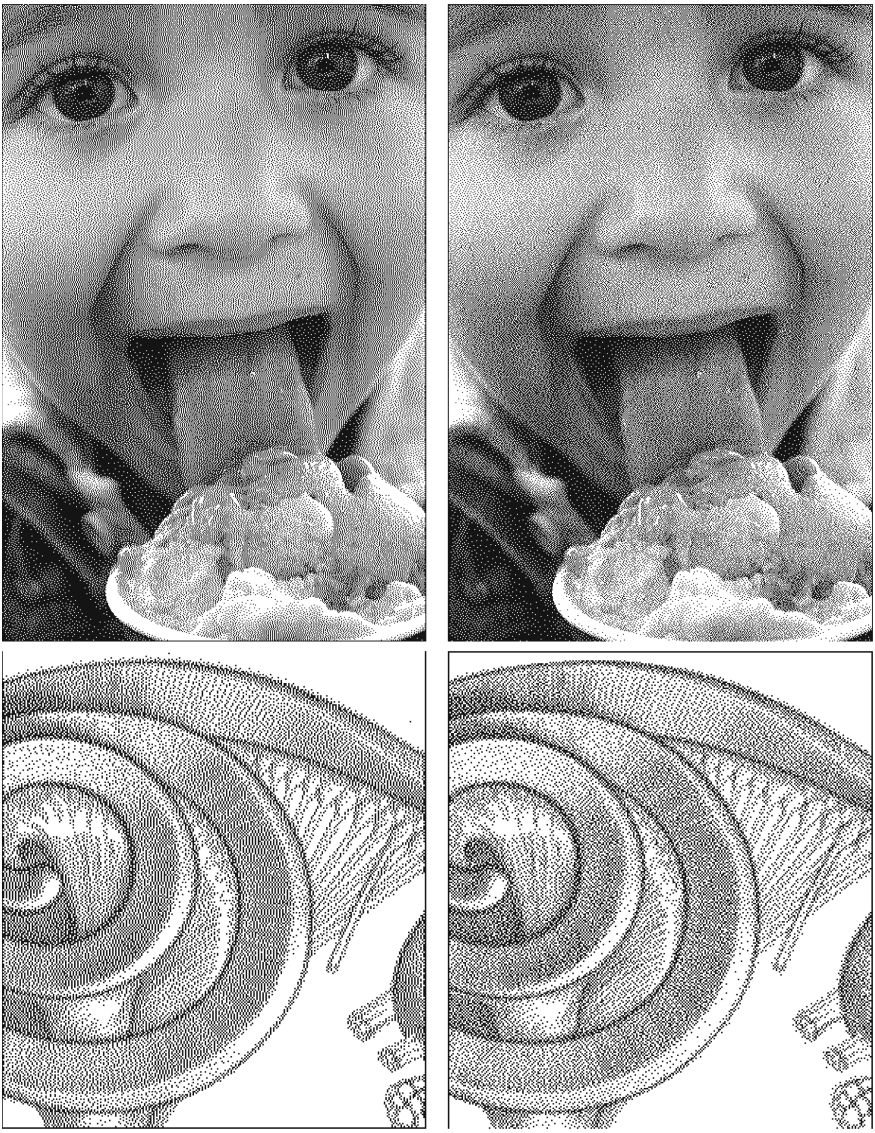


Fig. 8. (Left) Ostromoukhov - Tone-dependant weights (3 neighbour pixels - deterministic); (Right) Our Tone-dependent method (2 neighbour pixels - deterministic)

Remarks: Besides points sampling optimization, our deterministic method still presents diagonal, structured, and anisotropic textures, which likely originate from the use of only two neighboring pixels. However, the points distribution and density in the highlights and shadows have improved, as well as the processing time, since we no longer have to sample discrete values from a probability

distribution. It is now comparable with the methods of Ulichney and Ostromoukhov, as well as standard error-diffusion techniques.

4 Conclusions and Future Work

In this article, I presented two new methods for generating high-quality dithered images based on the principles of cellular automata and Pascal's Triangle as probabilistic and tone-dependency deterministic rules. Dithering and halftoning are well-established techniques, and these new methods offer valuable insights into potential enhancements in this field, as well as reverse engineering past techniques into new innovations. In their simplest form, these methods address challenges in simulating detailed textures and gradients for image processing. This new framework and its discrete singularity suggest future improvements, such as extending the neighbour scheme from multinomial coefficients for color and multilevel image processing. I've called it the "*The Blue Tip of the Iceberg*" as this article only presents a small and visible portion of a much larger and complex framework. The majority of its mass lies beneath the surface of the water, hidden from view.

The Python implementation can be found at https://github.com/GovYvanR/Dithering-with-Pascal-Triangle, this version already implement the possibility of modulating the threshold.

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From Gutenberg to Llamas: Print Optimization Through First Principles and AI

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Abstract. Optimization has been a thread running through the history of printing, from the motivation of Johannes Gutenberg developing moveable type to the present efforts to apply AI techniques to solving a broad range of printing challenges. The need for optimization here ranges from the solution of specific, well-defined challenges like the prediction of print color from as few measurements as possible, via the boosting of ink efficiency and printed image quality to the need for ease of use and automation necessitated by a desire to minimize human intervention in print manufacturing. While the former category of challenges is well suited to various flavors of machine learning, the latter becomes approachable thanks to the advent of large language models and other generative AI methods. A critical factor in this optimization spectrum is also the control domain in which printed output can be specified and therefore also adjusted. Here the HANS paradigm offers a clear benefit by making the space in which print is controlled linear, which allows for optimization processes performed on top of it to be directed at the specific challenges at hand, instead of also having to compensate for the ill-behavedness of conventional print control spaces. The paper concludes with a bit of blue-sky thinking about what the near future of print optimization may look like, where AI moves from particular, narrow application to providing an end-to-end infrastructure.

Keywords: AI · printing · imaging · pipeline · color · optimization · HANS

1 Introduction

Printing is called to be a transparent process that materializes content without altering or interfering with it. When someone looks at a printed surface, they see the content it depicts and only in exceptional cases, like that of fine art photography or the use of a new special effect, is any attention paid also to the print itself as an object.

The desire for a faithful rendering of content has led to a history of improvements to print quality, from increasing spatial resolution, to expanding color gamuts and to an ever more precise and reliable deposition of inks or other colorants onto substrates. Increasing capabilities have also put pressure on the control domain in which print output is specified, where historically the visually highly non-linear and varied relationships between material quantities and resulting appearance have been a challenge.

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The need for the printing process to be transparent has two flavors: a consumer and a commercial one. In the consumer context, the user of a printing system just wants it to work. They do not want to have to learn about how it works, why it works, how it is best configured and used. They just want to trigger printing and for the resulting print to materialize content in the way they expect and desire. In the commercial context, the picture is exactly the same! Additionally there is also a requirement of the operator being able to make high-level choices about trade-offs among attributes like quality, cost and speed and in rare cases also exercise more specific control over some parts of the process.

A third element also comes into play in printing, which are economic considerations. For different reasons, both consumers and commercial users of printing want to keep its cost at a minimum, given the required performance. Printing, after all, is not about print but about content. This need places growing pressure both on containing material costs and on making the printing operation as efficient as possible, minimizing the need for using materials for setting up, verifying or re-printing.

Finally, sustainability also plays an important role in printing and impacts it broadly, from the quantity of power consumed across substrate, ink and press manufacture, via the waste incurred during production to the recyclability and decomposability of printed materials.

All of the above places substantial pressure on optimizing the printing process and the present paper aims to review some examples of what can be optimized and how, ranging from reducing the waste incurred in configuring printing workflows via modelling their behavior accurately and efficiently, via improving the visual quality of print and thereby making it more faithfully embody content, to reducing the quantity of inks needed to deliver the printed output, which is good for both cost of operation and environmental impact. In the context of these optimizations, an emphasis will be placed on using a variety of AI techniques, in some cases also coupled with first-principles models. These topics of print optimization will be prefixed by a brief overview of the HANS print control domain, which is particularly suited for optimization, and the paper will conclude with some blue-sky thinking about what more recent developments in generative AI could bring to print optimization in the future.

2 Controlling Print for Optimization

When optimizing the output of an imaging system, such as a printing one, there is a host of considerations to be made, ranging from a clear definition of what is being optimized (e.g., color gamut, smoothness, ink use efficiency, etc.), via what the objective of the optimization is (e.g., maximizing or minimizing some property or aiming at delivering a specific balance among multiple ones) to what data is needed to perform it and what mechanism to employ.

A sometimes overlooked or implicit consideration is also the domain in which optimization is to be performed. In the context of colorimetry this can be in the form of choice of color space (e.g., CIE XYZ, CIE L*a*b* [1] or CIECAM16 [2]), when dealing with image capture the choice may be between whether a scene-linear [3] or log [4] representation of RGB is used, etc.

2.1 Print Control via Ink Quantities

When it comes to print optimization, whether it be for 2D or 3D printing, this has conventionally been done either in terms of ink (or other printing material) quantities [5] or device color space values like CMYK or RGB [6]. While good results can be achieved in this way, they are impeded by the inherent non-linearity and complexity of control domain versus print property relationships. These stem from how ink quantities relate to print color, which is the result of the combination of multiple and different non-linearities combining.

Considering a print's color, it is the result of multiple interactions:

1. Ink modulating light, where the relationship between ink quantity and light absorption is non-linear and where the non-linearity varies from ink to ink.
2. Ink interacting with the substrate it is deposited on, where the combined effect of ink and substrate properties is a non-linear combination of their individual color or spectral properties.
3. Ink interacting with other inks, where the color of multiple inks overlapping is a non-linear combination of their individual properties.
4. The combined spatial pattern formed by multiple inks, where per-ink pattern colors relate to combined pattern color in non-linear and complex ways, since variable interactions between patterns differently modulate incident light.
5. The spatial distribution of an ink pattern on a substrate, where the color of a pattern is not the linear combination of its component colors (i.e., those of variously-inked and un-inked parts), but subject to lateral scattering that introduces a non-linearity.

In other words, the relationship can be conceptually sketched out as follows:

$$T_c = f_5(f_4(f_3(f_2(f_1(q_i), s)))) \tag{1}$$

where T is a tristimulus value, $c \in \{X, Y, Z\}$, q_i is the quantity of the i-th ink, s is the substrate and $f_{1–5}$ are unknown, different and ink, ink combination and substrate dependent non-linear functions.

While these are key contributors to the complexity of the ink quantity versus print color relationship within a print, there are further complicating factors when the view zooms out and considers the digital inputs to a printing system, where differences arise also between those inputs' specified ink quantities and the actual ink quantities and spatial patterns materialized in print.

The combined effect of the above factors can be seen in Fig. 1, which provides one example in measured terms and one and in visual terms.

There are two key consequences here, when samples are taken of the ink quantity versus color (or spectral reflectance) relationship. First, that it is complex and variably non-linear and, second, that the non-linearity at the samples does not extend to the space between them. I.e., even a perfect characterization of the non-linearity inherent at sampled points in an ink quantity space is likely to be inaccurate in-between.

2.2 Print Control via Neugebauer Primary Coverages

An alternative to controlling print via ink quantities, which is subject to the multiple non-linearities reviewed in the previous sub-section, is to do so in terms of Neugebauer

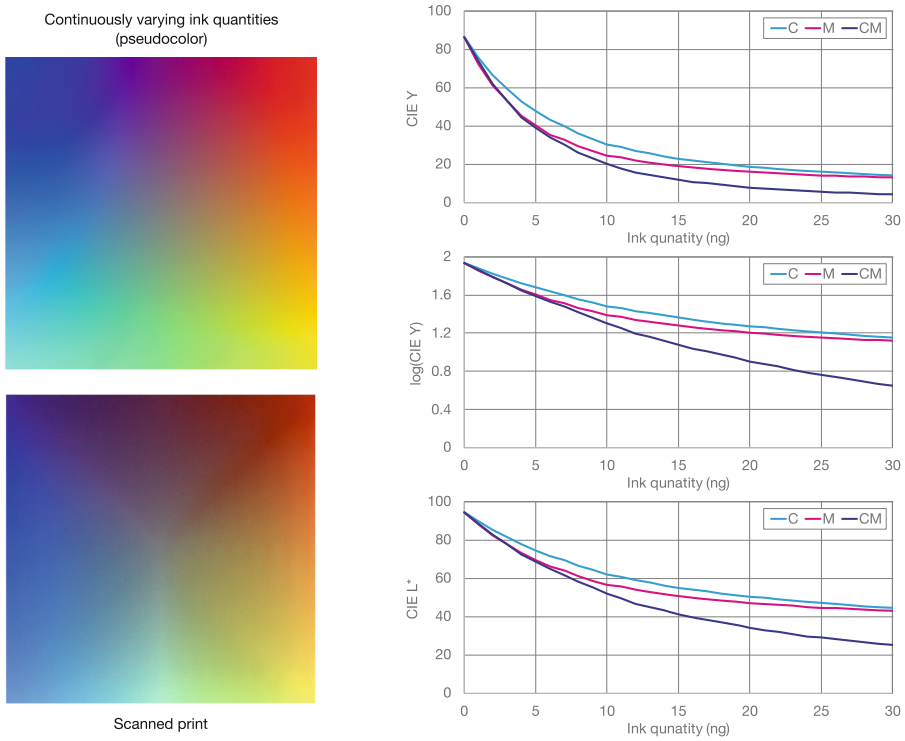

Fig. 1. (Left) (top) Smoothly varying ink quantities visualized in pseudocolor and (bottom) scan of a resulting print showing various discontinuities. (Right) Measurements of ramps of increasing quantities for cyan, magenta and combined cyan + magenta inks in terms of (top) CIE Y, (middle) the log of CIE Y, and (bottom) CIE L*.

primary area coverages [7, 8]. This paradigm is called HANS (Halftone Area Neugebauer Separation) and branded commercially as HP Pixel Control.

Here Neugebauer primaries (NPs) are all the possible ink combinations at a given print location. E.g., in a printing system with four inks, CMYK, and binary deposition (i.e., an ink either is or is not deposited at any given location), there are $2^4 = 16$ NPs: blank substrate, C, M, Y, K, CM, CY, CK, MY, MK, YK, CMY, CMK, CYK, MYK and CMYK. In general, for n inks that can be deposited at k levels there are k^n NPs.

Print can then be controlled by specifying a k^n-dimensional vector where each component expresses the relative local area covered by the corresponding NP and where the components sum to one (since they express the full NP composition of some area). E.g., [_, C, CM] = [0.1, 0.7, 0.2] specifies that 10% of some local neighborhood in a print be left blank, 70% be covered by the cyan ink only and 20% be covered by cyan ink overprinted by magenta ink.

Shifting control from ink quantities to NPacs (NP area coverages), or halftone pixel types in the digital domain, side-steps the first four non-linearities listed in Sect. 2.1, since these are a consequence of considering quantities of ink without taking into account their

interactions. The fifth non-linearity, derived from optical dot gain, can be addressed by a suitable choice the space in which NP colorimetries are represented, where the Yule-Nielsen modification [9] of CIE XYZ allows for the effect of lateral scattering to be taken into account by introducing a single, overall non-linearity.

In summary, the relationship can be expressed as follows:

$$T_c = \left(\sum_{p=1}^{k^n} T_{cp}^{\frac{1}{y}} a_p \right)^y \tag{2}$$

where y is the Yule-Nielsen exponent, T_{cp} is a tristimulus value of the p-th NP and a is that NP's area coverage. Note that in an optimization context it is possible to operate in the Yule-Nielsen-modified domain and to undo its non-linearity after optimization, which makes it possible to continue to operate in a linear space where convexity also holds.

Fig. 2. Effect of continuously varying ink quantities (left) versus Neugebauer primary area coverages (right).

The end result of the move from ink quantities to Neugebauer primary area coverages is that while the relationship between ink quantities and a print's colorimetry is highly non-linear in a complex, composite way (Eq. 1), the relationship between NPs and print colorimetry (in Yule-Nielsen-modified XYZ) is linear and convex, with area coverages being the weights (Fig. 2). The challenge of the correspondence between digital and materialized, physical quantities remains, but since it takes place on top of a vastly simpler relationship, its detrimental consequences are significantly lesser for NPac control than ink quantity control.

Beyond linearity and convexity, a move from controlling print via ink quantities to NP area coverages also result in an important and exponential expansion of the solution space over which optimization can operate. Considering the two domains, it can be seen that for n inks, the ink quantity domain is n-dimensional, since the basic control unit is an n-component ink quantity vector. In the NPac case, for n inks, the NPac vector is k^n dimensional, where $k \geq 2$ (since the fewest levels of ink quantity per local area are two: no ink and some ink).

For the simplest case of using only three, CMY inks in a binary deposition system, the ink quantity space is 3 dimensional while the NPac space has $2^3 = 8$ dimensions.

While this is already significant, the difference tends to be dramatically larger in actual printing systems. E.g., for a printer that uses CMYKcm inks (i.e., with a 6D ink quantity space) and can deposit up to 3 drops of ink per halftone pixel, the NPac space is $4^6 = 4096$ dimensional, while for a printer with 14 inks (i.e., with a 14D ink quantity space) and the same maximum of 3 drops per ink per halftone pixel, NPac space is $4^{14} = 268{,}435{,}456$ dimensional.

In other words, NPac spaces are exponentially related to the number of inks (and ink levels per halftone pixel), which means that they allow for the specification and control over printable patterns that are inaccessible and uncontrollable via ink quantities.

2.3 Impact of Control Domain on Optimization Performance

To illustrate the impact of the control domain on optimization performance, a supervised learning approach based on polynomial regression was applied to the challenge of modelling the color and break strength of multi-jet fusion [10] 3D-printed parts. Overall the objective function here was:

$$min_T \|\mathbf{T} * f(\mathbf{M}) - \mathbf{P}\| \tag{3}$$

where $\mathbf{P}$ is an N × O matrix of measured part properties (e.g., CIELAB L*, a*, b* and break strength in MPa), $\mathbf{M}$ is an N × X matrix of print control domain vectors used to make those parts, $f()$ is a transformation adding cross, higher-order or constant terms and $\mathbf{T}$ is the matrix minimizing the error between measured and predicted part properties.

Table 1. Model accuracy results for color and break strength predictions made from agent quantity or Mvoc vectors for prints made using a 6-agent multi-jet fusion system. Prediction errors from agent quantities show how much worse they are than predictions from Mvocs.

	Mvoc control			Agent quantity control		
	median	95th perc.	max.	median	95th perc.	max.
Color (ΔE00)	2.8	7.8	11.3	5.1 (1.8×)	9.3 (1.2×)	12.8 (1.1×)
Break strength (ΔMPa)	0.9	4.8	6.3	2.3 (2.6×)	7.9 (1.7×)	13.6 (2.1×)

Using color and break strength measurements of 143 dog bone test parts printed with a 6-agent system ("agent" being the preferred nomenclature in the 3D case for what would be referred to as "inks" in 2D, since they also include functional materials) and optimising their prediction either from agent quantity vectors of the 3D-equivalent of NPacs known as Movcs (material voxel contents) [11] using a 4th-order polynomial with cross terms yielded the results in Table 1. Overall, it can be seen that median errors are around twice as high when making predictions from agent quantities than from Mvocs.

3 Print Optimization via Machine Learning

Having introduced the topic of print control domain and the impact of its well-behavedness on optimization performance, the following sub-sections will review a variety of machine learning optimizations that improve both print properties and the efficiency of printing setup processes.

3.1 RONT – A Hybrid Print Color Model

One of the key features required of any optimization solution is the ability to predict the color of a print from the inputs to a printing system. This is the case both when color itself is being optimized, and when optimization is applied to other print attributes, where color is the space in which it is indexed.

Here two categories of approaches are available: a first-principles, analytical one and a supervised learning, general-purpose one. While the former has the potential to predict system output well from few parameters, when these are known or can be measured, the latter can be successful even in the absence of insights about how the system works. A key challenge for first-principles models though is precisely that it is difficult if not impractical to obtain its parameters. As a result, even the parameters of first-principles models in print are often computed via optimization rather than measured. This, in effect, turns the first-principles model effectively into the structure of a supervised learning model and the difference between the nominally different categories becomes one of model structure rather than kind.

A further evolution of the above, unified view of first-principles and supervised learning methods is the composition of supervised learning models that combine first-principles and general-purpose components, where the result benefits from model structure that mimics a system's behavior, supplemented by further general-purpose tuning.

An example here is the RONT model [12], which uses Kubelka Munk [13] and Yule-Nielsen modified [9] Neugebauer [14] model components, extended by a regression component. In effect the first-principles models, with supervised-learning-trained parameters yield a first approximation from which a general-purpose regression delivers final prediction accuracy performance (Fig. 3).

In terms of color accuracy, Table 2 shows results for four printing systems that use different ink technologies (aqueous versus latex), have different numbers of inks (4 or 6) and can deposit different numbers of ink levels per halftone pixel (3 or 4) and therefore have a variety of numbers of NPs ($3^4 = 81$ to $3^6 = 729$). As can be seen, model accuracy improves dramatically by incorporating a general-purpose T matrix component in addition to the first-principles ones (e.g., 95^{th} percentile errors going from 4.9–23.2 down to 2.2–4.8). A question may here be asked about whether there is a need for the general-purpose elements at all. To answer it, it is worth considering the dimensionality of the model, which goes from k^n dimensions for the NPac input to s dimensions for the output, where k^n is the number of a system's NPs, which can go from as low as 8 to hundreds of millions in actual printing system, and s the spectral sampling, which is typically 31 (i.e., 400 to 700 nm in 10 nm steps).

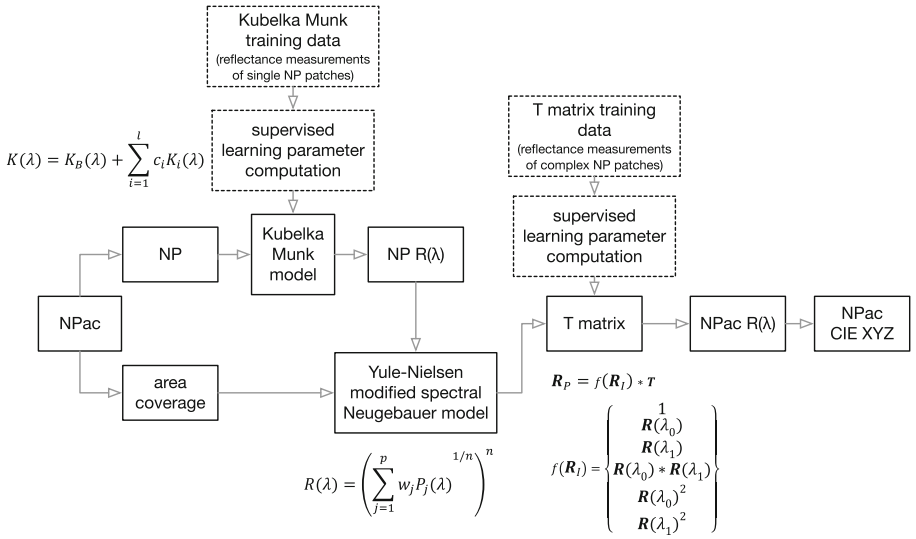

Fig. 3. The RONT model's hybrid, first-principles and general-purpose supervised learning structure. R() is reflectance, n is the Yule-Nielsen exponent, B the substrate, K is Kubelka Munk combined absorption, scattering coefficient, P is predicted NP reflectance and f() shows a simplified vector using constant, first, cross and second order terms.

Table 2. RONT model performance in terms of ΔE2000 color differences between measured and predicted colorimetries with and without T matrix. The System column shows the printer name followed by the number of inks and NPs.

system	model	median	95th perc.	max.
Z6200-4-81	RON	8.5	23.2	35.4
	RONT	1.5	4.8	8.2
PWXL4000-4-81	RON	3.2	8.1	15.6
	RONT	0.9	2.2	3.4
L310-4-256	RON	4.1	11.2	29.9
	RONT	1.3	2.8	6.5
L310-6-729	RON	1.7	4.9	10.6
	RONT	0.9	2.2	7.7

Taking even a very modest example like the last one shown in Table 2, which has 729 NPs, attempting a direct regression between NP area coverages, even just with second order terms and a very minimal 8 cross-terms per NP would yield a model with 729 * 9 * 31 = 225,990 parameters. Coupled with the impracticality of measuring more than a few thousand printed color patches and the need for the training data to represent the complex and composite non-linearity of the system, computing such parameters

would be hugely underdetermined. In comparison, the RONT model for this system would only have 7 * 31 = 217 (i.e., one wavelength-dependent vector each per ink and the substrate) parameters for the Kubelka Munk part, 1 for the Yule-Nielsen part (the exponent) and 93 * 31 = 2883 for the T matrix (constant, first order, second order and neighboring wavelength cross terms), resulting in a total of 3100. Here the use of neighboring wavelength cross-terms derives from an analysis of physical spectral reflectances that shows their high degree of correlation [15].

3.2 Color Prediction on Pre-colored Substrates

A common challenge in textile printing is the need to match colors on a variety of pre-colored substrates. Instead of printing mostly on near-white materials, textile printing often involves depositing inks on a pre-dyed fabric. This results in the need to set up color matching on a large number of substrates and using conventional approaches that involve measuring large color characterization sets in each substrate color is impractical.

Supervised learning can be applied here to make predictions of print color on a pre-dyed substrate, given measurements from another color of the same substrate material [16]. E.g., given measurements on an undyed polyester fabric, predict what color would result from depositing ink on, e.g., a gray, green, orange or blue dyed polyester (Fig. 4).

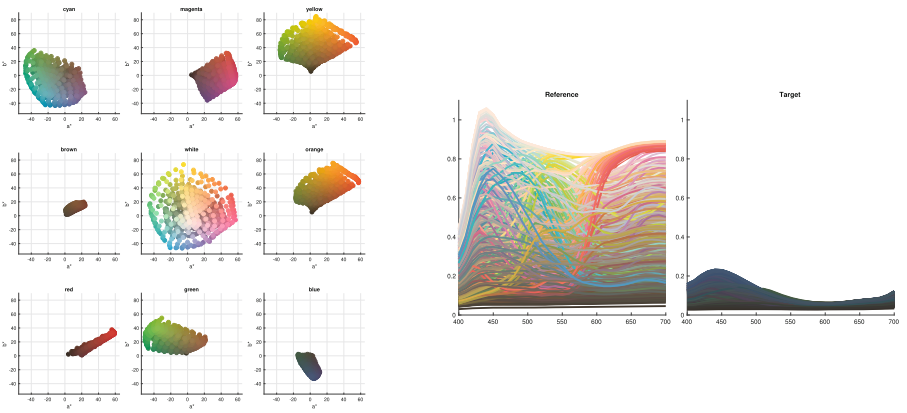

Fig. 4. (Left) colors of printing the same NPacs on different pre-colored substrates, (right) reflectances of NPac set on white and blue substrates. (Color figure online)

Taking the example of predicting reflectances on the blue substrate from reflectances on the white one on the basis reflectance measurements of 836 NPacs, a variety of supervised learning approaches can be applied. First, the same polynomial regression as applied when computing the T matrix in Sect. 3.1 can be applied here too.

Second, a simple multilayer perceptron (MLP) network [17] with reflectance inputs and outputs can also be trained. Here 5 layers were used with three hidden layers, a 31–62–124–62–31 nodes per layer structure with linear activation functions, mean squared error as the optimization criterion, the Adam optimizer [18] with a learning rate of 0.01

and a random 80:20 split between training and validation samples. The optimization converged after 1000 epochs.

Second, a Kolmogorov-Arnold Network (KAN) [19] was also used for predicting reflectances of prints on a blue substrate from those measured on a white one. A single hidden layer was used, resulting in a 31–31–31 node structure, cubic splines were the activation function and optimization was performed at 5 grid intervals in 20 steps, followed by model pruning, which reduced the number of nodes in the hidden layer from 31 to 9, and a subsequent 50 step optimization. Figure 5 shows both the overall network structure after pruning and the splines optimized for the first 10 wavelengths and the first of the 9 hidden nodes, which gives a sense of the kind of non-linearities that relate print reflectance between the two substrates.

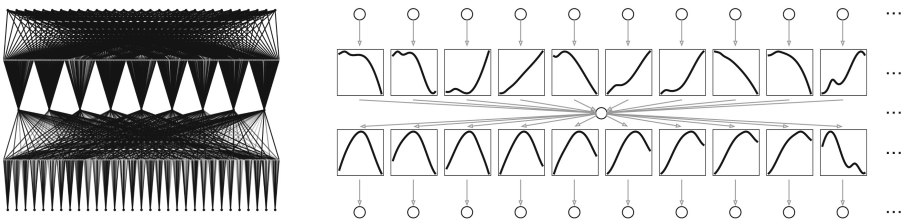

Fig. 5. (Left) Pruned KAN with (right) optimized splines for the first 10 wavelengths and the first hidden layer node.

Table 3. Model performance for predicting reflectance on a blue substrate from reflectance on a white one in terms of CIE ΔE2000 statistics.

model	training set	validation set	median	95th perc.	max.
Polynomial	836	0	0.3	0.7	1.9
MLP	836	0	1.2	3.5	6.4
	669	167	1.2	3.3	6.5
KAN	836	0	0.3	0.6	1.4
	669	167	0.3	0.7	1.8

The predictive accuracy of the above three approaches is shown in Table 3, where it can be seen that while the polynomial regression model performs very well, a simple application of an MLP network does not, which is likely due to the choice of polynomial used, which is informed by a first-principles understanding of the relationship to be modelled. Impressively, the level of accuracy delivered by KAN is effectively on par with the polynomial model, without having required any application-specific prior knowledge.

3.3 Unsupervised Learning for Print Property Optimization

While the above examples of using AI methods to optimize parameters of models that predict print properties all involve supervised learning, where for each input the correct

output is known and serves as training data, the more important optimization challenges in printing are those where the best output is unknown. This is the case when optimizing print properties such as low grain or high ink use efficiency, where the task is precisely to identify the optimal print control vectors for every color across a printing systems gamut.

A successful approach here consists of the following elements:

1. An accurate metric of the property to be optimized. This can in some cases be trivial, as in the case of ink use, which can be obtained directly and simply from the print control vector (e.g., for NPacs, knowing the drop weights or volumes if the inks, this is a simple area-coverage-weighted sum of NP ink weight or volume). In other cases, like that of print grain, the metric is more challenging to derive. E.g., a simple grain metric can be the standard deviation of an NPac's colors, which can be obtained directly from the NPac using a color model like RONT, described in Sect. 3.1: $g = \sum a_n \| NP_n - T \|$ where g is the grain metric, a_n is the area coverage of the n-th NP, NP_n is its color, T is the color of the NPac (i.e., the area-coverage weighted sum of its constituent NPs' colors) and $\| \|$ is a color difference metric. A further improvement to the metric can be had by rendering a patch corresponding to an NPac, assigning Yule-Nielsen-corrected XYZs to its pixels according to the NPs they contain, blurring it, to account for more complex ink-substrate interactions, and then computing the standard deviation of the result [20, 21]. Note that choices about patch size and blurring operator and windows size can be optimized using supervised learning.
2. An effective way to sample the domain of print control vectors. This is a particularly challenging task, since print control vectors occupy an impractically vast space and since any *a priori* intuitions about which parts of it may be more likely to yield optimal choices is likely to be wrong. Even in a very simple case of a CMYKcm ink set (where cm are lower-density variants of CM inks) with a maximum number of drops per halftone pixel of 3 there are $4^6 = 4096$ NPs and the space is therefore 4096D (each NP being addressable independently from the others). Then, considering even just an 8-bit encoding of area coverages, this yields $2^{8^{4096}} = 1.4 \times 10^{9864}$ possible print control vectors, i.e., NPacs. Given the intractable scale of the space in its entirety, a key strategy here is to prune it before sampling, where a simple solution is to exclude NPs that are highly unlikely to benefit any print property. E.g., in the example system here, where up to 3 drops per ink per pixel can be specified, one of the NPs would involve $6 \times 3 = 18$ drops (i.e., the maximum for each of the inks). Given that the average amount of ink that a substrate can hold over some local area is unlikely to be beyond 4–5 and even allowing for ink quantities at individual location to go above that local average, several of the NPs can be excluded from the sampling process. This strategy can bring the dimensionality of the relevant print control space down from 4096 in this case to low hundreds of dimensions. However, even considering a very modest 150 dimensions corresponds to $2^{8^{150}} = 1.7 \times 10^{361}$ possible NPacs. Good strategies to sample this kind of space involve exhaustive sampling of some, simple cases (e.g., the up to v vertex simplices formable by subsets of the vertices of the 150D hypergrid), random sampling of the whole space [7] (e.g., using RANSAC [22], where it is also possible to take advantage of rules like weighing the likelihood of NPs by lightness thus getting a more uniform distribution of NPac lightness), or

the use of linear programing to directly construct NPacs that optimally combine NPs given an attribute metric and a color model [23].
3. A printer color model (e.g., RONT) to connect print control vectors to colorimetries.
4. A sampling of the printing system's color gamut for which optimized print control vectors are identified. This is a combination of the NPac sampling of element 2, associated with the color sampling of the color gamut via the printer color model of element 3.

Given the above elements, unsupervised learning can be performed by sampling the NPac space, identifying the color of each sampled NPac, comparing each sample's print property level with that of other NPacs corresponding to the same color and finally for each color selecting the NPac that is optimal for it (Fig. 6). The end result of such a process is a set of colorimetries paired with optimized NPacs that can then be used to construct a look-up table that can be applied when a print job is processed for printing.

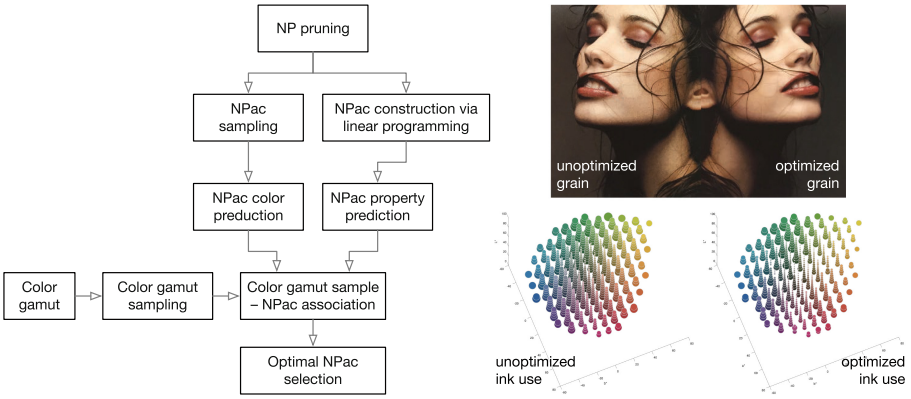

Fig. 6. (Left) overview of unsupervised learning for print property optimization with (right) examples of its impact on grain (showing scans of prints) and ink use (showing ink use across a color gamut).

4 An AI Infrastructure for Print Optimization

While the supervised and unsupervised learning methods from the previous sections have been shown to deliver solutions for a variety of printing challenges, the use of AI has until now been piece-wise and constrained to addressing individual, contained and well-delimited problems. With the advent of new approaches, such as generative pre-trained transformer (GPT) based large language models (LLMs) [24] and their text-to-image extensions [25], new opportunities arise to move towards an overall infrastructure where sensors, automation and AI enable efficiency and performance gains across the end-to-end printing workflow (Fig. 7).

Today, the printing industry faces principally two challenges [26]: making its operations more sustainable and struggling with a skills shortage. These in turn have a common

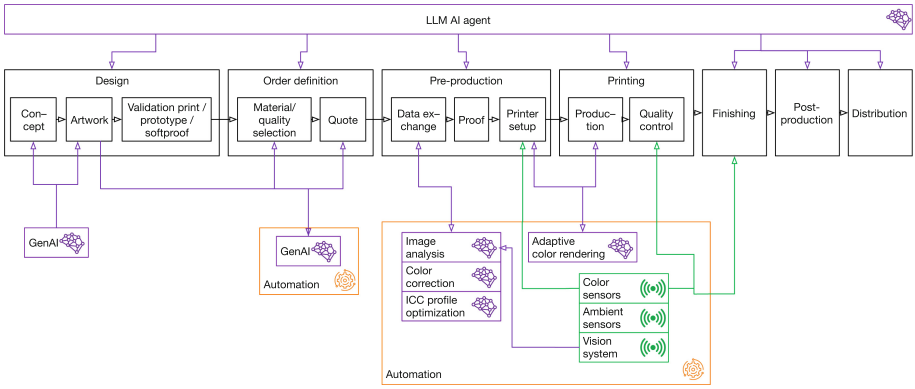

Fig. 7. An AI infrastructure for end-to-end printing workflows.

denominator, which is the reduction of waste from processes necessary to set up, maintain and verify saleable production and from the use of excessive material resources to deliver that saleable production. When it comes to sustainability, the link is direct, in that a printing operation can be characterized by the proportion of material used that does not constitute the print sold to a customer versus that which does and minimizing the former can be attempted via the kinds of methods covered in previous sections and extended by the addition of integrated sensor, automation, image analysis and job content correction that can be effected using methods like convolutional neural networks, reinforcement learning and generative adversarial networks. The skills shortage too can be related to waste, where less skilled operators are more error-prone and likely to follow trial-and-error strategies instead of applying a successful approach directly. Here LLM-powered AI agents can enable less skilled operators to more quickly and easily identify and be guided through performing successful solutions to the challenges they face, both during the earlier stages of design and order definition and then in pre-production and production.

5 Summary

In conclusion, the aim of the present paper was to provide an overview of how AI has been used successfully in printing to optimize both setup processes and the properties of printed output using supervised and unsupervised learning approaches. This has delivered direct benefits to customers by reducing their operational costs and enabling them to better meet their customers' needs. The advent of more recent, generative AI approaches opens up new possibilities to improve the printing experience further by taking AI from being a tool that is tuned to solve specific pre-defined challenges to becoming an underlying architecture that connects the end-to-end printing experience and allows users to interact with it in more immediate and natural ways.

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Instance-Based CycleGAN for Object Segmentation with Few Annotations

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Abstract. Nowadays deep networks provide excellent results in the context of object segmentation. Available models have been trained on common objects and are not designed to segment specific objects such as fruits or vegetables. In order to help breeders to accelerate and to modernize the process of agriculture products phenotyping, it is necessary to fine tune general models on specific species. Nevertheless, a minimum amount of annotations are required for this retraining step. In this paper, we propose a solution to minimize the annotation workload for each specie. The main idea consists in leveraging the annotations of one specie A in order to fine tune a model on a specie B with few annotations. For this purpose, we propose an Instance-based CycleGAN (ICG) that creates synthetic images of specie B along with corresponding annotations. By fine tuning a segmentation network with these synthetic images and annotations, we show that this network can obtain very good performance on the new specie B, without requiring to manually annotate a large amount of images for this specific specie B.

Keywords: CycleGAN · Instance segmentation · Unsupervised · Synthetic annotations · Plant phenotyping

1 Introduction

Proper detection and localization of agriculture products, such as fruits and vegetables, are critical for the growth of the economy in the Agri-food sector [1]. Particularly fruit detection is essential in agriculture products phenotyping due to the benefits it brings to crop breeding activities. Digital, automatic, and reliable phenotyping technologies are highly important to boost the advancement of genetic gain in breeding programs [2]. In fact, Computer Vision, Deep Learning, and Image-based Technologies applied to plants phenotyping aid to reduce the workload of scientific researchers by automatically measuring phenotypic indicators which accelerates the progress toward crop breeding optimization [2,3]. Moreover, plant phenotyping involves challenging computer vision tasks that

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require advanced image segmentation models that distinguish between diverse instances of objects in an image [2]. In recent years seeds companies started to develop Deep Learning models for precise fruit detection and segmentation of images acquired under uncontrolled conditions for the purpose of phenotyping. However, the annotation process required to build training datasets constitutes a bottleneck considering the change in varieties and typologies of species every year.

Since experts have already spent time to annotate images of few species, in this paper, we are trying to leverage this annotation to automatically annotate images of new species. Considering that we have a good amount of images and annotations for one specie, called A, and only few images and annotations for a second specie B, we propose a solution to create additional synthetic images of specie B along with accurate mask annotations, as illustrated on Fig. 1. And we show that this additional data can clearly boost the segmentation results on specie B.

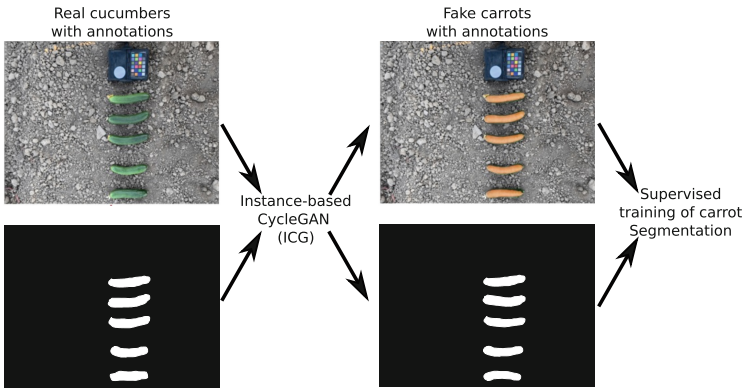

Fig. 1. Our Instance-based CycleGAN is able to independently transform each instance from one specie to another one while preserving the background. Thus, starting from a labeled dataset of cucumbers (specie A) and few carrot images (specie B), we can create many fake carrot images with accurate synthetic masks to train a new network.

Generative adversarial networks [4] are good solutions to create realistic synthetic images of a new specie, and specially to solve the problem of image-to-image translation [5,6]. In this case, it is possible to provide images from two species and learn to translate from one specie to another. While Pix2Pix requires paired images where the objects have to be located at the same position in the images [5], CycleGAN can be trained with unpaired data thanks to a cycle consistency process [6]. This network is more adapted to real applications where we have images of two different species acquired under uncontrolled conditions. Nevertheless, the classical training of a CycleGAN can't solve our problem of missing annotations for the second specie B. Indeed, Fig. 2 illustrates the image translation process when using a classical CycleGAN transforming a cucumber

(specie A) image to a carrot (specie B) image. The network is able to provide realistic carrot images but without any mask annotations for these fake images.

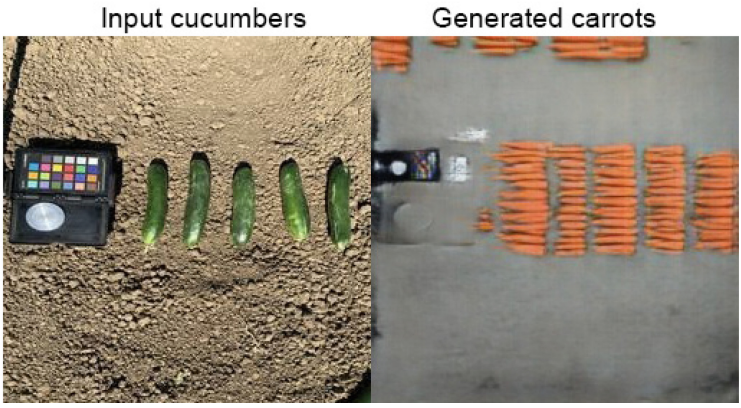

Fig. 2. Example of CycleGAN failed experiments.

In order to solve this problem, we propose a new approach called Instance-based CycleGAN (denoted ICG, hereafter) that preserves the background of the input image (specie A) and transforms individually and independently each instance of this image to the second specie B. With this solution, we can control the image location where we paste our new fake object and hence accurately deduce the mask annotations of each instance of specie B (see. Fig. 1).

The main contributions of this work are the following:

- We show that CyceGAN is a good solution to create synthetic images in the context of fruits/vegetables segmentation.
- We propose an Instance-based version of CycleGAN, that is able to transform independently each instance of one specie to a second specie.
- We run extensive experiments on real data showing that our solution helps in segmenting carrots in real images acquired under uncontrolled conditions.
- We fine tune and test the classical Mask R-CNN [7] on our synthetic images.

The rest of the paper is organized as follows: the related works are presented in Sect. 2 while Sect. 3 is devoted to our proposed CycleGAN pipeline. The experiments and results are described in Sect. 4 and finaly, we conclude and propose future works in Sect. 5.

2 Related Works

Generative Adversarial Networks (GANs). GANs were introduced in [4], and since their appearance in the field of Deep Learning they have achieved

outstanding results in many domains, such as image colorization [8], super-resolution [9], image inpainting [10], and image-to-image translation [5,6]. Conditional GANs (CGANs) were proposed to guide the data generation process by conditioning it to a given input variable [11]. In fact, CGANs were employed for Image-To-Image translation tasks, where the generation of output images is conditioned to a given input image, as in Pix2Pix [5] and CycleGAN [6]. While Pix2Pix works with paired training data, CycleGAN handles unpaired image-to-image translation, which gives much wider possibilities to experiment with translation models between domains and scenarios where paired correspondences are simply not available, such as when transforming zebras to horses, apples to oranges, or cucumbers to carrots. Given that there are no pair correspondences in CycleGAN, the generation process is regulated by incorporating a cycle consistency constraint in the translation, which is inspired by the idea that when translating from one domain to another, and then back to the original domain, the initial information must be preserved [6].

It has been reported that the cycle consistency loss in CycleGAN does not necessarily ensure structural consistency between input and generated image [12], which can be problematic in the case of data generation for segmentation purposes since the corresponding segmentation mask of the input image (from domain A) would not match the structure of the generated image (from domain B). In order to solve this, some approaches have been proposed in the literature, which include: SECLEGAN [13], CySGAN [14], and the model proposed by Yang et al. [12]. However, these previous approaches [12–14] do not change the shape of objects (object transfiguration) during translation, since they impose structural constraints to ensure consistency between input and generated images. In contrast, InstaGAN [15] does allow object transfiguration but it increases the complexity of the network by including a context preserving loss.

Our method is different since we leverage the capacity of CycleGAN to change the shape of instances during translation without increasing the complexity of the network. Furthermore, our method ensures that the network sees single instances of each domain during training, and thus is able to learn features on the instance level with higher details, which enhances the realism of the generated instances and allows to work with domains that highly differ in color, texture, and shape, and therefore to perform unpaired instance-to-instance translation. Moreover, we will show in Sect. 4 that by working on an instance level, the network generates instances of another domain with a clean background, which allows the estimation of the corresponding groundtruth segmentation mask.

Segmentation. Mask R-CNN was proposed in [7], it extends Faster R-CNN [16] by including an additional output branch to predict segmentation masks, in parallel with the existing branch for classification and bounding box regression. In our research work, we use the Detectron2 [17] framework developed by Facebook, which is a state-of-the-art library for object detection and segmentation algorithms; it includes different architectures such as Faster R-CNN, Mask R-CNN, RetinaNet [18], Panoptic FPN [19], and DensePose [20]. Recently, Detectron2 has been used in the agriculture sector in [21], where the authors worked on instances segmentation of olive trees, soil, and shadow areas.

3 Instance-Based CycleGAN

Our main task consists in creating synthetic images of carrots with mask annotations, so that we are able to train a carrot segmentation network. Assuming that we have a large image dataset of cucumbers with mask annotations and a small dataset of carrots with mask annotations, we propose to supplement the carrot dataset with fake images that are transformed versions of cucumber images. For this purpose, we train an instance-based CycleGAN (ICG) from a subset of the cucumber dataset and the full set of real carrot dataset. The idea is to train our ICG with balanced data between cucumbers and carrots. This is illustrated in Fig. 3, where we can see that the cucumber dataset is split in two subsets Cc1 and Cc2. Our ICG is trained with the set Cc1 and the real carrots RC1. Then, the trained ICG is used for inference on the Cc1 subset to create the fake carrots FC1, while the subset Cc2 is giving birth to the fake carrots FC2. These two sets of fake carrots (FC1 and FC2) are considered differently in the experiments, since FC1 is obtained from Cc1 which has been used for training ICG while FC2 is obtained from completely new cucumber images Cc2, not used for ICG training.

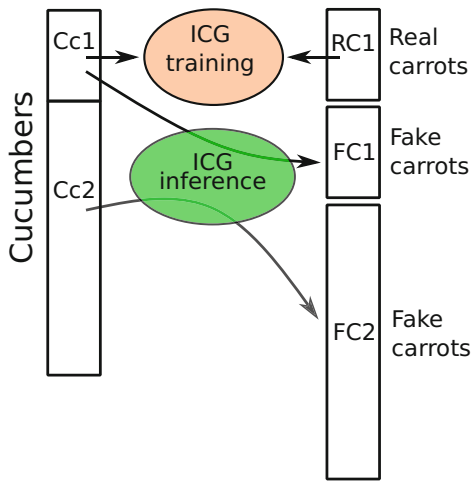

Fig. 3. The exploitation and creation of the cucumber and carrot images. All the cucumber images are real images with mask annotations.

The training step of our ICG is detailed in Fig. 4. The inputs are from both side cucumbers (Cc1) and carrots (RC1) along with the corresponding mask annotations. The provided masks are used to extract each instance of the images and to paste it in a single image with black background. Since the size of both datasets Cc1 and RC1 are low, we apply classical geometrical data augmentation to each single image before feeding the network. Then, we apply the clas-

sical CycleGAN training process with cycle consistency and unpaired instance images [6].

To be precise, let consider the species (*domains*) A and B (which are cucumbers and carrots, in our illustration). We have two generators G_A and G_B creating fake images of specie A and B, respectively, and two discriminators D_A and D_B which are trained to differentiate true from fake images of specie A and B, respectively. The complete loss we are minimizing while training our ICG is the following:

$$\begin{aligned}\mathcal{L}(G_A, G_B, D_A, D_B) = \mathcal{L}_{GAN}(G_A, D_A, A, B) \\ + \mathcal{L}_{GAN}(G_B, D_B, B, A) \\ + \lambda_{cyc}\mathcal{L}_{cyc}(G_A, G_B) \\ + \lambda_{id}\mathcal{L}_{id}(G_A, G_B),\end{aligned} \tag{1}$$

where the GAN losses $\mathcal{L}_{GAN}$ are the classical adversarial losses that control that the transforms from A to B and from B to A are correct. The cycle loss $\mathcal{L}_{cyc}$ controls that the model can transform one specie to another and back ($A \rightarrow B \rightarrow A$ and $B \rightarrow A \rightarrow B$). And the identity loss $\mathcal{L}_{id}$ regularizes the generators to be near identity mappings when real samples of the target domain are provided as the input to the generator. This regularization has been shown to improve color composition in the mappings. λ_{cyc} and λ_{id} control the balance between the losses.

The inference stage is detailed in Fig. 5. For this step we just consider the transform from cucumber to carrot, and not the inverse since our aim is to supplement the carrot dataset. Thus, the inputs are a cucumber image and the corresponding mask image. The first step consists in the extraction of the instances from the image and to paste it on a small image with black background. Then our trained ICG is used to transform each cucumber instance into a fake carrot. A simple thresholding on the result image can provide the corresponding mask annotation. Finally, given the input original image and annotations, the created fake carrots and their masks, we can paste the fake carrot on top of the original cucumber image and provide the corresponding mask annotation. Note that the masks for the fake carrots are different from the masks of the input cucumbers since ICG is free to change the shape of the vegetables, depending on the shapes of the trained images (cucumbers and carrots).

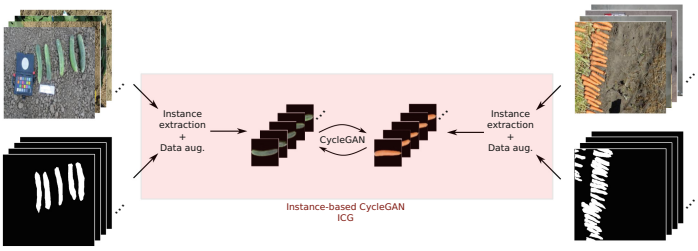

Fig. 4. Training step of our Instance-based CycleGAN.

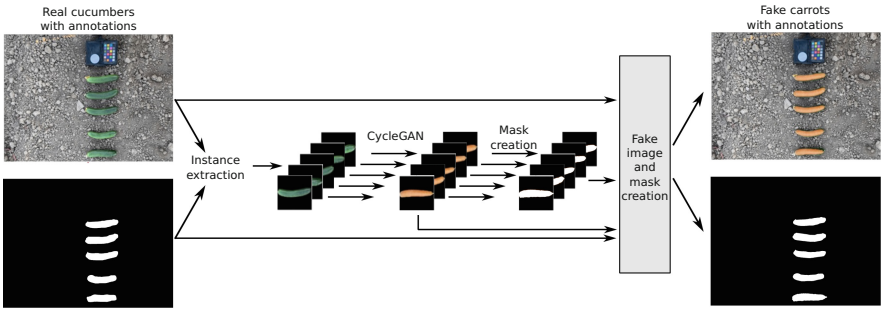

Fig. 5. Inference step of our Instance-based CycleGAN.

The fake carrots and their annotations are used to train a segmentation network in a supervised way.

3.1 Implementation

The images used in all the experiments were captured by breeders and varietal selection technicians. For the CycleGAN experiments, we used the CycleGAN Tensorflow 2 implementation available in [22]. Similarly as in [6], in all our experiments we used a batch size of 1, a learning rate of 0.0002 with a linear decay after the first 100 epochs, and a maximum total of 200 epochs. The weights of cycle consistency and identity losses were set to $\lambda_{cyc} = \lambda_{id} = 10$ (Eq. 1) for all the experiments.

For the segmentation experiments, we used Detectron2 [17] framework version "v0.6" and Mask R-CNN with feature pyramid network (FPN) built with Resnet-101 as the backbone for instance segmentation (RESNETXT-101-FPN-3X). We used this architecture in all our segmentation experiments. For the transfer learning we used the pre-trained weights on ImageNet [23] and then retrain the network for carrot segmentation. All segmentation models were trained for 1000 epochs, with a learning rate of 0.0025, and the evaluation period was set to every 100 epochs, where the evaluation during training is done with the test images. Moreover, during inference and test, we used a threshold of 0.5 as the minimum value for a prediction to be considered positive.

4 Experiments

4.1 Instance-Based CycleGAN

Creating Fake Carrots. Since our objective is to use a small amount of annotated data, we selected only 265 annotated carrots (RC1 dataset from Fig. 3) and 273 annotated cucumbers (Cc1 dataset from Fig. 3). We applied the preprocessing steps of our CycleGAN pipeline, and in this case, 3 random rotations were applied to each original image from each domain in order to achieve at least 1000 training examples per domain.

Figure 6 show examples of our CycleGAN pipeline, illustrating that it is possible to generate realistic carrots, with variety in texture and color. The shape is also modified during CycleGAN translation and the segmentation annotations of the fake carrots are estimated correctly. Moreover, the results of inference on cucumbers not seen during training are satisfactory, which shows the robustness of CycleGAN inference on new data.

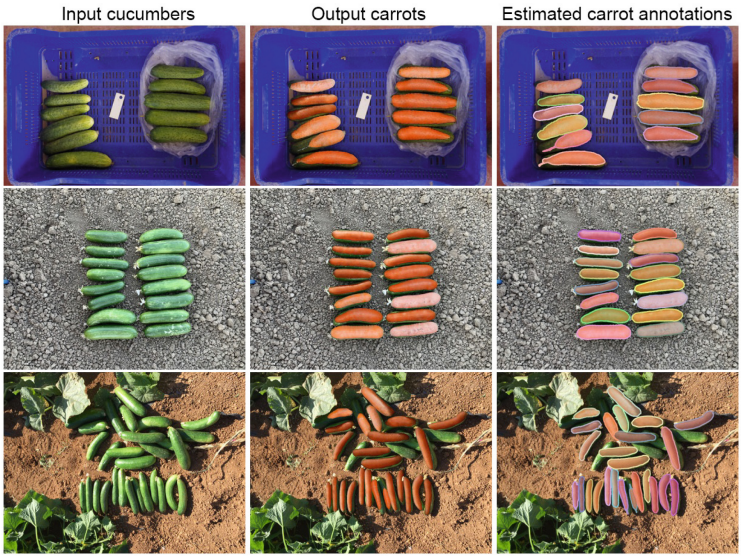

Fig. 6. Illustration of the creation of fake carrot images.

Squashes and Peppers. In order to show that this approach can be applied to many different vegetables, we also train an ICG between peppers and squashes. For this ICG, the difference between the shapes of the two species is more important than between carrots and cucumbers. We train the ICG with 133 original pepper and 240 original squash annotations, with random rotations to reach 1200 training images on each domain.

Figure 7 shows examples of transform between the two species. As we can see our ICG is able to generate fake squashes and fake peppers which are realistic in shape even with variations in texture and color. Figure 8 shows examples for the full pipeline, creating images with several fake instances and their corresponding annotations. As we can see, the instance segmentation annotations are estimated correctly in both cases.

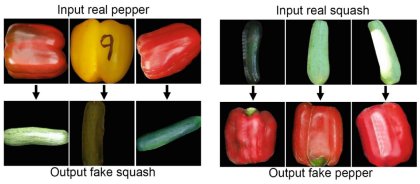

Fig. 7. Examples of Squash/Pepper instance transforms.

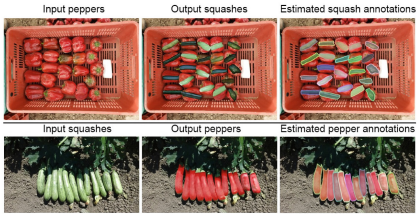

Fig. 8. Examples of Squash/Pepper images and annotations transform.

4.2 Segmentation

For the segmentation experiments, we used real and fake carrot data (RC1, FC1 and FC2 presented in Fig. 3) for training the segmentation network. For testing, we have created a dataset of 1928 annotated carrots, acquired with different cameras under uncontrolled environments, with diversity in illumination conditions and backgrounds. Moreover, the carrot images are also challenging for a segmentation task regarding the number of instances per image, with a minimum of 52, a maximum of 218, and an average of 122 carrots per image.

Evaluation Metrics. We measured the performance of the segmentation models with the following metrics: precision, recall, IoU [24], F1 score [25,26], and Average precision (AP). For COCO metrics [27], AP is the average across multiple IoU levels (the minimum IoU value to consider a prediction as positive), in fact, AP is measured as the average AP for IoU from 0.5 to 0.95 with a step size of 0.05. Moreover, AP_{50} and AP_{75} are measured at IoU of 0.5 and 0.75, respectively. In our research context, we use the COCO AP metric included in Detectron2. In this work, the main idea is to create synthetic data to train a network and not to provide realistic images. Consequently, we don't assess the realism and quality of the generated images.

Results. Table 1 shows the results of the evaluation on the real carrot images (1928 annotated carrots). Note that these images are from real carrots (no fake image) and have never been seen during the segmentation network training. They are real unseen test images. We provide the results on these images for three different sets of training data:

- Training on RC1: 265 annotated carrots, the ones used for training ICG.
- Training on RC1 + FC1. FC1 is constituted by 273 fake carrots, outputs of our ICG by transforming the cucumbers Cc1, which have been used to train ICG.
- Training on FC2: FC2 is constituted of 513 fake carrots, outputs of our ICG by transforming the cucumbers Cc2, which have not been used to train ICG.

The results in Table 1 clearly show the improvements provided by our approach. Indeed, by using only the few real annotated data (RC1) to train the

Table 1. Segmentation results on real carrot images. Note that the 273 fake carrots with an "*" are the transformations of the 273 cucumbers used to train our ICG, while the 513 carrots of the last row are the transformations of 513 cucumbers not seen during the ICG training.

Training data	AP	AP_{50}	AP_{75}	Precision	Recall	IoU	F1
Original carrots RC1 (265)	70.52	86.32	83.87	94.90	89.01	84.80	91.66
Original carrots RC1 (265) + Fake carrots FC1 (273)	72.46	86.50	84.17	**95.44**	90.09	86.39	92.58
Fake carrots FC2 (513)	**74.56**	**89.29**	**88.07**	93.79	**92.18**	**86.88**	**92.93**

segmentation network, we get an Average Precision (AP) of 70.52. If we supplement this training set with fake carrots provided by our ICG, the AP increases to 72.46, showing that fake carrots can be used to train a segmentation network and improve the results on real carrots. Finally, if we consider only fake carrots (FC2) for training the network, we outperform the two previous results and get 74.56 as AP. This shows that fake data is not only useful to supplement a small training set, but can be used to train a network without requiring any real data.

Consequently, we can conclude that our ICG has two main advantages:

- **Annotation reduction.** The experiments show that we can train a CycleGAN with only around 250 annotations from each domain and use real and fake images to improve the results over a network that has been trained only on real images. So without new annotations, the accuracy is boosted by adding fake data.
- **Robustness.** The experiments show that our ICG can be fed with new cucumbers (not seen during the ICG training) to provide fake carrots with annotations that can be used to train a segmentation network that outperforms all the other networks trained with real data. This shows that ICG can generalize to new images.

5 Conclusion

We have developed a CycleGAN pipeline that allows the network to perform unpaired instance-to-instance translation and create a synthetic dataset for instance segmentation. Thanks to this work we are able to leverage annotated data from one domain and use it for the benefit of a second domain with less available data. Furthermore, we created a carrot segmentation model of images acquired under uncontrolled conditions and increased the segmentation accuracy by including fake data during training.

In addition, we show that our ICG is able to generate realistic instances, with diversity in color, texture, and shape, both for cucumbers/carrots and peppers/squashes dataset.

During our experiments, we have pasted each fake instance on the same background and similar location than the input real image, but it's clear that every new created fake instance could have been pasted on any new background with geometric augmentation applied to the object and the corresponding annotation. This would open the door to infinite number of training data. This is the aim of our future works.

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ChromaPose: Robustness of 2D Pose Estimation Under Different Color Illuminations

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Abstract. This study investigates the robustness and precision of 2D human pose estimation techniques, particularly feature-based and 3D shape-based methods, against the backdrop of varying color illumination conditions simulated by chromatic adaptation, as well as varying spatio-temporal dynamics. While the AIST++ dance videos dataset serves as the primary data, the insights gained are pertinent to broader contexts like sports, highlighting the critical influence of lighting on the performance of pose estimation technologies.

Keywords: 2D Pose estimation · Chromatic Adaptation · Lighting Conditions · Feature-Based Pose Estimation · Shape-Based Pose Estimation

1 Introduction

ChromaPose aims to provide tools and methods for the analysis of dance and theatre performances, particularly from archival video materials. ChromaPose seeks to identify and address common challenges in archival material, such as poor color and light contrast and inadequate lighting, which can significantly impede the visibility of performers' movements and consequently, the precision of human body pose estimation. At the heart of ChromaPose's research is the AIST++ dataset [23], a rich collection of dance videos showcasing a variety of styles and viewpoints. Despite its breadth, the dataset does not fully represent the challenges typical in the digitization of performing arts, such as the effects of stage lighting, shadows, and costume colors on visual clarity.

To address these gaps, ChromaPose subjects the AIST++ dataset to specific illumination changes simulated by chromatic adaptations to explore how changes in color and brightness affect the performance of 2D pose estimation techniques. The project aims to identify the most effective 2D pose estimation methods that can withstand the dynamic lighting and chromatic shifts often found in archival dance and theatre footage, as illustrated in Fig. 1, and by developing

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a novel visualization technique to evaluate the accuracy and stability of these methods. This innovative visualization not only enables a comparative analysis of different techniques but also enhances our comprehension of their performance amidst the unique challenges of the performing arts. Furthermore, by delving into the complexities of temporal performance footage, the research expands the scope of digital archiving and human pose estimation, highlighting solutions to visual challenges and setting the stage for creating new strategies that take into account lighting and visual factors for future data collection initiatives.

Fig. 1. Sample challenging scenarios: strong colored illumination, complex poses, occlusion, low light and back lit conditions, in archive videos from PREMIERE project [20].

2 Related Works

The AIST++ Dance Motion Dataset [23], an extension of the AIST Dance Video Database, significantly advances dance movement analysis and human pose estimation by utilizing multi-view videos to estimate camera parameters, 3D human keypoints, and dance motion sequences. With over 10.1 million images of 30 subjects from 9 views, providing the largest collection of 3D human keypoint annotations, and encompassing 1,408 sequences across 10 dance genres, AIST++ supports diverse research tasks like multi-view keypoints estimation, motion prediction/generation, and cross-modal analysis with music. This dataset, featuring professional dancers in various styles, enhances human pose estimation models, showing marked improvements in accuracy and robustness, especially against complex movements and occlusions. Its real-world scenarios and detailed annotations make AIST++ invaluable for developing new methodologies in pose estimation and bridging computer vision with performing arts, thus serving as a key resource for advancing research in these fields.

2.1 Human Detection and Tracking

In the realm of human body detection and tracking, the integration of state-of-the-art detection algorithms with advanced tracking methodologies forms the backbone of current systems. YOLOv7 [24] exemplifies this integration, offering real-time, high-accuracy object detection. This version enhances the capabilities of its predecessors through architectural optimizations and algorithmic improvements, making it exceptionally suitable for dynamic and complex environments where rapid object detection is crucial. Paired with YOLOv7, ByteTrack [28]

emerges as a robust tracking system that exploits the detection efficiency of YOLOv7 to offer seamless tracking across video frames. ByteTrack effectively addresses the challenges of occlusions and rapid object movements by employing a sophisticated association strategy. This approach leverages spatial-temporal information to maintain consistent object identities, thereby significantly reducing identity switches and enhancing tracking continuity. StrongSORT++ [5] introduces filtering and Gaussian Smooth Interpolation (GSI) techniques to further refine tracking accuracy. The filtering relies on probabilistic models to predict future object states based on historical movement data, which is instrumental in handling the unpredictability of object trajectories. This predictive capability is crucial for maintaining the track of objects when they move swiftly or become momentarily obscured. GSI smooths the estimation of object trajectories. By assuming object movements follow a Gaussian distribution, it interpolates positions during periods of occlusion or lost detections, ensuring a coherent and accurate representation of object paths over time. This method not only aids in overcoming interruptions in object visibility but also contributes to a more stable tracking.

2.2 Feature Based 2D Pose Estimation

Feature-based 2D pose estimation is a pivotal process which aims to detect and track human body parts within images and videos, fostering advancements in interactive technologies and motion analysis. Human pose estimation methodologies bifurcate into two main approaches: bottom-up and top-down, each adopting a unique perspective in identifying and associating human body parts. Bottom-up Approaches approaches address pose estimation as a part of detection challenge. OpenPose [2] leverages a complex Convolutional Neural Network (CNN) architecture to detect individual body parts initially. It introduces Part Affinity Fields (PAF), innovatively predicting the connections between body parts to elucidate the human body's structure in imagery. Contrarily, top-down methods such as Convolutional Pose Machines (CPMs) [25] and AlphaPose [7] (or RMPE for Regional Multi-person Pose Estimation) begin with an aggregate human body model, refining it to align with observed image configurations. AlphaPose utilizes pre-established human body models to identify essential keypoints, laying the groundwork for pose reconstruction. While another approach implied by MIPNet [11] is notably proficient in managing occlusions and variations in human body scale, posture, and appearance, ensuring resilient pose estimation.

Amidst these established methods, ViTPose [26,27] emerges at top, distinguishing itself by integrating Vision Transformer (ViT) technology [4] with traditional pose estimation frameworks. Introduced in 2022, ViTPose revolutionizes the analysis of visual data by adopting transformers to meticulously model the spatial relationships among body parts. This approach excels in managing long-range dependencies, thereby enhancing the accuracy and efficiency of pose estimation. Demonstrating superior performance over predecessors like OpenPose [2] and AlphaPose, as demonstrated in Table 1(a), ViTPose sets new benchmarks in the field. Its architecture, optimized for deep and nuanced feature extraction,

outshines conventional CNN-based methods, proving to be more adaptable and precise across a variety of visual contexts.

2.3 Shape Based 2D Pose Estimation

The advent of shape-based 2D pose estimation represents a breakthrough in computer vision, using 3D models to analyze and project human forms onto 2D planes. This approach integrates human anatomy with mathematical models to reconstruct poses accurately, addressing challenges like occlusions and body size variations. Central to this method is the Skinned Multi-Person Linear (SMPL) [16] model, which simplifies capturing the human body's geometry from 2D images, enabling advances in pose estimation. 4DHumans [8] introduces a revolutionary approach to human mesh recovery and tracking over time. At its core is HMR 2.0 [10], a transformer-based network that sets new standards in analyzing and reconstructing human poses from single images, even those previously deemed challenging. When applied to video, it employs 3D reconstructions from HMR 2.0 [10] for advanced tracking capabilities, managing multiple individuals and maintaining their identities through occlusions. 4DHumans [8] exemplifies state-of-the-art tracking performance in monocular video analysis, also showcasing significant improvements in action recognition tasks. OSX [14] shifts the focus towards a holistic recovery of the 3D human body, including face and hands, from single images. By overcoming resolution challenges without resorting to segmented network processing for each body part, OSX presents a unified pipeline named Component Aware Transformer (CAT). This model employs a body encoder for overall parameter prediction and a local decoder for high-resolution face and hand details, ensuring cohesive and realistic human posture and movement representations. The introduction of the UBody dataset further underscores OSX's applicability to real-world scenarios, highlighting its adaptability and precision in expressive whole-body mesh recovery. SMPLer-X [1] ventures into Expressive Human Pose and Shape estimation (EHPS) by proposing a generalist foundation model that employs a vision transformer (ViT) architecture up to the ViT-Huge variant. It capitalizes on a massive training corpus from a wide array of datasets, demonstrating remarkable versatility and accuracy across numerous testing environments. SMPLer-X's [1] methodology includes rigorous data and model scaling, highlighting the critical role of dataset diversity and model capacity in enhancing EHPS performance. The model's success is further evidenced by its state-of-the-art results on several benchmarks, including AGORA [19] and UBody [14], underscoring the potential of vision transformers in complex human body pose estimation tasks.

These models underscore the transformative potential of shape-based 2D pose estimation. By bridging the gap between 2D imaging and 3D structural understanding, they pave the way for innovations across various domains. From animation and virtual reality to healthcare and sports analytics, the implications of accurately capturing the human body form in motion are profound and far-reaching.

3 Methodology

3.1 Human Detector and Pose Filtering

Within the scope of the PREMIERE project [20], we have crafted a pipeline that addresses various digital processing stages, with a key feature being the initial separation of the 'human detector' from subsequent stages like human body pose estimation and segmentation. This separation aims to enhance detection accuracy and ensure spatial-temporal consistency vital for video analysis, with additional algorithms discussed later. This approach also isolates the accuracy and error rates of human body detection from pose estimation, allowing for an impartial comparison of different pose estimation techniques on our chromatically adapted dataset. We employed YOLOv7 [24] E6E variant, selected for its support of higher resolution (1280×1280 pixels) and reliability in person detection. However, to mitigate false positives in challenging conditions such as low resolution, rapid movement, and insufficient lighting, which could erroneously classify humans as other entities (e.g., dogs, cats, chairs), we restricted the detection to the 'person' class exclusively.

To complement the human body detection framework, the integration of a robust tracking algorithm was deemed necessary. ByteTrack [28] was selected for its efficacy and compatibility with YOLOv7 [24], demonstrating reliable tracking performance across standard datasets like MOT17 [17]. However, its application to archival footage revealed limitations, in challenging scenarios like low-light, low resolution, occlusion with very low features, prompting the exploration of advanced tracking capabilities. This exploration led to the incorporation of algorithms from StrongSORT++ [5], which, while independently capable, showed that the default ByteTrack [28] tracking outperformed the base

Table 1. (a) Comparison of top-down and bottom-up state-of-the-art methods [18]; (b) Color illumination conditions settings.

(a)

Method	COCO		CrowdPose		OCHuman	
	val	test	val	test	val	test
Top-Down Methods, ResNet101 + YOLO-v3						
MaskRCNN [9]	-	64.8	-	57.2	-	20.2
AlphaPose [7]	-	70.1	-	61.0	-	-
CrowdPose [13]	-	70.9	-	66.0	-	-
OPEC-Net [21]	-	73.9	-	70.6	-	29.1
MIPNet [11]	72.2	74.2	63.4	68.1	32.8	35.0
Top-Down Methods, HRNet-W48 (384x288) + Faster R-CNN						
HRNet [22]	76.3	75.5	68.0	69.3	37.8	37.2
MIPNet [11]	76.3	75.7	68.8	70.0	42.0	42.5
Top-Down Methods, ViT based						
VitPose++-H [26,27]	**79.4**	-	-	-	**85.7**	-
Bottom-Up Methods						
OpenPose [2]	-	64.2	-	-	-	-
HigherHRNet [3]	67.1	70.5	-	67.6	-	-

(b)

Setting	Target(r, g, b)	Luminance
1	(1.0, 0.3, 0.3)	−10
2	(0.63137, 1.0, 0.4823529)	−10
3	(0.2, 0.3, 1.0)	−10
4	(1.0, 0.3, 0.3)	0
5	(0.63137, 1.0, 0.4823529)	0
6	(0.2, 0.3, 1.0)	0
7	(1.0, 1.0, 1.0)	−15

version of StrongSORT in the context of archival footage. Consequently, with some reference to previous works [6] in comparisons and combinations of these methods, a hybrid approach called ByteTrack++ was developed, combining ByteTrack's [28] default tracking capabilities with enhancements from StrongSORT++ algorithms, namely AFLink and GSI, to achieve superior tracking performance.

Furthermore, to achieve more accurate and stable tracking of human poses over temporal video sequences, we implemented and applied a pose filtering method to each estimated pose keypoints. This pose filtering method involves applying Kalman filter to remove high noise levels from the estimated poses generated by the pose estimators followed by applying the Gaussian-based interpolation method proposed in [28] for continuous and visually coherent trajectories of human poses over time.

3.2 Change of Color Illumination Conditions

Chromatic adaptation (CA) can be used for simulating strong changes of color appearance to suit different color lighting conditions. In this research, we employed the Bradford chromatic adaptation method [12] due to its ability to mimic the adaptive mechanisms of the human visual system. Furthermore, we investigated the impact of employing four different destination white points alongside three distinct luminance settings (refer Table 1(b)) using this method to augment AIST++ [23] video frames with various color illumination conditions.

Estimation of the Illuminant: The estimation of the illuminant for each frame relies on the principle that in scenes lacking dominant light sources, the average color tends towards gray. This estimation provides a crucial reference point for subsequent CA processes.

Fig. 2. Example of simulated color illumination shifts. From left to right, top to bottom: Original AIST frame [23], resulting color shifts for settings 1 to 7.

Chromatic Adaptation with Varying Destination White Points: To explore the effect of different destination white points on human body pose estimation, we simulated color illumination shits using various target white points.

This process involved several key steps: (a) Conversion of the source and destination white points from sRGB to XYZ color space, a tristimulus model representing human color perception. (XYZ values were normalized to ensure consistency in luminance); (b) Utilization of the Bradford CA matrix to transform XYZ values to LMS (Long, Medium, Short) color space; (c) Calculation of the LMS gain by scaling destination LMS with source LMS, then multiplying CA matrix to obtain the CA transform; (d) the CA transform was then applied on the source image in the XYZ color space.

Luminance Adjustment: Further processing involved the transformation of the resulting XYZ color space into CIELAB color space, effectively separating chromatic information (a*, b*) from luminance (L*). The luminance values in the CIELAB color space were subsequently adjusted to emulate variations in lighting conditions, enabling the simulation of different illuminants' effects on image appearance. Finally, following luminance adjustment, the CIELAB values were converted back into sRGB, completing the color illumination shift process and yielding simulated images adapted to diverse color and luminance conditions. See examples of simulated color shifts in Fig. 2.

3.3 Metrics

Object Keypoint Similarity (OKS): As proposed in the MS-COCO dataset [15], the Object Keypoint Similarity (OKS) measure serves as a fundamental metric for assessing the accuracy of 2D human pose estimation. It quantifies the similarity between predicted and ground truth keypoint locations. In our research, we employ OKS to evaluate the accuracy of various estimation methods. Specifically, we evaluate accuracy using both the average OKS across all joints and the per joint OKS. This dual assessment approach offers several advantages. By considering the average OKS across all joints, we obtain a holistic view and insights into the overall accuracy and robustness of the various pose estimation methods' across the entire pose. We can identify which methods are more resilient to changes in color illumination and which may exhibit inconsistencies.

In addition to the aggregate evaluation, we also assess performance at the individual joint level using per joint OKS. This allows us to pinpoint specific joints where a pose estimation method excels or struggles. Mean Absolute Error (MAE) is another commonly used metric to quantify the accuracy of joint location estimation in 2D human pose estimation. In our research, we utilize MAE to evaluate the accuracy and consistency of various pose estimation method in estimating both the x-coordinate MAE_x and y-coordinate MAE_y for each individual joint.

$$\text{MAE}_{xj} = \frac{1}{N}\sum_{i=1}^{N}|x_{ij} - x_{ij}^{GT}| \qquad \text{MAE}_{yj} = \frac{1}{N}\sum_{i=1}^{N}|y_{ij} - y_{ij}^{GT}|$$

The mean absolute error for the x-coordinate of the joint (resp. for the y-coordinate) is denoted by MAE_x (resp. MAE_y). Here, N is the total number

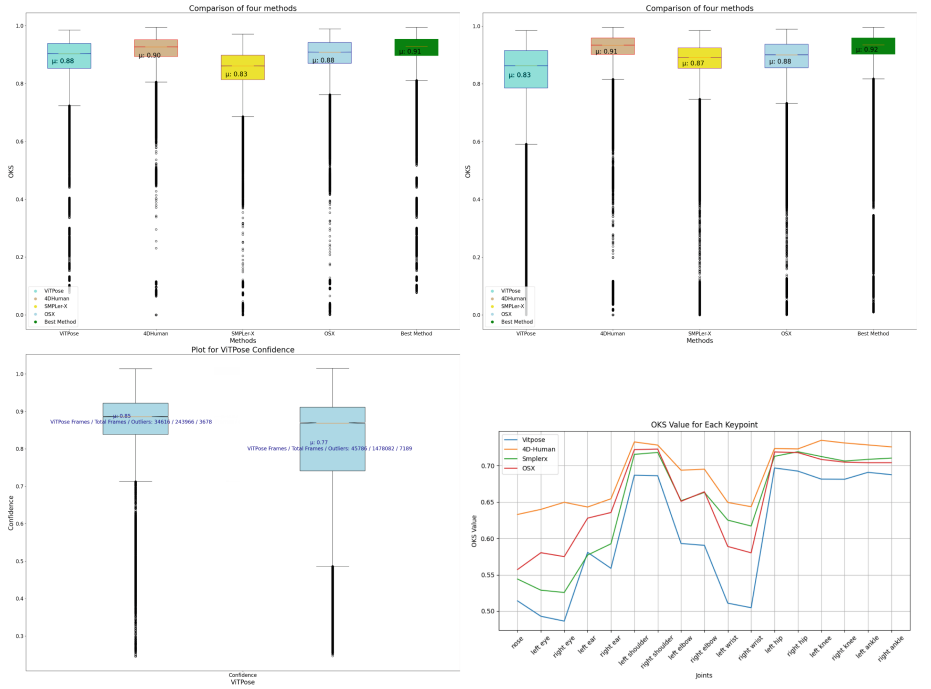

Fig. 3. Top: Comparison of different 2D Pose estimation methods: original videos (Left) and with chromatic adaptation settings 1 to 7 (Right); **Bottom:** ViTPose contribution to frames selected for best OKS (left); Visualization of OKS per joint with filtering (right).

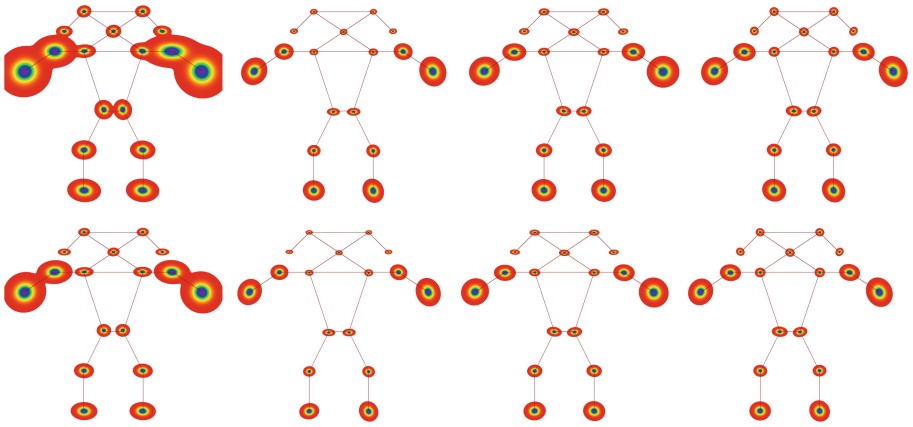

Fig. 4. Visualization of per joint stability of different methods without filtering (1st row) and with filtering (2nd row): Vitpose, 4D Human, OSX and SmplerX (From left to right)

of samples, x_{ij} and y_{ij} are the predicted x and y coordinates for joint j in the ith sample respectively, and x_{ij}^{GT} and y_{ij}^{GT} are the corresponding ground truth coordinates.

3.4 Visualization Methods

In the context of 2D human pose estimation in real world scenario, understanding the stability of individual joints is crucial for assessing the robustness and reliability of various pose estimation methods under different conditions. To address this, we adopt a methodological approach centered on a 2D Gaussian distribution to characterize the stability of each joint. This representation allows us to quantify the uncertainty or variability inherent in the estimated joint positions.

For each joint j, we compute the error between the predicted and ground truth coordinates. This error captures the deviations in both the horizontal (x) and vertical (y) directions. We computed the covariance matrix which encodes the relationship between errors in x and y dimensions, providing insights into the joint's stability in both directions. From the covariance matrix, we derive the standard deviations along the x and y axes. These standard deviations serve as measures of the joint's stability, indicating the extent of variability or dispersion in the estimated joint positions.

Utilizing the mean error and covariance matrix, we construct a multivariate normal distribution, representing the joint's positional uncertainty. By evaluating the Probability Density Function (PDF) of this distribution across a grid of coordinates spanning the image, we obtain a 2D Gaussian map. This map visualizes the likelihood of each joint being located at each point in the image, with higher probabilities indicating greater stability. The resulting 2D Gaussian maps provide valuable insights into the stability of various pose estimation methods under varying color illuminations. Additionally, the standard deviations along each axis offer quantitative measures of joint stability, facilitating comparisons across joints and pose estimation methods.

3.5 Choice of Performances Dataset

We initiated our study with the AIST++ dataset [23], comprising 12,670 videos that span a variety of dance genres, feature multiple dancers, that were captured from 9 distinct camera angles per performance, as highlighted in Sect. 2.1. Our methodology involves processing these video frames to detect individuals, apply filtering, track, and interpolate to ensure a consistent temporal sequence across complex and rapid movements. This forms the foundation for our comparative analysis of various human body pose estimation techniques using data captured under optimal lighting conditions.

Subsequently, we introduced variations in illumination conditions by simulating color shifts using the chromatic adaptation method outlined in Sect. 3.2. We randomly chose 50 dance sequences, ensuring each was represented by 9 different camera perspectives, same as original dataset so that there is no imbalance in the data due to skipped camera angles, totaling 450 videos. These were

then augmented with 7 distinct color shifts settings (refer Table 1(b)) previously described, yielding 3,150 videos tailored for assessing the precision and robustness of 2D pose estimation under varied illumination conditions. For both datasets -the original 12,670 videos and the 3,150 videos subjected to color illumination shift- we evaluated the performance of pose estimation models, both with and without the integration of supplementary filtering and interpolation algorithms. This comprehensive approach allows to gauge the effects of these additional processing on the accuracy and consistency of pose estimation, as well as to understand the influence of simulated lighting conditions.

4 Results and Discussions

4.1 Accuracy

Using the Object Keypoint Similarity (OKS) metric, we computed the average OKS across all estimated poses, for each pose estimation method, considering both the original videos and those with color illumination shift. Figure 3(top) illustrates the OKS plot for each method. The best feature-based pose estimation method, Vitpose, exhibited strong performance, with an OKS of 0.88 on the original video dataset, comparable to the best shape-based method, 4D-Human, with an OKS of 0.9. However, when evaluating performance on the color illumination shift dataset, Vitpose's accuracy declined to 0.83, whereas the accuracy of shape-based methods remained consistent. Further analysis, illustrated in Fig. 3(bottom left), delved into the distribution of frames utilizing Vitpose versus the best shape-based method per frame within both datasets. Intriguingly, in the original video dataset, the ratio of frames employing Vitpose versus the best shape-based method was 0.142. However, this ratio diminished to 0.031 when analyzing the dataset with color illumination shift.

These findings underscore the susceptibility of feature-based pose estimation methods, such as Vitpose, to variations in color illumination. The observed decrease in accuracy and the shift in the ratio of method usage between the datasets underlines the potential challenges in deploying feature-based pose estimation methods in real-world scenarios with diverse lighting conditions.

Furthermore, considering per-joint accuracy under color illumination shift, as depicted in Fig. 3 (bottom right), the per-joint plot for all methods exhibited a similar and consistent pattern across all methods. Although each method's per-joint plot varied in vertical position on the OKS scale, indicating differing overall accuracies, they all experienced similar fluctuations in accuracy at corresponding joints. Notably, joints around the face (eyes, nose, and ears) consistently exhibited the lowest accuracy, while joints at the shoulders, hips, knees, and ankles demonstrated the highest accuracy. This observation underscores the challenges posed by color illumination shift on pose estimation accuracy, particularly in facial regions.

Table 2. Comparison of OKS, Mean Error (x, y) and Standard Deviation (x, y) for Models: VitPose, 4D-Human (without filter and with filtering) for Joints J1 to J17 (nose, left eye, right eye, left ear, right ear, left shoulder, right shoulder, left elbow, right elbow, left wrist, right wrist, left hip, right hip, left knee, right knee, left ankle, right ankle) without filtering

Model	Metric	J1	J2	J3	J4	J5	J6	J7	J8	J9	J10	J11	J12	J13	J14	J15	J16	J17
ViTPose	OKS	0.50	0.48	0.48	0.58	0.55	0.69	0.68	0.60	0.60	0.54	0.53	0.70	0.69	0.68	0.68	0.70	0.69
	MAE_x	7.73	7.89	8.01	9.35	10.56	10.36	10.58	17.53	18.14	20.86	21.63	**8.95**	**9.53**	9.45	9.56	10.13	10.39
	MAE_y	6.50	6.51	6.73	5.45	5.75	6.31	6.44	12.28	12.42	19.17	20.08	13.81	14.02	10.08	9.89	7.37	7.54
	SD_x	9.77	9.04	9.60	10.40	11.59	14.50	15.31	28.51	38.91	33.87	34.65	11.86	11.67	15.64	15.85	21.05	20.76
	SD_y	8.08	7.60	7.41	6.92	7.13	8.44	10.56	20.61	20.77	31.99	33.34	12.13	12.99	12.05	11.71	14.63	14.66
4D-Human	OKS	**0.64**	**0.65**	**0.65**	**0.65**	**0.66**	**0.74**	**0.74**	**0.70**	**0.70**	**0.66**	**0.65**	**0.73**	**0.73**	**0.74**	**0.74**	**0.74**	**0.73**
	MAE_x	**4.38**	**4.26**	**4.05**	**5.11**	**5.21**	**4.44**	**4.83**	**7.34**	**7.01**	**9.01**	**9.66**	10.27	10.64	**3.98**	**4.18**	**5.36**	**5.64**
	MAE_y	**3.69**	**3.14**	**3.03**	5.17	4.51	**5.11**	5.71	**6.48**	**6.39**	**9.31**	**9.94**	7.75	7.51	**4.97**	**5.68**	**5.98**	**6.80**
	SD_x	**4.88**	**4.80**	**4.52**	**5.02**	**4.97**	**5.59**	**5.89**	**12.03**	**11.80**	**16.34**	**16.88**	**8.14**	**8.43**	**8.28**	**8.61**	**13.66**	**13.42**
	SD_y	**3.90**	**3.50**	**3.54**	**3.43**	**3.49**	**4.04**	**4.58**	**10.15**	**10.39**	**17.18**	**18.61**	**4.70**	**4.87**	**7.11**	**8.00**	**12.68**	15.06
ViTPose(f)	OKS	0.51	0.49	0.49	0.58	0.56	0.69	0.69	0.59	0.59	0.51	0.50	0.70	0.69	0.68	0.68	0.69	0.69
	MAE_x	6.74	7.04	7.17	8.49	9.45	9.43	9.51	15.83	16.26	18.76	19.25	**7.97**	**8.53**	8.72	8.78	9.46	9.75
	MAE_y	5.96	6.00	6.23	5.05	5.33	5.80	5.93	10.69	10.77	17.07	17.67	13.03	13.28	9.37	9.21	6.61	6.81
	SD_x	7.68	7.36	7.57	8.13	8.54	11.66	11.54	22.25	22.60	26.72	27.02	9.01	9.19	12.47	12.46	16.68	16.39
	SD_y	5.42	5.16	5.21	4.25	4.30	5.93	6.32	14.85	15.03	25.34	26.25	7.98	7.95	9.30	8.84	11.22	11.34
4D-Human(f)	OKS	**0.63**	**0.64**	**0.65**	**0.64**	**0.65**	**0.73**	**0.73**	**0.69**	**0.70**	**0.65**	**0.64**	**0.72**	**0.72**	**0.73**	**0.73**	**0.73**	**0.73**
	MAE_x	**4.24**	**4.14**	**3.86**	**5.11**	**5.08**	**4.36**	**4.71**	**7.17**	**6.77**	**8.85**	**9.35**	10.30	10.56	**3.90**	**4.05**	**5.12**	**5.41**
	MAE_y	**3.57**	**3.06**	**2.93**	5.19	4.49	**5.10**	5.69	**6.28**	**6.18**	**9.17**	**9.68**	7.68	7.43	**4.75**	**5.50**	**5.53**	**6.38**
	SD_x	**4.63**	**4.58**	**4.29**	**4.90**	**4.80**	**5.28**	**5.53**	**11.21**	**10.90**	**15.33**	**15.67**	**8.08**	**8.25**	**7.75**	**7.88**	**12.40**	**12.04**
	SD_y	**3.15**	**2.81**	**2.80**	**2.67**	**2.84**	**3.76**	**4.39**	**9.54**	**9.82**	**16.31**	**17.68**	**4.51**	**4.70**	**6.49**	**6.96**	**10.19**	12.68

4.2 Stability

To analyse the stability of individual joints under color illumination shift, we employed our 2D Gaussian distribution visualization method (detailed in Sect. 3.4) to conduct both quantitative and qualitative analyses. Figure 4 (top) and Table 2 showcase the stability of various pose estimation methods when our filtering technique is not applied.

From Fig. 4 (top), it is evident that the feature-based method, Vitpose, exhibits the highest instability across all joints when no filtering method is employed, while 4D-Human emerges as the most stable method. This qualitative observation is further supported by the quantitative analysis provided in Table 2. Specifically, 4D-Human demonstrates the lowest standard deviation along the horizontal axis across all joints, indicating superior stability in the horizontal direction. Additionally, 4D-Human boasts the lowest standard deviation along the vertical axis across all joints except for Joint 17 (right ankle). These findings highlight 4D-Human's exceptional stability across both horizontal and vertical axes under color illumination shift followed by OSX (refer supplementary material), which shows the best stability along the vertical axis for Joint 17. This stability trend reinforces the notion that shape-based methods consistently outperform feature-based methods over a temporal video when no filtering is applied.

The observed instability in feature-based methods such as Vitpose can be attributed to their reliance on local features within the image for detecting keypoints, which may lead to sensitivity to noise or errors in feature detection. Feature-based methods may lack sufficient global contextual information about the overall body pose, making them more susceptible to high noise or instability in pose estimation, particularly in challenging lighting conditions or noisy environments. This limitation becomes evident in cases of mismatched joint errors under low lighting, especially involving symmetrical joints such as left-right hips, knees, and ankles flipping with each other. Upon implementing our filtering technique to mitigate such noise, notable improvements in stability were observed across all methods. This refinement is depicted in Fig. 4 (bottom) and Table 2, where significant stability enhancements are evident for Vitpose, alongside increased stability across shape-based pose estimation methods.

5 Limitations and Future Works

One promising direction for future exploration involves the development of more sophisticated relighting techniques. Chromatic adaptation generally performs well for opaque surfaces in human pose estimation, but its effectiveness diminishes for translucent and fluorescent materials, which can be part of the human subject such as clothing items. For example, under blue lighting, distinguishing facial features may become challenging while high-frequency details like pores and fine lines become more noticeable. Enhancing the capability to simulate diverse lighting conditions with greater accuracy and realism than chromatic adaptation, could help in more accurate experimentation and analysis, which

will further enrich our understanding of pose estimation robustness under varying environmental factors.

In addition, there is a compelling opportunity to propose and explore novel 2D methods for pose estimation in challenging scenarios. We advocate for a rethinking of traditional approaches and propose a new method that combines the strengths of feature-based and shape-based techniques. This hybrid approach entails training a model to estimate pose keypoints while simultaneously capturing the contour and shape of the human body. By integrating these elements, the proposed method aims to deliver enhanced accuracy, stability, and reliability across varying environmental conditions such as low light.

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Perceptual Quality, Medical Imaging, and Art Applications

Investigation of Image Features for Perceptually Equivalent Gloss Reproduction Through Comparison of Real Objects and Images

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Abstract. In this study, we focus on glossiness that changes when reproducing colorimetric images of real objects. We experimentally explore image reproduction that is perceptually equivalent to the real object in terms of glossiness, which is necessary to retain the glossiness that can change. In addition to direct comparison of perceptual glossiness of real objects and colorimetrically reproduced images, psychophysical experiments were conducted to directly compare edited images and real objects to reproduce perceptual glossiness. Because direct comparisons of real objects and colorimetric reproduction images showed that the perceptual gloss decreased due to imaging for most stimuli, psychophysical experiment was conducted using processed image to reproduce the perceptual glossiness. The results of both experiments were analyzed using GLCM features, a type of texture analysis, suggesting that contrast, dissimilarity, and entropy are more reproducible than colorimetric reproduced images, whereas homogeneity is less reproducible than colorimetric reproduced images. When homogeneity and ASM are smaller than the colorimetric reproduced image, the image tends to be reproduced with a glossy appearance more similar to the original image.

Keywords: shitsukan management · appearance reproduction · glossiness

1 Introduction

Humans judge the quality of objects and their behavior based on visual information such as color and shitsukan [1]. Shitsukan is a Japanese term that describes the comprehensive sensations that humans experience from physical stimuli. In recent years, research on shitsukan information has been actively conducted across various fields such as information engineering, psychophysics, and neuroscience. In particular, glossiness that is directly related to the attractiveness and value of an object has been considered very important in the design of appearance and product development in the industrial world. Many studies focusing on gloss perception have been actively discussed in the fields of psychophysics and neuroscience [2]. For example, Beuckels et al. [3] attempted to construct a glossiness space by adding haze of the object surface to the existing glossiness space for surface gloss perception, which consists of the two axes of distinctness-of-image (DOI) and contrast gloss for real-world objects. Their study suggested that the

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haze of an object affects its perceived glossiness and sharpness. In addition, Chen et al. [4] conducted a survey on the display features required to reproduce the glossiness of a real object with an image on a display. The results of glossiness matching using real objects and images on an HDR stereoscopic display suggested that the luminance and dynamic range of the display affect the glossiness match of real objects and images.

We have conducted studies [5–7] on shitsukan perception of glossiness, translucency, and roughness using real objects and images, including an analysis of the relationship between perceived shitsukan and physical quantities obtained through psychophysical experiments. For example, Tanaka et al. [5] investigated the perception of shitsukan for real objects and images using 34 real samples consisting of 10 different materials such as stone, paper, and glass, and colorimetrically reproduced images with colors equivalent to those of the real samples. The results of this investigation revealed that the perceived shitsukan of real objects and images could differ; however, the aspect of image reproduction of perceptually equivalent shitsukan of real objects was not discussed.

Therefore, this study investigates the aforementioned issue by focusing on the image reproduction of glossiness, which is one of the most important and actively discussed topics among the shitsukan. It is very important to develop a technology that perceives glossiness equivalently to real objects on displays that have many physical limitations on the presented information. Although many studies have focused on gloss perception [1], there is insufficient research on the relationship between colorimetric and perceptual image reproduction with the goal of preserving the glossiness of the real object and the image. In addition to direct comparison of perceived glossiness of real objects and colorimetrically reproduced images, this study will conduct psychophysical experiments in which direct comparison between edited images and real objects will be conducted to determine perceptual gloss reproduction. By analyzing the results of these experiments, we aim to address the relationship between the real object, the colorimetrically reproduced image, and the perceptually reproduced image. As a first step toward creating perceptually reproduced images with a glossiness equivalent to that of the real object, we apply the glossiness editing algorithm developed in our previous research [8] to change the shitsukan of the images. Using this method, we attempt to obtain a perceptually reproduced image by conducting an experiment to select an image with glossiness equivalent to that of a real object from a group of edited images of experimental sample images.

2 Experiments

In this study, two types of psychophysical experiments were conducted. In Experiment A, perceptual gloss was directly compared to verify whether the gloss perception of real objects made of different materials was the same as that of their colorimetrically reproduced images. If the results of Experiment A showed that the gloss of the real and colorimetrically reproduced images differed, it is desirable to reproduce images in which the gloss perception is equal. Therefore, in addition to Experiment A, a psychophysical experiment, Experiment B, was conducted to select images with equivalent perceptual gloss by comparing real and processed images for the purpose of reproducing perceptual gloss.

2.1 Experiment A

To prepare the colorimetrically reproduced images, the display used in the experiment (ColorEdge CG3146, EIZO Corp.) was calibrated in advance. This allows the display to reproduce an image with the desired CIE_XYZ by mutually converting CIE_XYZ and RGB using the look-up table (LUT). The average value of color difference (CIE 1976) for all 24 colors in a color checker for colorimetric image reproduction using LUT was 1.2. This accuracy is approximately half of the Just Noticeable Difference of 2.3 for color differences, implying that an image that reproduces the colors of the real object in a colorimetric method has been reproduced. The dynamic range of the display was relatively wide: 0.00076 cd/m^2 (black) - 962.48202 cd/m^2 (white).

Experimental Stimuli

Real samples. In this study, a total of 67 stimuli with different surface textures, colors, and shapes but the same size was used to include a wider range of mundane materials. In addition to the 34 stimuli used in the previous study [5], 33 new stimuli were added, including two shell samples. Figure 1 shows the examples and physical gloss of the stimuli used in this study, which is categorized into 11 specific materials, including 16 fabric, 13 glass, 7 paper, 7 stone, 5 resin, 5 wood, 4 leather, 4 metal, 2 rubber, 2 ceramic, and 2 shell real stimuli. Some metals and finished papers have a high physical gloss, while other materials have a low to moderate physical gloss. Note that objects with high transparency, such as glass and resin, are not really suitable for measuring physical gloss with a gloss meter (Rhopoint IQ-S, Konica Minolta, Inc.) and should only be used as a reference. In the experiment, each real stimulus was 50 $\times$ 50 mm in size and placed individually in a viewing box perpendicular to the observer's eyes and illuminated using a mixture of diffuse and spot lighting equivalent to D65.

Images. Image stimuli were generated by photographing and placing digitally imaging real stimuli in a viewing box with a D65-equivalent LED light source to reproduce a colorimetrically equivalent image. A 2D spectroradiometer (SR-5100, Top-con Technohouse Corp.) was used as the camera for photography. The CIE_XYZ image data obtained from the measurement was converted using LUT and represented on the display to obtain a colorimetrically reproduced image.

Experimental Procedure. This evaluation experiment was conducted to investigate the relationship between gloss perception in direct comparison to real objects and colorimetrically reproduced images. In this judgment experiment, real stimuli placed in a viewing box were compared with image stimuli placed on a display, and the glossiness of each stimulus was judged using the magnitude estimation method. The magnitude estimation method is a technique in which the evaluation value of a reference object is set as a reference value with a positive constant, and the evaluation value of the evaluation target is linearly evaluated with two indices, 0 and the reference value. In this case, the upper limit of the evaluation value is not set. In this judgment experiment, the glossiness of the image stimulus was judged by setting the reference value of the real stimulus to 100. For example, when rating the glossiness of the presented image stimulus, observers responded "0" if they did not perceive the glossiness of the image stimulus to be any higher than that of the real stimulus, and "200" if they perceived the glossiness of the image stimulus to be twice as high as that of the real stimulus.

To adapt to the lighting environment, the observer first waited for 5 min, and then placed his chin on the chin rest so that his eyes were perpendicular to the center of the image stimulus on the display. The real stimuli were placed side by side in close proximity at such a distance that they appeared to be the same size as the image stimuli with a viewing angle of 8°, allowing the observers to compare the two stimuli by changing their line of sight only while resting their chin on the rest. The viewing distance at which pixels become indistinguishable on the retina, calculated from the resolution specification of this display, is 58 cm. In this experiment, the distance between the display and the observer was 110 cm, so the subject was not considered to be affected by the resolution of the display. The observer examined the presented real stimulus with both eyes and answered the rating value of the glossiness of the image stimulus shown on the display. After the observer completed the rating, the stimulus was replaced by the experimenter and the same rating was repeated. This rating experiment consisted of two trials with 67 stimuli.

2.2 Experiment B

In this evaluation experiment, with the goal of reproducing images with a perceptually equivalent gloss to that of real objects, a group of edited colorimetrically reproduced images of each stimulus is used to obtain an image with a perceptually equivalent gloss to that of the real object.

Experimental Stimuli. This section describes the method used to generate the edited images used in Experiment B. The glossiness editing method used was developed by Manabe et al. [8]. In this method, the glossiness of an image is changed by following three steps: gloss detection, gloss editing, color space conversion of the input image. In gloss detection, the image is clustered in advance, and then divided into separate images for each object with the same hue value, followed by gloss detection for each object. Gloss detection is performed in the saturation-intensity plane by converting the RGB values of each pixel in the input image to the HSI color space. This property can be used to determine the minimum square value of the intensity plane and to derive an approximate curve of non-glossy pixels using the least squares method. The distance between the approximate curve and the group of glossy pixels is expressed as the gloss intensity. Finally, the gloss is edited using the gloss intensity. After editing the gloss, an inverse transformation from the HSI color space to the RGB color space is performed to obtain an image with edited image gloss. An example of image gloss processing is shown in Fig. 2. In the experiment, the real stimuli placed in a viewing box are compared to the edited image stimuli shown on the display, and an image is selected from a group of images, which is equivalent to the real object in terms of glossiness. Prior to the experiment, a set of gloss-edited images was prepared. A total of 31 gloss-edited images were created by editing the glossiness of the colorimetrically reproduced images for each stimulus.

Experimental Procedure. The evaluation procedure is described below. The evaluation environment is the same as in Experiment A, where real and image stimuli can be directly compared. The experimental procedure is also the same, except that the images to be

compared are processed images. After confirming the glossiness of the presented real stimuli and the image stimuli on the display with both eyes, the experimenter used the pre-prepared keyboard to swap the presented images and selected the image whose glossiness he judged to be equal to that of the real stimulus. Observers were not told how the images were changed through the keyboard operation. After the observer selected a glossy manipulated image for a stimulus, the experimenter changed the stimulus and repeated the same rating, selecting an image for each stimulus whose glossiness matched that of the real stimulus in a trial.

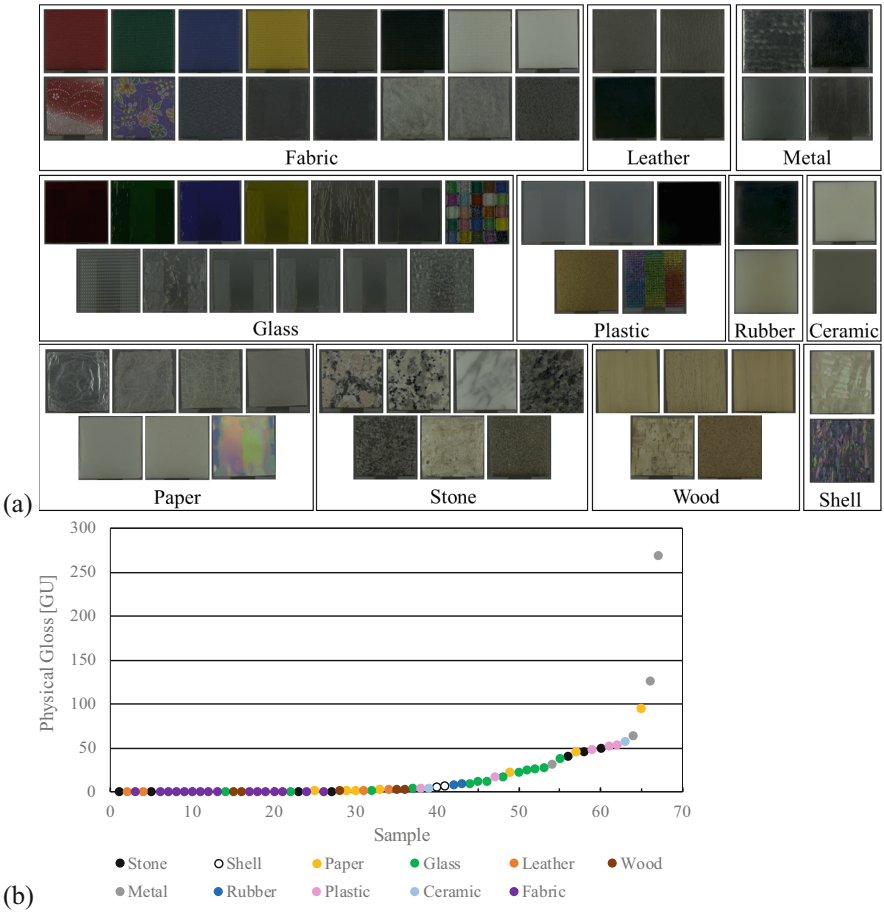


Fig. 1. Experimental samples of real objects. (a) Sample images and their material categories (b) Physical gloss measured by the Gloss Unit (60°) according to ISO 2813.

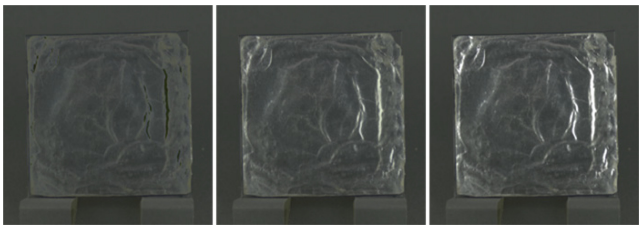


Fig. 2. Example of image gloss processing (from left to right: gloss suppression, original, gloss enhancement).

3 Results and Discussion

The same five observers in their 20s participated in this evaluation experiment for both Experiments A and B. The observers had previously undergone Ishihara-type color vision and visual acuity testing, and they all had normal color vision and visual acuity (20/20 or better).

3.1 Experiment A

Using the results of Experiment A, we determine whether the glossiness was maintained by both colorimetrically reproduced images and real ones. After removing outliers from the rating groups using the Smirnov-Grubbs test, the averaged ratings were calculated. Figure 3 shows the averaged ratings with error bars for the standard deviations. A total of 61 of 67 stimuli have lower averaged gloss ratings for the colorimetric reproduction image stimuli than for the real stimuli. This trend has been confirmed in previous research [5] and may be due to the influence of the presence or absence of binocular disparity in the viewing conditions of the real object and the image. Six stimuli that were rated equal to or higher than the real stimuli were two-colored crepes with patterns, yellow canvas, white canvas, white silicone, and green glass. Compared to the other stimuli, there were no striking similarities. The length of the error bars confirms that the variability of the observers' judgments tends to increase as the difference in glossiness from the real stimuli increases. Overall, the results indicate that the perceived glossiness of the stimuli differs from the mere colorimetric image reproduction of the real stimuli.

3.2 Experiment B

In Experiment B, image stimuli from the edited image set (hereafter referred to as perceptually reproduced images), which the observers perceived to have a glossiness equivalent to that of the real stimulus, were selected. To investigate which image features were used by the observers to compare the glossiness to the real objects, the analysis here is based on the gray level cooccurrence matrix (GLCM) feature [9], a texture analysis method that uses image features extracted from 2D luminance images. In this analysis, we used contrast, dissimilarity, homogeneity, angular second moment (ASM), and entropy, which are a type of GLCM features. The reason for using these GLCM features is that previous studies have found them useful in modeling glossiness with physical indicators

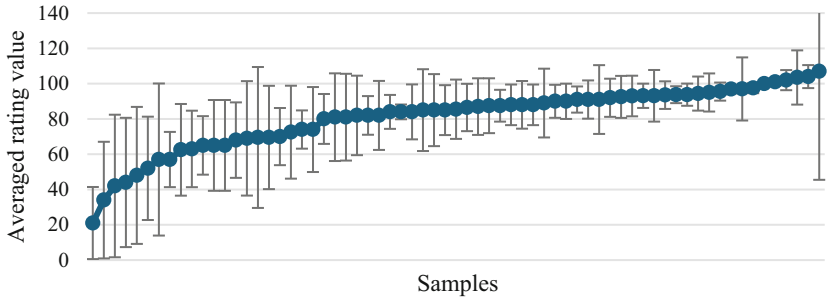


Fig. 3. Averaged rating value for colorimetrically reproduced images in Experiment A.

[7]. In particular, it has been suggested that contrast and entropy are useful as features of glossiness and other shitsukan factors [10].

Because the real object features are unknown without imaging, we alternatively analyzed the image features of the colorimetric reproduction images used in Experiment A. On the other hand, the image features of the perceptually equivalent glossiness selected by the observer in Experiment B were obtained, and the results of both experiments were compared using GLCM features.

Figure 4 shows the results for each feature, where the x-axis is the GLCM of the colorimetrically reproduced image and the y-axis is the GLCM of the perceptually reproduced image, and each plot is the feature for each stimulus. The values of the axes here are normalized values with the maximum value in the calculation of each GLCM feature as 1. The approximate curve of the linear approximation (intercept 0), its approximation equation, and the correlation coefficient are shown. If the slope of the approximation equation is 1 (the 45-° line indicated by the solid gray line), the feature indicates that the colorimetric and perceptually reproduced images may not need to be changed overall. For each feature, the slope is not 1, but increases or decreases, and the distribution of plot points shows the characteristics of the perceptually reproduced image with larger or smaller features, respectively. For example, if a plot point is above the 45-° line, it indicates that image processing that increases the value of the GLCM feature will bring the perceived glossiness closer to that of the real image. The further away the plot from the approximation formula or the 45-° line, the greater the difference between the colorimetrically reproduced image and the perceptually reproduced image.

These results suggest that the present image processing method may also be able to reproduce images that feel perceptually equivalent in glossiness. Specifically, it was confirmed that when contrast, dissimilarity, and entropy are reproduced to be larger than in colorimetric reproduction, and when homogeneity is reproduced to be smaller than in colorimetric reproduction, there is an overall tendency to reproduce images that are perceptually closer to glossy images. However, because the distribution of plot points is spread out, it is difficult to edit gloss perception and image reproduction using a disciplined and unique algorithm when the target is a rich stimulus consisting of different materials, colors, and textures. In the future, a glossy image reproduction method tailored to the image content will be required.

Examples of image stimuli for which there were significant differences between colorimetric and perceptually reproduced images for each GLCM feature are shown in Fig. 5. As can be seen from the figure, the differences in the images are so minute that the reader would have to observe them on the actual display rather than on paper to see the differences, but these are stimuli for which contrast, dissimilarity, and entropy were significantly higher and homogeneity was smaller than for the colorimetric reproduction images.

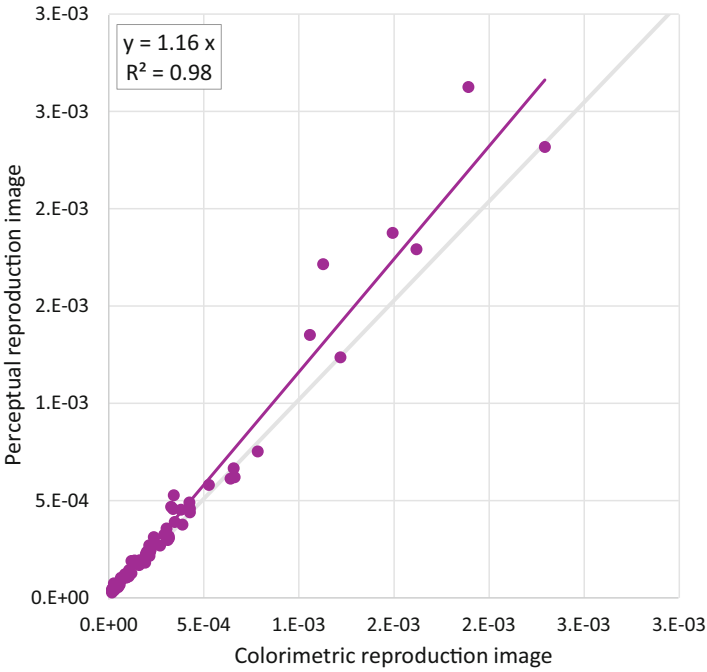
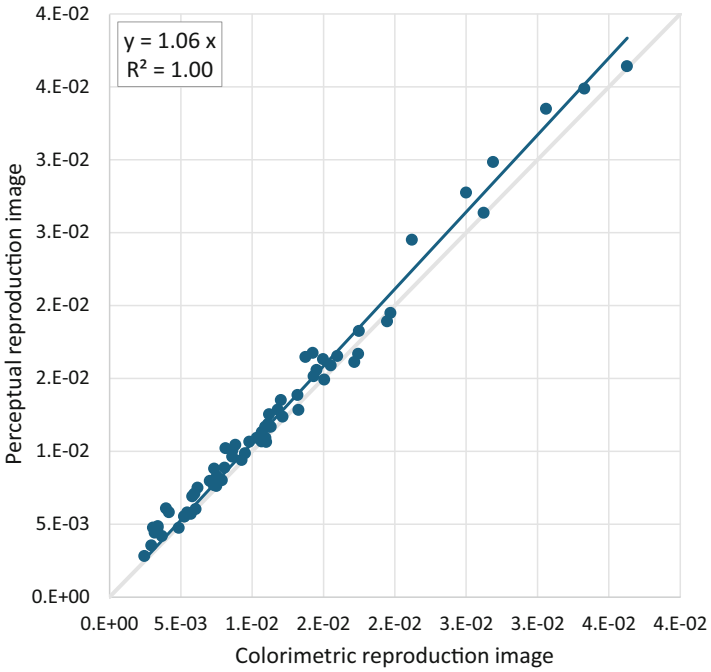


Fig. 4. GLCM features for colorimetric and perceptually reproduced images (from top to bottom: contrast, dissimilarity, homogeneity, ASM, and entropy).



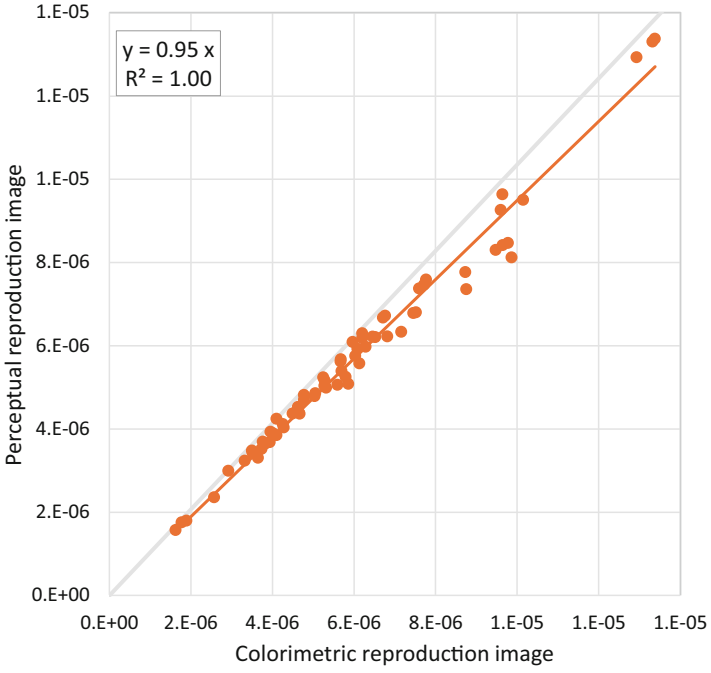
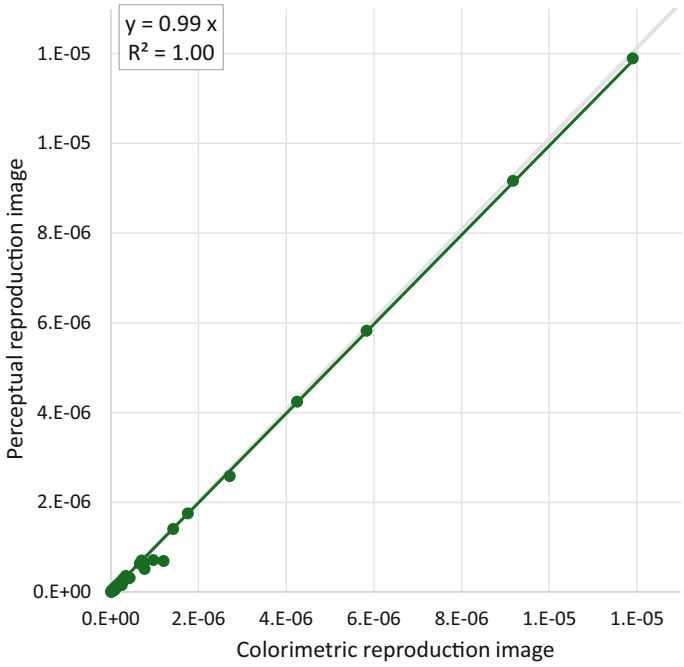


Fig. 4. (*continued*)



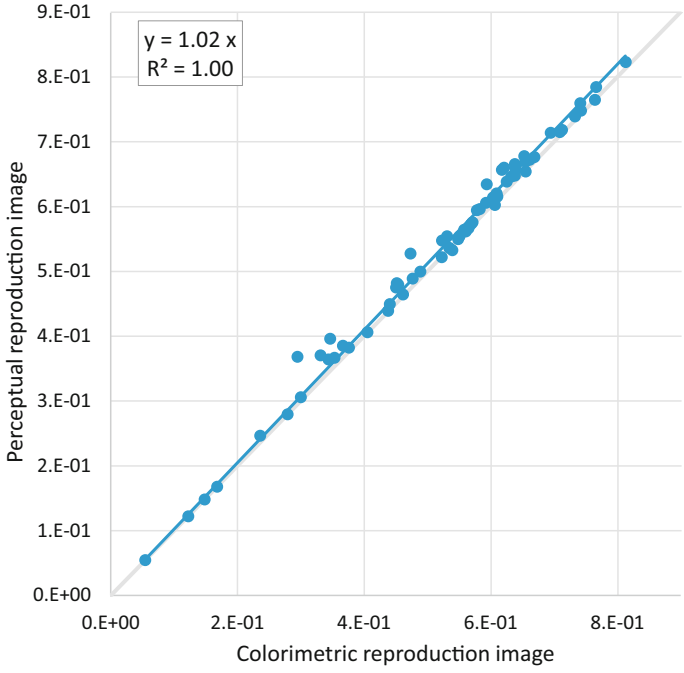


Fig. 4. (*continued*)

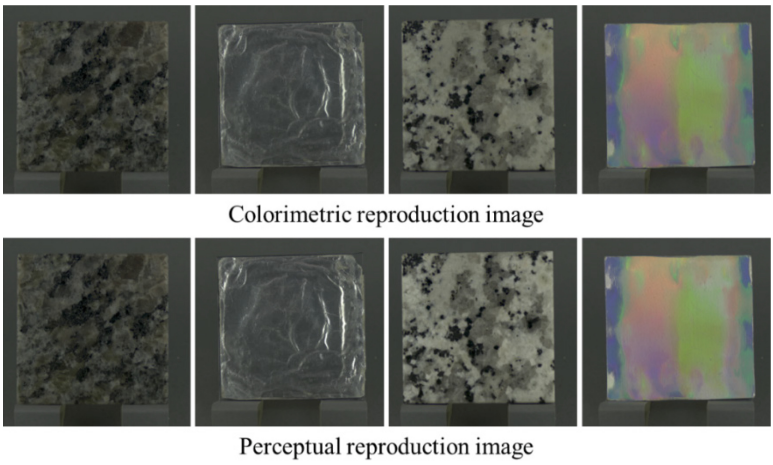


Fig. 5. Examples of image stimuli with significant differences between colorimetric and perceptually reproduced images for each GLCM feature.

4 Conclusion

In addition to the direct comparison of perceptual gloss of real and colorimetrically reproduced images, this study conducted a psychophysical experiment in which manipulated images were directly compared to real images to reproduce equivalent sensations of perceptual gloss. When the real object and the colorimetrically reproduced image were directly compared, it was observed that the perceived glossiness decreased due to imaging for most of the stimuli. Because this makes it desirable to realize a perceptually reproduced image with a glossiness equivalent to that of the real object, psychophysical experiments were conducted to directly compare the processed image with the real object to reproduce the perceived glossiness. The results of the experiment were analyzed using GLCM features, a type of texture analysis, and it was found that contrast, dissimilarity, and entropy were reproduced to make the image larger than the colorimetrically reproduced image, while homogeneity was reproduced to make the image smaller than the colorimetrically reproduced image. However, because adjustments need to be made for each individual stimulus, it is desirable to conduct more detailed studies in the future.

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Disclosure of Interests. The authors have no competing interests.

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Classification of Lung Nodules on CT via Pseudo-colour Images and Deep Features from Pre-trained Convolutional Networks

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Abstract. Developing accurate methods for differentiating benign vs. malignant pulmonary nodules on CT is crucial for the correct management of patients referred for suspicious lung cancer. In this context deep learning by convolutional neural networks (CNN) has been gaining ground as an alternative to conventional methods based on feature engineering, although the use of CNN is often hampered by the lack of sufficiently large datasets for training. Herein we explore the effectiveness of deep features from pre-trained convolutional networks 'off-the-shelf' to discriminate benign vs. malignant lung nodules from CT images. To this end we compare three approaches (two classic, one novel) for pseudo-colour image generation which allow the grey-scale CT data to be fed into CNN models designed for and trained on colour images. The classification performance was estimated by internal and cross-validation using two independent datasets (LIDC-IDRI and LUNGx), giving four experimental conditions altogether. Conventional radiomics features were used as baseline reference. The best accuracy achieved by deep features in the four experimental conditions was respectively 88.5%, 69.6%, 68.9% and 65.2%, that obtained through conventional features 85.5%, 60.9%, 65.2% and 63.8%.

Keywords: Lung cancer · Convolutional neural networks · Pseudo-colours · Transfer learning

1 Introduction

The American Cancer Society [2] ranks lung cancer (including small cell and non-small cell types) as the second most frequently diagnosed oncological disorder in the United States for males and females (excluding skin cancer). The

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survival in people with lung cancer depends on a number of factors, including the histological type and the stage at which the disease is first detected. Early diagnosis and treatment are therefore crucial for improving the overall outcome.

At an initial stage lung cancer usually presents as a small, rounded opacity within the lung interstitium frequently discovered incidentally while doing imaging tests like Computed Tomography (CT) or chest X-rays. However, though some pulmonary nodules may indicate early-stage lung cancer, the majority of them are benign and frequently result from infection, scarring, or autoimmune illnesses [8]. As a consequence the management of patients with suspicious lung nodules may pose significant challenge to the clinician. The diagnosis relies on the evaluation of anamnestic data such as age, history of smoking, and exposure to carcinogenic agents as well as on radiographic variables like size, density, margins, wall thickness, anatomical collocation and stability over time [13,23].

In recent years computerised analysis of medical scans based on the extraction of quantitative features (*radiomics*) has attracted widespread research interest as a potential tool to improve the diagnosis of indeterminate lung nodules detected on CT [7]. The idea behind radiomics is to take advantage of the portion of picture data that is invisible to the human eye–yet may be clinically significant [10].

There are two approaches to radiomics: *conventional radiomics*, which relies on feature engineering (hand-crafted features) and deep learning radiomics [17,21]. Deep learning by convolutional neural networks (CNN) is generally considered superior in accuracy, but its use is often hampered by the lack of sufficiently large datasets for training. Yet, previous studies indicate that features by pre-trained CNN used off-the-shelf can be effective even to to tackle recognition tasks different than those for which the networks have been trained [11,24], and this includes the classification of suspicious lung nodules [12].

The off-the-shelf strategy is facilitated by the huge availability of pre-trained models: by the time of writing PyTorch offers as many as twenty classification models and over a hundred combinations model/pre-trained weights [27]. The missing link here is that CT images are intensity-based (grey-scale) whereas the large majority of pre-trained CNN work with colour images. Common approaches to bridge this gap consist of generating a pseudo-colour RGB image for instance by repeating the original grey-scale on the three colour channels (we refer to this method as GS in the remainder), or by stacking three orthogonal slices into the red, green and channel (we denote this approach as as PCL in the rest of the paper) and feeding it to the network [1].

In this work we propose a pseudo-colouring scheme based on principal component analysis (PCA) as an alternative to GS and PCL for extracting deep features by pre-trained CNN from CT scans. We compared the effectiveness of the three methods for lung nodule classification by combining them with three pre-trained CNN models (ConvNeXT, ResNet50 and Swin V2), four classifiers (1-NN, Gaussian naïve Bayes, linear classifier and logistic regression) and three feature normalisation methods (none, min-max and Z-score). Two independent datasets (LIDC-IDRI and LUNGx) were used for the investigation yielding

four testing conditions: internal validation on each dataset separately and cross-validation. Conventional radiomics features were also considered for a baseline comparison.

2 Materials

Two independent datasets containing solid and part-solid lung nodules from CT scans, denoted as LIDC-IDRI and LUNGx in the remainder, were considered in this study. For both datasets the common criterion for inclusion was a nodule size (defined as the length of the largest side of the axis-aligned bounding box [6]) between 10.0 mm and 50.0 mm. Dataset-specific inclusion criteria and further details are described below.

2.1 LIDC-IDRI

This dataset contains 633 nodules (261 benign, 372 malignant) collected from the open-access Lung Image Database Consortium image collection (LIDC-IDRI [4,9]). Every nodule comes with manually delineated margins and annotated scores by at least one radiologist for subtlety, calcification, sphericity, margin, lobulation, spiculation, texture, and estimated degree of malignancy.

The inclusion criteria were: a) presence of annotated scores by at least two radiologists, b) solid or part-solid nodules defined as average annotated texture score $\geq$3.0 (where 1 indicates non-solid/ground-glass and 5 solid) and c) average malignancy score either $\leq$2.5 or $\geq$3.5 (where 1 stands for highly unlikely to be malignant and 5 highly suspicious).

A region of interest (ROI) based on the 50% majority consensus rule was generated for each nodule – i.e., any voxel was considered in the ROI if it was marked as such in at least half of the manual delineations available for that nodule, not in the ROI otherwise. Ground truth for malignancy was based on the manually-annotated score: all nodules with average malignancy score $\leq$2.5 were considered benign; all those with average score $\geq$3.5 malignant.

2.2 LUNGx

This collection of 69 nodules (32 benign, 37 malignant) is a subset of the dataset proposed for the LUNGx challenge for computerized lung nodule classification [3]. Benignity/malignancy was in this case determined either by histological evaluation or stability over time. The selection of solid and part solid nodules was carried out by the authors based on visual assessment, and lesion delineation was performed manually by two experts as described in [6].

3 Methods

3.1 Conventional Radiomics Features

A total of 109 IBSI-compliant [14] conventional radiomics features including 12 morphological features, 19 intensity-based features, 23 intensity-histogram features, 23 features from grey-level co-occurrence matrices (GLCM), 11 features

from grey-level run-length matrices (GLRLM), five features from neighbourhood grey-tone difference matrices NGTDM and 16 features from grey-level size zone matrices (GLSZM) were extracted from each region of interest using LIFEx v 7.4.0 (LITO-Curie, SHFJ-CEA, CNRS, Univ. Paris Sud, University Paris Saclay, Orsay, France [22]). The complete list and the feature values are available in the supplementary material.

Image pre-processing involved signal windowing between −1000 HU and 500 HU (same settings used in [1,5]) and discretisation into 256 levels. Spatial resampling to uniform cubic voxels of dimension 0.6 mm × 0.6 mm × 0.6 mm was also applied.

3.2 Features from Pre-trained CNNs

Image Preparation. To keep the settings as close as possible to those used for the extraction of the conventional features (Sect. 3.1) the CT scans were preliminary resized to isotropic voxels of size 0.6 mm × 0.6 mm × 0.6 mm, then the signal intensity was again clamped between −1000 HU and 500 HU and discretised into 256 levels. Next, a cubic tensor $\mathbf{X}_{ijk}$ (Fig. 1) aligned with the sagittal (i), coronal (j) and axial (k) direction containing 91 × 91 × 91 isotropic voxels (approx side length = 55 mm) was extracted from the CT scans around the centroid of each nodule's ROI and saved in NIFTI format.

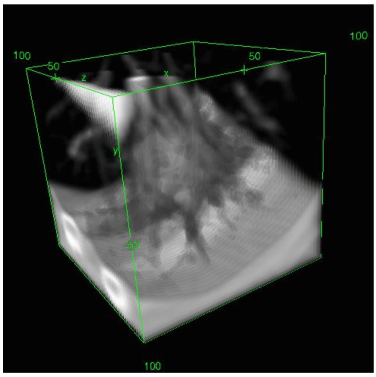


Fig. 1. Sample of a cubic tensor extracted around the region of interest (taken from the LIDC-IDRI dataset). The denser area in the middle represents the nodule. Rendered with ImageJ + 3D Viewer [25].

Pseudo-colouring Methods. Three approaches for generating planar, pseudo-colour RGB images from the cubic tensor $\mathbf{X}_{ijk}$ – respectively denoted as GS, PCL and PCA – were considered in this study.

Grey-Scale Image from the Central Axial Slice. (GS) In this method axial slices are converted to pseudo-colour RGB images just by repeating the the grey-scale input image on each colour channel [20]. Alternatively, one could use triples

of axial, sagittal and coronal patches, as for instance proposed in [28], but the problem with this solution is that the image would undergo non-isotropic rescaling when fed to the network. This is best to avoided when it comes to the classification of lung nodules, as form factors are known to be strong predictors of benignity/malignancy [6].

Pseudo-colour RGB from Three Central Orthogonal Slices. (PCL) This method follows the approach proposed in [26,30]. It involves selecting three orthogonal slices (axial, sagittal and coronal) passing through the central point of the tensor and stacking the three resulting grey-scale images into one single RGB (pseudo-colour) image as shown in Fig. 2.

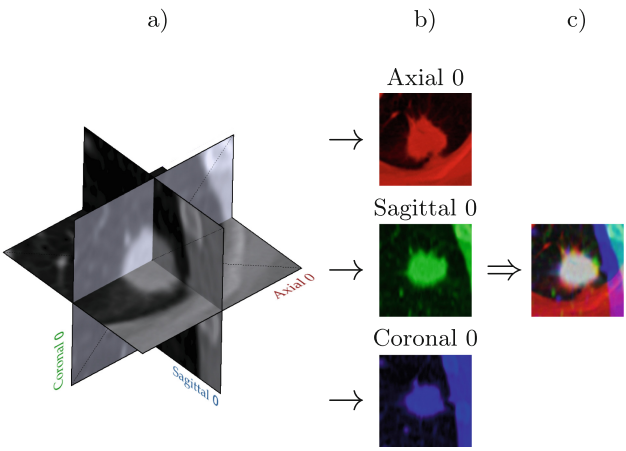


Fig. 2. Generation of a pseudo-colour image from three orthogonal slices passing through the center of the nodule. From left to right: a) three grey-scale images are obtained from the axial, coronal and sagittal plane (numbering denotes the slice position within the tensor, where 0 indicates the central slice); b) each grey-scale image is assigned to one colour channel, and c) the channels are merged into the pseudo-colour RGB image.

Pseudo-colour RGB by Principal Components Analysis. (PCA) In this approach each tensor is preliminary reshaped into a two-dimensional matrix $\mathbf{M}$ of size $(91\times91)\times91$ where each row represents a one-dimensional, end-to-end slice taken along the axial direction k for any position i, j on the trans-axial plane. Next, dimensionality reduction is obtained by projecting $\mathbf{M}$ via PCA onto the first three principal axes (eigenvectors) resulting in a $(91\times91)\times3$ matrix $\mathbf{P}$. Each row of $\mathbf{P}$ is then normalised to unit length (L_1) and the matrix $\mathbf{P}$ eventually reshaped to a three-channel pseudo-colour RGB image of size 91×91 where the red, green and blue channel respectivelky represent the relative strength of the 1^{st}, 2^{nd} and 3^{rd} principal component (Fig. 3).

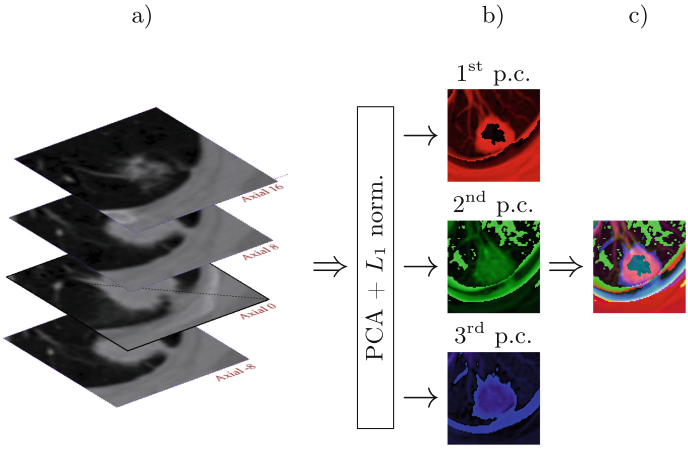


Fig. 3. Generation of a pseudo-colour image via principal component analysis. From left to right: a) stack of trans-axial slices (numbering denotes the slice position within the tensor; 0 indicates the central slice); b) PCA is carried out after projecting the data along the trans-axial direction c) the L_1-normalised first three principal components are respectively assigned to the red, green and and blue channels of the pseudo-colour RGB image.

Background Removal. For all the pseudo-colour generation methods described above we considered two variants: one in which no background removal was applied, meaning that both the ROI and the contextual information around the ROI were retained (Fig. 4, top row); and the other where the area outside the ROI was removed by setting the corresponding voxel/pixel values to zero (Fig. 4, bottom row).

Deep Feature Extraction. Deep features were computed by taking the L_1 normalised output of the last '`avgpool`' layer from pre-trained ConvNeXT [19], ResNet50 [15] and Swin V2 models [18]. Feature extraction was carried out with all the networks frozen (in '`eval`' mode) with the best available weights as provided by PyTorch's '`DEFAULT`' option [27].

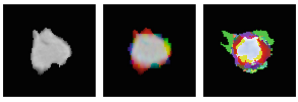
Background removal	Pseudo-colouring method		
	GS	PCL	PCA
No			
Yes			

Fig. 4. Background removal.

3.3 Classification and Accuracy Estimation

Classification models for predicting benignity/malignancy were built by combining the conventional and deep learning feature sets with four classifiers (nearest neighbour with L_2 distance, Gaussian naïve Bayes, linear classifier and logistic regression) and three feature scaling methods: no scaling, min-max and Z-score. To prevent data leakage the scaler's parameters were always fit on the training data, then the trained scaler was used to transform the train and test data separately. Overall accuracy (fraction of the nodules classified correctly) was the primary figure of merit; sensitivity and specificity were also recorded.

3.4 Experimental Plan

We carried out a full-factorial experiment based on the factors and levels summarised in Table 1, resulting in a total of 920 experimental conditions.

Table 1. Experimental plan: factors and levels.

Factor	Levels
Pseudo-colour method*	GS, PCA, PCL
Background removal	Yes, No
Feature extraction	conventional, ConvNeXT, ResNet50, Swin V2
Feature normalisation	None, Min-max, Z-score
Classifier	1-NN, Gaussian NB, Linear classifier, Logistic regression

*Does not apply to conventional radiomics features

4 Results

Tables 2, 3, 4 and 5 show the best performance (by accuracy) achieved by each pseudo-colouring method (GS, PCL and PCA) in the four train/test set combinations considered in the study; the performance of conventional radiomics features is also reported at the bottom of each table as the baseline reference. Each row indicates the combination of levels that determined the optimal response, in terms of accuracy, for each of the pseudo-colouring methods. Where that was

Table 2. Results of internal validation on the LIDC-IDRI dataset. Boldface indicates the highest accuracy, italics the baseline value (conventional features). Key to abbreviations: LC = linear classifier, LR = logistic regression.

Pseudo-colour method	Feature extraction	Background removal	Classifier	Normalisation method	Acc. (%)	Sens. (%)	Spec. (%)
GS	ResNet50	No	LC	Z-score	59.7	33.3	78.2
PCA	ConvNeXT	Yes	LR	Min-max	84.7	80.1	87.9
PCL	ConvNeXT	Yes	LR	Min-max	**88.5**	87.0	89.5
–	conventional	–	LR	Min-max	*85.5*	77.4	91.1

Table 3. Results of internal validation on the LUNGx dataset. Boldface indicates the highest accuracy, italics the baseline value (conventional features). Key to abbreviations: NB = Gaussian naïve Bayes classifier, LC = linear classifier, LR = logistic regression.

Pseudo-colour method	Feature extraction	Background removal	Classifier	Normalisation method	Acc. (%)	Sens. (%)	Spec. (%)
GS	Swin V2	No	NB	Z-score	66.7	62.5	70.3
GS	Swin V2	No	NB	Min-max	66.7	62.5	70.3
PCA	ConvNeXT	No	LR	Min-max	65.2	59.4	70.3
PCL	ConvNeXT	Yes	NB	Min-max	**69.6**	65.6	73.0
PCL	ConvNeXT	Yes	NB	Z-score	**69.6**	65.6	73.0
PCL	ConvNeXT	Yes	NB	None	**69.6**	65.6	73.0
–	conventional	–	LC	Min-max	*60.9*	34.4	83.8

Table 4. Results of cross validation (train set = LUNGx, test set = LIDC-IDRI). Boldface indicates the highest accuracy, italics the baseline value (conventional features). Key to abbreviations: LC = linear classifier, LR = logistic regression.

Pseudo-colour method	Feature extraction	Background removal	Classifier	Normalisation method	Acc. (%)	Sens. (%)	Spec. (%)
GS	ResNet50	No	LR	None	58.8	0.0	100
GS	ResNet50	No	LC	None	58.8	0.0	100
GS	ResNet50	Yes	LR	None	58.8	0.0	100
GS	ConvNeXT	No	LR	None	58.8	0.0	100
GS	ConvNeXT	Yes	LR	None	58.8	0.0	100
GS	ConvNeXT	Yes	LC	None	58.8	3.1	97.8
GS	Swin V2	No	LR	None	58.8	0.0	100
GS	Swin V2	No	LC	None	58.8	0.0	100
GS	Swin V2	Yes	LR	None	58.8	0.0	100
GS	Swin V2	Yes	LC	None	58.8	0.0	100
PCA	Swin V2	Yes	LC	Min-max	**68.9**	77.4	62.9
PCL	Swin V2	Yes	LC	Min-max	68.1	59.8	73.9
–	conventional	–	LR	None	*65.2*	28.7	90.9

achieved by more than one combination all of them were reported. The complete results for all the experimental conditions are provided as supplementary material.

We observe that deep features from pseudo-coloured images outperformed conventional radiomics features in all the four train/test conditions. We also note that among the pseudo-colouring methods PCL granted the best accuracy in three combinations out of four (Tables 2, 3 and 5), PCA in the remaining one. Finally, deep features extracted after background removal proved the best option in all the conditions – only in one case features from the full images achieved an ex-aequo (Table 5).

In absolute terms the results of internal validation on the LIDC-IDRI dataset are comparable with those reported in previous studies on untrained models [12,16,29]. As for LUNGx, this is a notoriously difficult dataset, which partly explains the lower performance obtained with internal validation, although the figures are again in agreement with previously published works [3,6]. The cross-validation conditions (LIDC-IDRI/LUNGx and LUNGx/LIDC-IDRI) could not be compared to the literature as we are not aware of prior research with the same settings.

Table 5. Results of cross validation (train set = LIDC-IDRI, test set = LUNGx). Boldface indicates the highest accuracy, italics the baseline value (conventional features). Key to abbreviations: NB = Gaussian naïve Bayes classifier, LC = linear classifier, LR = logistic regression.

Pseudo-colour method	Feature extraction	Background removal	Classifier	Normalisation method	Acc. (%)	Sens. (%)	Spec. (%)
GS	ConvNeXT	Yes	LC	Z-score	62.3	40.6	81.1
GS	ConvNeXT	Yes	LR	Z-score	62.3	43.8	78.4
GS	ResNet50	Yes	NB	Z-score	62.3	37.5	83.8
GS	ResNet50	Yes	NB	None	62.3	37.5	83.8
GS	ResNet50	Yes	NB	Min-max	62.3	37.5	83.8
PCA	ResNet50	No	NB	None	63.8	50.0	75.7
PCA	Swin V2	No	LR	Min-max	63.8	34.4	89.2
PCL	ResNet50	Yes	NB	Min-max	**65.2**	46.9	81.1
PCL	ResNet50	Yes	NB	Z-score	**65.2**	46.9	81.1
PCL	ResNet50	No	LR	Z-score	**65.2**	40.6	86.5
PCL	Swin V2	Yes	LC	None	**65.2**	37.5	89.2
–	conventional	–	LR	None	*63.8*	28.1	94.6

5 Conclusions

In this work we have investigated the ability of deep features from pre-trained convolutional networks 'off-the-shelf' to distinguish benign from malignant lung nodules on CT scans. Specifically, we have evaluated three methods (two traditional, one novel) for creating pseudo-colour images that enable the grey-scale CT data to be used as input for CNN models designed for colour images. The proposed approach displayed potential, as the results indicate that deep features were more accurate than traditional radiomics features. Of the three pseudo-colouring approaches, PCL produced the best accuracy in three train/test combinations, and PCA in the other.

We also discovered that removing the background improved the performance, implying that contextual information may not be advantageous to classification, and could actually represent a confounding factor. This effect and its potential implications for future developments should be further addressed in future studies. Furthermore, parallel extraction of features from the ROI and the background only (dual pathway) could also be investigated as a means to improve accuracy.

6 Limitations and future work

This study is not exempt from limitations, among which the relatively small sample size of one of the two datasets (LUNGx, $n = 69$) and the retrospective nature of the data. The results should be therefore validated on larger, ideally perspective cohorts of patients. In future studies it would also be interesting to investigate whether other diagnostic features, such as the presence/degree of spiculation, lobulation or the nodule's location – where available – may have a bearing on the results. Another direction for future research could be to determine whether the information provided by conventional radiomics features and by

features from pre-trained CNN complement one another, and the potential ways to integrate them.

Supplementary Material Complete results for all the factors and levels of the experimental plan, features and metadata are available at https://github.com/bianconif/cciw_2024.

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Ethical Statement. The present study was carried out on de-identified data from publicly available datasets, therefore does not qualify as research on human subjects.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

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Painter Verification Using Color Palettes: An Exploratory Study

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Abstract. Forgery detection in fine art necessitates collaboration among art historians, conservators, scientists, and forensic experts. Traditional methods, which rely on expert visual assessment and scientific analysis, are effective but often time-consuming and costly. Recent advancements in technology have introduced image analysis and machine learning techniques, offering efficient and precise alternatives for detecting art forgeries. Excluding material and chemical-based clues, assessing the authenticity of paintings can be broadly modeled along three visual dimensions: color, brushstrokes, and contents. This paper examines the efficacy of using color as a feature for determining the authenticity of paintings. We utilize machine learning algorithms to analyze the color palettes of over 100,000 digital images from approximately 1,500 artists. We compactly represented paintings through their color palettes and different other color-based features. Our experiments, designed as verification tasks, explore the potential of color-based features to verify the authorship of the artworks

Keywords: Color palette · Color-based features · Painter verification · Forgery detection

1 Introduction

Art forgery remains a persistent challenge in the world of fine art, casting doubt on the authenticity and integrity of countless masterpieces throughout history. With the advancement of technology, forgers have become increasingly sophisticated in their methods, making it ever more difficult for art experts to distinguish between genuine works and skillful imitations. The stakes are high, not only in terms of monetary value but also in preserving cultural heritage and artistic legacy [25].

Effective forgery detection often requires collaboration between art historians, conservators, scientists, and forensic experts. By pooling their expertise and resources, interdisciplinary teams can conduct thorough investigations and cross-validate findings using multiple methods and perspectives. Traditional methods of forgery detection often rely on the keen eyes of art historians and conservators, supplemented by painstaking chemical analysis and historical research.

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One of the oldest and most traditional methods of detecting forgery involves the expertise of art connoisseurs and scholars. These individuals possess extensive knowledge of artistic styles, techniques, and historical context. By closely examining the painting's composition, brushwork, color palette, and overall aesthetic, connoisseurs can often discern subtle differences that may indicate a forgery. Scientific techniques such as spectroscopy, radiography, and microscopy play a crucial role in forgery detection. These methods allow conservators and scientists to analyze the materials and techniques used in the painting's creation. Art historians often conduct stylistic analysis to compare the painting in question with authenticated works by the same artist or period. This involves studying elements such as composition, perspective, subject matter, and brushstroke technique. Anomalies or deviations from the artist's typical style may indicate a forgery or imitation [19,26].

While effective, these methods are time-consuming, expensive, and not always foolproof. Moreover, the growing sophistication of forgeries demands equally sophisticated countermeasures. In recent years, the intersection of art and technology has opened new avenues for assessing the authenticity of arts. Some promising approaches rely on the application of image analysis and computer vision techniques. By leveraging the power of machine learning algorithms, researchers have begun to develop automated tools capable of discerning subtle inconsistencies and anomalies. The use of these techniques in forgery detection offers several advantages over traditional methods. Firstly, it can process vast amounts of visual data rapidly, allowing for the efficient analysis of large collections of artwork. Secondly, it can detect patterns and features that may elude the human eye, thereby enhancing the sensitivity and accuracy of the detection process.

Image processing and computer vision techniques have been exploited for automatic artist recognition [14,18,27]. Johnson et al. [17] exploited Wavelet filters to analyze and model the brushstrokes of the paintings. Features are extracted from image patches and several classification strategies are investigated. Similarly, Polatkan et al. [24] use supervised machine learning on features derived from Hidden Markov-Tree-modeling of the paintings' wavelet coefficients to distinguish copies from originals. [28] collected about 1000 drawing from four famous illustrators and design a recognition algorithm using color histogram, GIST, HOG, BoW Dense SIFT and BoW Color Dense SIFT features. Cetinic et al. [6] observed that using multiple features improve the accuracy of the recognition of the artists. In recent years, deep learning has been demonstrated to provide superior recognition results than traditional approaches [4]. Leveraging a ResNet50 betwork under a transfer learning framework, Uguail et al. [29] achieved 98% in image-based classification tasks during validation using a test set of authentic paintings by Raphael. Similarly, Dobbs et al. [12] are able to authenticate contemporary art from of 100 artists with an accuracy of 91.23% using a variant of a ResNet model. Brushstrokes are also analyzed using deep learning techniques. In [16] paintings are scanned using a chromatic confocal optical profilometer to produce surface height data. Using an ensemble of con-

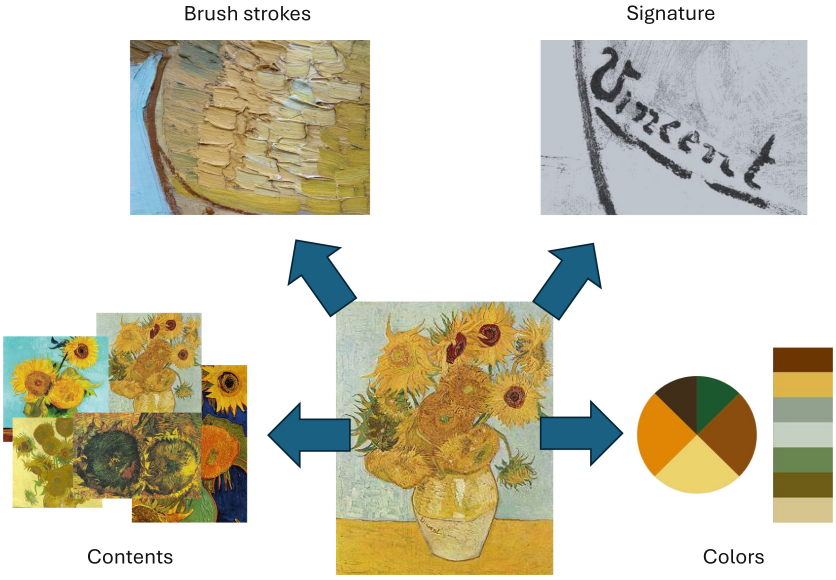

Fig. 1. Visual clues for attribution of paintings.

volutional neural networks (CNNs) for attribution an accuracy up to 96% is obtained.

Here we consider the problem of determining the authenticity of fine arts in the context of paintings. Following the works in the literature, the visual analysis (we do not consider chemical analysis of the materials and paints) required to assess the authenticity of paintings can be broadly modeled along several dimensions (see Fig. 1): color (i.e. verifying that the palette is in accordance with the artist's aesthetic), brushstrokes (i.e. verifying that the texture of the paint is coherent with the "hand of the painter"), artist's signature, and contents (i.e. the subject and composition are coherent with the period and style of the artist).

In this paper we focus our analysis on the efficacy of color as a feature to assign authorship to paintings. Specifically, we evaluate the use of color palette and machine learning algorithms on a dataset of more than 100,000 digital images of paintings belonging to about 1,500 artists. The color palette is used to compactly represent the paintings and possibly characterize the artist who painted them.

Just as writers have distinctive styles and vocabularies, painters develop unique visual languages that include their use of color. While it may not always be possible to definitively identify a painter based solely on their color palette, their use of color is certainly an important aspect of their style and can contribute significantly to their recognizability as an artist. For example, Vincent van Gogh is associated with his vibrant yellows and blues. in this work we do not claim that the author of a painting can be always recognized by its color

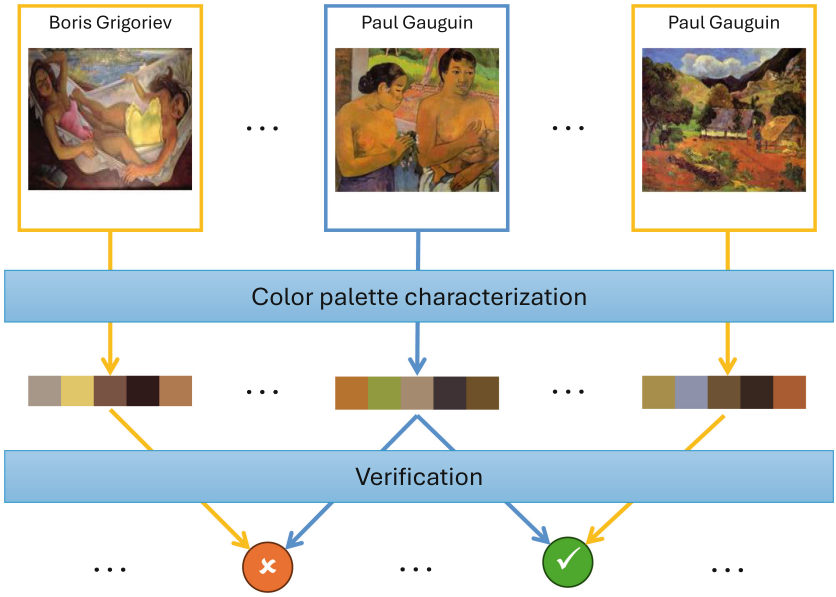

Fig. 2. Schematic representation of the verification approach investigated in this paper: given two paintings we want to assess the performance of their color palette in the disambiguation of their authors.

palette, but that color palette may provide valuable information to a system that analyze and combine different characteristics of a painting.

To assess the performance of the color palettes in the disambiguation of the authorship of the paintings, we performed several experiments designed as verification tasks. Figure 2 illustrates the verification approach studied in this work. The method proposed in this work should not be considered as an alternative solution to existing methods for painting verification, but as a complementary approach to be used in conjunction with existing systems, as for example [3].

The paper is structured as follows. Section 2, describes the materials used and method applied for the experiments. Section 3, presents and comments the results of the experiments. Finally, Sect. 4, concludes the paper with considerations for future works.

2 Materials and Methods

2.1 Dataset

The dataset used in this work has been obtained from the Painter by Numbers Kaggle competition [13]. The dataset contains 103,250 images of paintings obtained by a publicly available providers of digital artworks (mainly from WikiArt.org). The images are organized in a hierarchical way. Painings are categorized by genres (41 classes), styles (125 classes) and artists (1,508 authors). The

Fig. 3. Examples of paintings in the dataset from different authors.

dataset has been designed to predict if a pair of images are artworks made by the same artist or not. We previously used a modified version of ths dataset to build and evaluate a multitask deep multibranch neural network to predict the paintings categories [3]. Differently from the previous work, here we are interested in using only color attributes to recognize the authors of the paintings (Fig. 3).

2.2 Color Palette Extraction

To compute the color palettes we take inspiration from our previous work [9]. Formally, we define a color palette $\mathcal{P}$ as a set of m colors defined in a n-dimensional color space $\mathcal{P} = \{c(1), c(2), \ldots, c(m)\}$. The images in the dataset are defined in the $sRGB$ color space so $n = 3$ and a color $c(i)$ is represented by the triplet $[c_r(i), c_g(i), c_b(i)]$ denoting the red, green, and blue components. To extract a color palette for an image $\mathcal{I}$, $\mathcal{P}(\mathcal{I})$, we used unsupervised k-means clustering algorithm. Before computing the color palette, all the colors are converted into the Lab color space using CIE D65 as the reference white. As a distance between colors we use the L2 distance in the Lab space, thus corresponding to the ΔE_{76}. Cluster centers initialization is performed by selecting m seeds with the k-means++ algorithm [2]. The centers of the clusters constitute the final color palette.

We also investigate if mapping the colors in the color palettes to a common small set of colors may affects the recognition process. To this end, we adopt the color names mapping defined by the ISCC-NBS system by the Inter-Society Color Council and the National Bureau of Standards (NBS, now NIST) [1]. The system defines more than 260 colors subdivided into 13 main categories and

Fig. 4. Examples of color palettes. Original: the color palette extracted from the image with the k-means clustering approach. ISCC: the same color palette mapped to the ISCC color system of 267 color names.

several other derived subcategories [5]. Each color in the original color palette is mapped to the ISCC one having minimum distance from it.

2.3 Characterization of the color palette

There are several algorithms that may default to producing color palettes with around five colors for various reasons, including simplicity, efficiency, and visual appeal. For example 5 colors seems to be the generally accepted number of colors in a representative color theme [22,30]. However, this is not a universal rule, and the number of colors in a palette can vary widely depending on the algorithm and its parameters. We limited in this work our analysis to color palettes composed by five colors, however we are aware that some painting could require more (or less) colors to faithfully represent the artist style [20,21].

Color palette and Color Histogram Following the procedure described in Sect. 2, from each painting image we extracted a palette with 5 colors. For analysis, the color palette is coded in the CIELab color space. We also converted the color palette into the ISCC color space to evaluate the effect of a limited color range. For completeness we also considered computing the histogram using the

in-gamut set of colors of the ISCC color system corresponding to 260 colors. All the colors are coded in the CIELab space. Figure 4 shows some examples of extracted color palettes.

Two color palettes $\mathcal{P}_1$ and $\mathcal{P}_2$ are compared by using a distance inspired from the Hausdorff distance: we compute the average ΔE_{76} distance among the colors in the two palettes after having associated each color in the first palette to the most similar one in the second palette. Formally, let $\sigma_i = \{\sigma_{i_1}, \sigma_{i_2}, \sigma_{i_3}, \sigma_{i_4}, \sigma_{i_5}\}$ be the $i-$th permutation of the color order in the second palette $\mathcal{P}_2$. The color associations between the two palettes is then induced by the permutation σ resulting in the minimum average ΔE_{76} distance among the colors, i.e.:

$$\sigma = \arg\min_{\mathrm{i}} \frac{1}{5} \sum_{\mathrm{j}=1}^{5} \Delta \mathrm{E}_{76} \left(\left[\mathrm{c}_{\mathrm{r}}^{'}(\mathrm{j}), \mathrm{c}_{\mathrm{g}}^{'}(\mathrm{j}), \mathrm{c}_{\mathrm{b}}^{'}(\mathrm{j}) \right], \left[\mathrm{c}_{\mathrm{r}}^{''}(\sigma_{\mathrm{i_j}}), \mathrm{c}_{\mathrm{g}}^{''}(\sigma_{\mathrm{i_j}}), \mathrm{c}_{\mathrm{b}}^{''}(\sigma_{\mathrm{i_j}}) \right] \right) \quad (1)$$

where $c^{'}(\cdot) \in \mathcal{P}_1$ and $c^{''}(\cdot) \in \mathcal{P}_2$. Our distance can thus be written as:

$$d(\mathcal{P}_1, \mathcal{P}_2) = \text{mean } \Delta \mathrm{E}_{76}(\mathcal{P}_1, \sigma(\mathcal{P}_2)) \quad (2)$$

The color histograms are normalized and compared using two different distances: the L2 distance, and the weighted distance introduced by Hafner [15] that takes into account also the cross-talk between colors.

Universal Color Harmony and Color Emotions. The computational models for color harmony and color emotions used in this work are based on the work by Ou and Luo in [23]. The models are defined in terms of hue, chromaticity and lightness similarities. Color harmony is derived from psychophysical data collected around the world and the obtained Universal Color Harmony model that correlate well across different population. The color emotion models characterize colors in terms of "warm/cool", "heavy/light" and "active/passive" scales.

Given two $CIELab$ colors (L_1, a_1, b_1) and (L_2, a_2, b_2), the harmony score CH_U between them takes into account four elements: the hue similarity ($CH_{\Delta H}$), the chroma similarity ($CH_{\Delta C}$), the lightness difference ($CH_{\Delta L}$) and the high lightness contribution (CH_{Lsum}):

$$CH_U = CH_{\Delta H} + CH_{\Delta C} + CH_{\Delta L} + CH_{Lsum} \quad (3)$$

with

$$CH_{\Delta H} = -0.7 \tanh(-0.7 + 0.04 \Delta H_{ab}) \quad (4)$$

$$CH_{\Delta C} = -0.3 \tanh(-1.1 + 0.05 \Delta C_{ab}) \quad (5)$$

$$CH_{\Delta L} = 0.4 \tanh(-0.8 + 0.05 \Delta L) \quad (6)$$

$$CH_{Lsum} = 0.3 + 0.6 \tanh[-4.2 + 0.028(L_1 + L_2)] \quad (7)$$

where Δ_L is the CIELab lightness difference; ΔC_{ab} is the CIELab chroma difference; ΔH_{ab} is the CIELab hue difference.

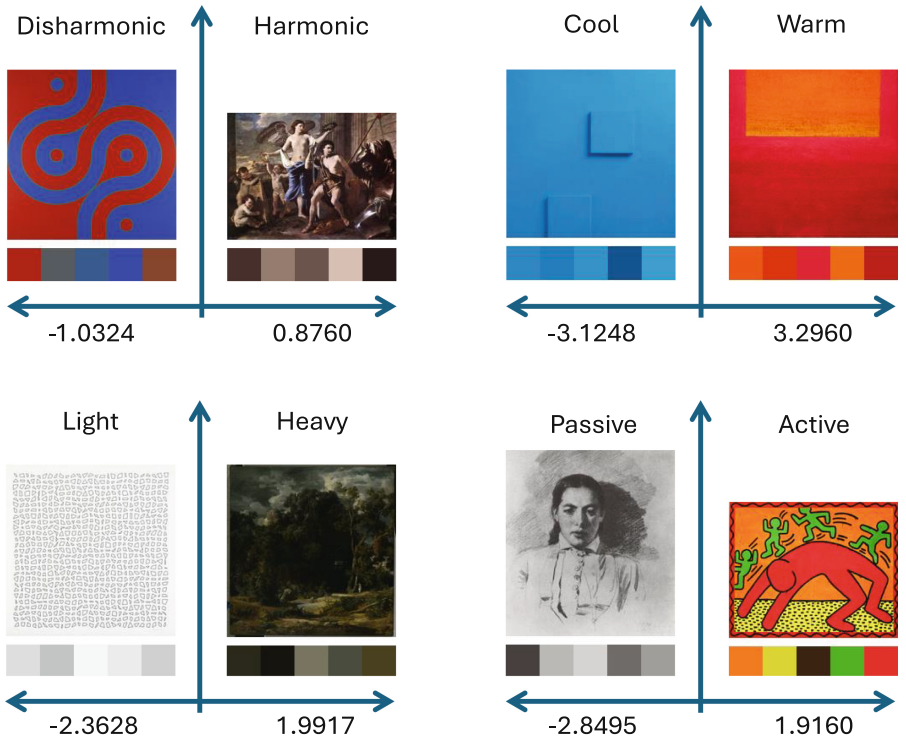

Fig. 5. Examples of color emotion attributes.

Since the harmony value is defined between two colors, we extend it for a set of colors following [10]. The color emotion scales on a (L, a, b) are defined as follows:

$$WC = -0.89 + 0.052C_{ab}[\cos(h_{ab} - 50°) + 0.16\cos(2h_{ab} - 350°)] \quad (8)$$

$$HL = 3.8 - 0.07L \quad (9)$$

$$AP = -3.4 + 0.067\{(L-50)^2 + 1.93(a+1)^2 + 1.05(b-9)^2\}^{\frac{1}{2}} \quad (10)$$

where WC, HL and AP are the "warm/cool", "heavy/light" and "active/passive" attributes respectively. The color emotion for a set of color is the average of the attribute of all colors in the set. Figure 5 shows some examples of color emotion attributes.

Dynamic Spatial Chromatic Histogram. In order to consider information about the spatial distribution of the colors, we also investigate the use of the Spatial Chromatic Histogram (SCH) [7,8]. This histogram is a vector of tuples, one for each color. Each tuple is composed of three elements: the ratio of pixels of the i-th color $h(i)$, the baricenter of all the regions of the same color $\mathbf{b}(i)$,

and the spread of the pixels from their baricenter $\sigma(i)$. We compute the SCH feature on the histogram of the ISCC colors. Given two SCH feature vectors, $S_1 = [h_1(i), \mathbf{b_1}(i), \sigma_1(i)]$ and $S_2 = [h_2(i), \mathbf{b_2}(i), \sigma_2(i)]$, they are compared using a the following color similarity:

$$f(S_1, S_2) = \frac{1}{2} \sum_{i}^{m} \min(h_1(i), h_2(i)) * \\ * \left(\frac{\sqrt{2} - ||\mathbf{b_1}(i) - \mathbf{b_2}(i)||}{\sqrt{2}} + \frac{\min(\sigma_1(i), \sigma_2(i))}{\max(\sigma_1(i), \sigma_2(i))} \right) \tag{11}$$

We also experimented with a variation of the SCH, namely the Dynamic Spatial Chromatic Histogram (DSCH) [11]. This features is inspired by the SCH but it can be computed on any set of colors. The similarity of two DSCH feature vectors is performed by taking into account also the similarity between color pairs. In this work, we used a variant of the original similarity defined in [11]. Here, the color similarity between colors, $\mathrm{Sim(C(i), c(j))}$, is computed in the CIELab color space using the ΔE_{76} measure. Given two DSCH feature vectors, $D_1 = [c_1(i), h_1(i), \mathbf{b_1}(i), \sigma_1(i)]$ and $D_2 = [c_2(j), h_2(j), \mathbf{b_2}(j), \sigma_2(j)]$, they are compared using the following color similarity:

$$g(D_1, D_2) = \frac{1}{3m^2} \sum_{i}^{m} \sum_{j}^{m} \mathrm{Sim(c_1(i), c_2(j))} * \\ * \left(\frac{\min(h_1(i), h_2(j))}{\max(h_1(i), h_2(j))} + \frac{\sqrt{2} - ||\mathbf{b_1}(i) - \mathbf{b_2}(j)||}{\sqrt{2}} + \frac{\min(\sigma_1(i), \sigma_2(j))}{\max(\sigma_1(i), \sigma_2(j))} \right) \tag{12}$$

Figure 6, shows some examples of how DSCH and SCH features are computed. For each block of image, the top row shows the original image, the middle row shows the five color distributions, and the third row shows the color histogram computed on the ISCC color system.

3 Experiments

In this section we report the results of the painter verification performance obtained by the different features described in Sect. 2.3. The verification performance is measured in terms of the Area Under the Curve (AUC) of the Receiver Operating Characteristic curve (ROC), that is the plot of the true positive rate (TPR) against the false positive rate (FPR) at different threshold settings.

The verification pairs are randomly generated from the test set of our modified version [3] of the Painter by Numbers dataset: for a random half of the paintings in the test set a matching pair is created by randomly selecting from the test set a target painting from the same author; for the second random half of the paintings in the test set a non-matching pair is created by randomly selecting from the test set a target painting from a different author. Therefore the

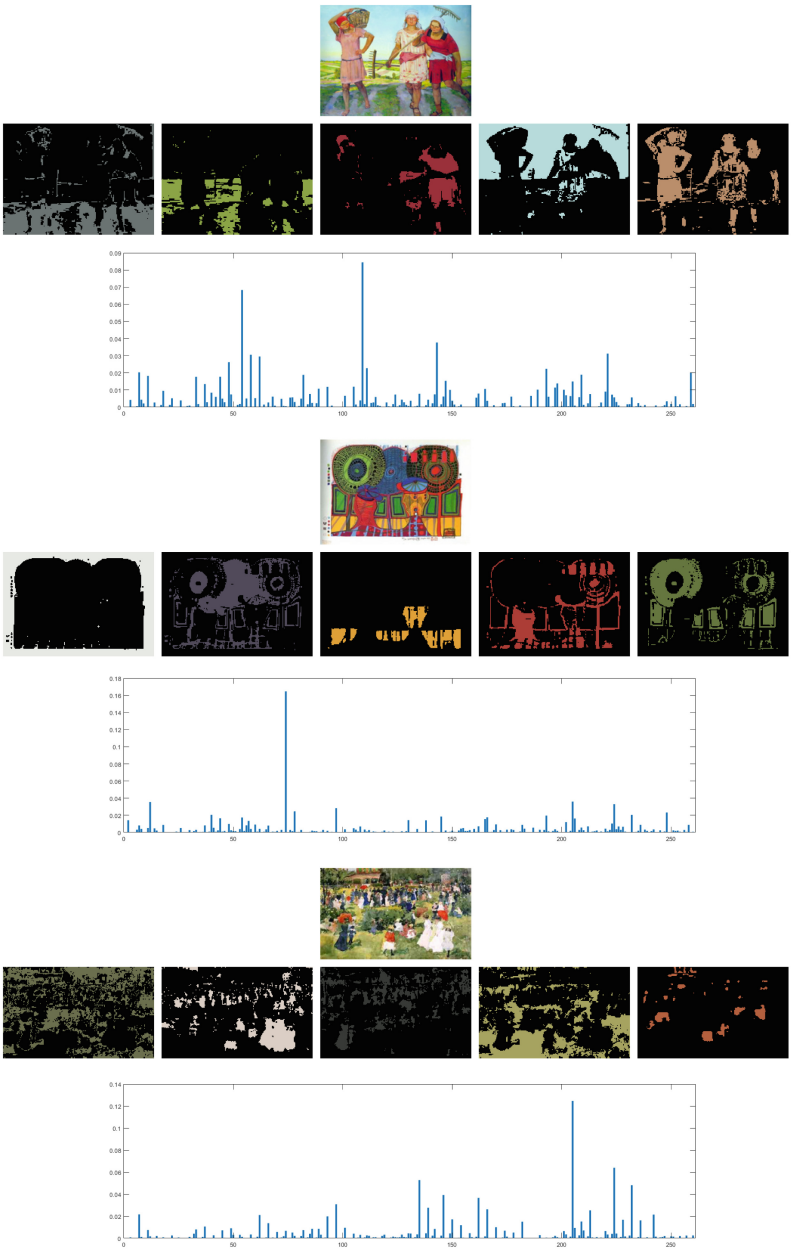

Fig. 6. Examples of how DSCH and SCH features are computed. For each block of images, the top row shows the original image, the middle row shows the five color distributions, and the third row shows the color histogram computed on the ISCC color system.

Table 1. Painter verification results. For each method, in addition to the verification performance (AUC), we report the distance used (XTalk: color crosstalks), the quantization technique used to obtain the palette (KMS: k-means), the eventual mapping of the obtained colors in ISCC, and the eventual use of additional features given as input to the linear SVM classifier. The best result is reported in bold, the second one is underlined. The palettes contain 5 colors. The histograms contain 267 bins.

Method	Distance	Quantization	Mapping	Extra	AUC
Palette	Eq. 2	KMS	None	None	61.0
Palette	Eq. 2	KMS	None	Harmony, Emotion+SVM	63.0
Palette	Eq. 2	KMS	ISCC	None	60.3
Palette	Eq. 2	KMS	ISCC	Harmony, Emotion+SVM	62.8
Histogram	XTalk [15]	None	ISCC	None	52.0
Histogram + Clip	XTalk [15]	None	ISCC	None	53.2
Histogram	L2	None	ISCC	None	63.9
Histogram	L2	None	ISCC	Harmony, Emotion+SVM	64.4
DSCH Palette	Eq. 12	KMS	ISCC	None	61.7
DSCH Palette	Eq. 12	KMS	ISCC	Harmony, Emotion+SVM	<u>64.5</u>
SCH Histogram	Eq. 11	None	ISCC	None	64.1
SCH Histogram	Eq. 11	None	ISCC	Harmony, Emotion+SVM	**64.6**

cardinality of the generated verification pairs is the same of the test set, which is about 30,000.

In addition to the distance between the individual features, we also stack it with the distance between harmony and color emotions, resulting therefore in a 5-dimensional feature vector. These feature vectors are then used to train a linear SVM together with the matching/non-matching labels. The verification results for this configuration are reported using a two-fold cross-validation in order to be comparable with those obtained by individual features.

The verification results are reported in Table 1. For each method, in addition to the verification performance, we report the distance used, the quantization technique used to obtain the palette, the eventual mapping of the obtained colors in ISCC, and the eventual use of additional features given as input to the linear SVM classifier. From the results it is possible to see how the single method giving the best verification performance is the ISCC-mapped SCH histogram with an AUC of 64.1. We can also observe how the use of harmony and color emotions improves the verification results for all the individual features considered with an average AUC improvement of about 1.7%: this improvement is higher for palette-like features (average AUC improvement of about 2.28%) than for histogram-like features (average AUC improvement of about 0.5%). Considering also harmony and color emotions, the best method is again the ISCC-mapped SCH histogram with an AUC of 64.6.

4 Conclusions

In this paper we conducted an exploratory study on the use of color as the solely feature for the task of painter verification.

The verification pairs, i.e. pairs of paintings belonging to the same artist and pairs belonging to different ones are generated from the Painter by Numbers dataset which consists of more than 1,500 different authors.

For each painting we compute a compact representation by extracting the color palette, the color histogram, the Spatial Chromatic Histogram, the Dynamic Spatial Chromatic Histogram, and other color-based features related to the concepts of harmony and emotions (e.g., cool vs warm, light vs heavy, and passive vs active). For each pair of paintings to be verified, the verification score is computed for each individual feature by considering the distance defined for each of them, as well as stacking it with the difference between harmony and color emotions and feeding them to a linear SVM. The experimental results showed that the use of the additional harmony and color emotions features improves the verification performance of all the single features considered, with the best configuration resulting in an AUC of about 64.6.

As future work we plan to include other simple features that we know are important for painter verification, e.g. texture that is able to characterize the brushstroke style of the painter.

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Color Constancy

Illuminant Equivariant Networks for Computational Color Constancy

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Abstract. Neural networks are now the standard solution to many computer vision problems. Their generalization ability enables them to successfully address various tasks in computational photography, such as enhancement, restoration, and color constancy. However, their performance is highly dependent on the illumination conditions of the training images. When faced with test images under different illuminant conditions, these networks often struggle to perform their tasks correctly. In this paper, we investigate the efficacy of illuminant equivariant neural networks for the illuminant estimation task, which is crucial for computational color constancy. These networks are equivariant to the photometric transformations that characterize changes in lighting conditions. They achieve this capability through mathematical derivation rather than specific augmentation during training. We implemented the equivariant versions of state-of-the-art neural networks for illuminant estimation and tested them on the NUS dataset. The results demonstrate that the equivariant networks maintain stable performance even with significant changes in illumination, whereas the original standard networks exhibit a serious degradation in their accuracy.

Keywords: Illuminant Estimation · Illuminant Equivariant Networks · Computational Color Constancy · Computational Photography

1 Introduction

A common requirement for computer vision systems is the robustness to changes in illumination. Many systems are targeted to scenarios with limited control over lighting conditions, and testing for every possible illumination configuration (i.e. number and color of light sources) is unrealistic.

Deep learning models, in particular, are known not only for their amazing performance in dealing with a large variety of computer vision tasks but also for the lack of interpretability of their outputs, which, in turn, makes it hard to predict how they could react to a change in illumination.

There are a few ways to build robust deep learning systems for computer vision:

M. Cotogni: Work performed while at University of Pavia.

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- normalize input images to discard the contribution of unwanted variations in the input;
- use data augmentation to train the model to tolerate certain kinds of perturbations;
- make the model *equivariant* to a suitable set of transformations.

This paper explores the third option to build robust convolutional networks for illuminant estimation.

Briefly, a system f is *equivariant* with respect to a set of transformations T when applying one of such transformations to the input x causes a predictable change in the output $f(x)$:

$$f(t(x)) = t'(f(x)), \quad t \in T,\ t' \in T', \tag{1}$$

where t' can be derived from t. A special case is that of *invariance*, which occurs when t' is the identity. For instance, fully convolutional neural networks are equivariant to translations, and graph neural networks are (often) equivariant to permutations of the vertices. Equivariance to photometric transformations was first proposed in [10], where we focused mainly on the problem of robust image recognition. This work addresses using equivariant networks for illuminant estimation and expands the experiments to different neural architectures.

Here, we will address the problem of illuminant estimation, which consists of predicting the color of the light source that illuminated the input scene. Illuminant estimation is often used for computational color constancy, that is, to render the image as if it was taken under a neutral illuminant. Equivariance will ensure that a change in the light source's color will correspond to an equal change in the prediction. Therefore, in this case, $T = T'$ will be the set of color changes, and $t = t'$ in Eq. 1. An equivariant illuminant estimator would behave as depicted in Fig. 1.

The rest of the paper is organized as follows: Sect. 2 briefly reviews the literature on equivariant networks and illuminant estimation; Sect. 3 introduces illuminant equivariant networks; Sect. 4 presents and discusses the experimental results. Finally, Sect. 6 summarizes the findings of this work and introduces possible directions for future research.

2 Related Works

Illuminant estimation methods can be divided into two categories: statistics-based and learning-based. The former category makes assumptions about the statistical properties of natural scenes to estimate the color of the illuminant [30]. Learning-based methods estimate the illuminant's color by exploiting the inner knowledge embedded in a training set of annotated data. In the last decade, learning-based methods have prevailed because of their superior accuracy. Many of these methods exploit especially designed features [5,7,12,14,26], others work by using neural networks to directly process the input images [1,3,4,18,23,28].

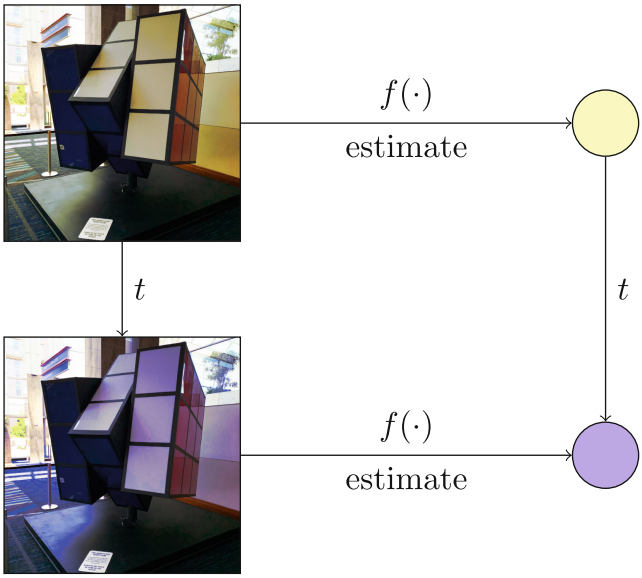

Fig. 1. Equivariance to a global change in the color of the light source: the estimated illuminant color on the transformed image $f(t(x))$ (on the bottom), is identical to $t(f(x))$ representing the estimate for the original image (on the top) subjected to an equal color transformation.

A more thorough presentation of the problem has been given in the seminal works [15,17].

Equivariance is a crucial property in neural networks as it allows for the imposition of an inductive bias [25]. In convolutional networks, equivariance enables the exploitation of the knowledge that objects and patterns can appear in different locations within an image [20,21]. For graph neural networks, it ensures that the model's performance is independent of the vertex labeling order in graphs [24]. In 3D processing, equivariance facilitates the generalization of convolutions to 3D meshes [11]. Various convolutional network models have been proposed in computer vision to achieve equivariance with respect to geometric transformations, such as translations, rotations, scalings, and reflections [8,9]. However, achieving equivariance with respect to photometric transformations, which include variations caused by changes in lighting conditions or the acquisition device is still an open issue addressed by few works [10,22,29]. These photometric variations are typically addressed using normalization techniques (e.g., color balancing), which can lead to the loss of potentially valuable information [2].

3 Illuminant Equivariant Networks

An illuminant equivariant network can be achieved by combining a logarithmic transformation with an offset equivariant network, that is, a network preserving uniform additions to the color channels. The logarithmic transformation of global illumination is needed to transform the multiplicative effect of a change in illumination in the Von Kries Model [19] into the addition of an offset. In the following, we assume that images are in the logarithmic $R'G'B'$ space defined as follows:

$$\begin{aligned} R' &= -\log(\max\{R, \epsilon\}), \\ G' &= -\log(\max\{G, \epsilon\}), \\ B' &= -\log(\max\{B, \epsilon\}), \end{aligned} \tag{2}$$

with $R, G, B \in [0, 1]$ and $\epsilon = 10^{-4}$ used to prevent a null argument for the logarithm. In this color space, we can model a global color change as the addition of an offset $\Delta \in \mathbb{R}^3$ to all pixels in an image:

$$(R', G', B') \to (R' + \Delta_R, G' + \Delta_G, B' + \Delta_B), \tag{3}$$

with $\Delta = (\Delta_R, \Delta_G, \Delta_B)$.

In this context an offset equivariant function $f : \mathbb{R}^m \to \mathbb{R}^n$ would behave as follows for all global color offsets $\Delta \in \mathbb{R}^3$:

$$f(x + G_m \Delta) = f(x) + G_n \Delta, \tag{4}$$

where $G_m \in \mathbb{R}^{m \times 3}$ (and, similarly, $G - n$) is a matrix assigning features to one of the three color channels and is defined as the 3×3 identity matrix repeated $m/3$ times:

$$G_m = \begin{pmatrix} I \\ I \\ \vdots \\ I \end{pmatrix}. \tag{5}$$

In this case, we have the following transformations: $t(x) = x + \Delta G_m$ and $t'(x) = x + \Delta G_n$.

Note that offset equivariance is closed under functional composition. Therefore, we can design an offset equivariant neural network by combining multiple offset equivariant layers. Here, we will discuss the layers needed to implement the architecture used in the experiments. We omit those, like pooling layers, which are already offset equivariant without any modification.

3.1 Linear Layers

For linear layers, Eq. (4) becomes:

$$W(x + G_m \Delta) + b = Wx + b + G_n \Delta, \quad \forall \Delta \in \mathbb{R}^3, \tag{6}$$

which results in the following linear constraint:

$$WG_m = G_n, \tag{7}$$

which can be enforced by redefining the linear layer as follows (G_m^+ denotes the pseudo-inverse of G_m):

$$\text{linear}(x) = (W - (WG_m - G_n)G_m^+)x + b. \tag{8}$$

3.2 Convolutional Layers

Convolutions are a special kind of linear operator. Therefore, the method used to make a convolutional layer equivariant is similar to that described in the previous section. Here, the weights W form a four-dimensional array $k \times k \times n \times m$ and the linear constraint analogous to (7) is:

$$\widetilde{W}G_m = G_n, \quad \widetilde{W}_{ij} = \sum_{l=1}^{k}\sum_{h=1}^{k} W_{lhij}. \tag{9}$$

The convolutional operator enforcing the constraint is:

$$\text{conv}(x) = \frac{1}{k^2}(W - (WG_m - G_n)G_m^+)) \star x + b, \tag{10}$$

where $\star$ denotes convolution between the four-dimensional weight array and the three-dimensional x (height $\times$ width $\times$ m). Note that the row-by-column products like WG_m have been extended to act on the last dimension:

$$(WG_m)_{lhij} = \sum_{s=1}^{m} W_{lhis}(G_m)_{sj}. \tag{11}$$

3.3 Activation Function

As activation function, we used a modified ReLU, defined as:

$$\text{activation}(x) = \max(x, \text{avg}(x)), \tag{12}$$

where avg is the global average pooling, and where the maximum is repeated for each component in x. The resulting $\mathbb{R}^m \to \mathbb{R}^m$ operator is equivariant:

$$\begin{aligned} \text{activation}(x + G_m\Delta) &= \max(x + G_m\Delta, \text{avg}(x + G_m\Delta)) \\ &= \max(x + G_m\Delta, \text{avg}(x) + \Delta)) \\ &= \max(x, \text{avg}(x))) + G_m\Delta \\ &= \text{activation}(x) + G_m\Delta. \end{aligned} \tag{13}$$

3.4 Training

In summary, to build an offset equivariant network for illuminant estimation, we need a conventional network processing (log) RGB images and yielding triplets of values that can interpreted as illuminant estimates. Then, we replace each layer with its equivariant counterpart. Training the equivariant network does not require any special care. Any variant of backpropagation and gradient descent can be used to minimize a suitable loss function, such as the angular error between the target and the estimated illuminants.

4 Results

We tested the effectiveness of the illuminant equivariance networks in the illuminant estimation task, which is a crucial step for color constancy. In this task, the objective is to predict the color of the light sources in an image and then compensate it to obtain an output image that shows natural illumination (as perceived by the human visual system).

We used the NUS Dataset [6] as a reference benchmark in our experiments. This dataset, specifically designed for the computational color constancy task, comprises 1853 images captured with nine cameras. The images in the dataset include both indoor and outdoor scenes. The coordinates of the color checkerboard inserted in each image are provided, along with the coordinates of the small region masks for every color patch. The ground truth illuminants were obtained by using the difference between the two brightest achromatic patches containing no saturated values, eliminating the effects of dark levels and sensor saturation.

As the illuminant estimator, we use the Color Cerberus Network [27]. This neural model takes a 64×64 image as input and provides an illuminant estimate as output. As shown in Fig. 2, this neural network is composed of pooling layers followed by a cascade of four blocks, each consisting of a 3×3 convolutional layer, a ReLU activation, and a 2×2 Max pooling layer. Finally, a convolutional layer followed by a linear head provides the illuminant estimation.

We instantiate an original Cerberus model following the implementation described in the original paper, and an equivariant version where the original layers have been replaced by their equivariant counterparts. Figure 2 highlights the differences between these two versions. We trained both models according to the procedures in the original paper and evaluated them using three-fold cross-validation on the NUS dataset. The metric used as reference was the Reproduction Angular Error (RAE) [13], defined as:

$$RAE(\hat{I}, I) = \arccos\left(\frac{\hat{I}_r/I_r + \hat{I}_g/I_g + \hat{I}_b/I_b}{\sqrt{3\left(\hat{I}_r^2/I_r^2 + \hat{I}_g^2/I_g^2 + \hat{I}_b^2/I_b^2\right)}}\right), \tag{14}$$

where $\hat{I}$ and I are the estimated and the target illuminant, respectively.

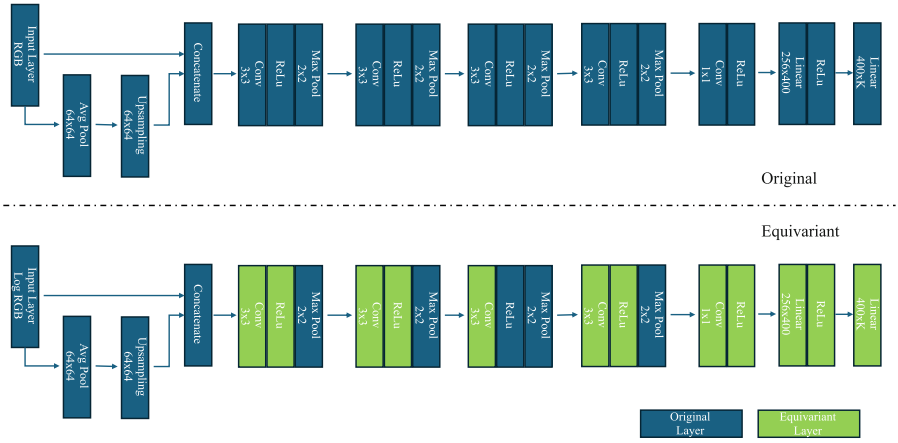

Fig. 2. Color Cerberus Architecture. The upper part of the figure shows the original architecture, while the lower part presents its equivariant version. Green blocks indicate that the original layers have been replaced with their equivariant versions.

Table 1. Comparison of original and offset equivariant networks on NUS images. Performance is measured in terms of median reproduction angular error and computed by three-fold cross-validation on the whole dataset. Three levels of distortions are considered: no distortion ($S = 0.0$), moderate ($S = 0.5$), and strong distortion ($S = 0.9$).

			Median RAE (deg)		
Architecture	Variant	# par.	S 0.0	S 0.5	S 0.9
Cerberus	Original	206 345	**1.93**	11.5	30.3
	Equivariant	196 668	2.03	**2.30**	**3.44**
Resnet	Original	265 566	1.81	12.98	32.20
	Equivariant	265 566	2.71	5.14	8.84

We report the results in Table 1. The two trained versions, the original and the equivariant show similar results for the Median RAE computed across the three-fold cross-validation on the NUS dataset ($S = 0$). This represents the standard scenario, with the original illuminant found in the dataset. We then evaluated the behavior of the two trained models by introducing a distortion in the test images. More precisely, we applied an artificial illuminant with a random hue and variable saturation:

$$I = I_{\text{NUS}} \odot \text{HSV-TO-RGB}(H, S, 1), \tag{15}$$

where I_{NUS} is the original illuminant, as taken from the dataset, H is randomly taken in the range $[0° - -350°)$, and $S \in [0, 1]$ is a parameter tuning the amount of distortion ($S = 0 \implies I = I_{\text{NUS}}$).

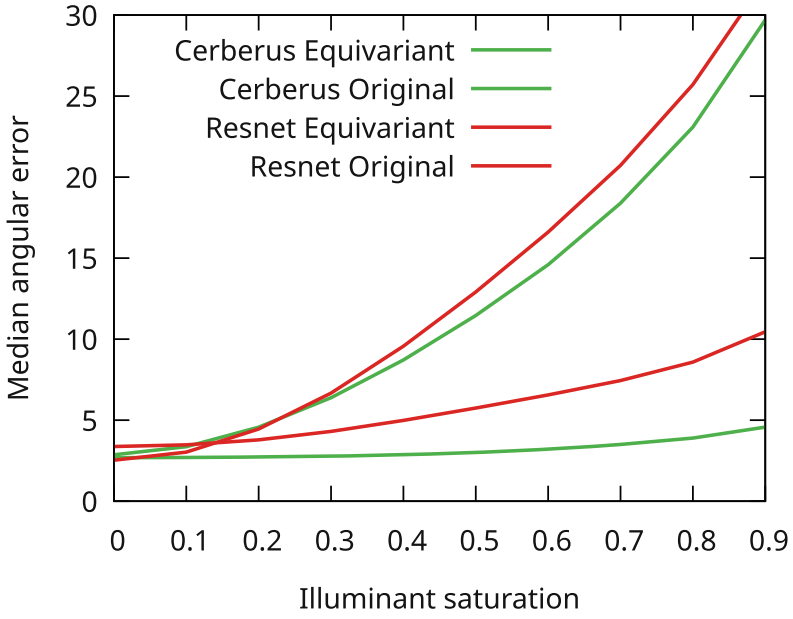

Fig. 3. Median Angular Error contained by the original models and their equivariant versions on the illuminant estimation task on the NUS Dataset, varying the saturation of the illuminant

Results are shown in Table 1 in the columns labeled $S = 0.5$ and $S = 0.9$. As observed, the error of the original model increases significantly and becomes very high when the saturation of the illuminant reaches 0.9. This is also illustrated in Fig. 3, where the error grows more than linearly. In contrast, the equivariant model shows remarkable stability, and the error increases only slightly, even at very high saturation levels. This is also visible in Fig. 4: the equivariant model can correctly estimate and compensate for the illuminant even when the image is highly distorted (the hue in the images is due to the random hue applied as a distortion factor). This result confirms that our illuminant equivariant version of the neural network can handle images with significant illuminant distortions without augmentation or pre-processing simply by training the network as with the original model on non-distorted images.

To show the generality of the approach, we extended the experiment to a different architecture, the ResNet-20. Table 1 presents a comparative analysis of the performance of the original and equivariant versions of the ResNet-20 architecture [16]. The original ResNet-20 model outperforms the equivariant version in the baseline scenario ($S = 0.0$). However, as the saturation increases, the original model's performance degrades significantly. At moderate saturation ($S = 0.5$), the original ResNet model shows a Median RAE of 12.98, whereas the equivariant model maintains a much lower error. This trend continues at high saturation levels ($S = 0.9$), where the original model's error rises sharply

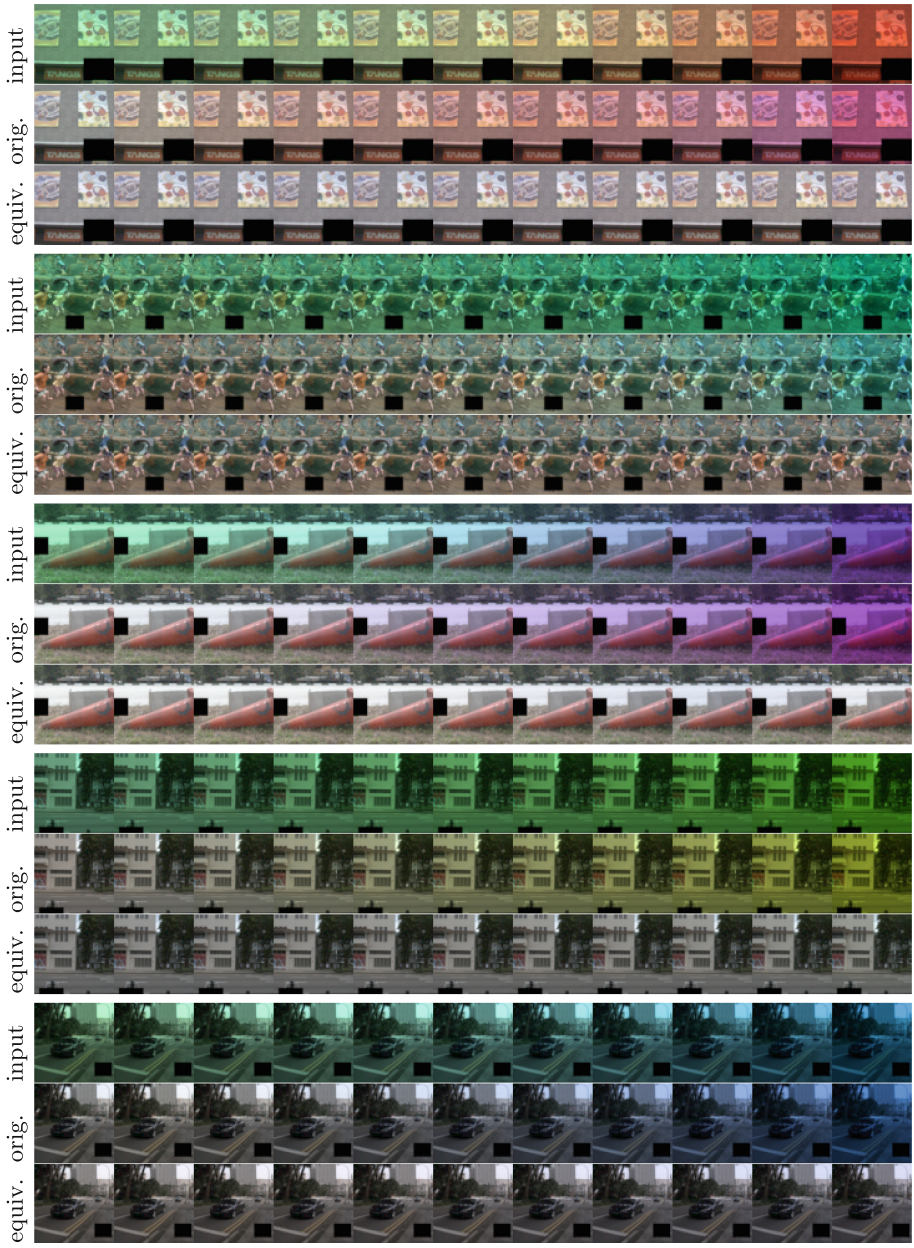

Fig. 4. Examples of images balanced using the estimates found by the original and the equivariant Cerberus networks. The input images have been distorted by introducing an extra illuminant with variable saturation.

while the equivariant model's error remains relatively low. In Fig. 3, we also compare the two versions of the ResNet-20. The original ResNet model's error increases dramatically with higher saturation, indicated by the steep slope of the dashed red line. In contrast, the equivariant ResNet model exhibits a much more gradual increase in error, depicted by the solid red line, demonstrating its robustness to illuminant distortions. These results demonstrate that while the original ResNet model performs well under standard conditions, its performance deteriorates with increasing illuminant saturation. In contrast, the equivariant ResNet model exhibits robust performance across all tested conditions, confirming the effectiveness of equivariant networks in handling images with varying illuminant color.

The comparison of equivariant versions of ResNet-20 and Cerberus shows that both models exhibit robust performance under varying illuminant saturation levels. However, the equivariant Cerberus model consistently outperforms the equivariant ResNet-20 across all tested saturation levels. This indicates that while both equivariant models improve stability and accuracy under challenging conditions, the Cerberus architecture, specifically designed for this task, offers superior robustness to illuminant distortions compared to ResNet-20.

5 Limitations

The proposed offset-equivariant framework is very versatile. However, it cannot be easily applied to all components in modern neural networks. For instance, we found two kinds of layers problematic: normalizations and residual connections. Fixing these was possible (see [10] for the details) but at the cost of contrasting their original purpose. The result is that our equivariant models require a longer training time than their non-equivariant counterparts.

Another limitation of this work is the use of artificially distorted images. Using images with real variations in the lighting conditions is extremely challenging. We plan to address this point in future works.

6 Conclusions

In this paper, we investigated the effectiveness of illuminant equivariant neural networks for illuminant estimation. We introduced and implemented the equivariant versions of two state-of-the-art neural networks, specifically the Cerberus and ResNet-20 architectures. These networks are designed to be equivariant to photometric transformations, enabling them to handle changes in lighting conditions without requiring specific data augmentation during training. Instead, they achieve this through mathematical derivation.

We evaluated the performance of these equivariant networks on the NUS dataset. The results show that the equivariant models consistently maintain stable and low error rates across different levels of illuminant saturation compared to the original architectures. In particular, the equivariant Cerberus model

demonstrated superior robustness, significantly outperforming its original counterpart and the equivariant ResNet-20 model, especially under high saturation conditions. These findings confirm the robustness and generalizability of the equivariant neural networks in handling images with varying illuminant distortions.

In future work, we plan to extend this study in different directions. In particular, we seek to extend equivariance to other photometric transformations, like those related to non-diagonal illumination models, or local changes in illumination. We also plan to explore the definition of equivariant networks involving alternative architectures, such as Vision Transformers, and different domains like 3D and hyperspectral images.

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Revisiting Color Constancy Using CNNs: Including Recent Observations

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Abstract. In recent studies, alongside introducing new approaches for color constancy, we have also focused on improving existing techniques and introducing new perspectives. Our motivation is the idea that investigating different strategies, concepts, and their combinations that have not been analyzed in this field in detail yet, might help us to find simple, effective, and cost-efficient solutions. Thereupon, we utilized observations we obtained from our algorithms to analyze how we can enhance the performance of well-known learning-free methods. We demonstrated why using *salient* pixels, performing block-based operations, and carrying out scale-space computations benefit color constancy approaches significantly and provide a notable performance increase. In this study, we make use of our recent observations on learning-free algorithms to analyze if they are also beneficial for enhancing the performance of a learning-based color constancy model. According to our evaluation, all of our observations contribute to the performance of a convolutional neural network model and increase its effectiveness in estimating the illuminant. Thus, the contribution of these strategies in learning-based models should be further investigated to improve their performance with simple yet effective solutions.

Keywords: Color constancy · illuminant estimation · white-balancing

1 Introduction

Our visual system has evolved in such a way that it can discount the illuminant of the environment, and perceive the true colors of the objects [61]. For instance, when we enter a room illuminated by purple light, we are able to recognize the color of a white object as white. This ability of unconsciously discounting the light source of a scene is called *color constancy*, and it is usually also referred to as a phenomenon since the biological mechanism behind this ability is not yet fully understood. While we can perform color constancy unintentionally, machine vision systems require guidance to perform this task since as a result of the interaction between the sensitivity of the camera sensors and the light source of the scene, a white object illuminated by purple light would be captured as purple [55]. The field analyzing this problem from the perspective of computer science is called *computational color constancy*. It benefits both digital photography and various computer vision tasks including but not limited to underwater image enhancement and image dehazing.

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In computational color constancy, we utilize an image formation model where we mostly assume that *(i)* the surfaces are Lambertian, i.e., the surface is equally reflecting the light into all directions, and *(ii)* there is a point light source illuminating the scene. Furthermore, we presume that the camera capturing the scene consists of three sensors that detect the energy of incoming light by reacting to a particular segment of the visible spectrum, e.g., short-, middle-, and long-wavelengths. By taking these assumptions into account, we can mathematically express an image I based on the irradiance E falling onto the sensors of the capturing device, and the sensor sensitivity function S of the camera representing the responses of the sensors to light at a particular wavelength as follows:

$$I(x, y) = \int_w E(x, y; \lambda) S_i(\lambda) d\lambda, \tag{1}$$

where (x, y) is the spatial location, λ corresponds to the wavelength of the visible spectrum w, and $i \in \{\text{long, middle, short}\}$.

We can formulate the irradiance falling onto the sensors of the camera by making use of its relationship with the reflectance $R(x, y; \lambda)$, the scaling factor $G(x, y)$, and the point light source $L(x, y; \lambda)$ as follows:

$$E(x, y; \lambda) = G(x, y) R(x, y; \lambda) L(x, y; \lambda), \tag{2}$$

where $G(x, y)$ can be taken as $cos\ \alpha$, where α is the angle between the surface normal vector and a vector pointing in the direction of the illumination source.

We can model an image by utilizing Eqs. 1 and 2 as follows:

$$I(x, y) = G(x, y) \int_w R(x, y; \lambda) L(x, y; \lambda) S_i(\lambda) d\lambda. \tag{3}$$

In color constancy, we aim at computing L from a color cast image I by taking advantage of Eq. 3. Yet, despite the aforementioned assumptions the ill-posed nature of the problem cannot be overcome due to the fact that the image is affected by both the lighting conditions and the often unknown sensor characteristics in real-world scenarios. Therefore, to relax the problem, we generally make additional assumptions such as a single light source is illuminating the scene, G does not affect the computation of the illuminant, and the responses of the capturing device are narrow-band, e.g., they can be approximated by Dirac's delta functions [21]. Consequently, a color-cast image can be represented as the Hadamard product of the (shaded) reflectance R and the single light source $\mathbf{L}$ as follows:

$$I(x, y) = R(x, y) \odot \mathbf{L}. \tag{4}$$

Over the last decades, various single-illuminant color constancy algorithms have been introduced which depend on different strategies [21]. These algorithms can be categorized as traditional and data-driven methods. The former utilizes image statistics to compute the color vector of the light source. The most well-known traditional methods are white-patch Retinex [39] and gray world [13] whose assumptions lay the foundations of several other classical color constancy algorithms [10,16,24,29,35,45,55,57]. Traditional methods are mostly cost-efficient and easy to implement, however, their ability of computing the color vector of the light source may decrease significantly

if only a limited number of different colors is available in the scene [14]. Close-up shootings and scenes with dominant sky or grass regions can be given as examples of scenes that are challenging for classical color constancy algorithms. On the other hand, data-driven methods generally achieve higher accuracy on images having a uniform color distribution. Data-driven models can be further grouped as gamut-based methods [9,22,23,25,30], Bayesian strategies [11,12,27], and neural network-based models [1,8,17,18,34,37]. Neural networks-based models estimate the illuminant by learning from the features present in their large-scale training sets [15]. Early neural network models make use of convolutional neural networks [4,7] which are still widely used in this field [1,6,18,34].

In our previous studies, we investigated the effects of utilizing block-based operations, *salient* pixels, and conducting scale-space computations in the field of color constancy. We showed that these operations can improve the performance of existing traditional color constancy algorithms, and we demonstrated that we can introduce new classical methods based on these strategies [54–58]. In this study, we utilize all our previous observations on learning-free algorithms and apply them to a convolutional neural networks (CNNs) model to ascertain whether they might also be useful for learning-based strategies. For our investigation, we adopt the CNNs model introduced by Bianco *et al.* [7] as our baseline architecture since *(i)* the simple structured model contains a low number of parameters, and *(ii)* it is specifically designed for estimating the color vector of light sources based on the image's locally varying statistics, i.e., image patches, which overlaps with one of our previous observations. We modify this model with a mechanism to guide the model to only consider the salient pixels since not all regions are informative while estimating the illuminant [35,55]. Moreover, we extend this model to multiple scales since estimating the illuminant with a scale-space approach improves the performance [57,58]. These two observations also coincide with experiments on the human visual system which we often try to mimic in computer vision, i.e., the human visual system may benefit from regions with the highest luminance while discounting the effects of the light source [20,42,43,50], and the illuminant estimate is available in the stimulus at the proper scale [48,49]. In our experiments, we demonstrate that our simple yet effective operations are beneficial for a learning-based model as well.

This paper is organized as follows. In Sect. 2, we revisit the color constancy model introduced by Bianco *et al.* [7], while also introducing our simple modifications and detailing the training phase. In Sect. 3, we introduce our experimental setup and discuss the performance of the model. Finally, in Sect. 4, we summarize our work.

2 Modifications Based on Recent Observations

As we discussed in Sect. 1, the modifications we proposed are based on our recent observations on learning-free color constancy algorithms. In order to investigate whether the observations are also beneficial for a learning-based approach, as our foundation model, we utilize the approach introduced by Bianco *et al.* [7]. In this section, we revisit the architecture, introduce our modifications, and detail our training procedure. A flowchart of the baseline model, together with our modifications can be seen in Fig. 1.

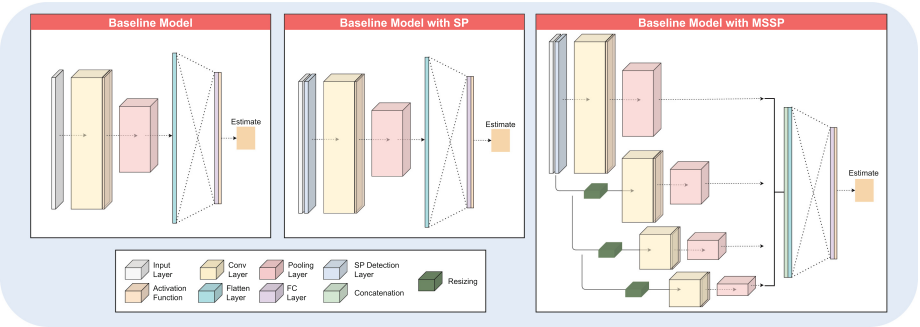

Fig. 1. The flowchart of the baseline model along with our modifications. The abbreviations SP and MSSP correspond to salient pixels and multi-scale with salient pixels, respectively.

2.1 Revisiting Color Constancy Using CNNs

The network consists of only 5 layers. The input layer takes in $32 \times 32 \times 3$ RGB image patches. Afterwards, a convolutional layer utilizes a bank of 240 $1 \times 1 \times 3$ kernels with a stride of 1 to filter the input. This operation results in a feature map with a size of $32 \times 32 \times 240$ which is reduced to $4 \times 4 \times 240$ by using a max-pooling layer with a 8×8 kernel. The obtained feature maps are then reshaped into a 3840 dimensional vector and sent to a fully connected layer of 40 nodes. Subsequently, a ReLU activation function is utilized, and a 40×3 dense layer is added to obtain the illuminant estimate. Consequently, the model consists of a total 154,723 trainable parameters.

2.2 Adjusting the Baseline Model

We prefer to maintain the input requirements of the network since Bianco *et al.* [7] stated that a block size of 32×32 is sufficient while smaller ones are not, and since we recently investigated the impact of block sizes on color constancy and demonstrated that there is indeed a certain block size range to effectively perform color constancy [55, 57, 58]. Adjusting the block size appropriately is important since when we choose large image blocks we might neglect the varying surface orientations present at local regions which decreases the considered amount of information [55]. On the other hand, if we select small block sizes the possibility of obtaining uniformly colored inputs increases, thus, the information that can be extracted from a block decreases [55]. Therefore, to respect the local information varying throughout the scene, and to keep the number of parameters of the model adequately low, we utilize image blocks with the size of $32 \times 32 \times 3$.

Salient Pixel Detection Layer. We first add a mechanism to the model to assign larger weights to the brightest pixels in the image and to decrease the impact of the non-informative elements. We can explain the reason behind the choice of this mechanism based on biological findings and from the perspective of computational photography [59]. According to biological studies the human visual system might be discounting

the illuminant of the scene by relying on the regions having the highest luminance rather than the darkest areas [40,42,43,50]. Furthermore, in digital photography it is known that estimating the illuminant can be easier when achromatic regions are used rather than colored ones [19,35,44,45,54,58]. In this study, to utilize the salient pixels, i.e., brightest pixels, we obtain the luma of the image block and determine the pixels having highest brightness by selecting the top brightest pixels in the block. It is important to stress that while selecting the brightest pixels, we exclude the over-saturated pixels to reduce possible noise since these elements negatively affect the performance of the color constancy algorithms [55]. According to our experiments, choosing 50% or more of the brightest pixels is sufficient to estimate the color vector of the light source. We select this threshold since in our previous experiments we determined that approximately 3% of the brightest pixels for a whole image is adequate to effectively estimate the illuminant [55]. Since we consider much smaller image regions in this study, i.e., blocks with the size of $32 \times 32 \times 3$, we adjust the number of the brightest pixels accordingly. After highlighting the top brightest pixels, we mask the darkest regions and weigh the input block with the resulting saliency map. It is worth noting that achromatic pixels have already been used in learning-based models as salient pixels [6]. However, the model used in that study requires a significantly high number of trainable parameters due to the nature of the U-Net architecture [47]. On the other hand, our simple weighting process allows us to avoid raising the complexity of the model while providing a significant performance increase which we demonstrate in Sect. 3.

Scale-Space Computations. Lastly, we consider scale-space computations in our approach since they allow us to capture local details and global context more effectively, and enable us to highlight discriminative patterns that are not apparent at an individual scale. Furthermore, we prefer to perform operations at multiple levels since for studies utilizing the color feature, the scale-space computations are beneficial thanks to their sensitivity to low-level features [28,41,51]. As color is a low-level feature, utilizing scale-space computations naturally benefits the task of illumination estimation [57,58]. This is also not surprising, considering that in the human visual system, the information we require for accurate local estimates of the illumination might be available in the stimulus at the appropriate scale [48,49]. To mimic the scale-space computations in the model, we downsample either the input block or the input block weighted by the salient pixels. We consider only the first 4 levels of the image pyramid since the spatial information degrades significantly at further coarser scales [53]. After forming the image pyramid, at each scale, we apply a convolutional layer with 240 filters followed by a max-pooling layer with distinct kernel sizes to pool the feature maps. For instance, we pool the features coming from the finest scale via a max-pooling operation with a kernel size of 8×8, while for the consecutive scale, we select a kernel size of 4×4. This ensures that we maintain a balance between the preservation of spatial information and the reduction of the dimensions of the feature maps. Thus, we optimize the feature extraction process for each scale and enhance the model's ability to capture various patterns across different levels of the pyramid. Subsequently, we combine the features into a vector and forward them to the fully connected layer. The scale-space computa-

tions increase the number of trainable parameters to approximately four times that of the baseline model since we are considering the first four levels of the image pyramid.

2.3 Training Phase

Data Augmentation. For the training process, we utilize the ColorChecker RECommended dataset [27], which is detailed in Sect. 3.1, to stay consistent with the prior study [7]. However, we employ a different data augmentation strategy from the prior work based on the observations from our previous study in which we demonstrated that the performance of learning-based models tends to decrease significantly when only a limited number of illumination conditions are considered in their training sets [56]. Motivated by this observation, and the fact that 568 images and illumination conditions alone are inadequate for the effective training process, we augment this dataset as follows. First of all, by using the linear input images and their corresponding ground truth color vector of the light sources, we obtain ground truth white-balanced images via the inversion of Eq. 4. Then, to augment the dataset by considering various single-illuminant light source conditions, we collect additional color vectors of the illuminant that are present in well-known benchmarks, i.e., the INTEL-TAU [38], and Cube+ [3] datasets. In case there is a repetitive illuminant vector after combining the ground truths of benchmarks, we discard the color vectors having the same RGB triplet [52]. Thus, we obtain an illumination set with unique color vectors of the light source. Afterwards, for each ground truth image, we randomly select 3 illuminants from this set, and by using Eq. 4, we render a total of 1704 new scenes. It is important to note that, while rendering the images, we remove the randomly selected illuminant from our illumination set to prevent rendering the scenes with the same color vector of the light source. Lastly, we extract $32 \times 32 \times 3$ non-overlapping blocks from the images and remove blocks containing a color calibration object, consequently, we obtain a total of 1307943 blocks. Through utilizing image blocks rather than the whole image, we obtain a high number of inputs that satisfy the requirements of learning-based models [7].

Training Parameters Our model learns its parameters by minimizing the Euclidean loss. We train our model using the Adam optimizer whose exponential decay rate for the first and second-moment estimates is assigned as 0.9 and 0.999, respectively. We set the number of epochs to 100 and the learning rate to $10e-7$. We clip the gradient of each weight individually to avoid that its norm is higher than 1. Furthermore, we consider to add a weight decay of $10e-8$ for better generalization. It is important to note that we did not fine-tune the model according to the angular error as in the prior work [7].

3 Experiments and Discussions

In this section, we detail our experimental setup, while briefly introducing the datasets and the evaluation strategy that is widely adopted in color constancy. To report the results of algorithms, we either run the original codes without any optimization, utilize the outcomes of the original works, or use the reported results from recent comprehensive publications [58].

3.1 Datasets

In our experiments, we utilize 2 publicly available, comprehensive single-illuminant benchmarks covering different lighting conditions and camera specifications. To evaluate our model on these datasets, if necessary, we perform preprocessing, i.e., masking the calibration objects, substracting the black level, and clipping the under- and over-saturated pixels according to the masks provided in the benchmarks.

The ColorChecker RECommended dataset is the processed version of the Gehler-Shi dataset [27,32]. The benchmark contains a total of 568 distinct scenes: 254 indoor, 85 outdoor, and 229 close-up images [14], and their corresponding ground truths along with the masks to handle under- and over-saturated regions, and the position of the calibration object. Two cameras, i.e., Canon 1D and Canon 5D, are utilized to capture the scenes. The INTEL-TAU dataset is introduced by Lakoom *et al.* [38], and it includes a total of 7022 images: 1466 indoor, 2327 outdoor, and 3229 close-up scenes [14]. The images are captured with 3 different cameras, namely, Nikon D810, Canon 5DSR, and Sony IMX135. All the scenes in INTEL-TAU are preprocessed, i.e., images have a linear response, and their black level is calibrated. Moreover, all the sensitive data, i.e., faces, and license plates, is masked. During our experiments, we only utilized the scenes from the sets of "*field1*" and "*field3*" since some of the images in the other sets, i.e. "*lab printouts*", and "*lab realscene*", include an unmasked calibration object.

3.2 Evaluation Metric

To statistically investigate the model's performance, we adopt a widely used error metric in color constancy, namely, angular error [33]. We report the mean, median, and mean of the best and worst $\%25$ of the angular errors.

The angle (ε) between the color vector of the estimated of the light source (L_{est}) and the ground truth illuminant (L_{gt}) can be calculated as follows:

$$\varepsilon(L_{est}, L_{gt}) = cos^{-1}\left(\frac{L_{est}L_{gt}}{\|L_{est}\|\|L_{gt}\|}\right). \tag{5}$$

3.3 Discussion

We report our statistical analysis for both benchmarks in Table 1. The first noticeable result is that modifying the baseline model with our strategies improves the performance significantly. Each stage, i.e., guiding the network with information coming from the salient regions, and carrying the model into multiple scales, enhances the overall performance of the baseline method. Furthermore, the worst cases almost always improve significantly through the modifications which is a valuable outcome since in color constancy it is known that improving the algorithms' performance for the worst cases is important.

We obtain the best outcomes when we either modify the baseline model with the salient pixels or consider salient pixels with the scale-space computations. If we compare the contribution of the salient pixels and the scale-space computations, it is clear that considering the salient pixels benefits more in general. This result can be explained

Table 1. Statistical results on 2 benchmarks. The top results are highlighted using color coding as follows, first: blue, second: cyan, third: green, forth: yellow, and fifth: orange. The abbreviations SP, MS and MSSP correspond to the salient pixels, multi-scale, and multi-scale with salient pixels, respectively.

		ColorChecker RECommended				INTEL-TAU			
	Algorithms	**B-25%**	**W-25%**	**Mean**	**Med.**	**B-25%**	**W-25%**	**Mean**	**Med.**
Learning-free	max-RGB [40]	1.49	17.47	7.78	5.43	1.70	19.24	10.49	11.14
	GW [13]	0.93	10.44	4.71	3.54	0.93	10.59	4.90	3.85
	1^{st} - GE [60]	0.93	14.17	5.79	3.68	0.94	13.79	5.89	4.07
	2^{nd} - GE [60]	1.03	14.70	6.09	3.97	1.00	14.13	6.09	4.25
	wGE [31]	0.78	15.57	6.08	3.33	0.80	14.89	5.99	3.63
	DOCC [26]	0.79	18.04	7.23	4.26	0.80	16.97	7.18	4.66
	MSGP [46]	0.76	8.35	3.81	2.96	0.64	8.23	3.57	2.56
	GI [45]	0.44	8.02	3.19	1.90	0.56	8.03	3.32	2.18
	BIO-CC [54]	0.86	9.84	4.40	3.30	0.76	9.42	4.14	3.05
	BB-CC [55]	1.06	7.37	3.48	2.71	0.79	7.25	3.37	2.63
	MSCC [58]	0.62	7.32	3.16	2.16	0.59	7.47	3.23	2.23
Learning-based	Quasi-U CC [6]	-	-	3.46	2.23	0.60	7.28	3.12	2.19
	SIIE [2]	0.55	6.53	2.77	1.93	0.73	7.80	3.42	2.42
	C3AE [37]	0.80	4.00	2.10	1.90	0.90	7.00	3.40	2.70
	One-Net CCC [18]	-	-	-	-	1.10	5.90	3.30	3.20
	FFCC [5]	0.57	6.75	2.95	2.19	0.70	7.96	3.42	2.38
	FC4 [34]	0.34	4.29	1.77	1.11	0.70	5.50	2.60	2.00
	C5 [1]	0.53	5.46	2.50	1.99	0.52	5.96	2.52	1.70
	BoCF CC [36]	-	-	-	-	0.90	6.10	2.90	2.40
	Baseline	0.92	6.54	3.23	2.64	0.92	6.55	3.20	2.56
	Baseline w/ SP	0.62	5.98	2.70	2.00	0.79	6.11	2.90	2.27
	Baseline w/ MS	0.75	6.32	3.00	2.34	0.84	6.56	3.16	2.51
	Baseline w/ MSSP	0.73	5.92	2.82	2.26	0.82	6.10	2.93	2.29

by the fact that not all regions are informative for color constancy, hence reducing the effect of the non-informative elements allows us to improve the overall performance. It is important to stress that while guiding the network to give more importance to the salient regions, our modification does not increase the number of trainable parameters. Hence, without increasing the complexity of the model, this modification allows us to enhance the performance significantly.

We provide visual investigations in Fig. 2 where we compare different modifications on the baseline model with other algorithms. We provide results chosen from the best and worst cases randomly. The baseline model modified with the salient pixels and/or the scale-space can output visually pleasing results with very low angular errors for the best cases while for the worst cases, an additional analysis should be conducted to enhance the outcomes even further.

As a final note, we would like to mention that in this study, we do not aim to design a state-of-the-art model that outperforms all other color constancy algorithms. We intend to present that with our simple yet effective modifications, we can improve the performance of a network, which we aimed as future direction in our previous works. As shown in Table 1, the baseline model alone cannot compete with the state-of-the-art models, but with our modifications, we can increase its performance so that this model

Fig. 2. Comparison of random results on best and worst cases for each stage of the modification. The angular errors are given on the bottom-right side of the images. Gamma is adjusted for better visualization.

performs well while providing competitive results. Thus, the effects of block-based strategies, salient pixels, and scale-space operations should be investigated thoroughly to improve the performance of learning-based color constancy algorithms.

4 Conclusion

Color is an important feature not only for us but also for a wide range of computer vision applications. Many image processing tasks, such as underwater image enhance-

ment and image dehazing, use color constancy at the first stage of their pipelines. Computational color constancy is framed as a computational challenge that has been widely studied for more than five decades due to its significance. Researchers have developed numerous methods, and have aimed at improving the existing algorithms by exploiting the effectiveness of different approaches to provide simple yet effective solutions to the ill-posed nature of the problem. Motivated by this aim, alongside developing new methods, we also investigate how we can improve the performance of algorithms in estimating the illuminant. In our previous studies, we explicitly demonstrated that the performance of learning-free methods can be significantly enhanced by using the following strategies: *(i)* considering varying local statistics of the scenes, i.e., block-based approaches, *(ii)* utilizing the salient pixels, i.e., the pixels having the highest luminance, and *(iii)* multi-scale operations. We mentioned that we will use our observations to investigate if they are beneficial for a learning-based model. Thereupon, in this study, we use our observations to modify a convolutional neural network designed for color constancy. According to our experiments, guiding the network to give more weight to the salient regions and carrying out scale-space computations improves the performance of the model, and allows it to compete with the state-of-the-art. Consequently, we can conclude that one can use the observations obtained from traditional algorithms to modify learning-based models with simple yet effective approaches to improve their effectiveness without significantly increasing their computational complexity. As future work, we will investigate the effects of our observations on different models, and further improve the algorithms' effectiveness in the worst cases.

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Enforcing Temporal Consistency for Color Constancy in Video Sequences

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Abstract. This paper focuses on enhancing temporal color constancy in video sequences, ensuring that the result not only achieves color accuracy frame-by-frame but is also consistent over time. Our approach consists of a three-step process: per-frame illuminant estimation and correction, video stabilization to ensure temporal consistency, and consensus-based illuminant correction. By employing consensus-driven illuminant estimation over the result of temporal stabilization, we effectively mitigate spatial artifacts and concurrently enhance the overall stability of the sequence. Our method is tested using the Gray Ball and BCC datasets, showing the potential of integrating temporal stabilization with color correction processes to enhance the visual continuity of video content. While our primary objective is to reduce temporal flickering, a significant side effect of our approach is the improvement of color constancy accuracy across frames.

Keywords: Color constancy · Temporal consistency · Temporal stabilization · Automatic white balance. · Video sequences

1 Introduction

Color constancy is the perceptual property of the visual system that ensures the colors of objects remain relatively constant under varying illumination conditions [11]. This process is crucial in digital imaging and is typically composed of two major steps: illuminant estimation and illuminant correction [7]. Illuminant estimation involves determining the dominant light source color in an image, while illuminant correction adjusts the colors in the image to appear as they would under a neutral light source.

Historically, the application of color constancy has been primarily focused on still images. Today, with the increasing availability of devices able to acquire video, the application of color constancy to video sequences, known as temporal color constancy, presents both new opportunities and challenges.

The principal opportunity presented by temporal color constancy is that the abundance of frames in video sequences allows the problem to be more

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effectively constrained. Single frame color constancy is inherently an ill-posed problem, characterized by the potential for multiple viable solutions. Typically, additional information is exploited to constrain this set of solutions, using for example assumptions on the scene content [18], knowledge derived from training data [6], or the physical plausibility of different solutions [9]. A straightforward assumption often utilized is the "gray world hypothesis", which posits that the average reflectance in a scene is achromatic. In the case of temporal color constancy, the additional information crucial for determining illuminant conditions in a given input often comes from the availability of multiple adjacent frames depicting a very similar scene content within a video sequence. This inter-frame assistance enables a more robust estimation of the lighting conditions of individual frames, as frames with clearer illuminant indicators can compensate for neighboring frames where such cues are sparse, ambiguous, or even misleading.

As previously noted, transitioning to the domain of video introduces significant challenges, particularly regarding the consequences of incorrect illuminant estimation and correction in the individual frames. In video sequences, the impact of such errors is, in fact, compounded: not only does each frame suffer from inaccurate color correction, but inconsistencies in the corrections applied across adjacent frames can also lead to flickering. This phenomenon disrupts the visual continuity and creates a distracting and unnatural appearance in the video stream. Avoiding flickering is particularly important in scenarios such as social video sharing, film production, surveillance systems, and other multimedia applications where changes in color perception can be annoying, distracting or even misleading.

Image color constancy is a problem clearly defined as involving a single input image, that is analyzed and adjusted to produce one output image with corrected colors. This approach operates under the principle of one input corresponding to one output, focusing solely on the colors within that specific image. Temporal color constancy, on the other hand, is a much less mature field of research, and is consequently less clearly-defined. In this paper, we distinguish between two possible sub-categories of temporal color constancy, resulting from the different nature of existing datasets [8,17]:

- Video Color Constancy: In this scenario, each frame of a video is processed in consideration of its preceding frames. This method requires adjusting each frame potentially with a different illuminant, i.e. multiple inputs each with its own corrected output. For specific applications such as offline video processing, it is also possible to include subsequent frames in the elaboration of the current one, thus breaking the causality constraint.
- Burst Color Constancy: Unlike video color constancy, burst color constancy treats a sequence of frames as a cohesive whole. All frames collectively inform the correction process, aiming to achieve a uniform illuminant correction across the entire burst. This model considers multiple inputs but produces a singular, consistent output for the whole sequence.

Figure 1 offers a visual representation of the aforementioned concepts, with "per-frame color constancy" bridging the gap between image color constancy and temporal color constancy, at the expense of introducing flickering artifacts.

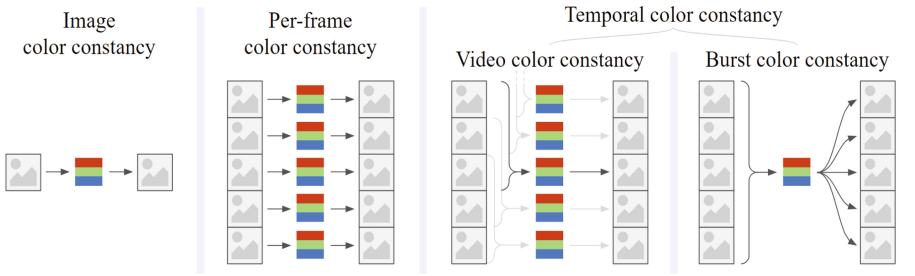

Fig. 1. Visual representation of different applications of computational color constancy to images and video sequences.

There have been some methodologies developed to address temporal color constancy, discussed in Sect. 2. By considering several frames together, these approaches inherently manage the aforementioned issue of flickering.

Alternatively, another approach involves applying image color constancy techniques to each individual frame of a video sequence, followed by a stabilization procedure to process the results and to enforce temporal consistency. Buzzelli and Erba [4] conducted research that delved into the applicability of image color constancy in the temporal domain. They characterized various single-frame methods with respect to their temporal stability. Building on this foundational work, we implement stabilization as a post-processing step to enforce temporal consistency. We evaluate the effectiveness of this approach on the Gray Ball and Burst Color Constancy datasets, and provide several considerations based on the outcomes of our analysis. The advantages of this approach are twofold. First, by decoupling the illuminant correction process from the video stabilization process, it is possible to select, and combine, the best algorithms from both tasks. This includes well-established methods for single-frame color constancy. Secondly, if we resort to a task-independent (agnostic) method for temporal consistency, the same method may also be used to stabilize other aspects of the processed video, correcting for example any artifacts also introduced by per-frame contrast enhancement algorithms.

2 Related Works

In this section we explore the state of the art for temporal color constancy methods, which explicitly account for the temporal dimension of video sequences to produce per-frame or per-sequence illuminant correction.

Wang et al. [19] enhanced illumination estimation for video sequences by exploiting correlations between adjacent frames. Their method segments videos

into scenes, assuming a consistent illuminant for all frames within a scene. This scene-based approach aggregates illuminant data across frames to derive a more stable and accurate chromaticity estimate for each scene. Their experimental results show that this method outperforms traditional single-frame algorithms by effectively using multi-frame information to maintain illuminant consistency across video shots.

Barron et al. [2] introduced the Fast Fourier Color Constancy (FFCC), a novel algorithm that significantly improves upon the traditional methods of illuminant estimation by transforming it into a spatial localization task on a torus in the frequency domain. Specifically for temporal color constancy, they have adapted FFCC to handle video sequences more effectively. They incorporate a smoothing model inspired by the Kalman filter to mitigate errors in individual frame predictions. This enhancement allows FFCC to maintain consistency and accuracy across frames, addressing the common challenge of flickering in video caused by frame-to-frame illuminant estimation variability.

Qian et al. [15] introduced a novel approach to temporal color constancy by considering multiple preceding frames for illuminant estimation. Their method utilizes an end-to-end trainable network, RCC-Net, which incorporates convolutional LSTMs to capture compositional representations over time and space effectively. By adapting the SFU Gray Ball Dataset for a temporal context, they demonstrate that RCC-Net consistently surpasses both the traditional single-frame algorithms and their temporal adaptations in terms of performance. This success is attributed to the network's ability to exploit sequential frame information, enhancing the accuracy of color constancy across video sequences.

Later, Qian et al. [17] expanded the scope of color constancy research by introducing a benchmark for Burst Color Constancy (CC), with a method that uses multiple frames from a sequence to estimate the illumination color of a shot frame, challenging the traditional single-frame approach. Their benchmark comprises 600 real-world sequences captured with a high-resolution mobile phone camera, a fixed train-test split for consistent evaluation, and a baseline method that demonstrates high accuracy across this new and previous datasets. Their study also reports results for over 20 well-known color constancy methods, including recent state-of-the-art developments. Additionally, the method has been enhanced by integrating a more robust backbone network for semantic feature extraction and a 2D LSTM for more effective spatial recurrent processing.

Buzzelli and Erba [4] addressed the limitations of traditional evaluations of color constancy algorithms, which typically rely on angular error analysis of static images. Recognizing the growing use of video in consumer technology, they proposed an expanded evaluation framework that includes temporal and spatial stability. This approach assesses how well these algorithms perform under varying scene conditions unrelated to illuminant changes, like motion of subjects or the camera. Using stable sequences from the Gray Ball and Burst Color Constancy datasets, their analysis revealed that while some algorithms excel in minimizing angular errors, they may falter on stability metrics. This work

emphasizes the need for multi-faceted evaluation criteria in the realm of color constancy, particularly for video applications.

Zini et al. [21] introduced COCOA, a novel strategy for combining color constancy algorithms through a compact neural network architecture. This architecture processes and amalgamates the illuminant estimations from various algorithms, each based on different assumptions about the input scene content. COCOA is versatile, applicable both to single-frame images and video sequences, the latter utilizing a Long Short-Term Memory (LSTM) module to manage varying-length sequences. The approach is tested using only learning-free algorithms that rely on simple image statistics, experimenting on the Shi-Gehler and NUS datasets for still images, and the Burst Color Constancy dataset for videos. Their results indicate that COCOA surpasses other combination strategies in performance, achieving illuminant estimation accuracy on par with more complex and computationally intensive methods. Additionally, the effectiveness of COCOA is maintained even with fewer training instances, and the study includes an assessment of each contributing method's impact on the final estimation accuracy.

3 Proposed Method

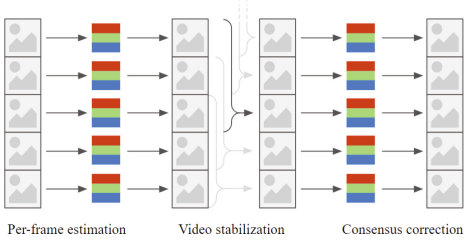

Fig. 2. Visual representation of the proposed three-step method to enforce temporal consistency for color constancy in video sequences.

Let $S = \{x_1, x_2, ..., x_i, ...\}$ be a sequence of frames x. By targeting video color constancy, our objective is to produce an illuminant estimation y_i for each frame, in the form of an RGB triplet. This relationship can be expressed through the function f, mapping the sequence of frames to their corresponding illuminant estimations:

$$E = f(S) = \{y_1, y_2, ..., y_i, ...\}. \tag{1}$$

We address the task through a three-step process: per-frame correction, temporal stabilization, consensus-based correction, as represented in Fig. 2.

In the first step, we generate a preliminary estimation y'_i for each frame, using a single-frame method g, and then we apply the corresponding illuminant correction with function h:

$$x'_i = h(x_i, g(x_i)) = h(x_i, y'_i) \quad \forall x_i \in S, \tag{2}$$

which produces a potentially-unstable sequence $S' = \{x'_1, x'_2, ..., x'_i, ...\}$. We refer the reader to Sect. 4.2 for the specific selection of methods g used in our experimental setup. As described by Eq. 2, before moving to the second step it is essential to apply the estimation as a correction on the original input x, because the temporal stabilization methods used in this work are designed to work with images (frames x'_i) rather than direct estimations (RGB triplets y'_i). The correction function h includes a von Kries-like diagonal transform [13], as well as gamma delinearization. These adjustments help produce images that more closely resemble sRGB outputs, which are better suited for the general-purpose temporal stabilization methods employed.

In the second step, we stabilize the sequence of corrected frames x'_i using a video stabilization method m:

$$S'' = m\left(S'\right) = \{x''_1, x''_2, ..., x''_i, ...\}. \tag{3}$$

We refer the reader to Sect. 4.3 for the specific selection of methods m used in our experimental setup. Theoretically, this process yields a stabilized sequence of frames. However, practical challenges may arise:

1. Local artifacts could be introduced by enforcing temporal consistency, potentially affecting the visual quality of the sequence.
2. The output from this two-step process cannot be directly compared against the estimations from the initial single-frame method using standard color constancy metrics.

To address these issues, in the third and final step we extract a per-frame stabilized illuminant by consensus [5]: we calculate the average illuminant for each frame by taking the ratio of the corrected frame x''_i to the original frame x_i and then averaging this ratio:

$$y_i = \frac{1}{|x_i|} \sum_{p \in x_i} \left(\frac{x''_i(p)}{x_i(p)} \right) \quad \forall x_i \in S. \tag{4}$$

Each original frame x_i can then be re-corrected globally using the stabilized illuminant y_i with the application of an adjusted von Kries diagonal transform.

4 Experimental Setup

4.1 Selected Datasets

Gray Ball. The Gray Ball dataset [8] is one of the few datasets potentially suitable for video color constancy. It comprises 11,346 images organized into 15 sequences, with many shots captured at close intervals. The dataset was recorded using a Sony VX-2000 digital video camera. Although it is an older dataset and lacks raw images, we follow Buzzelli and Erba's approach by processing a linear version of the dataset for illuminant estimation and correction. Concerning the presence of video cuts, the original sequences were manually curated into 337

smaller sequences to ensure only smooth transitions in scene content. From these, 168 sequences deemed stable were selected for further analysis, based on the extraction of MIC and STD metrics on the ground truth illuminants (see Eqs. 5 and 7). According to Buzzelli and Erba, the most and least stable color constancy methods on the Gray Ball dataset are found to be Second-order Gray Edge (GE2) [18] and the Grayness Index (GI) [16], respectively, from a benchmark of 11 methods. For the purposes of this paper, we will focus on evaluating the performance of these two extremes.

Burst Color Constancy (BCC). The BCC dataset [17], also known as the Temporal Benchmark dataset, was introduced by Qian et al. for burst color constancy. It comprises 600 sequences of varying lengths, ranging from 3 to 17 frames. These are divided into 400 sequences for training and 200 sequences designated for the test set, the latter of which we utilize in our analysis. The images were captured using a Huawei Mate 20 Pro mobile phone and are stored in a proprietary 16-bit RAW format. Additionally, the dataset includes 8-bit PNG images for each sequence. A SpyderCube calibration target was placed in the scene immediately after the sequence acquisition to provide an out-of-sequence reference shot, representing the entire video sequence. According to Buzzelli and Erba, the most and least stable methods for color constancy on the BCC dataset are Sensor-Independent illumination Estimation (SIIE) [1] and White Point (WP) [18], respectively. For the purposes of this paper, we will focus on evaluating the performance of these two extremes.

4.2 Selected Color Constancy Methods

Based on the preliminary experiments reported in Sect. 4.1, we have focused our experimentation on the following algorithms for single-frame computational color constancy.

White Point (WP). The White Point method for color constancy is based on the assumption that the brightest color present in a scene, is white under a neutral illuminant. Although this method is straightforward and computationally efficient, its performance can vary significantly depending on the presence of truly neutral colors in the scene.

Second-order Gray Edge (GE2). Developed by Van de Weijer et al. in 2007 [18], the Second-order Gray Edge method extends the gray world hypothesis by considering higher-order derivatives of color edges in the image. This approach posits that changes in edge colors in a scene are predominantly due to variations in the scene illumination. By analyzing the second-order derivatives, this method aims to more accurately estimate the global illuminant than simpler methods.

Grayness Index (GI). Introduced by Qian et al. in 2019 [16], the Grayness Index method focuses on identifying pixels within an image that appear to be achromatic or'gray' under a wide range of lighting conditions. These gray pixels are assumed to reflect the true color of the illuminant, as they exhibit minimal hue but varying intensities. By aggregating the colors of these gray pixels, the method estimates the scene's illuminant.

Sensor-independent Illumination Estimation (SIIE). Proposed by Afifi and Brown in 2019 [1], this method aims to provide a robust solution to the color constancy problem that is independent of camera sensors. The approach uses a deep neural network trained on a dataset comprising multiple cameras with diverse spectral sensitivities. By doing so, it learns to generalize illumination estimation across different imaging conditions without being biased towards any specific camera sensor characteristics.

4.3 Selected Temporal Consistency Methods

For the purpose of this paper, we will focus on two well-known algorithm for task-agnostic (blind) temporal consistency, whose publicly-available code allows us to experiment their applicability in the domain of color constancy.

Lai et al. 2018 Lai et al. [14] introduced an end-to-end solution utilizing a deep recurrent network to enforce temporal consistency across video sequences. Their method operates by processing both the original unprocessed video and the individually processed frames to output a video with enhanced temporal consistency. This approach is designed to be algorithm-agnostic, meaning it does not depend on the specifics of the image processing techniques applied to the original video. The network is trained to minimize both short-term and long-term temporal losses as well as perceptual loss, balancing temporal stability with perceptual similarity to the processed frames. Notably, their method operates without the need for optical flow computations, allowing for real-time performance even on high-resolution videos. The approach was tested on various tasks such as artistic style transfer, enhancement, colorization, image-to-image translation, and intrinsic image decomposition.

TDMS-Net. Zhou et al. [20] framed temporal consistency as a temporal denoising problem aimed at mitigating flickering in previously unstable pre-processed frames. To tackle this, they introduce a novel model called the Temporal Denoising Mask Synthesis Network (TDMS-Net). This network is designed to synthesize temporally consistent frames by jointly predicting a motion mask, soft optical flow, and a refining mask. The approach taken by Zhou et al. learns temporal consistency directly from the original video and its temporal features, which are then used to refine the output frames.

5 Results

5.1 Assessing Temporal Stabilization

Buzzelli and Erba [4] refer to a temporally stable sequence as a sequence that:

1. Does not contain video cuts.
2. Does not involve abrupt illuminant changes.
3. Does not span a wide set of illuminants (even if gradually changing).

Point 1 is addressed at dataset selection level: it involves manual processing for the Gray Ball dataset, as described in Sect. 4.1, while it is valid by assumption for the BCC dataset.

The identification of abrupt illuminant changes is achieved by quantifying the"Maximum Illuminant Change" (MIC) in the estimated illuminant $y = (y^{(R)}, y^{(G)}, y^{(B)})$ between consecutive frames. More precisely, for each pair of consecutive frames in a sequence S, we measure the angular error between their ground truth illuminants and then we select the maximum of such errors:

$$\mathrm{MIC}\,(S) = \max\left(\mathrm{err}\,(y_i, y_{i+1})\right), \quad i = 1...|S| - 1. \tag{5}$$

As angular error, we refer to the recovery error [12], which computes the angle between the RGB vector representing the ground truth illuminant U with the RGB vector representing the estimated illuminant V:

$$\mathrm{err}\,(U, V) = \arccos\left(\frac{U \cdot V}{||U||||V||}\right), \tag{6}$$

where "$\cdot$" is the dot product, and "$||...||$" the Euclidean norm. Alternatively, the reproduction error [10] may be also used.

To identify sequences spanning a large range of illuminants, the chromaticity Standard Distance (STD) is used as a metric for scatteredness. Specifically, we first convert the ground truth illuminants into Angle-Retaining Chromaticity (ARC) [3]: a 2-dimensional representation where Euclidean distances correspond to angular distances in the original RGB space. Then, we compute a 2-dimensional generalization of the concept of standard deviation, defined as:

$$\mathrm{STD}(S) = \sqrt{\sum_{i=1}^{|S|} \frac{(\alpha_{xi} - \overline{\alpha_x})^2}{|S|} + \sum_{i=1}^{|S|} \frac{(\alpha_{yi} - \overline{\alpha_y})^2}{|S|}}, \tag{7}$$

where $(\alpha_{xi}, \alpha_{yi})$ are the ARC coordinates of the i-th illuminant of sequence S, and $(\overline{\alpha_x}, \overline{\alpha_y})$ is the average of each ARC coordinate for the sequence.

5.2 Quantitative Results

In Table 1 we report aggregate results for the enforcement and assessment of temporal consistency in the domain of temporal color constancy. Specifically, we focus on three groups of images: the Gray Ball dataset (split into stable and unstable sequences following the first criterion from Sect. 5.1), and the BCC dataset. For each, we analyze the effect on the most stable algorithms for computational color constancy as highlighted in Sect. 4.1 (GE2 [18] and GI [16] respectively for Gray Ball, and SIIE [1] and WP [18] respectively for BCC). We first extract statistics on these algorithms, labelled as "Original", measuring both color constancy accuracy via the angular error ('err' from Eq. 6), and baseline temporal consistency via STD and MIC from Eq. 7 and Eq. 5 respectively. Focusing only on temporal consistency, in fact, would provide an incomplete

Table 1. Average statistics, expressed in degrees, for the impact of temporal consistency in temporal color constancy. The lower, the better. Best results in bold.

		Most stable algorithm			Least stable algorithm		
		GE2 [18]			GI [16]		
		err	SID	MIC	err	SID	MIC
Gray Ball(stablesequences)	Original	**5.098**	**1.139**	**1.896**	7.542	4.178	10.117
	Lai et al. [14]	5.464	1.737	3.040	7.266	3.779	7.364
	TDMSNet [20]	5.194	1.373	2.241	**7.235**	**3.589**	**7.347**
	(Δ Lai et al.)	(+0.366)	(+0.598)	(+1.144)	(-0.276)	(-0.399)	(-2.752)
	(Δ TDMSNet)	(+0.096)	(+0.234)	(+0.345)	(-0.307)	(-0.590)	(-2.770)
		GE2 [18]			GI [16]		
		err	SID	MIC	err	SID	MIC
Gray Ball(unstablesequences)	Original	**6.628**	**2.258**	**3.550**	7.208	4.984	12.969
	Lai et al. [14]	7.201	2.668	5.009	6.990	4.795	10.506
	TDMSNet [20]	6.646	2.539	4.141	**7.205**	**4.616**	**10.194**
	(Δ Lai et al.)	(+0.573)	(+0.410)	(+1.459)	(-0.219)	(-0.189)	(-2.463)
	(Δ TDMSNet)	(+0.018)	(+0.281)	(+0.591)	(-0.003)	(-0.368)	(-2.775)
		SIIE [1]			WP [18]		
		err	SID	MIC	err	SID	MIC
BCC	Original	4.494	2.561	4.971	6.986	5.128	11.253
	Lai et al.	4.483	2.594	4.817	6.432	4.227	7.961
	TDMSNet	**4.389**	**2.433**	**4.479**	**6.098**	**3.767**	**7.081**
	(Δ Lai et al.)	(-0.011)	(+0.033)	(-0.154)	(-0.554)	(-0.901)	(-3.292)
	(Δ TDMSNet)	(-0.105)	(-0.127)	(-0.492)	(-0.888)	(-1.361)	(-4.172)

view of the overall performance, since an hypothetical "Do-nothing" algorithm would produce results that are perfect in terms of temporal consistency, but useless in terms of actual color rendition. From this baseline, we then evaluate the effect of enforcing temporal consistency through the method by Lai et al., and TDMSNet, following the methodology defined in Sect. 3.

Considering that, for all involved measures, a lower result is better, we can observe how temporal consistency is always improved for color constancy algorithms that were originally found to be extremely unstable. On the other hand, already-stable algorithms appear to be negatively affected (on average) by enforcing temporal consistency in the Gray Ball dataset, and produce mixed results on the BCC dataset. Finally, it may be observed that, despite not actively working to improve the angular error, temporal consistency methods Lai et al. and TDMSNet also have a beneficial impact on angular error itself as a byproduct of temporal stabilization: this is due to the inherent removal of outliers in the sequence of per-frame estimated illuminants.

5.3 Qualitative Results

In addition to the quantitative analysis, we report in the following some visualizations for a qualitative assessment of the enforcement of temporal consistency.

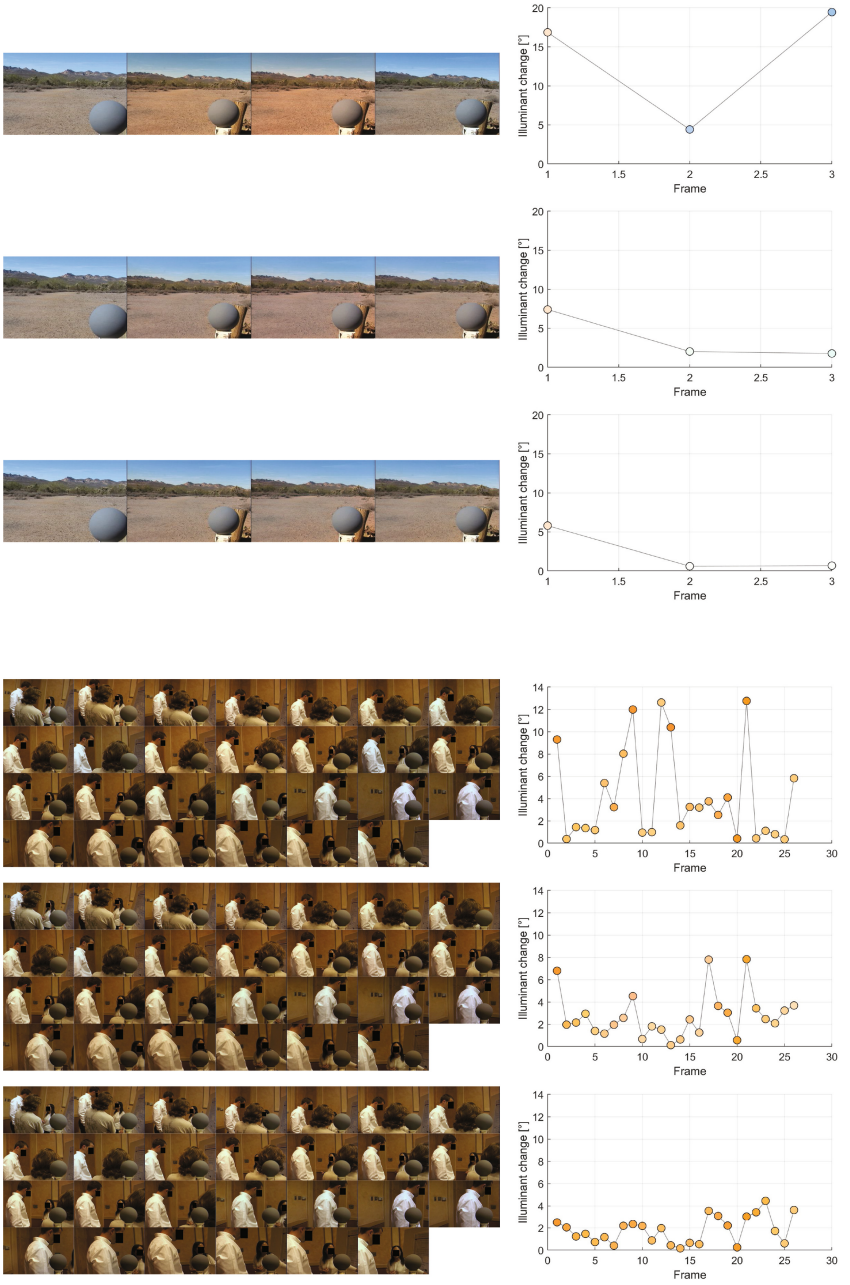

Fig. 3. Temporal stabilization on sample sequences. For each, we show the original sequence (top), the output as stabilized with Lai et al. (middle), and the output as stabilized with TDMSNet (bottom). We show the sequences on the left, and the plots of illuminant change (Eq. 5) on the right.

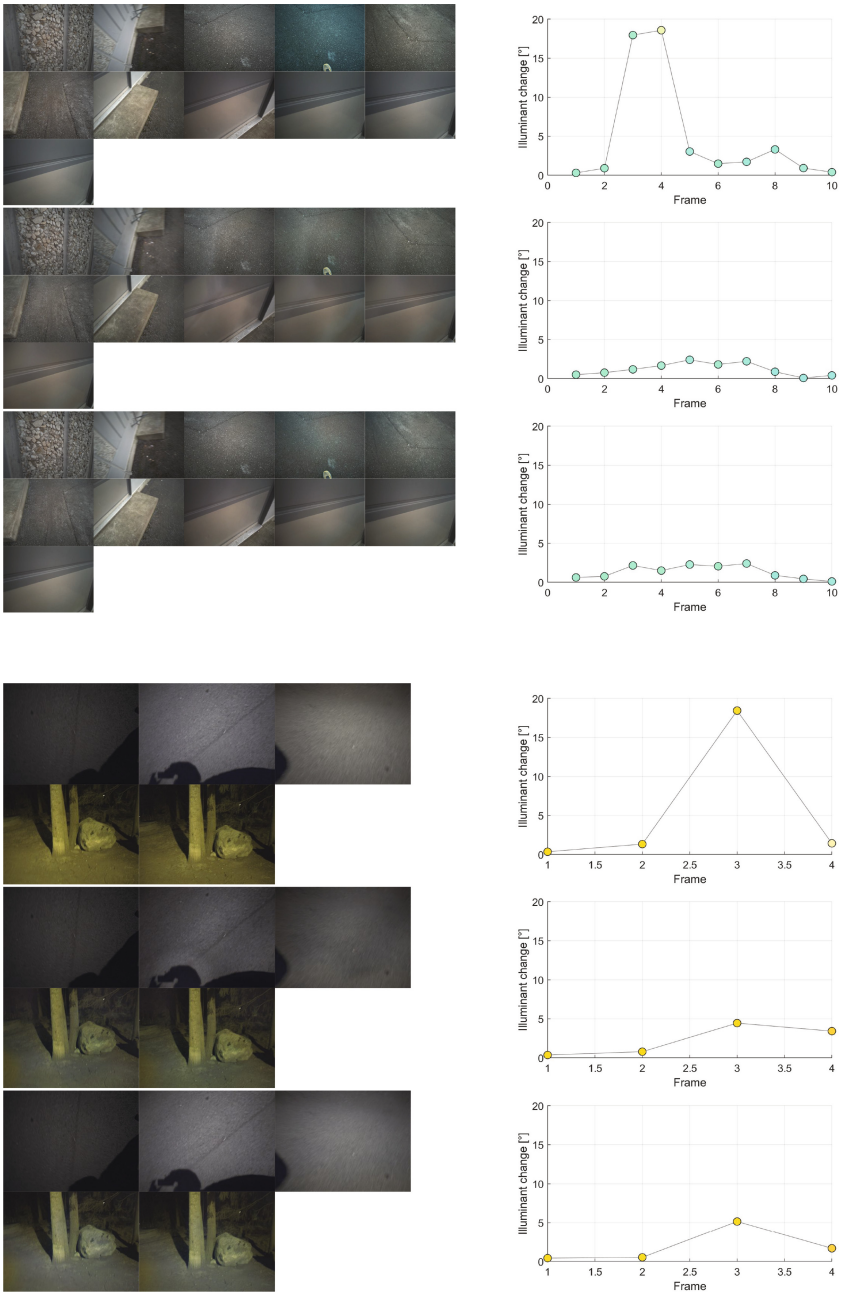

Fig. 4. Temporal stabilization on sample sequences. For each, we show the original sequence (top), the output as stabilized with Lai et al. (middle), and the output as stabilized with TDMSNet (bottom). We show the sequences on the left, and the plots of illuminant change (Eq. 5) on the right.

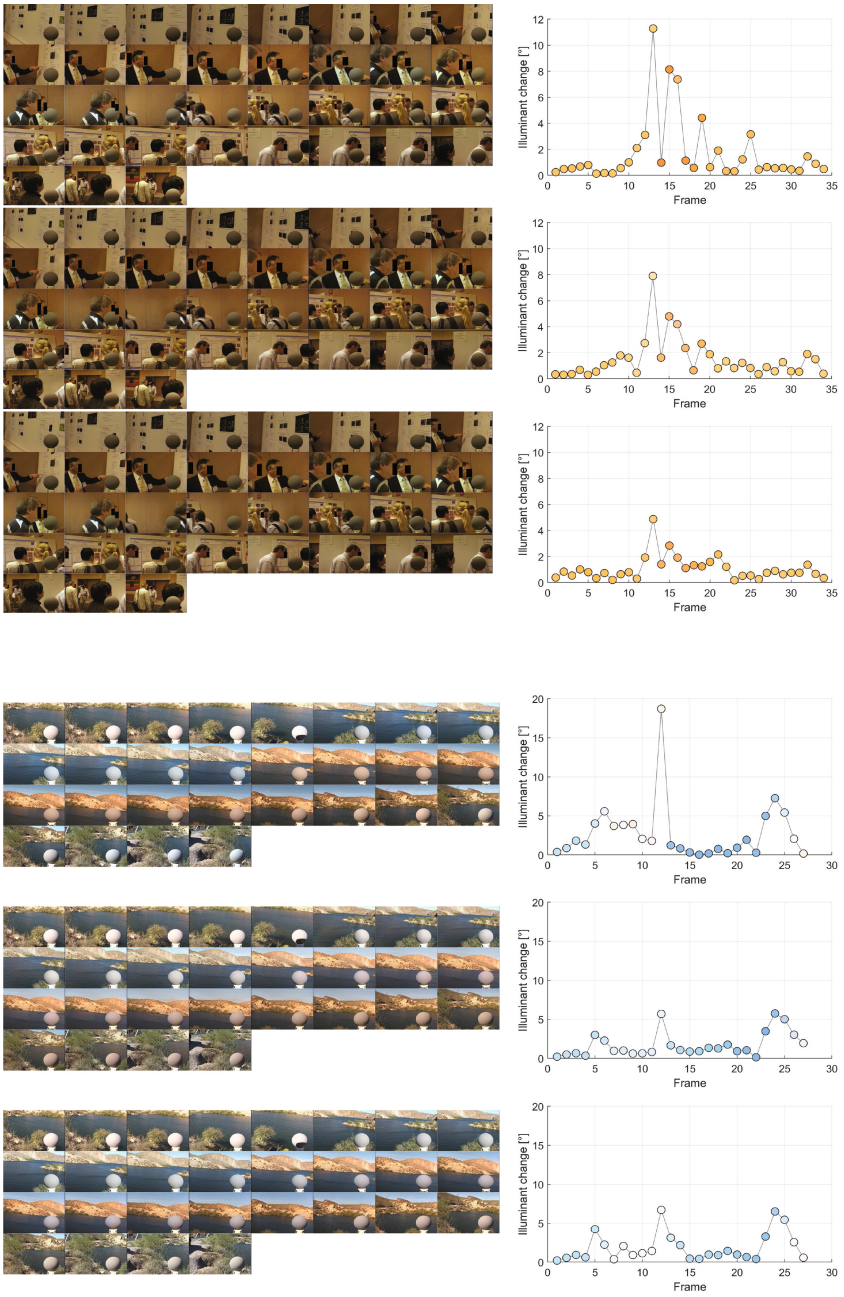

Fig. 5. Temporal stabilization on sample sequences. For each, we show the original sequence (top), the output as stabilized with Lai et al. (middle), and the output as stabilized with TDMSNet (bottom). We show the sequences on the left, and the plots of illuminant change (Eq. 5) on the right.

Figures 3, 4 and 5 present a selection of sample sequences on the left, and the corresponding plots of illuminant change between consecutive frames on the right (whose maximum corresponds to MIC). For each, we show the original sequence (top), the output as stabilized with Lai et al. (middle), and the output as stabilized with TDMSNet (bottom). The second sequence in Fig. 3 showcases a clear result where temporal stability is improved with respect to the original sequence, since the tint on the main subject's shirt is more stable. Nonetheless, some variability is still observable, as also documented by the corresponding illuminant change plot.

6 Conclusions

In this study, we have explored the challenges of achieving temporal color constancy in video sequences, a critical area in the advancement of automatic white balance and color correction. Our proposed method, which combines per-frame color correction with advanced temporal stabilization techniques, addresses the inherent variability and instability in video illuminant estimation. Through experiments conducted using the Gray Ball and BCC datasets, our approach demonstrated improvements in maintaining color consistency across video frames compared to traditional single-frame methods, localized to sequences that were originally very unstable. The use of consensus-driven illuminant correction ensured that no spatial artifacts are introduced by the temporal stabilization step.

Future work will focus on refining these techniques and exploring their applications in more diverse scenarios, including real-time video processing and integration into consumer video devices.

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